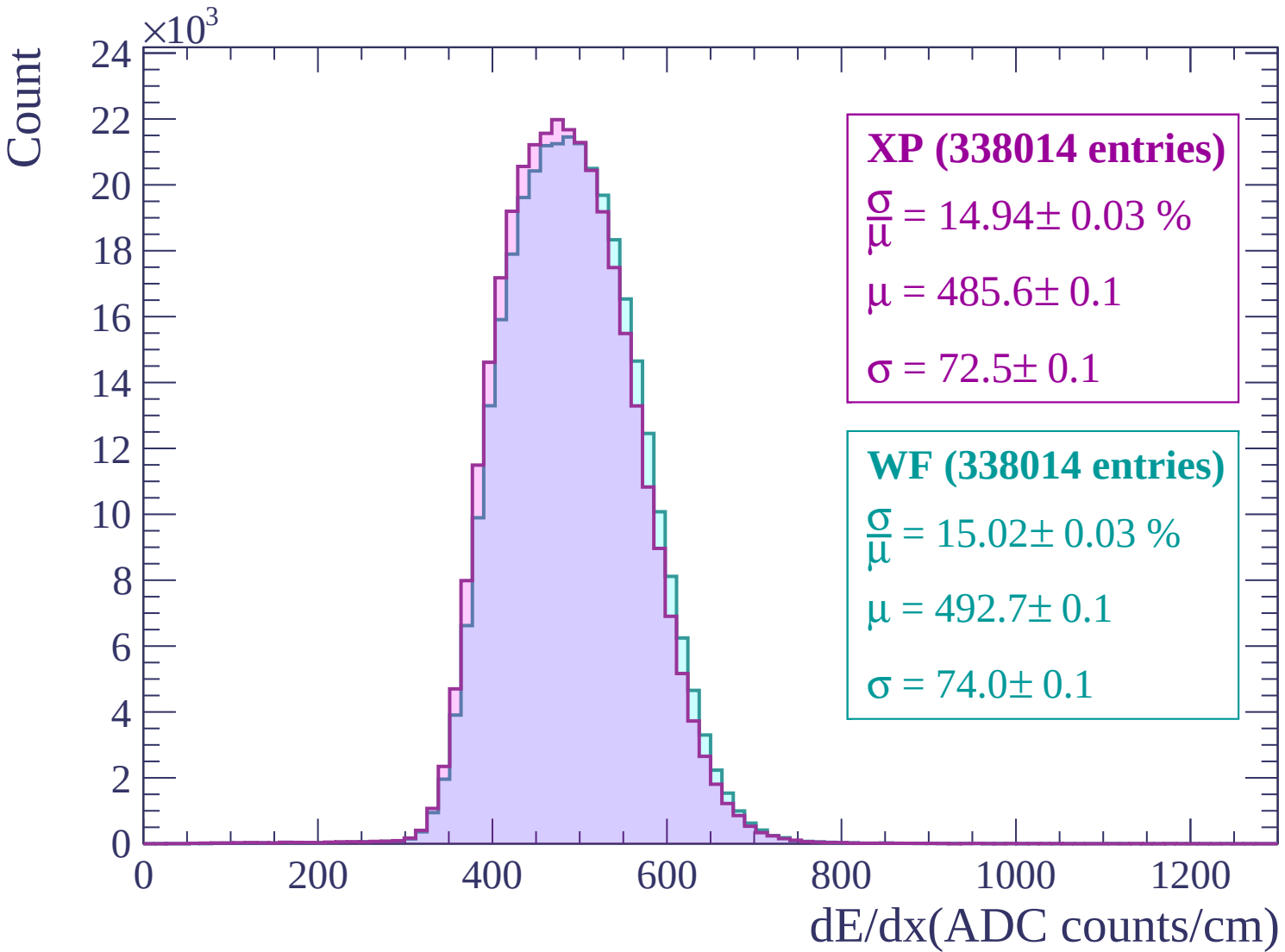
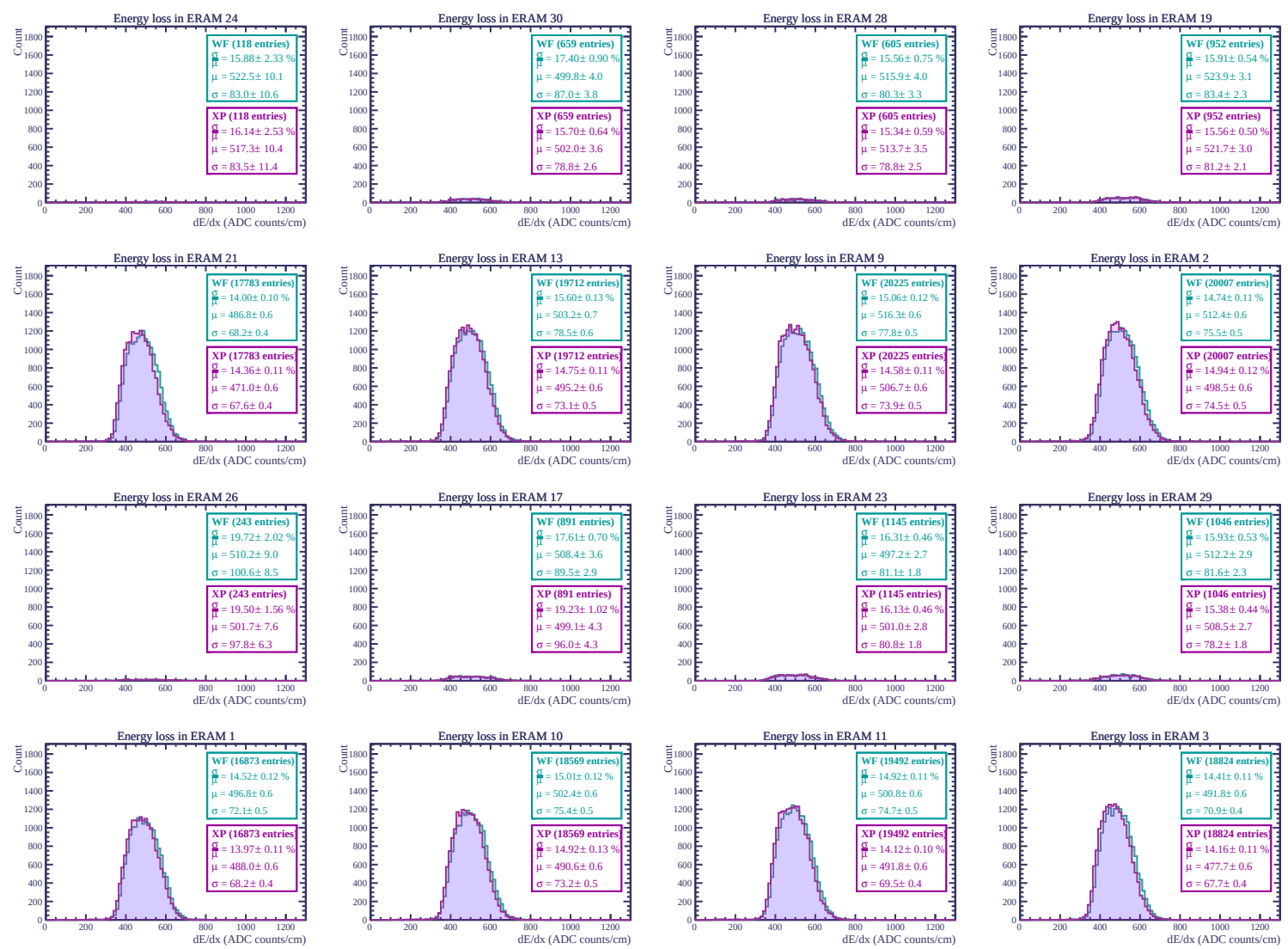
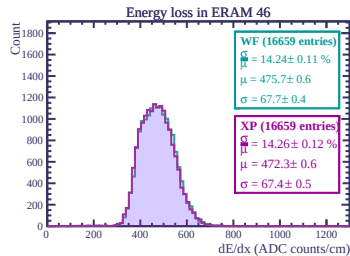
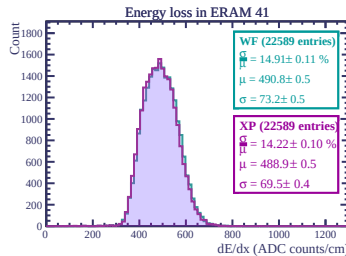
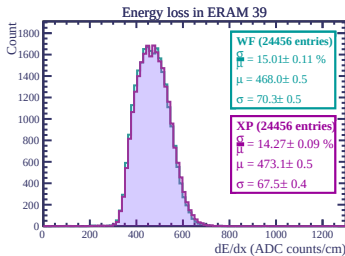
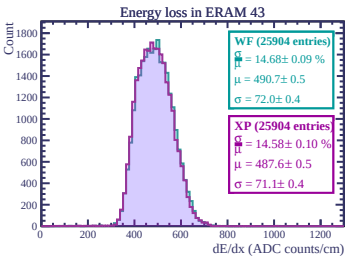
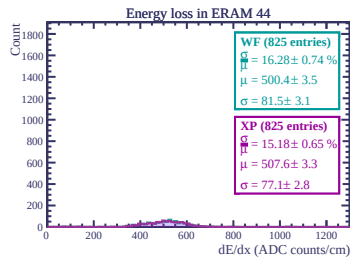
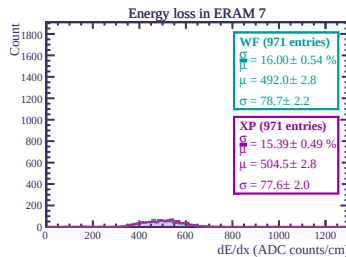
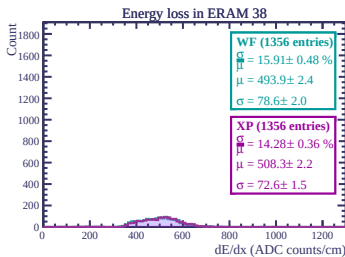
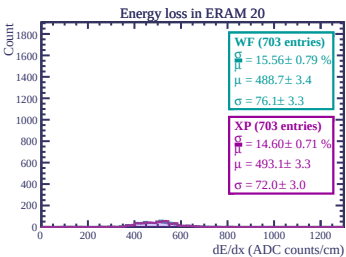
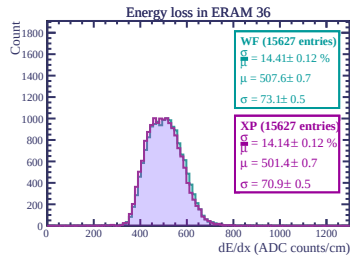
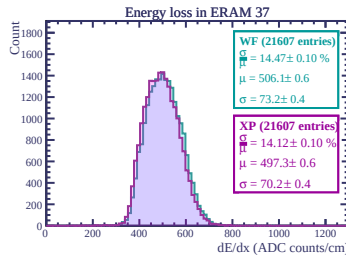
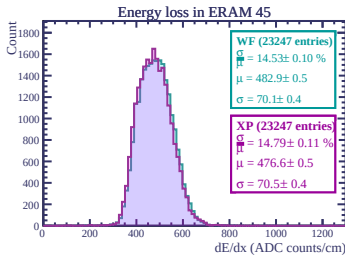
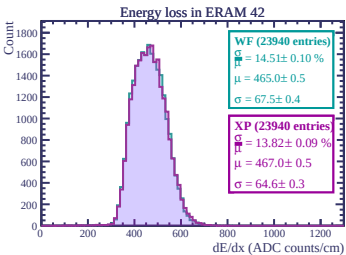
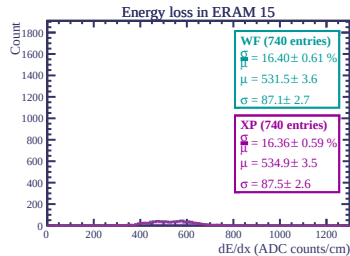
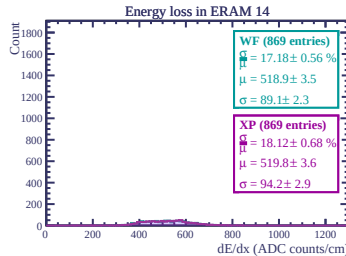
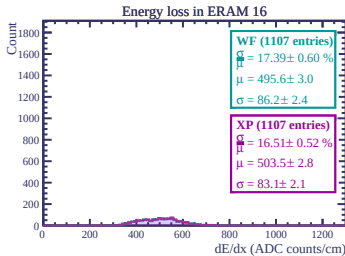
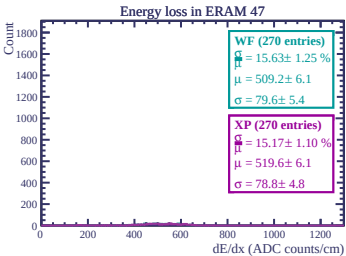
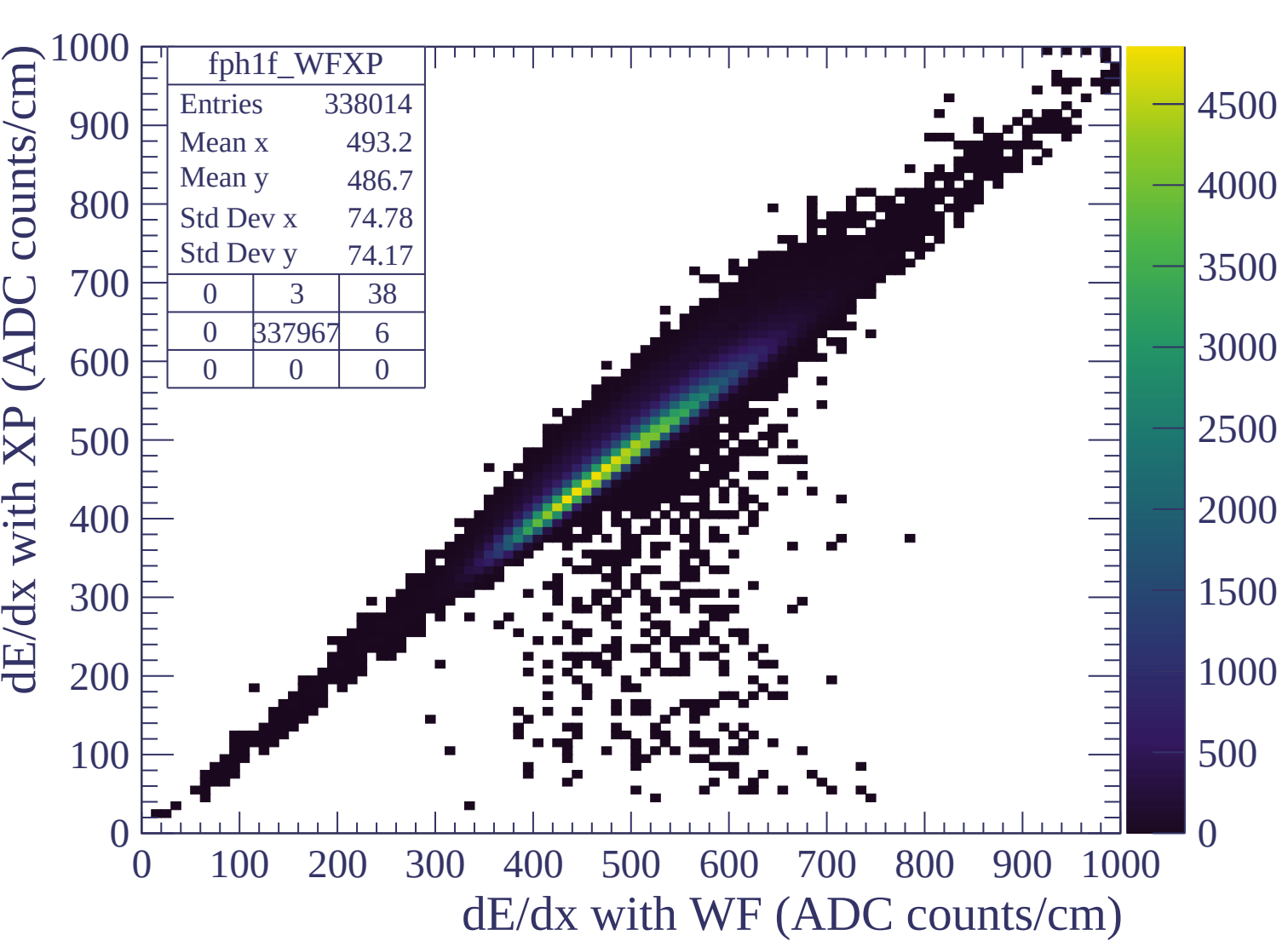
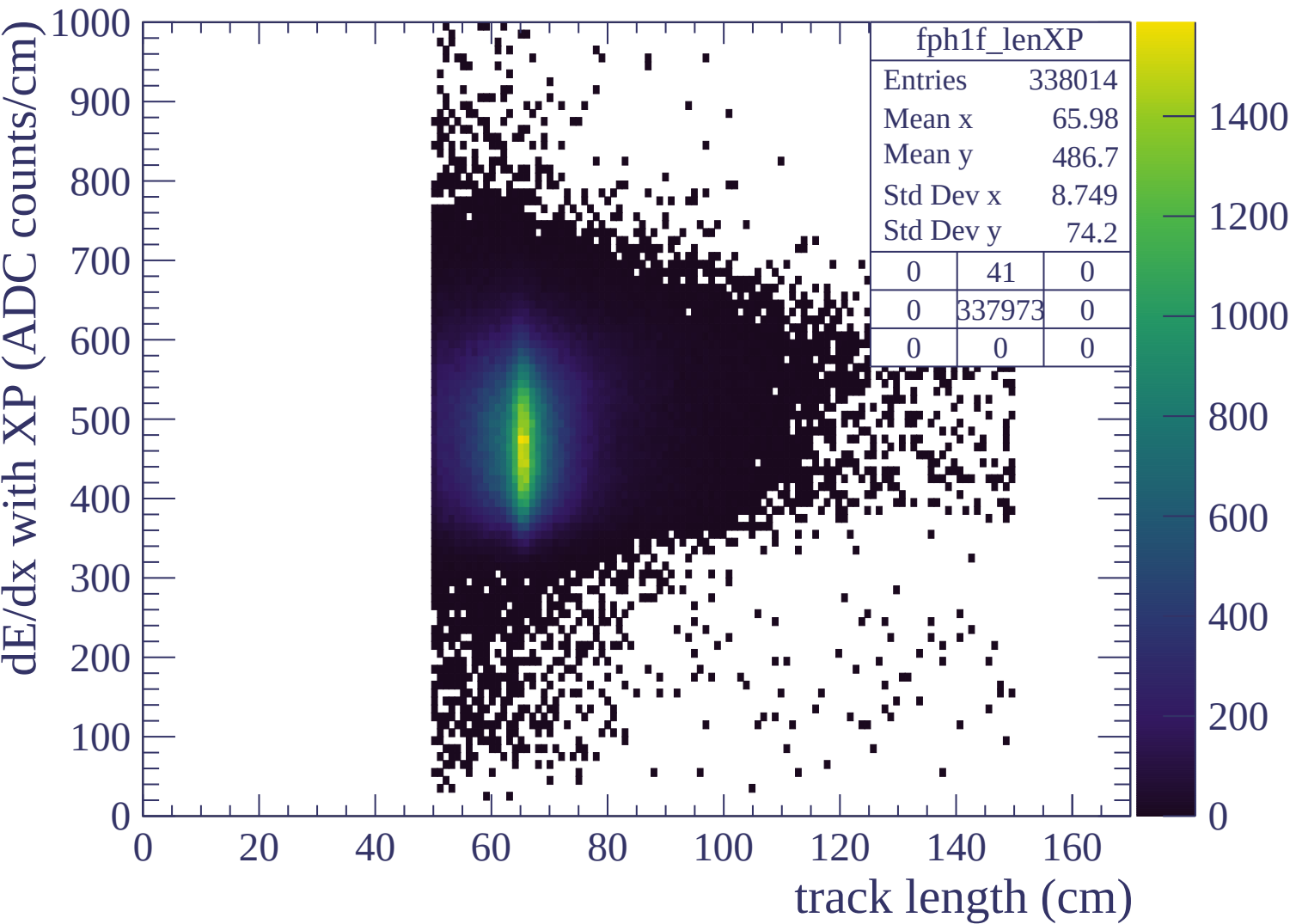


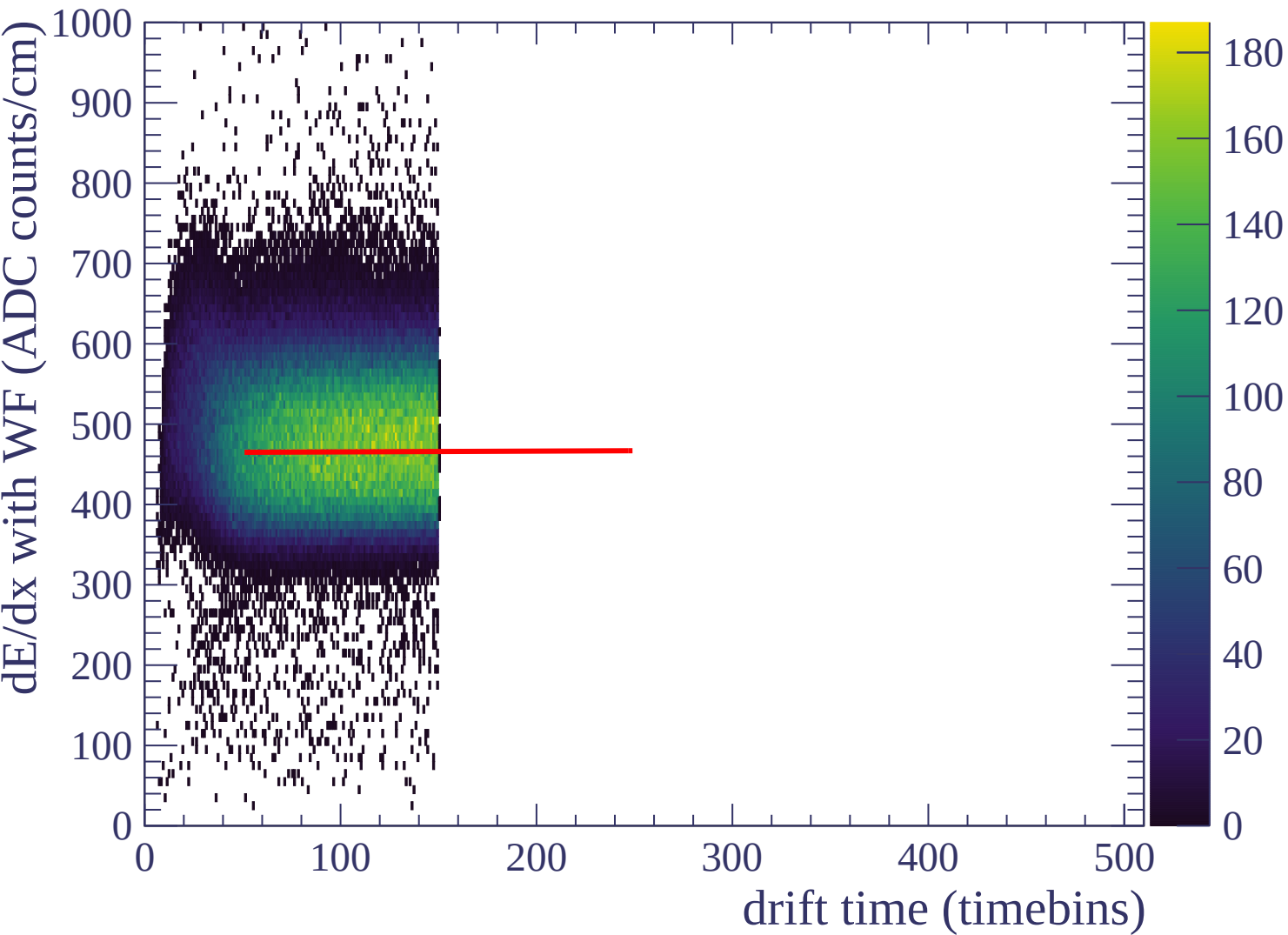


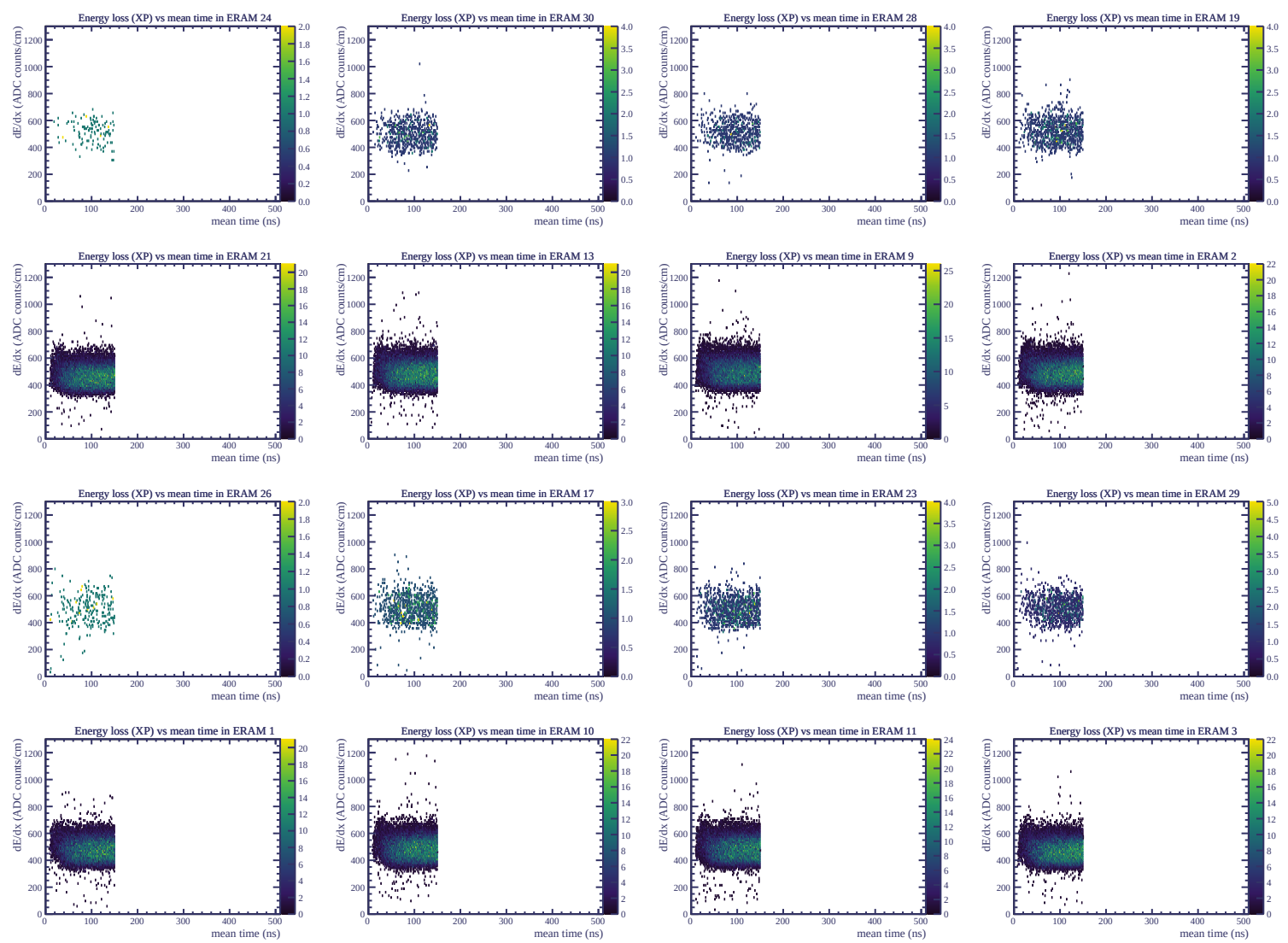


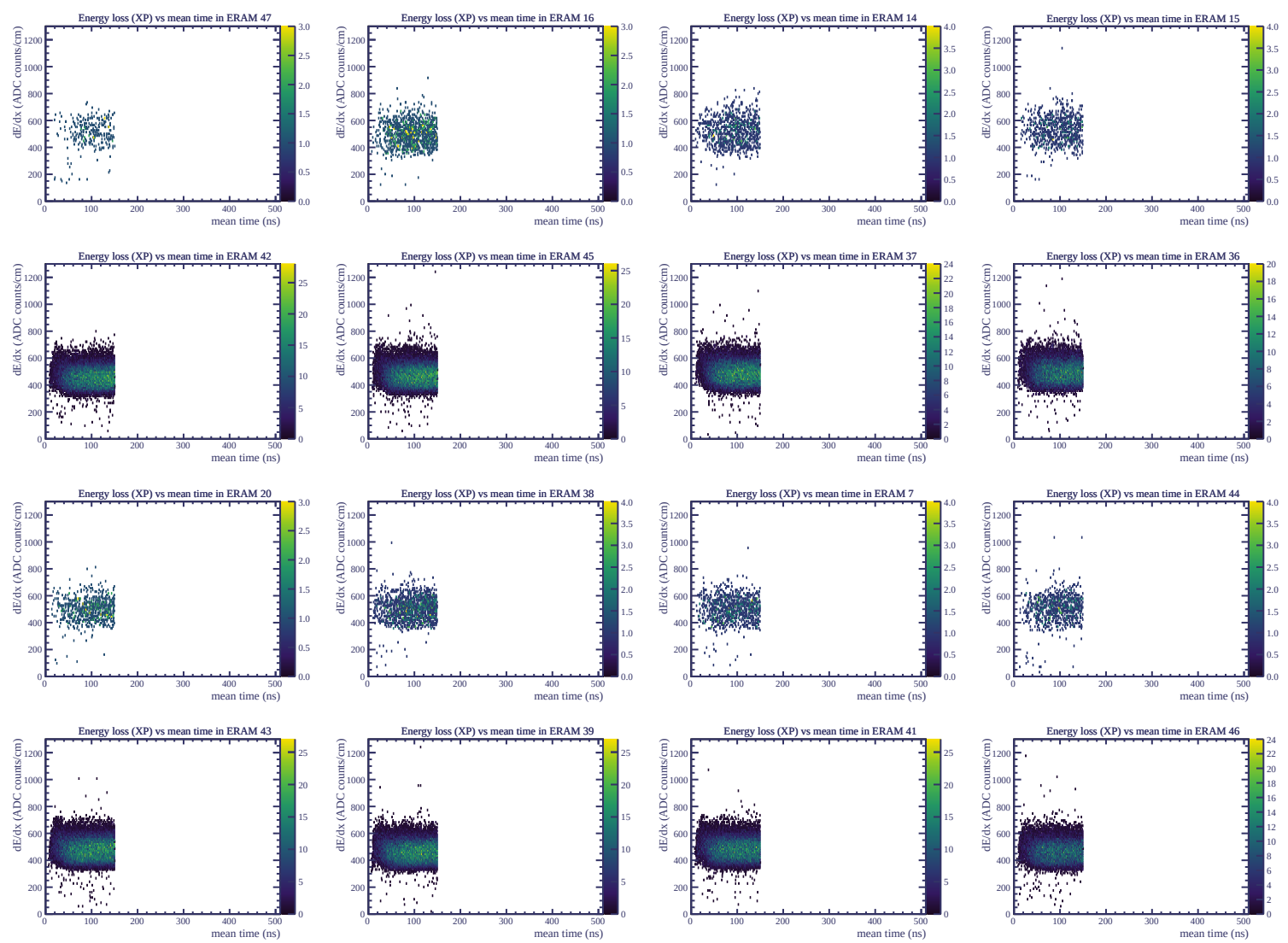




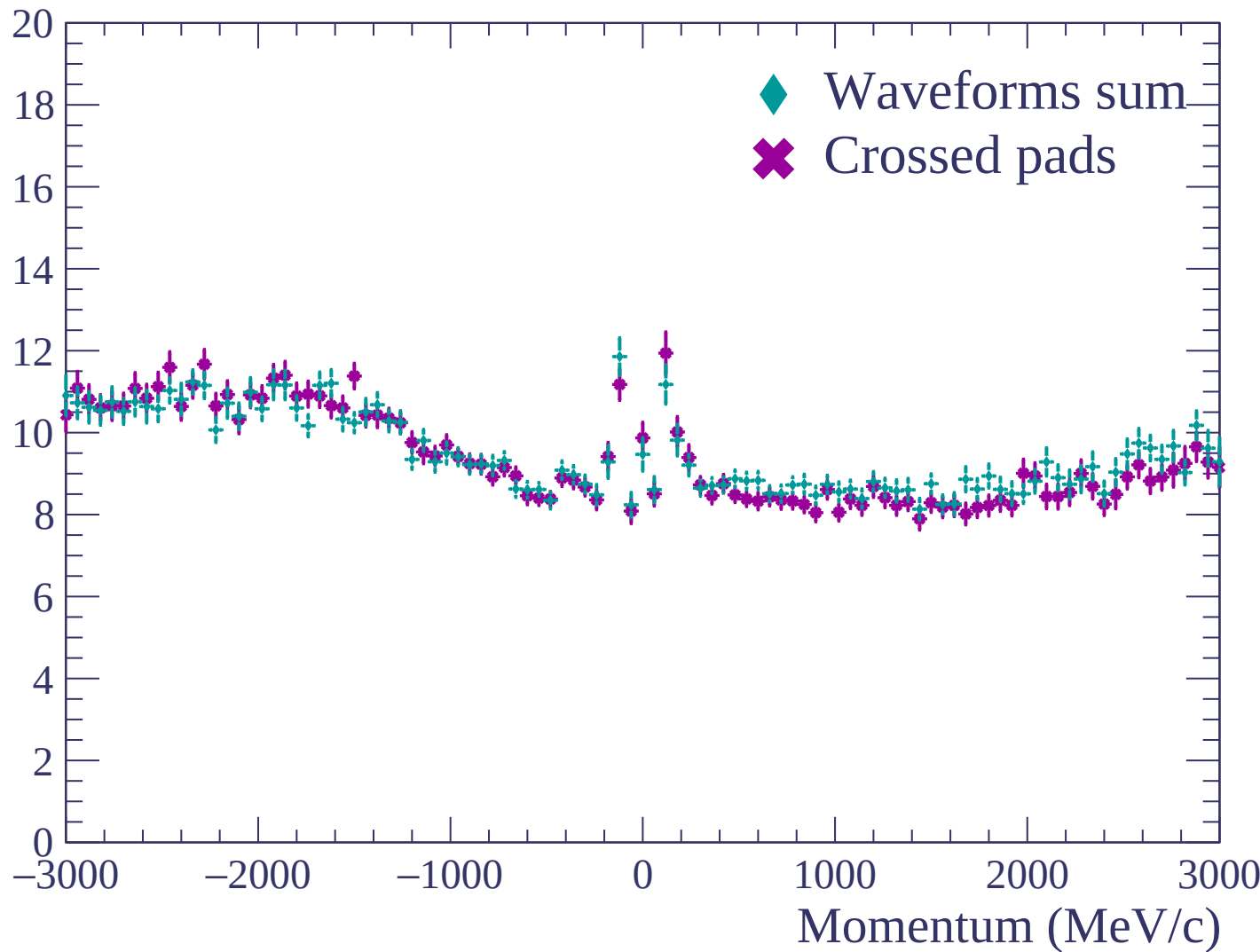


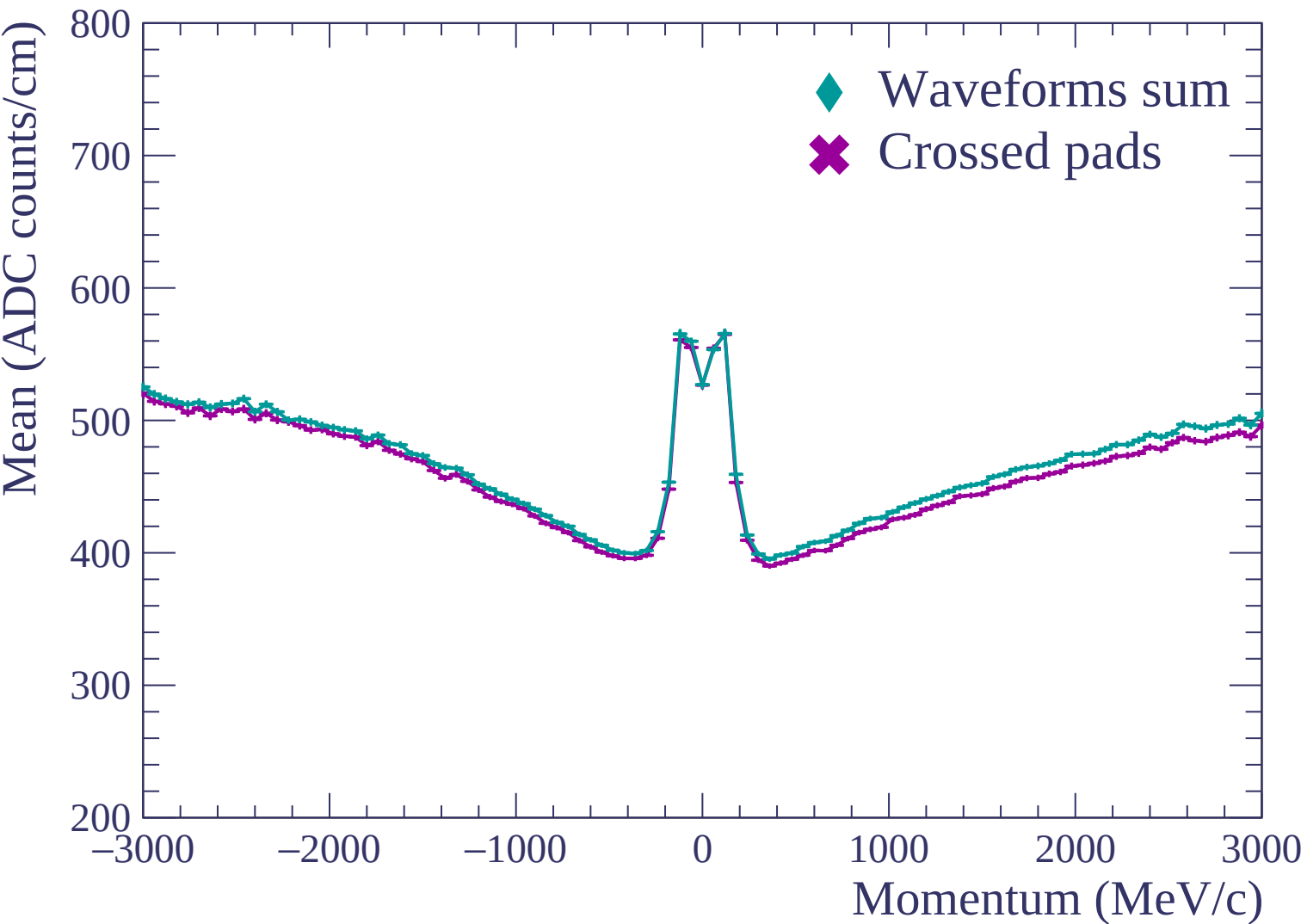


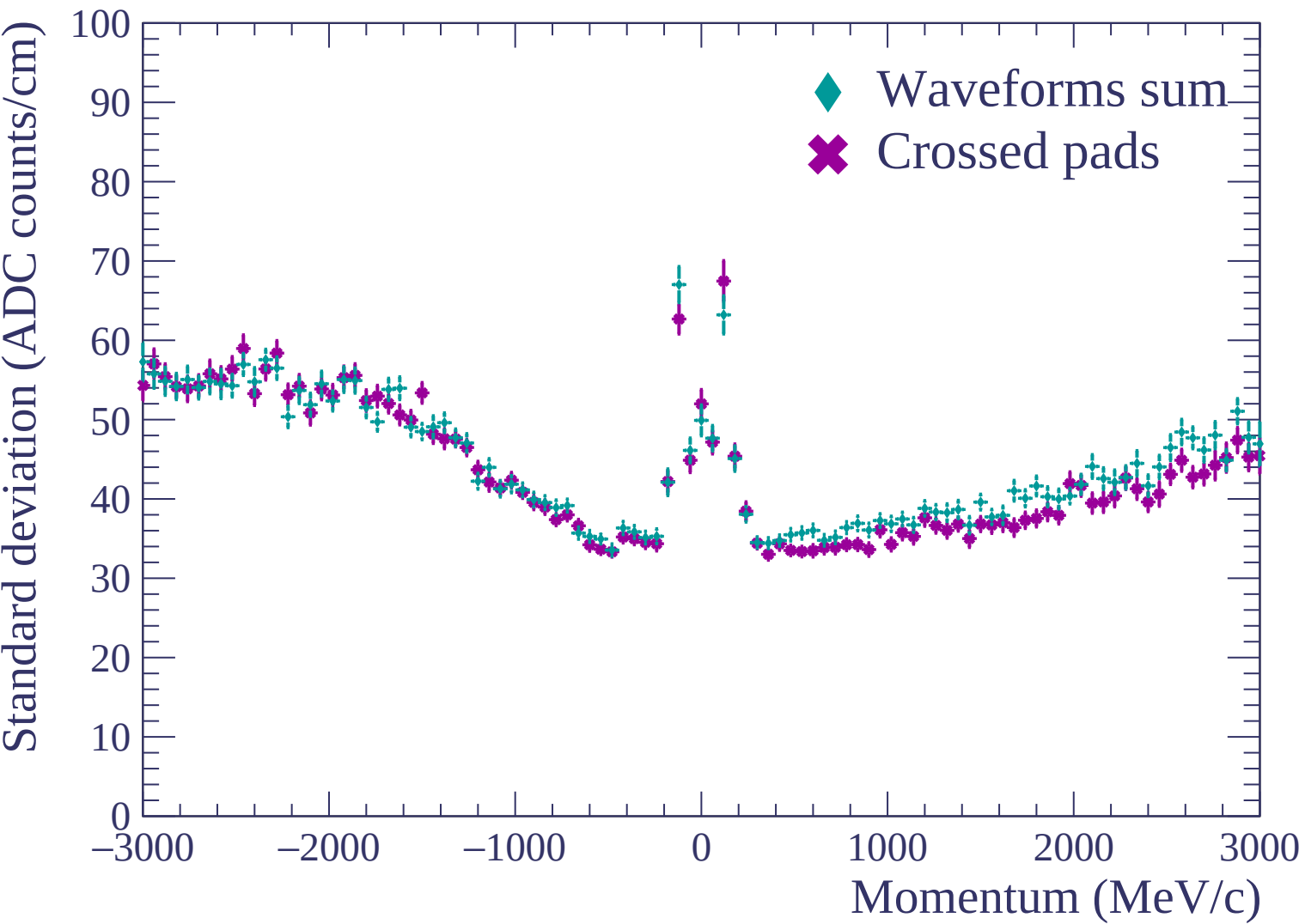


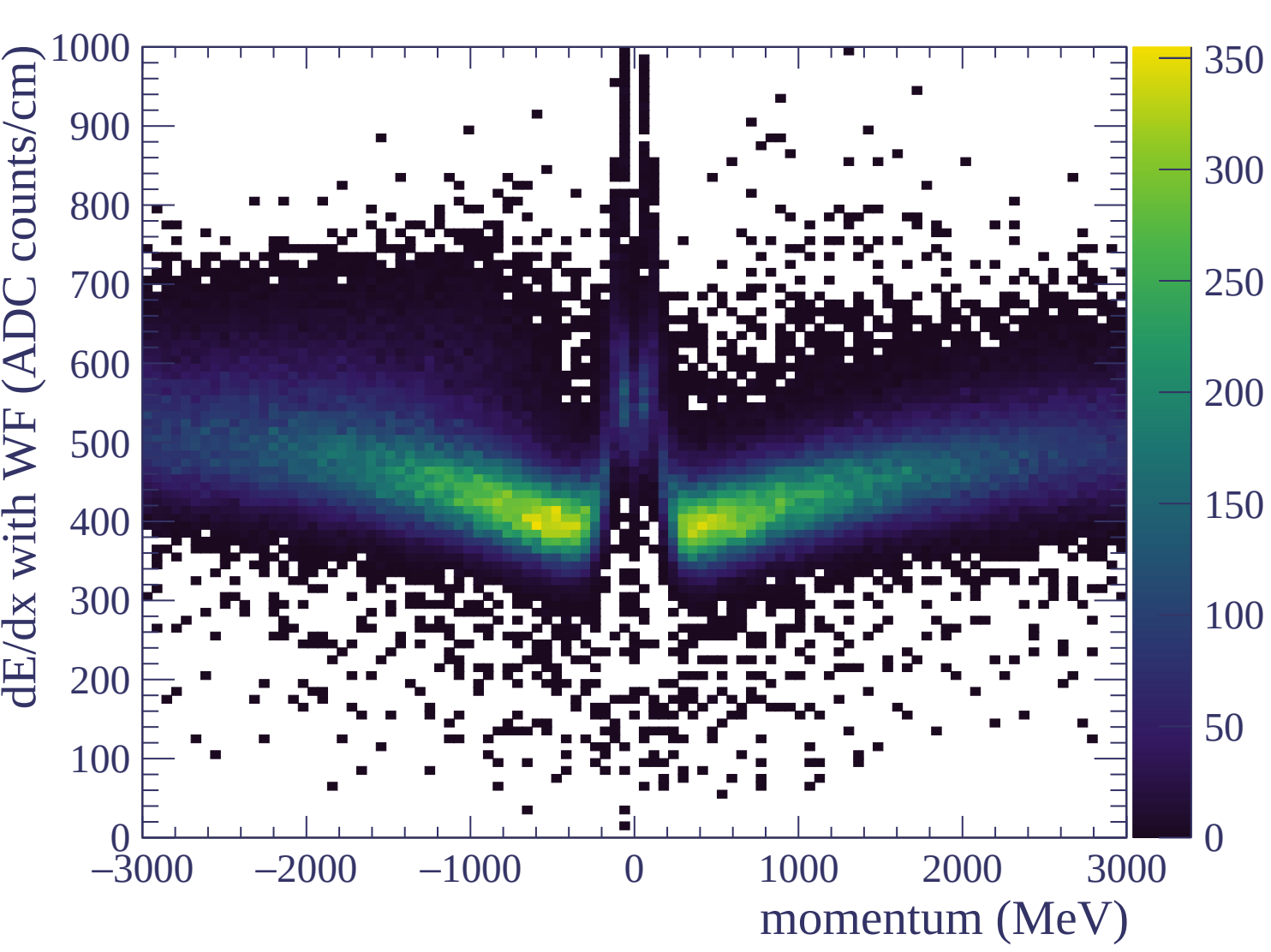


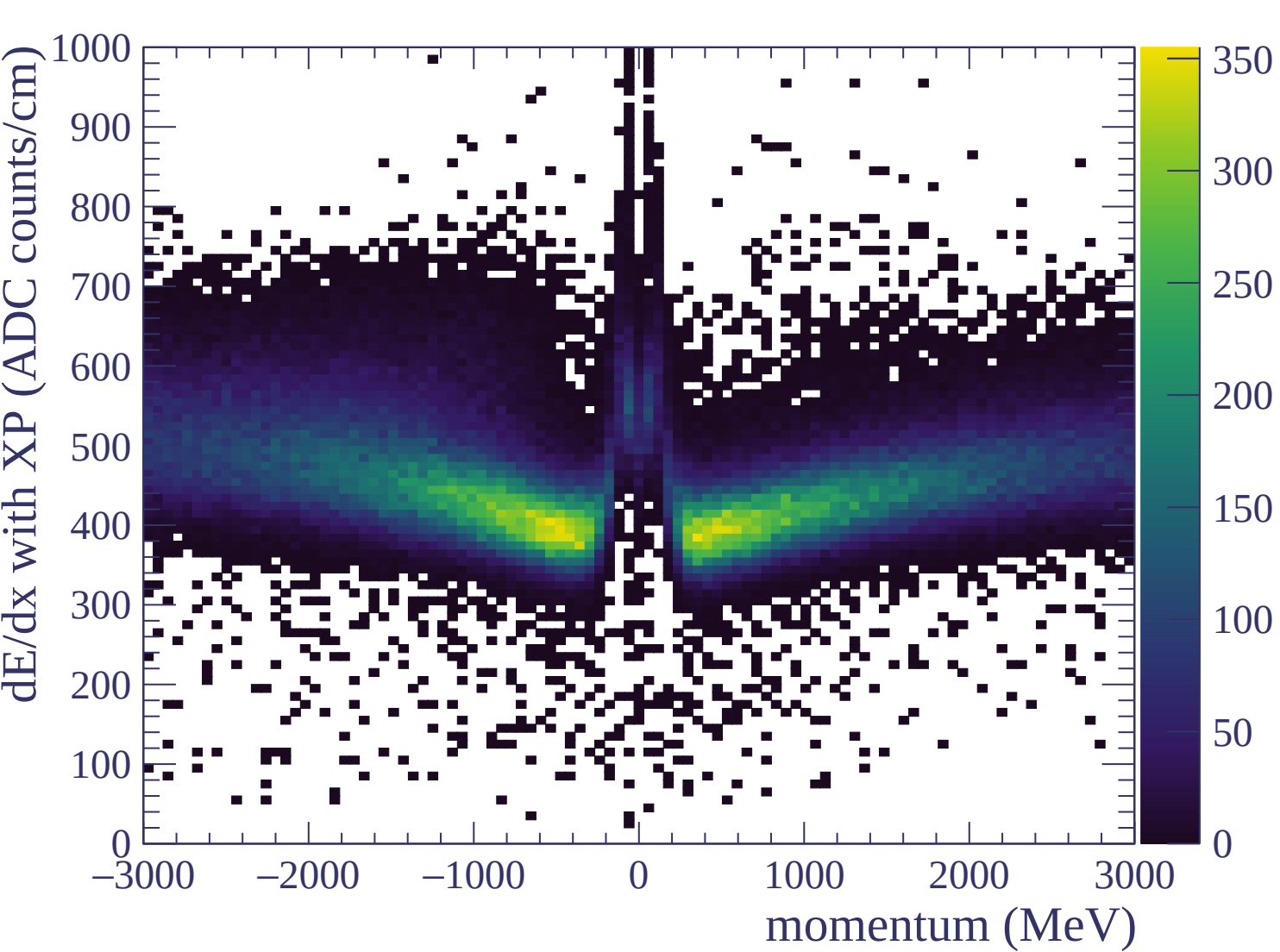
Resolution (%)

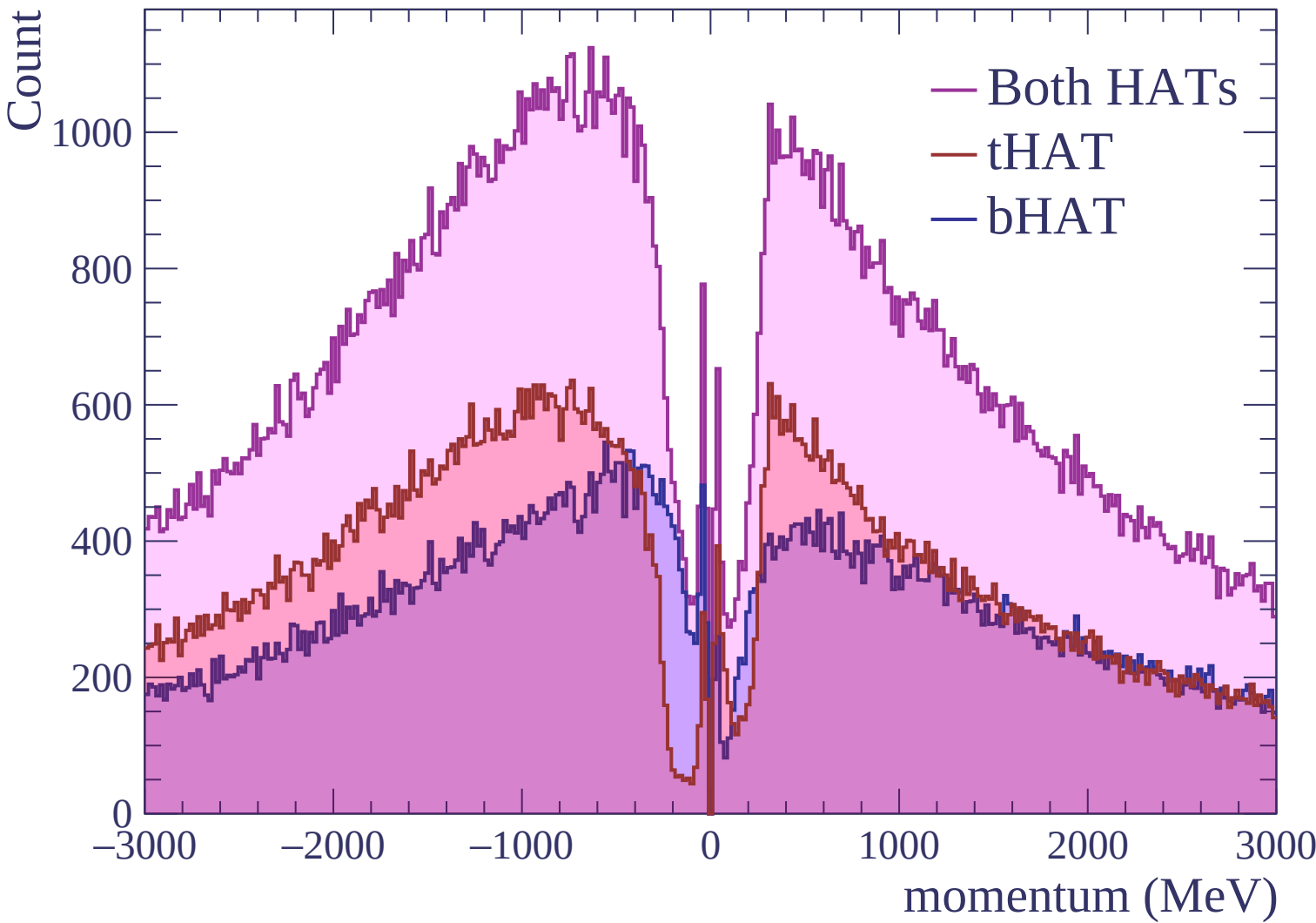


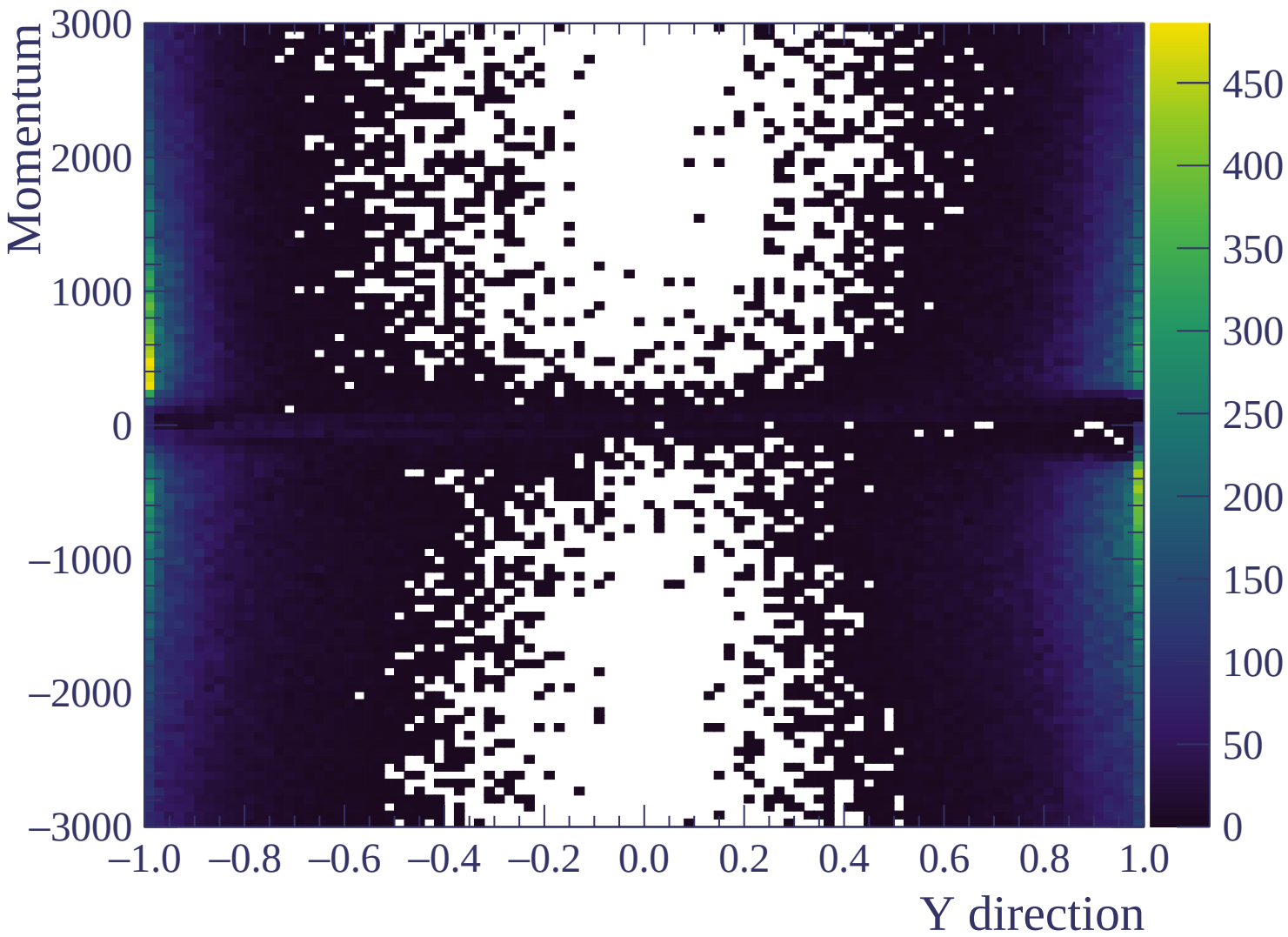


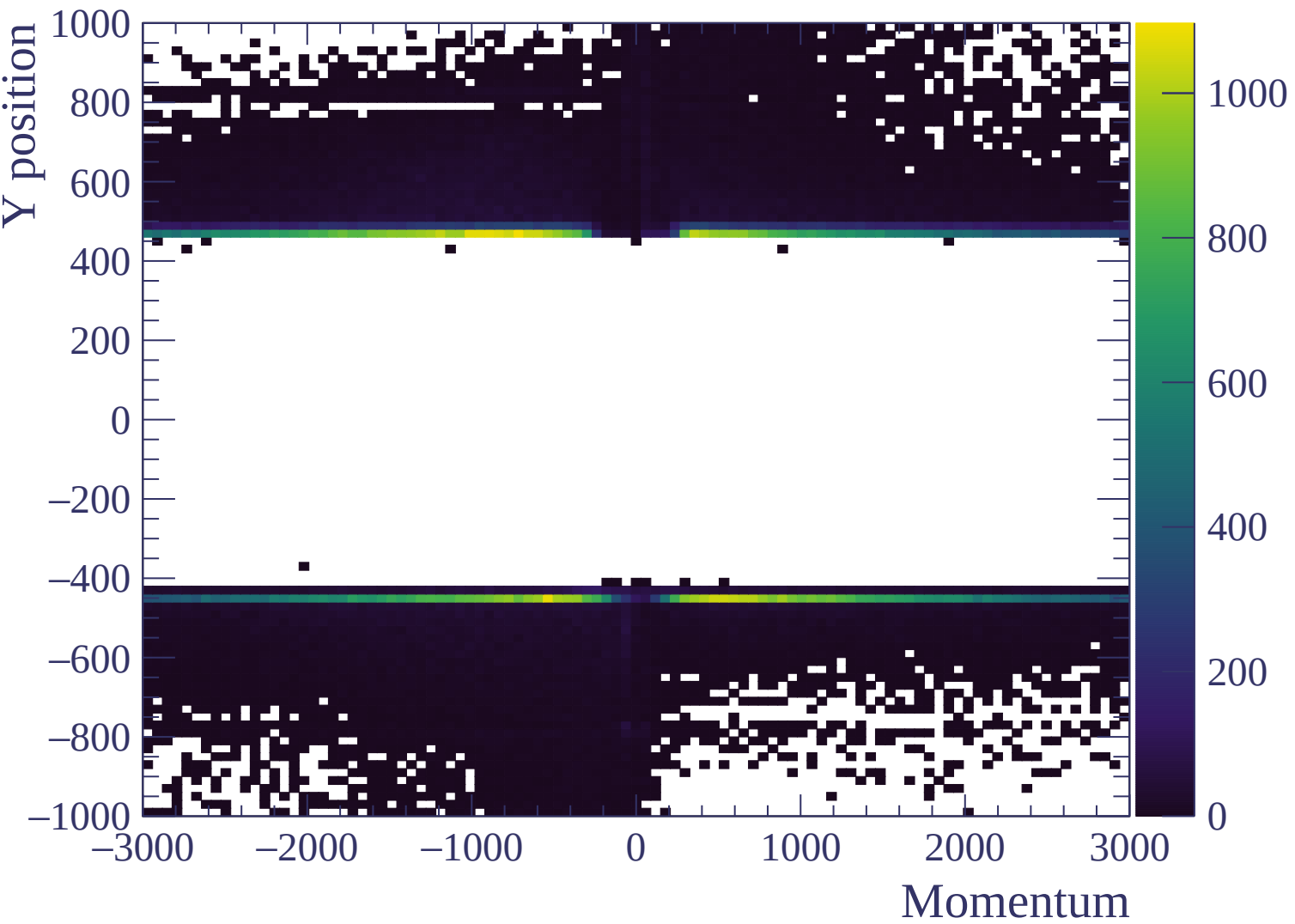


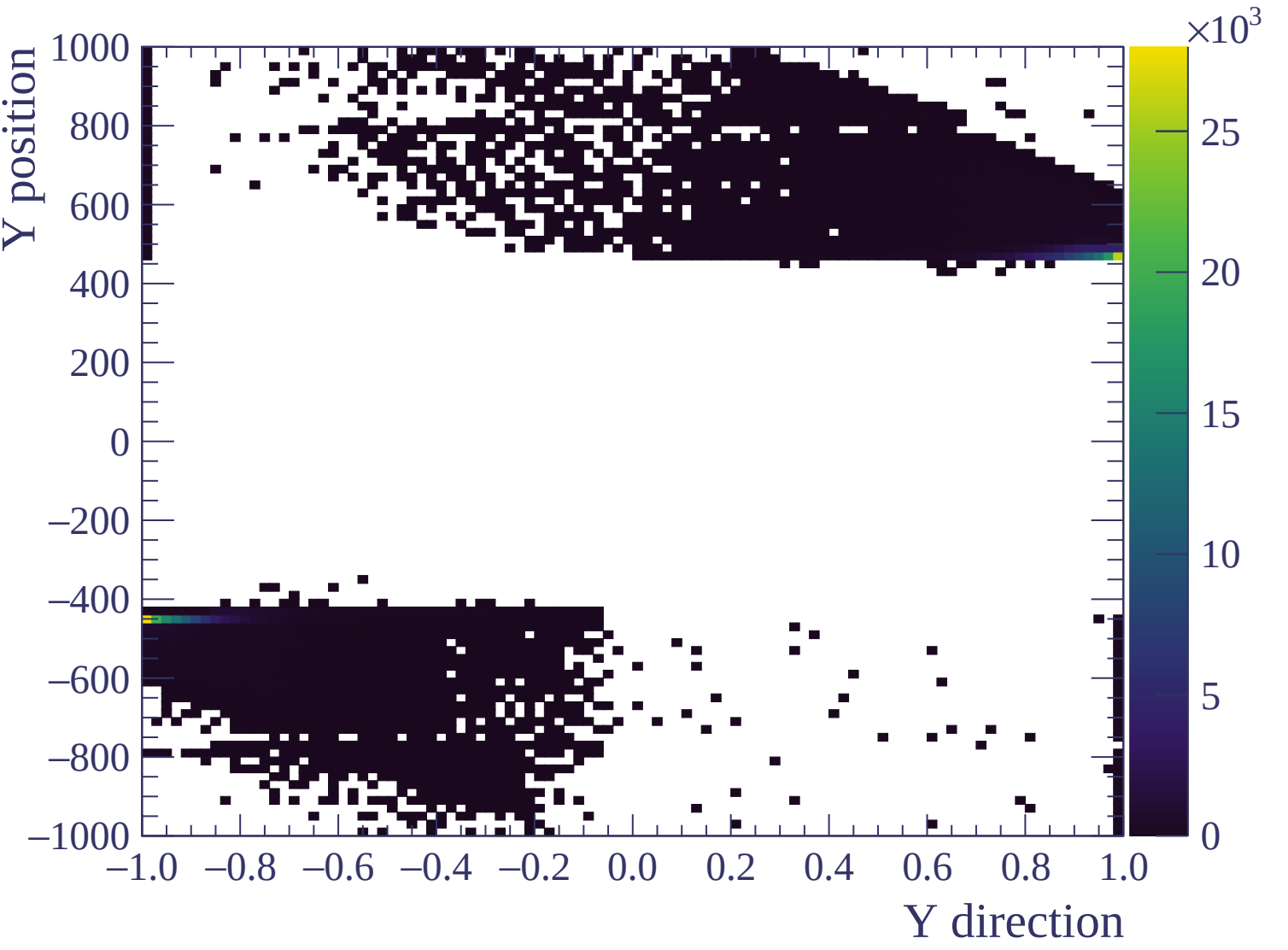






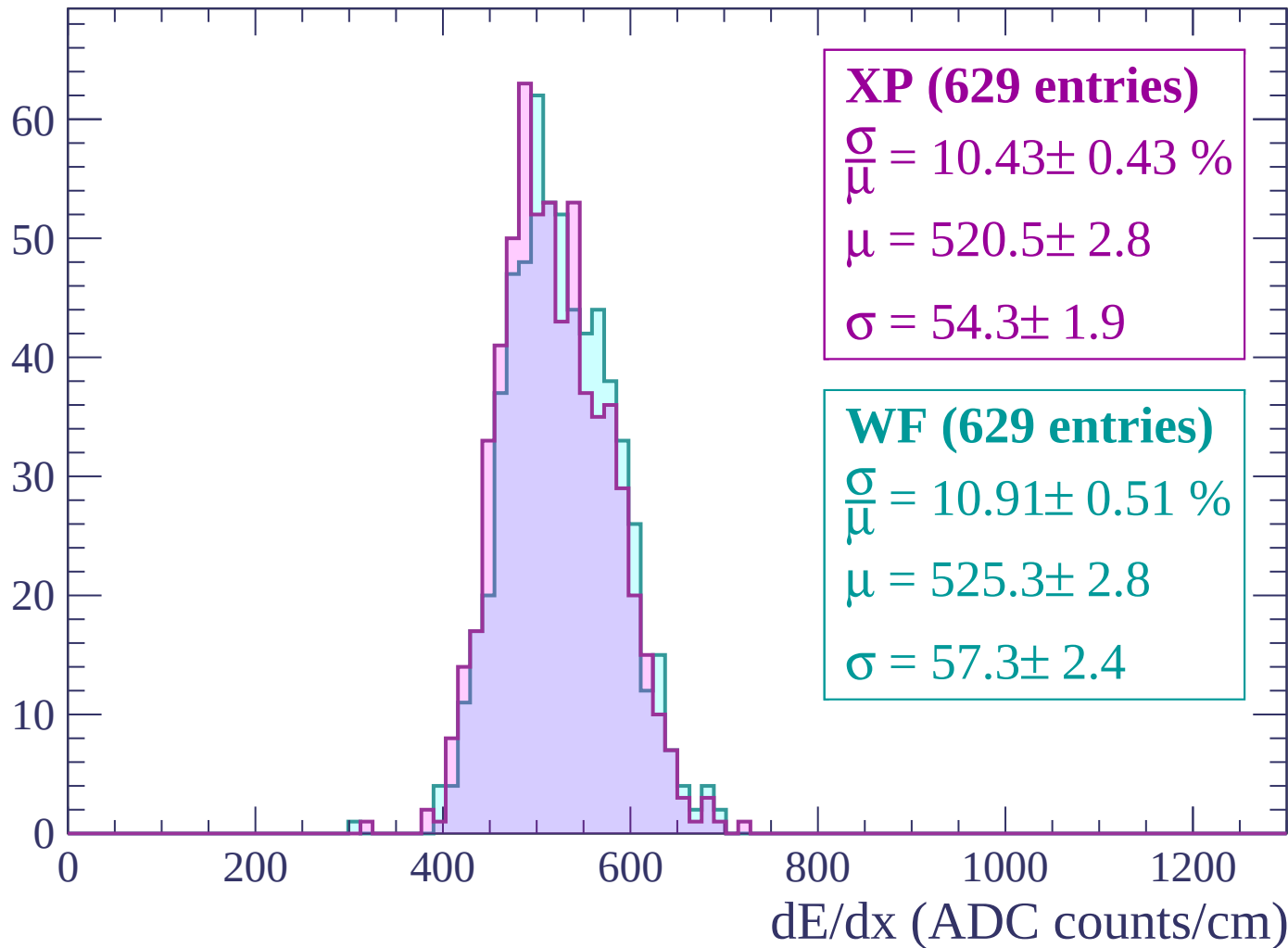






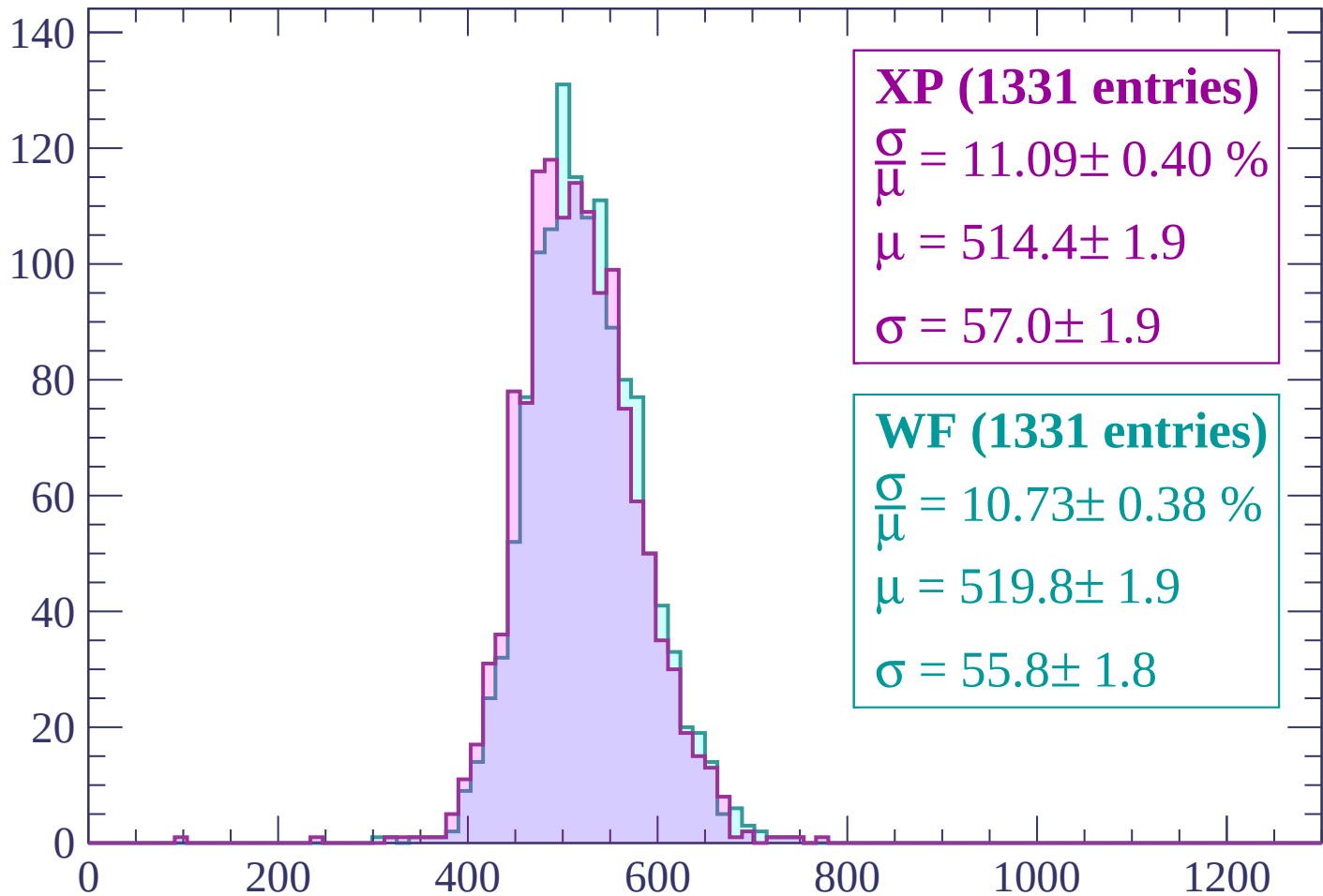
Energy loss | -3000 < p < -2940

Count



Energy loss | $-2940 < p < -2880$

Count



XP (1331 entries)

$\frac{\sigma}{\mu} = 11.09 \pm 0.40 \%$

$\mu = 514.4 \pm 1.9$

$\sigma = 57.0 \pm 1.9$

WF (1331 entries)

$\frac{\sigma}{\mu} = 10.73 \pm 0.38 \%$

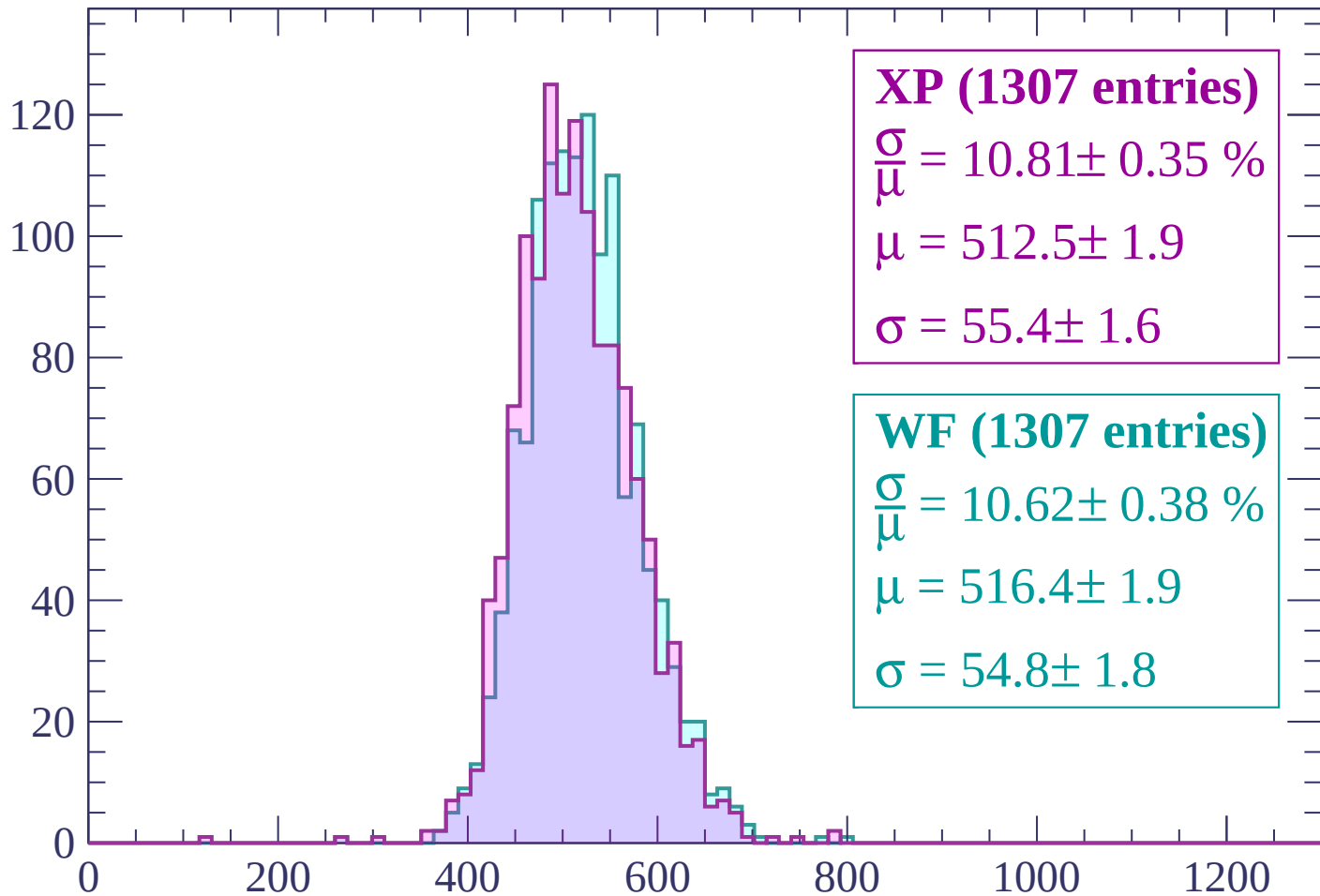
$\mu = 519.8 \pm 1.9$

$\sigma = 55.8 \pm 1.8$

dE/dx (ADC counts/cm)

Energy loss | $-2880 < p < -2820$

Count



XP (1307 entries)

$\frac{\sigma}{\mu} = 10.81 \pm 0.35 \%$

$\mu = 512.5 \pm 1.9$

$\sigma = 55.4 \pm 1.6$

WF (1307 entries)

$\frac{\sigma}{\mu} = 10.62 \pm 0.38 \%$

$\mu = 516.4 \pm 1.9$

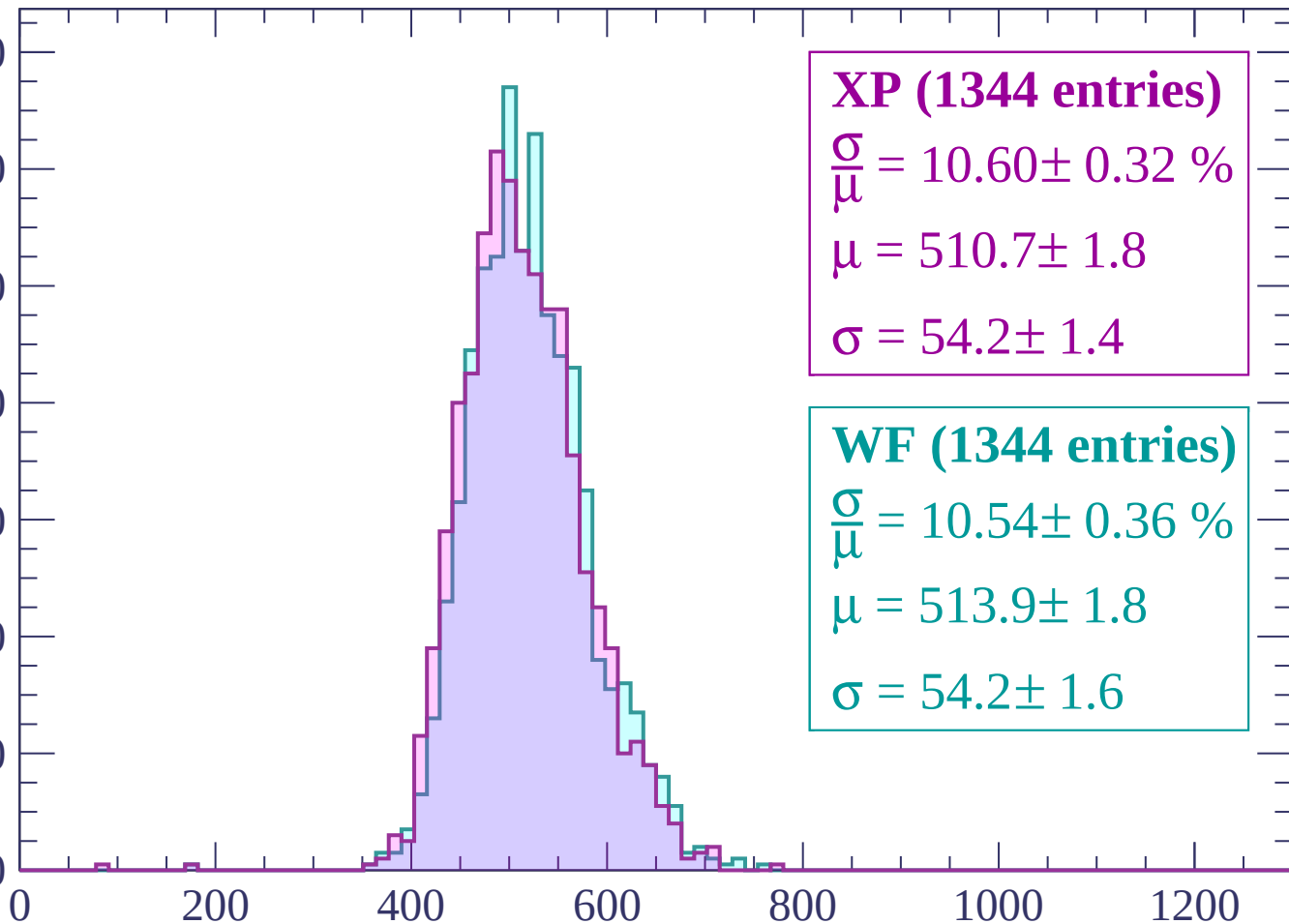
$\sigma = 54.8 \pm 1.8$

dE/dx (ADC counts/cm)

Energy loss | -2820 < p < -2760

Count

140
120
100
80
60
40
20
0



XP (1344 entries)

$\frac{\sigma}{\mu} = 10.60 \pm 0.32 \%$

$\mu = 510.7 \pm 1.8$

$\sigma = 54.2 \pm 1.4$

WF (1344 entries)

$\frac{\sigma}{\mu} = 10.54 \pm 0.36 \%$

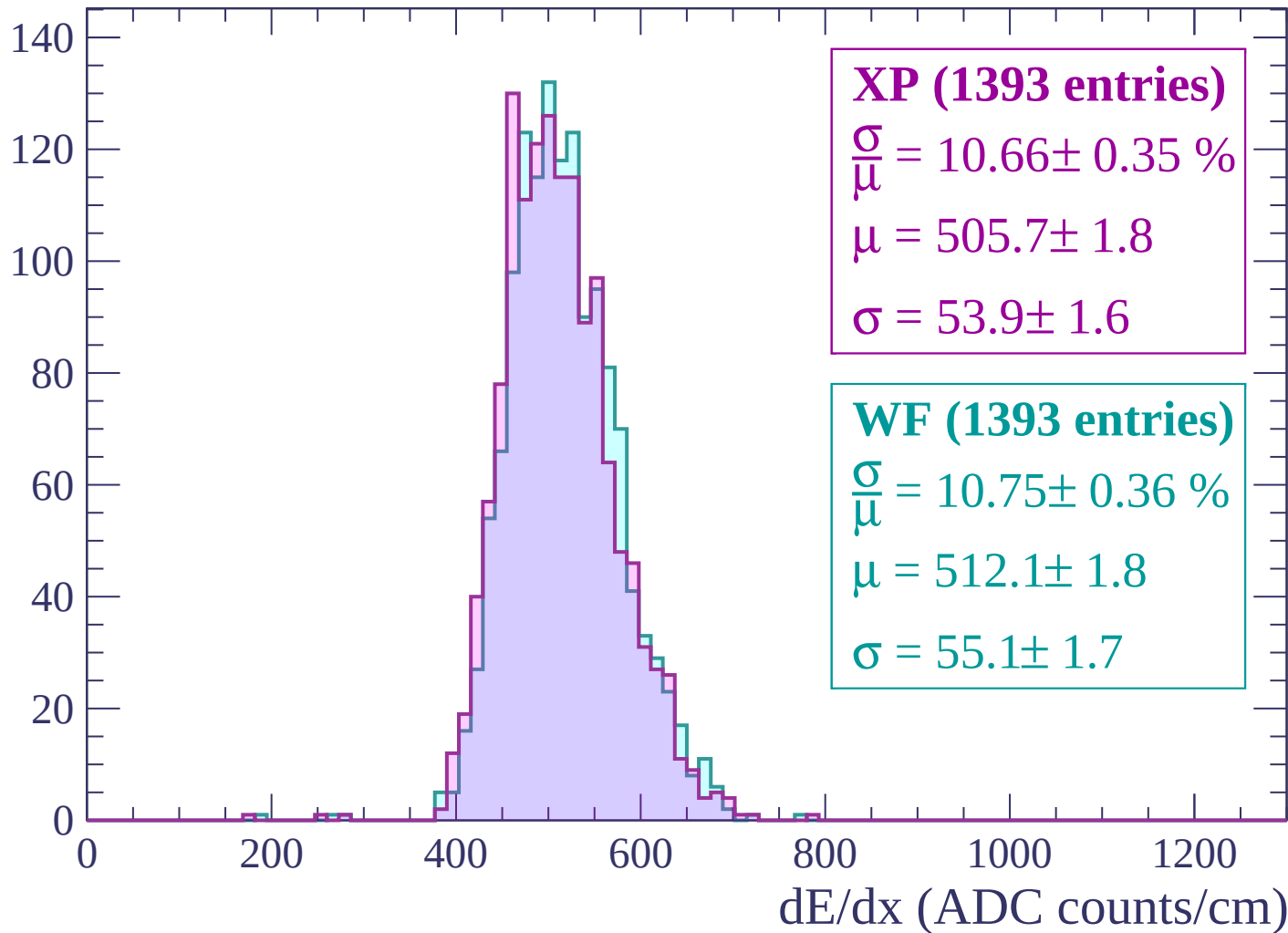
$\mu = 513.9 \pm 1.8$

$\sigma = 54.2 \pm 1.6$

dE/dx (ADC counts/cm)

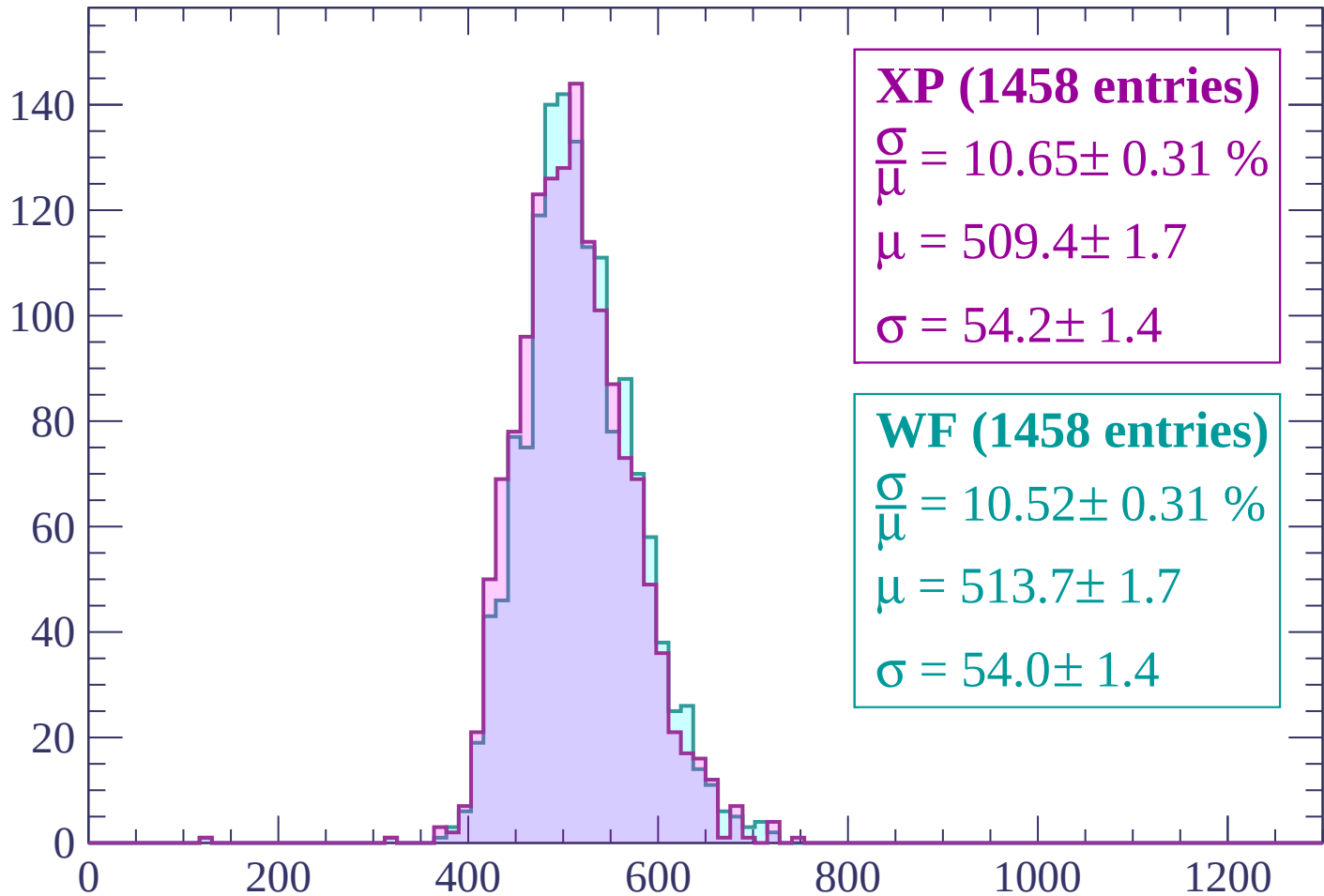
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP (1458 entries)

$$\frac{\sigma}{\mu} = 10.65 \pm 0.31 \%$$

$$\mu = 509.4 \pm 1.7$$

$$\sigma = 54.2 \pm 1.4$$

WF (1458 entries)

$$\frac{\sigma}{\mu} = 10.52 \pm 0.31 \%$$

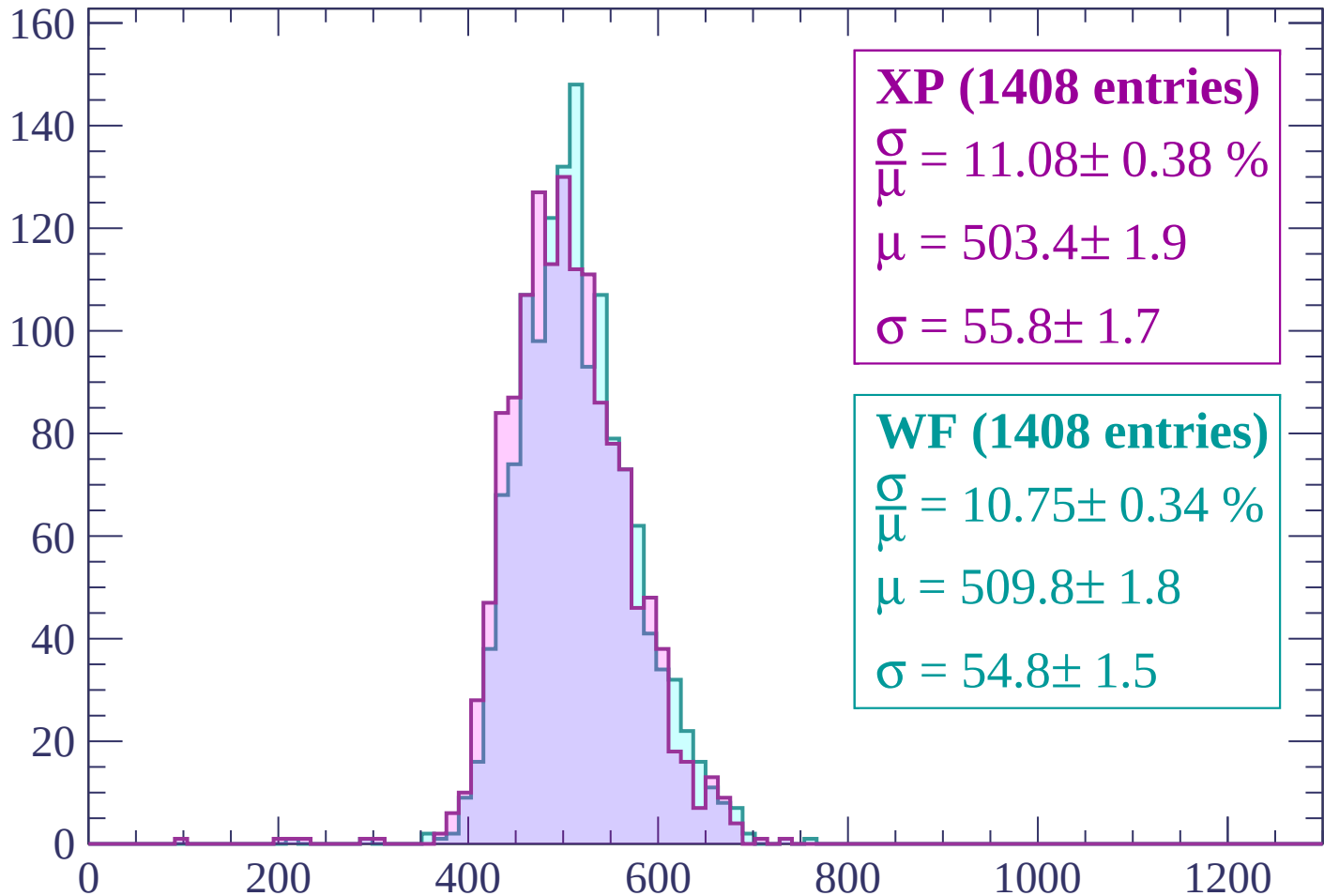
$$\mu = 513.7 \pm 1.7$$

$$\sigma = 54.0 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-2640 < p < -2580$

Count



XP (1408 entries)

$$\frac{\sigma}{\mu} = 11.08 \pm 0.38 \%$$

$$\mu = 503.4 \pm 1.9$$

$$\sigma = 55.8 \pm 1.7$$

WF (1408 entries)

$$\frac{\sigma}{\mu} = 10.75 \pm 0.34 \%$$

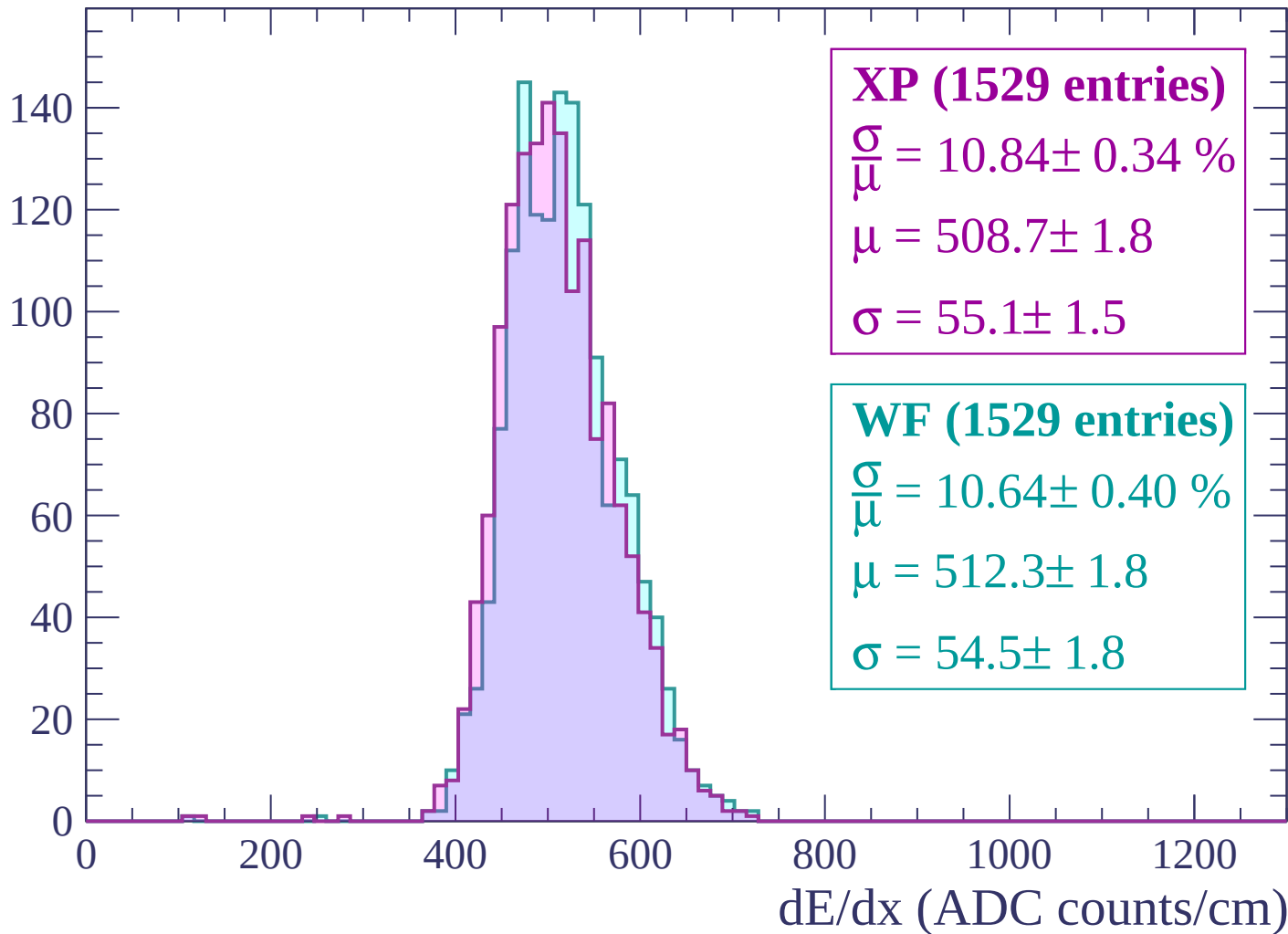
$$\mu = 509.8 \pm 1.8$$

$$\sigma = 54.8 \pm 1.5$$

dE/dx (ADC counts/cm)

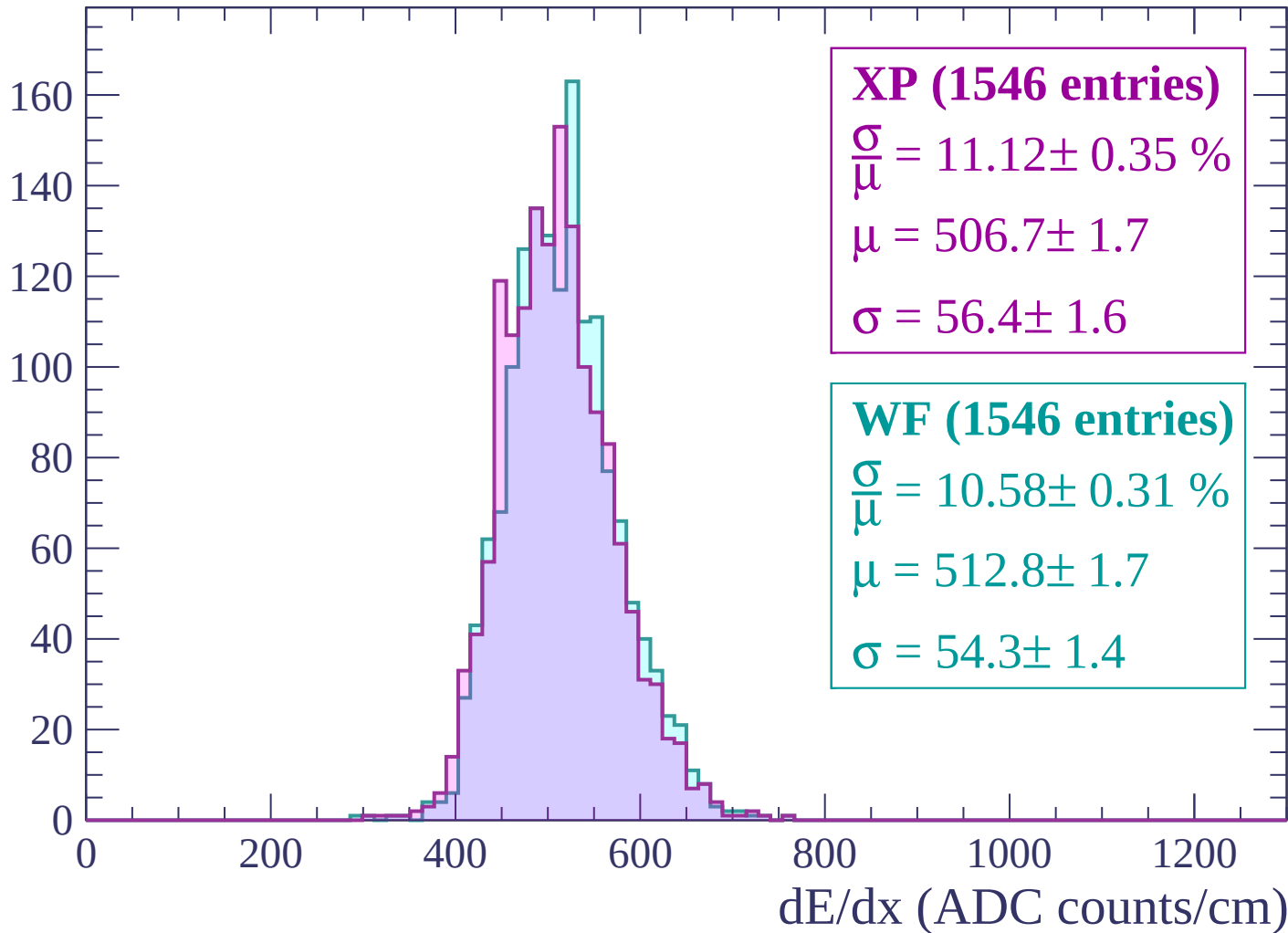
Energy loss | $-2580 < p < -2520$

Count



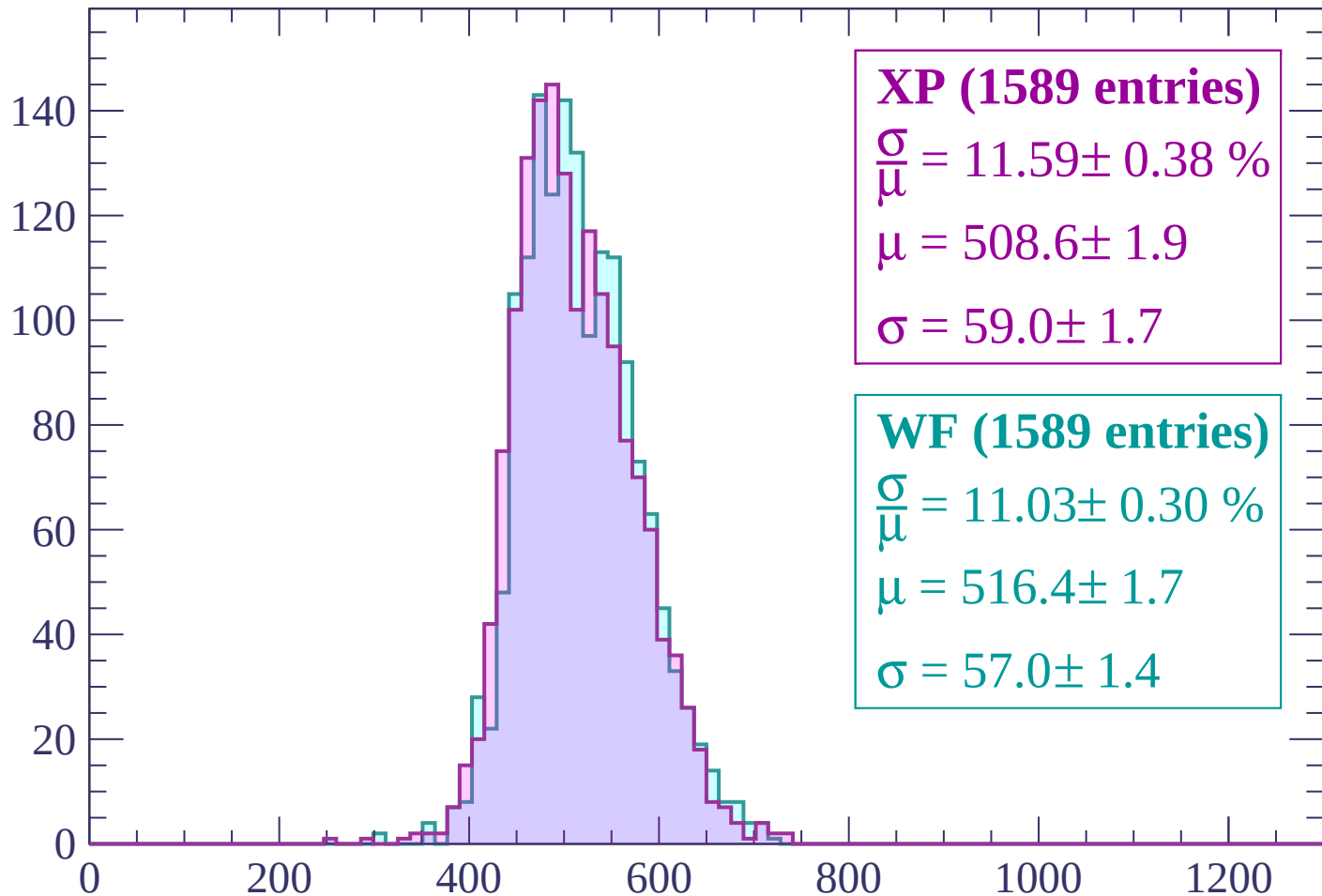
Energy loss | -2520 < p < -2460

Count



Energy loss | $-2460 < p < -2400$

Count



XP (1589 entries)

$$\frac{\sigma}{\mu} = 11.59 \pm 0.38 \%$$

$$\mu = 508.6 \pm 1.9$$

$$\sigma = 59.0 \pm 1.7$$

WF (1589 entries)

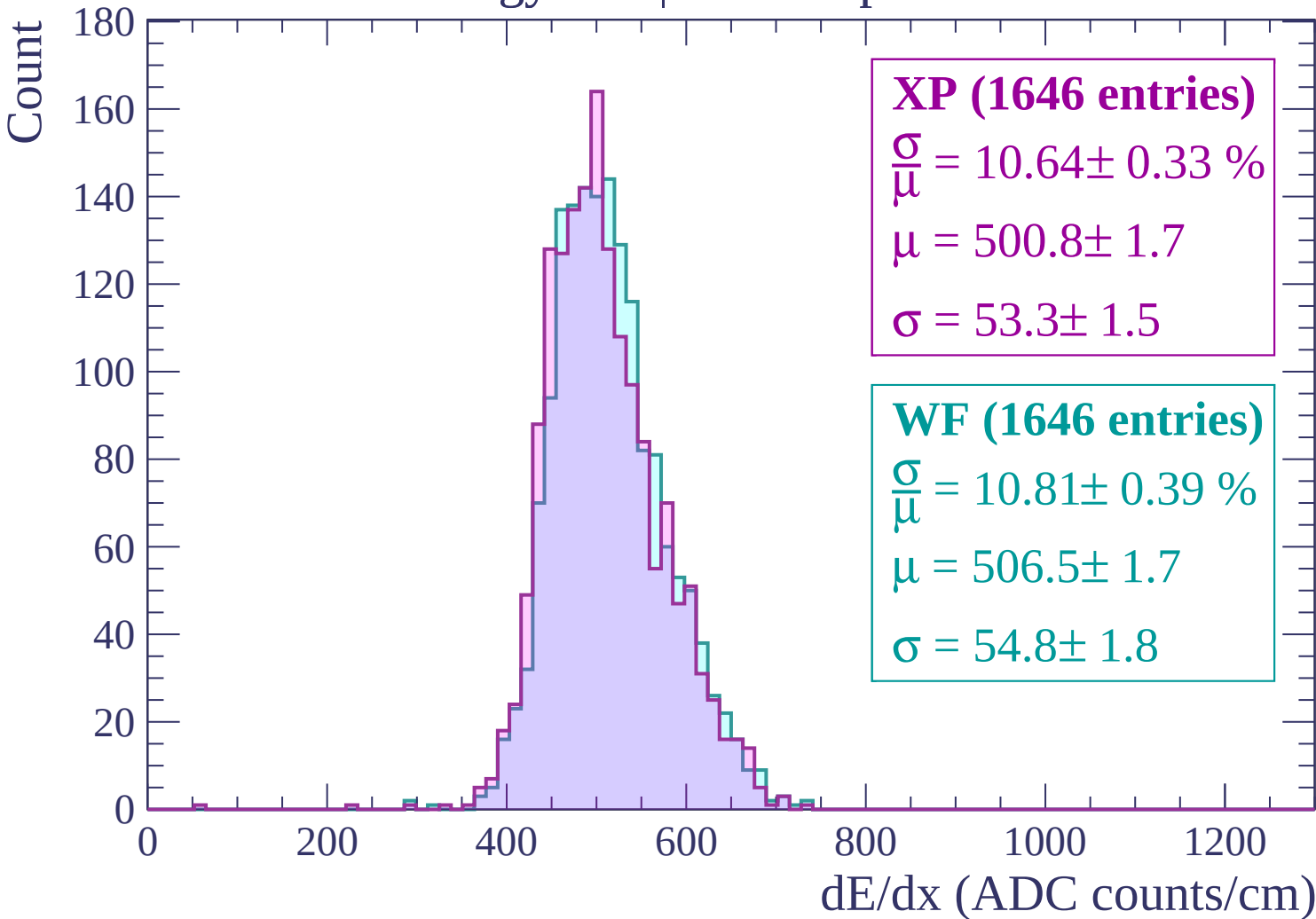
$$\frac{\sigma}{\mu} = 11.03 \pm 0.30 \%$$

$$\mu = 516.4 \pm 1.7$$

$$\sigma = 57.0 \pm 1.4$$

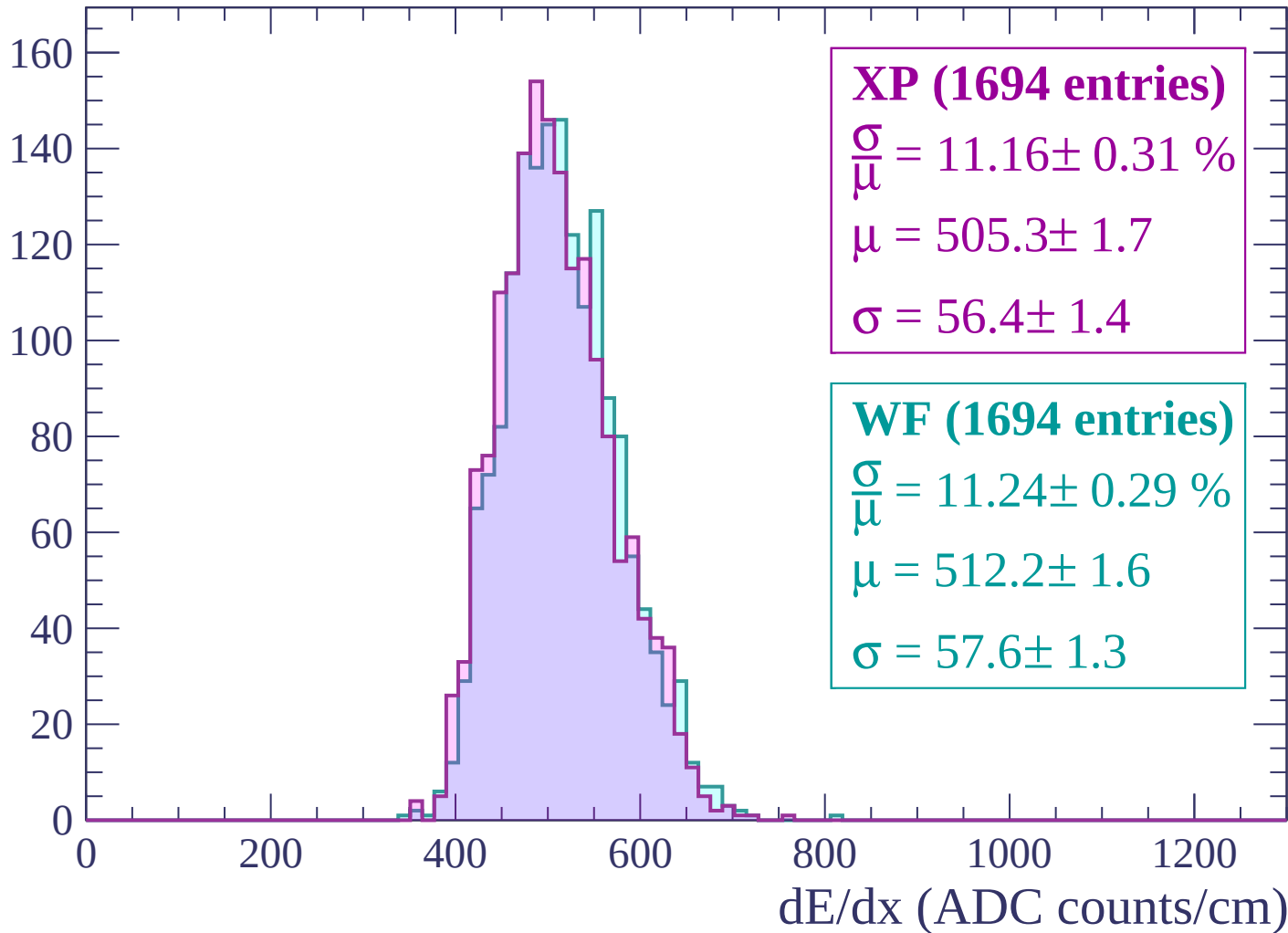
dE/dx (ADC counts/cm)

Energy loss | $-2400 < p < -2340$



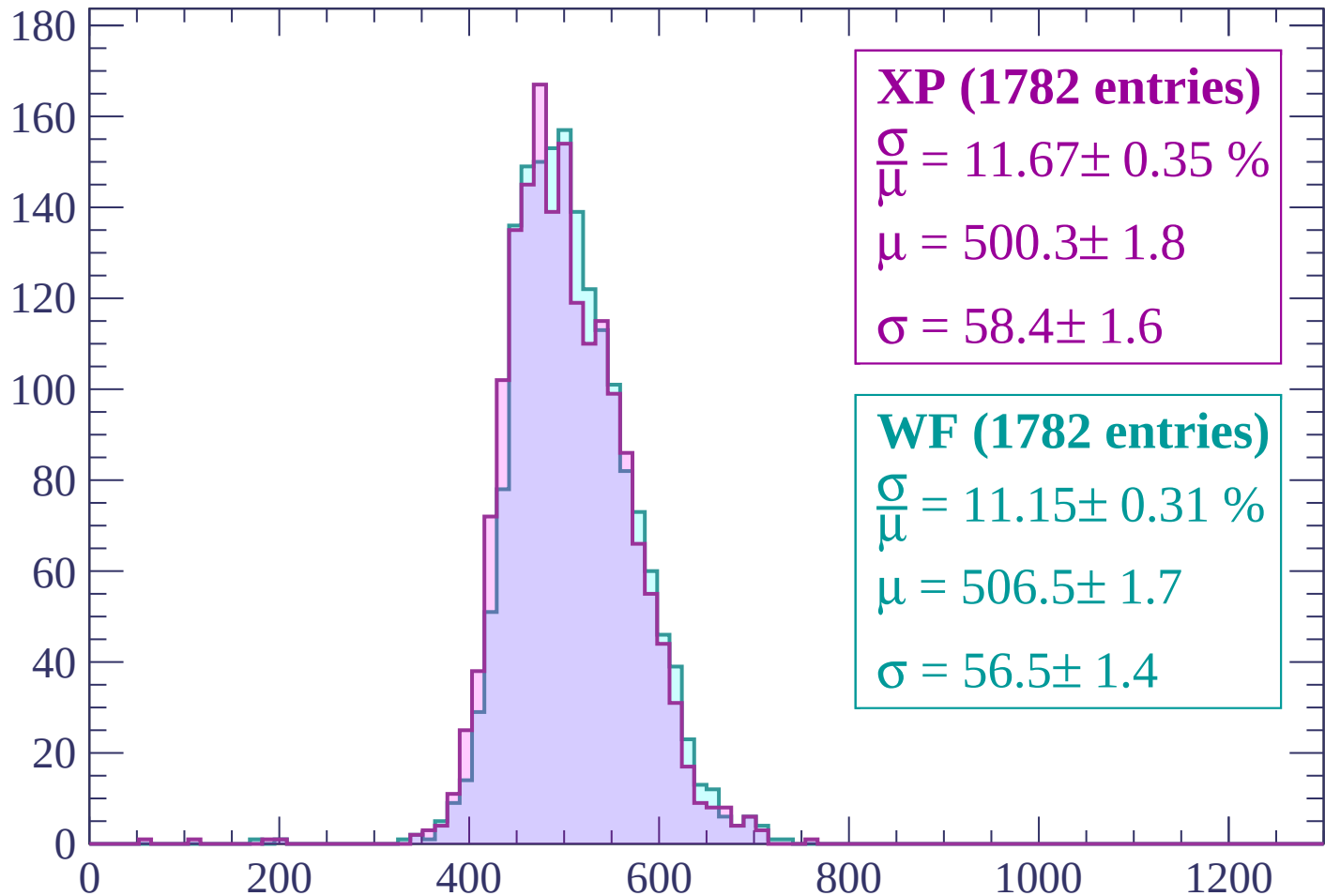
Energy loss | -2340 < p < -2280

Count



Energy loss | -2280 < p < -2220

Count



XP (1782 entries)

$$\frac{\sigma}{\mu} = 11.67 \pm 0.35 \%$$

$$\mu = 500.3 \pm 1.8$$

$$\sigma = 58.4 \pm 1.6$$

WF (1782 entries)

$$\frac{\sigma}{\mu} = 11.15 \pm 0.31 \%$$

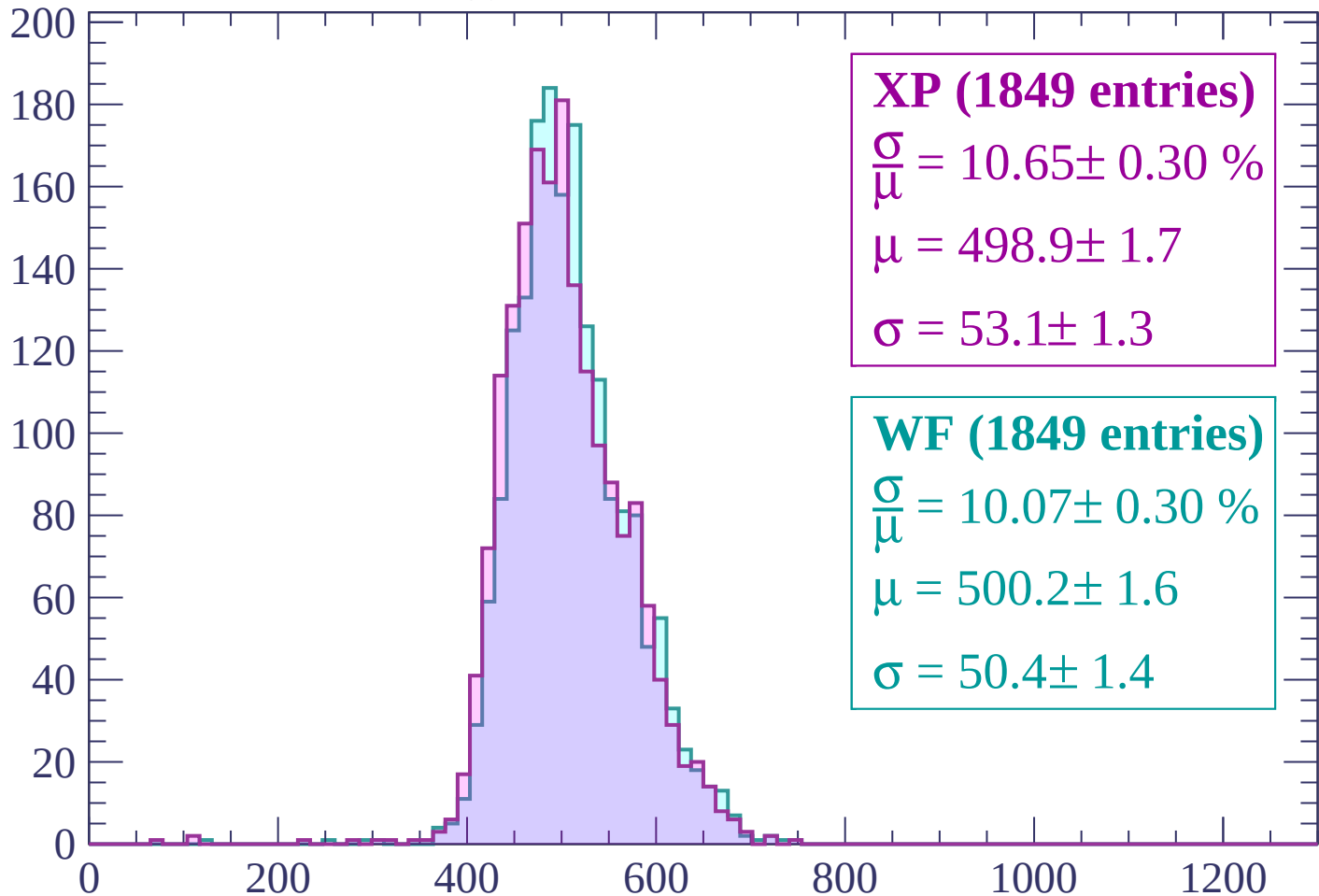
$$\mu = 506.5 \pm 1.7$$

$$\sigma = 56.5 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -2220 < p < -2160

Count



XP (1849 entries)

$$\frac{\sigma}{\mu} = 10.65 \pm 0.30 \%$$

$$\mu = 498.9 \pm 1.7$$

$$\sigma = 53.1 \pm 1.3$$

WF (1849 entries)

$$\frac{\sigma}{\mu} = 10.07 \pm 0.30 \%$$

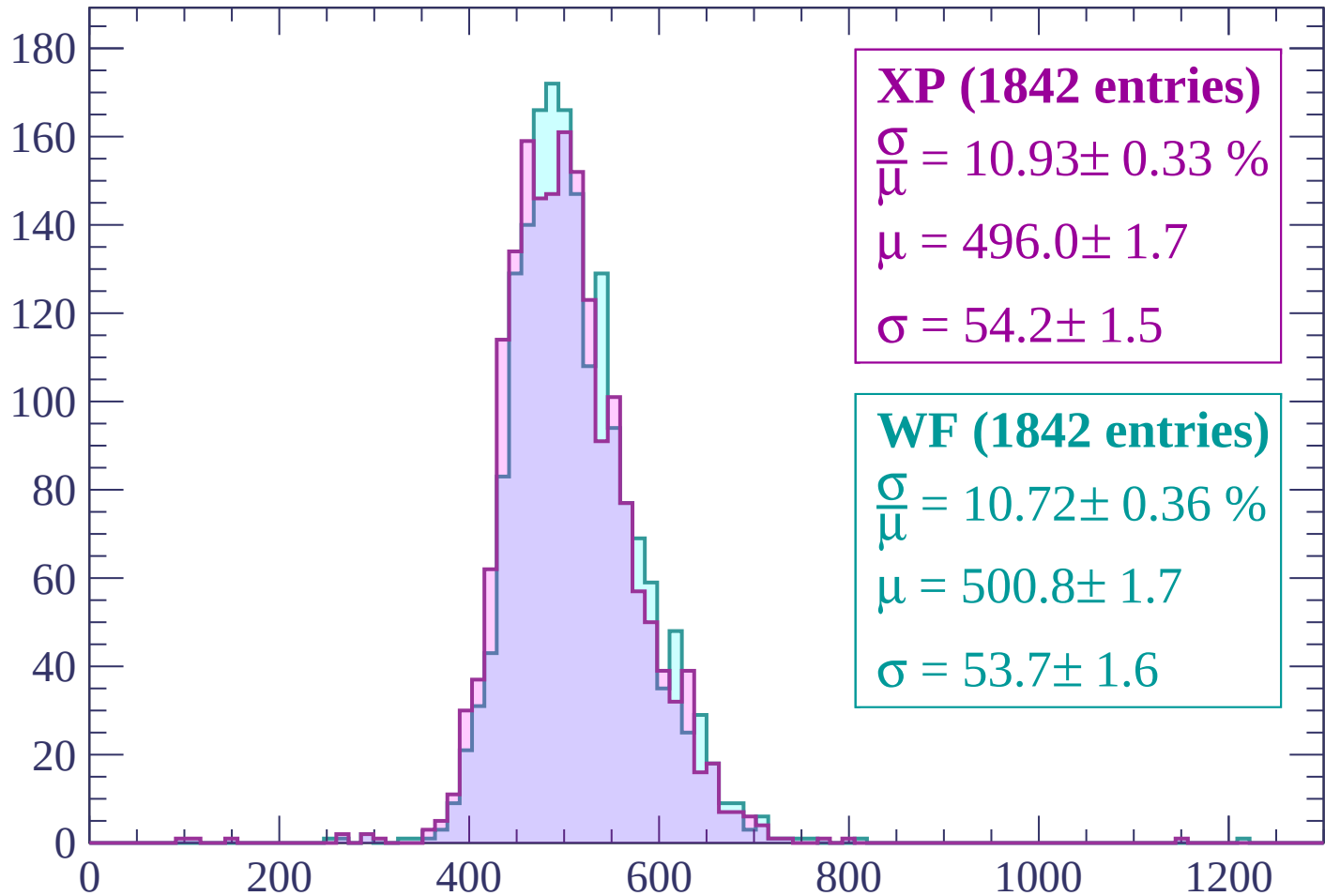
$$\mu = 500.2 \pm 1.6$$

$$\sigma = 50.4 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -2160 < p < -2100

Count



XP (1842 entries)

$\frac{\sigma}{\mu} = 10.93 \pm 0.33 \%$

$\mu = 496.0 \pm 1.7$

$\sigma = 54.2 \pm 1.5$

WF (1842 entries)

$\frac{\sigma}{\mu} = 10.72 \pm 0.36 \%$

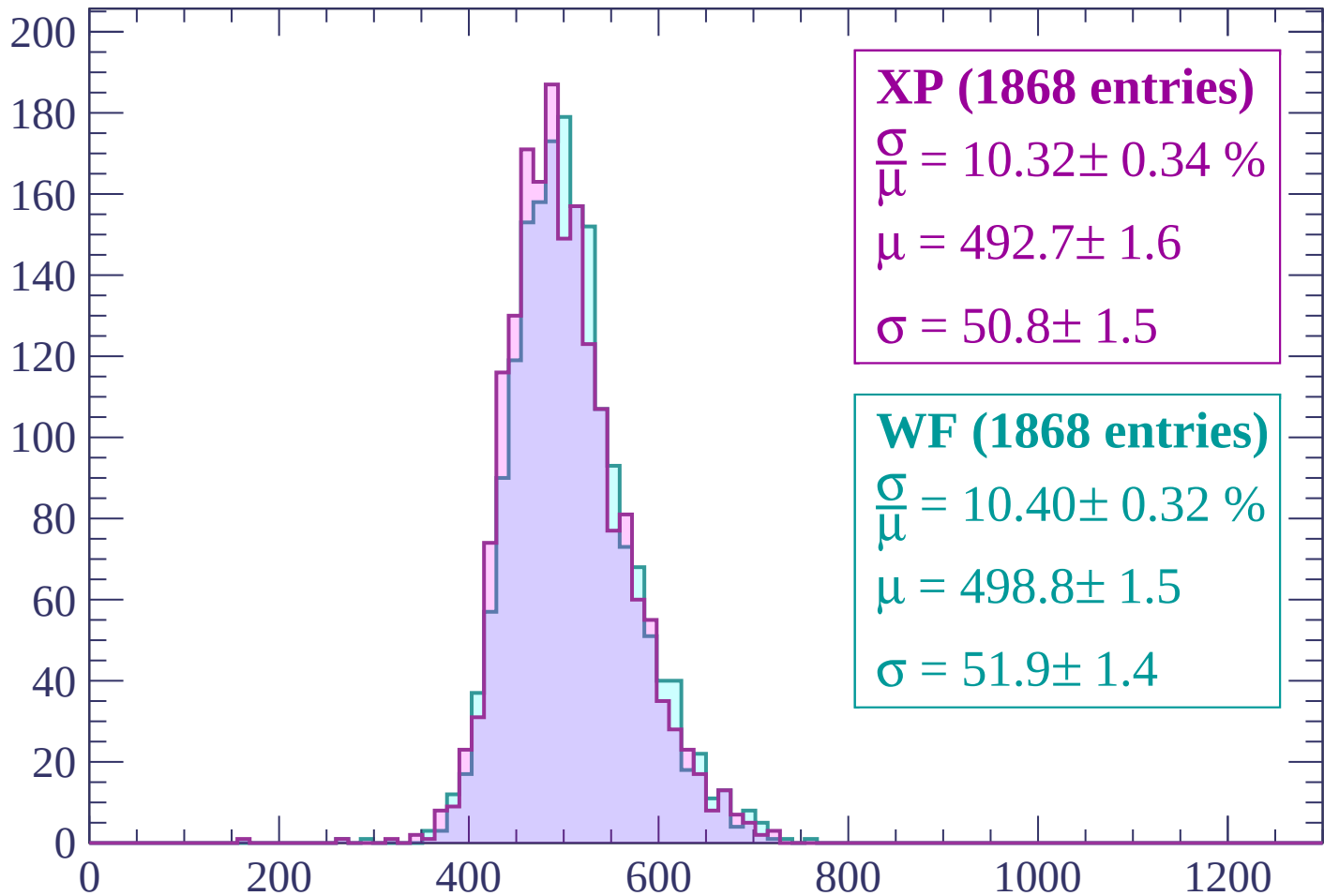
$\mu = 500.8 \pm 1.7$

$\sigma = 53.7 \pm 1.6$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count



XP (1868 entries)

$$\frac{\sigma}{\mu} = 10.32 \pm 0.34 \%$$

$$\mu = 492.7 \pm 1.6$$

$$\sigma = 50.8 \pm 1.5$$

WF (1868 entries)

$$\frac{\sigma}{\mu} = 10.40 \pm 0.32 \%$$

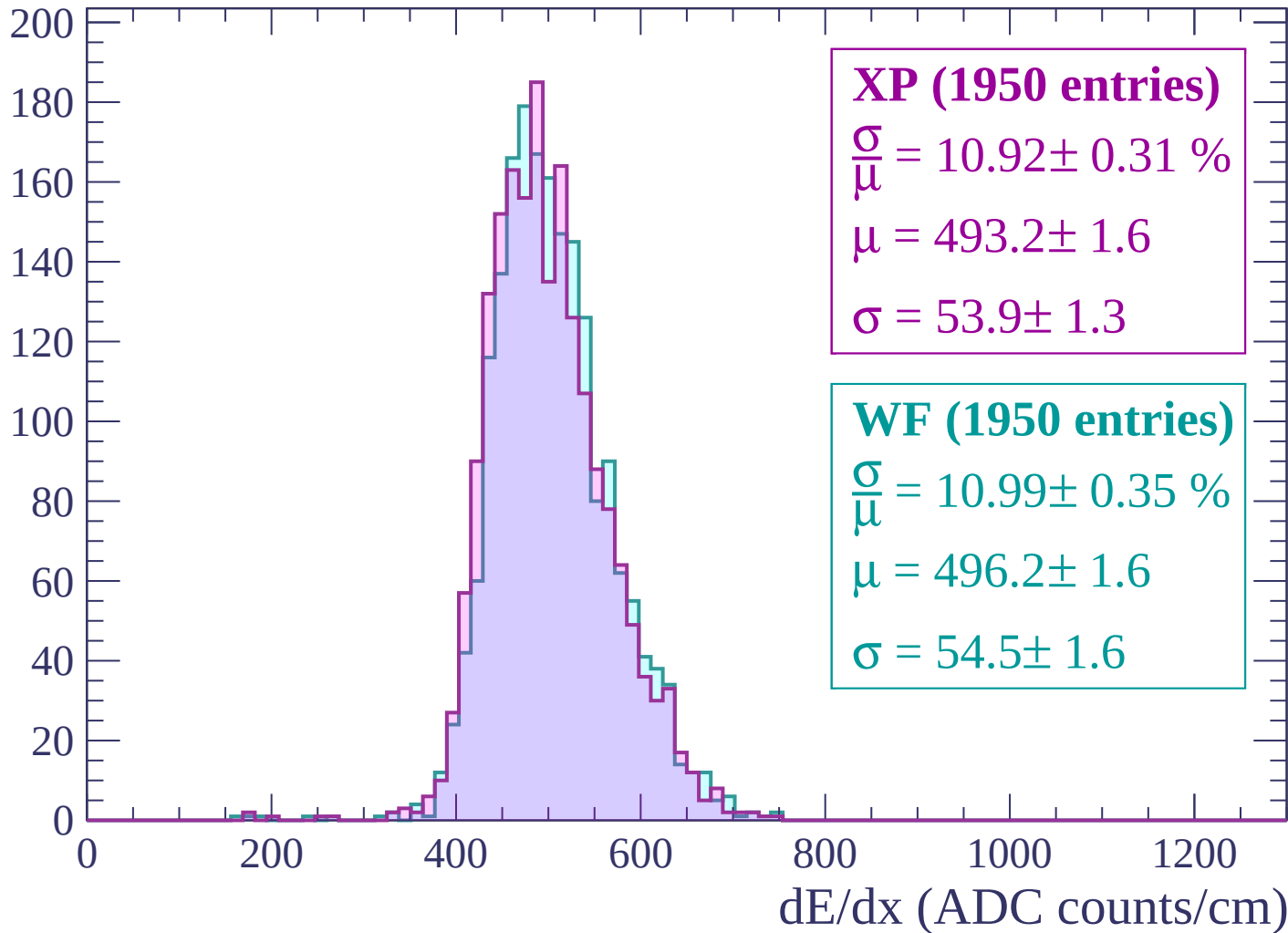
$$\mu = 498.8 \pm 1.5$$

$$\sigma = 51.9 \pm 1.4$$

dE/dx (ADC counts/cm)

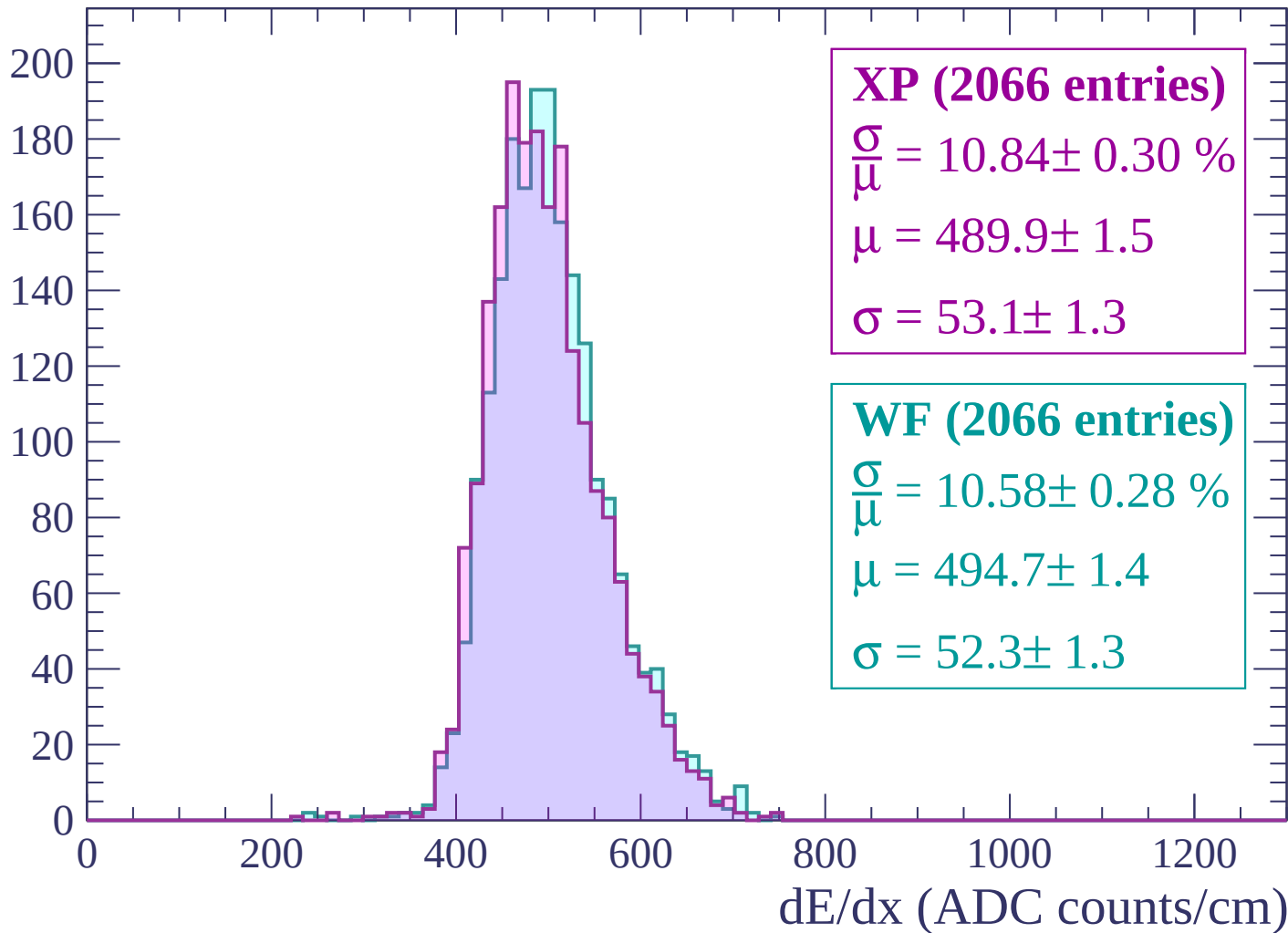
Energy loss | $-2040 < p < -1980$

Count

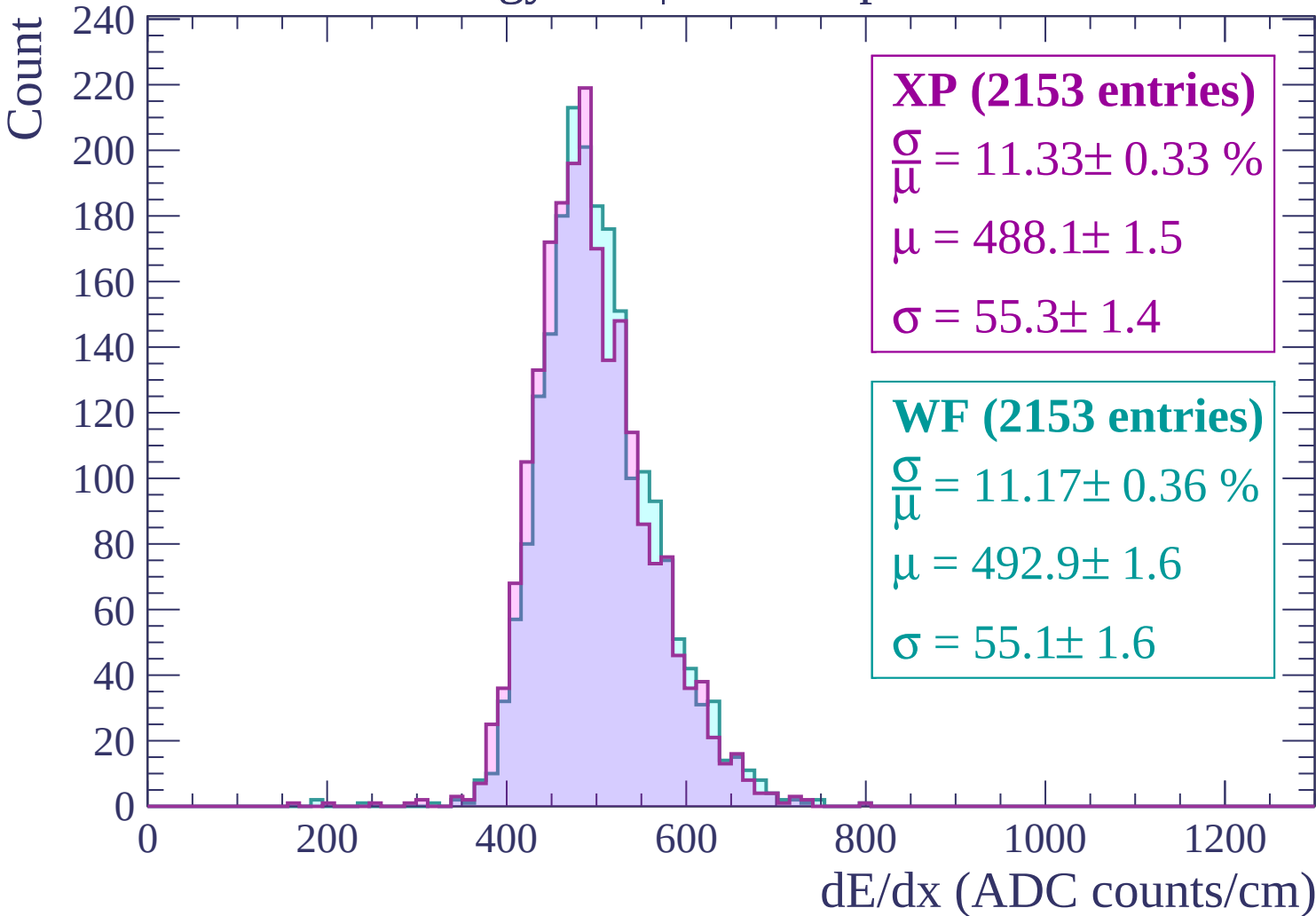


Energy loss | $-1980 < p < -1920$

Count

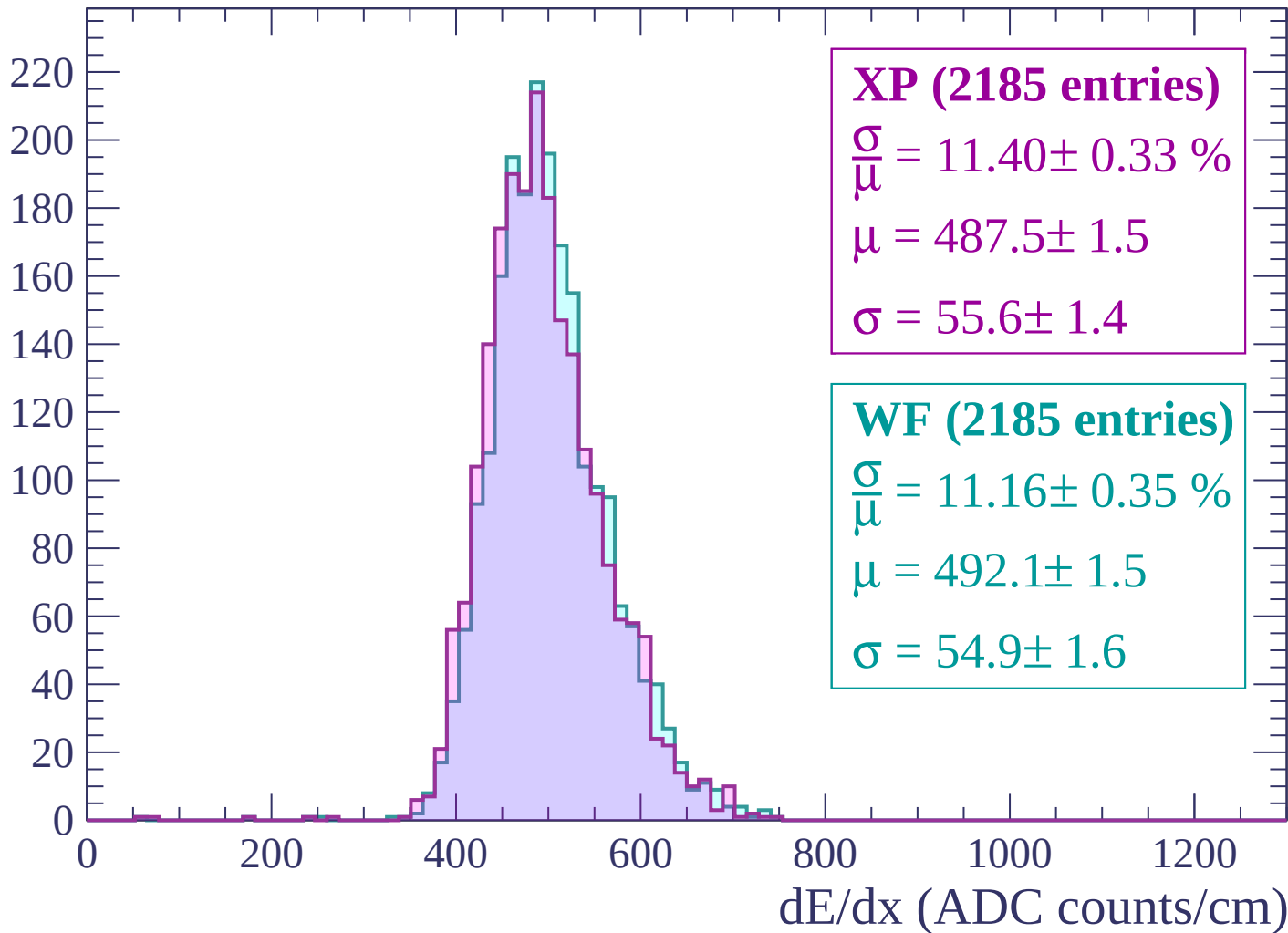


Energy loss | -1920 < p < -1860



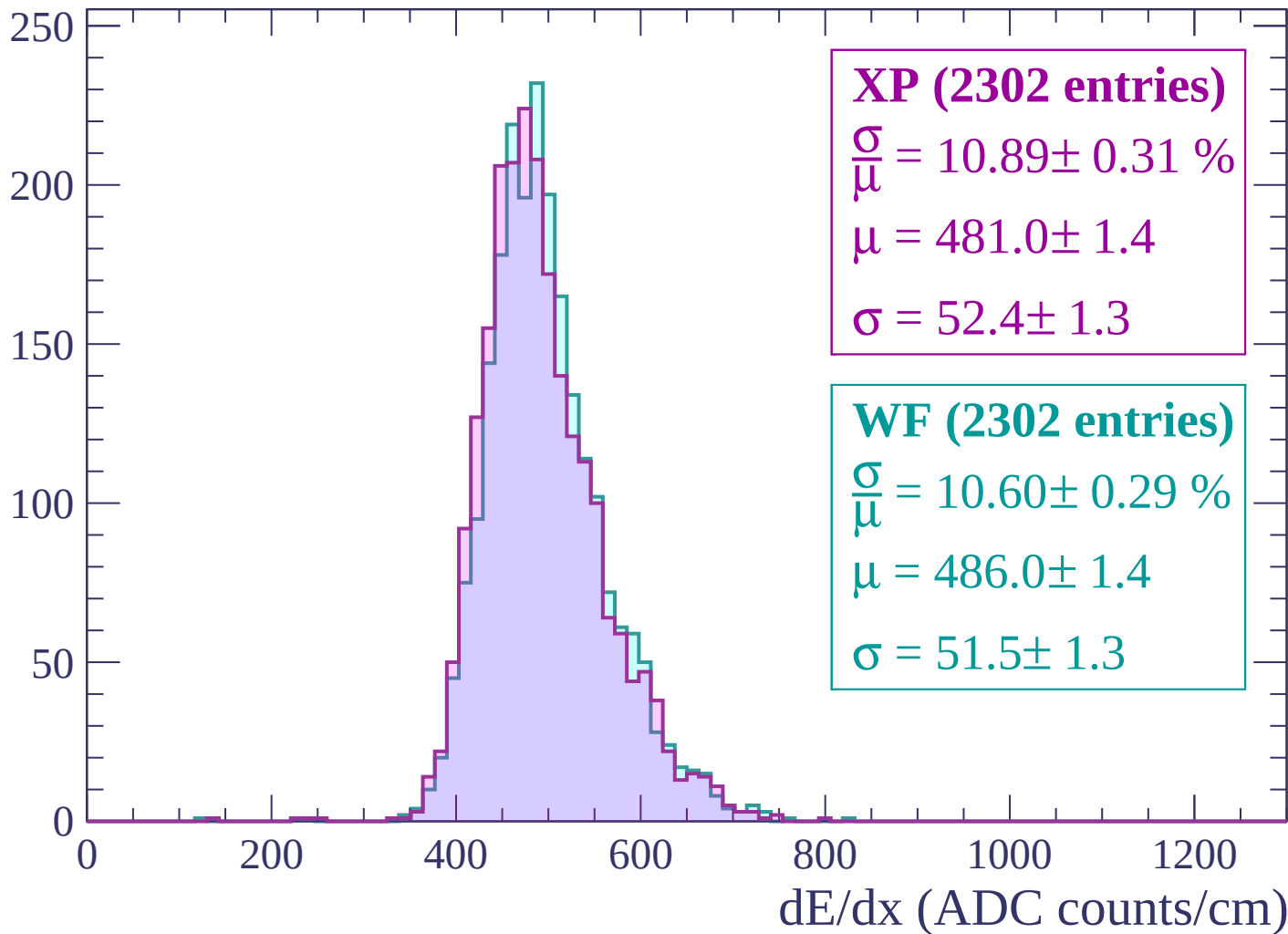
Energy loss | $-1860 < p < -1800$

Count



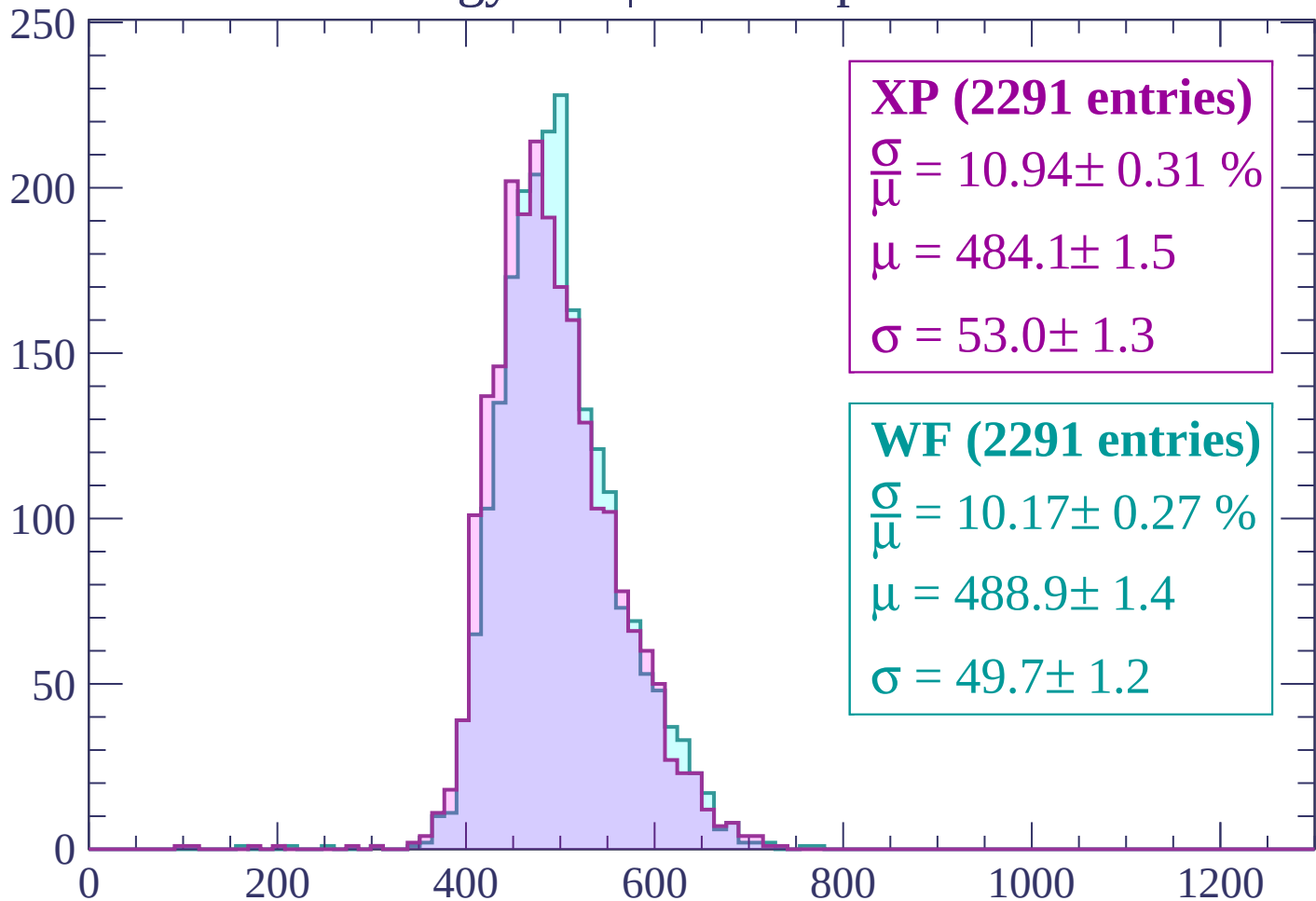
Energy loss | -1800 < p < -1740

Count



Energy loss | $-1740 < p < -1680$

Count



XP (2291 entries)

$$\frac{\sigma}{\mu} = 10.94 \pm 0.31 \%$$

$$\mu = 484.1 \pm 1.5$$

$$\sigma = 53.0 \pm 1.3$$

WF (2291 entries)

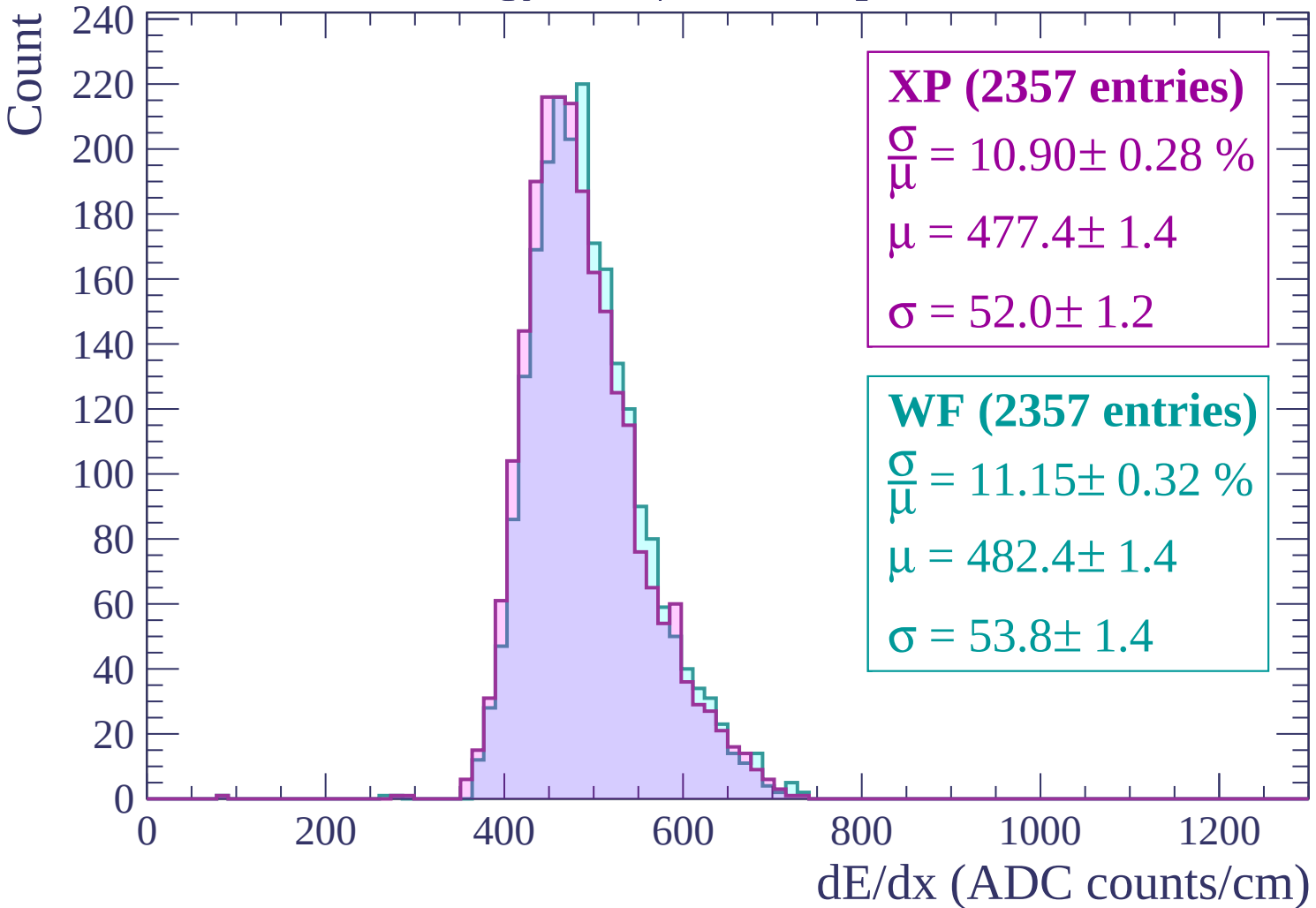
$$\frac{\sigma}{\mu} = 10.17 \pm 0.27 \%$$

$$\mu = 488.9 \pm 1.4$$

$$\sigma = 49.7 \pm 1.2$$

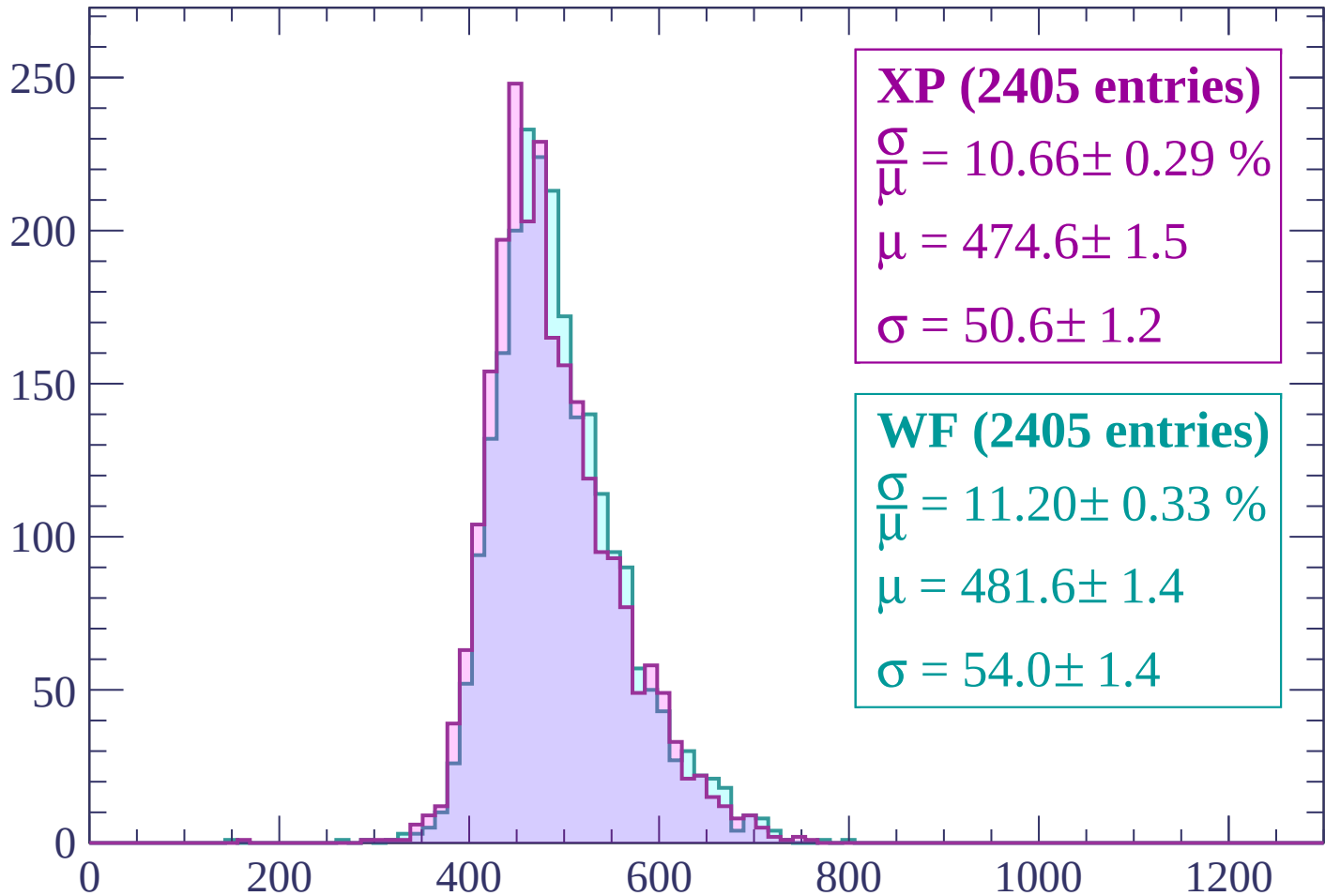
dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1620$



Energy loss | -1620 < p < -1560

Count



XP (2405 entries)

$$\frac{\sigma}{\mu} = 10.66 \pm 0.29 \%$$

$$\mu = 474.6 \pm 1.5$$

$$\sigma = 50.6 \pm 1.2$$

WF (2405 entries)

$$\frac{\sigma}{\mu} = 11.20 \pm 0.33 \%$$

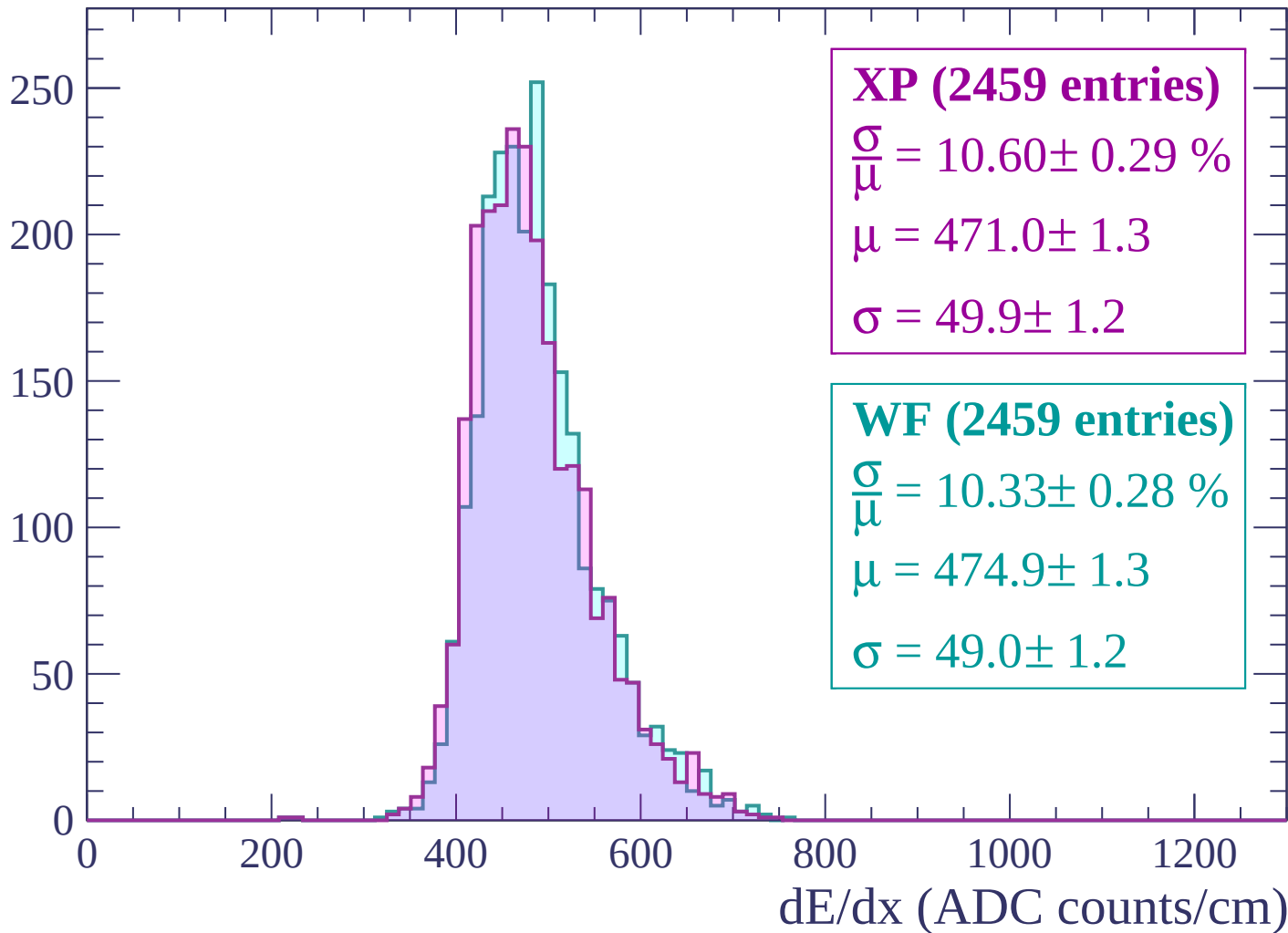
$$\mu = 481.6 \pm 1.4$$

$$\sigma = 54.0 \pm 1.4$$

dE/dx (ADC counts/cm)

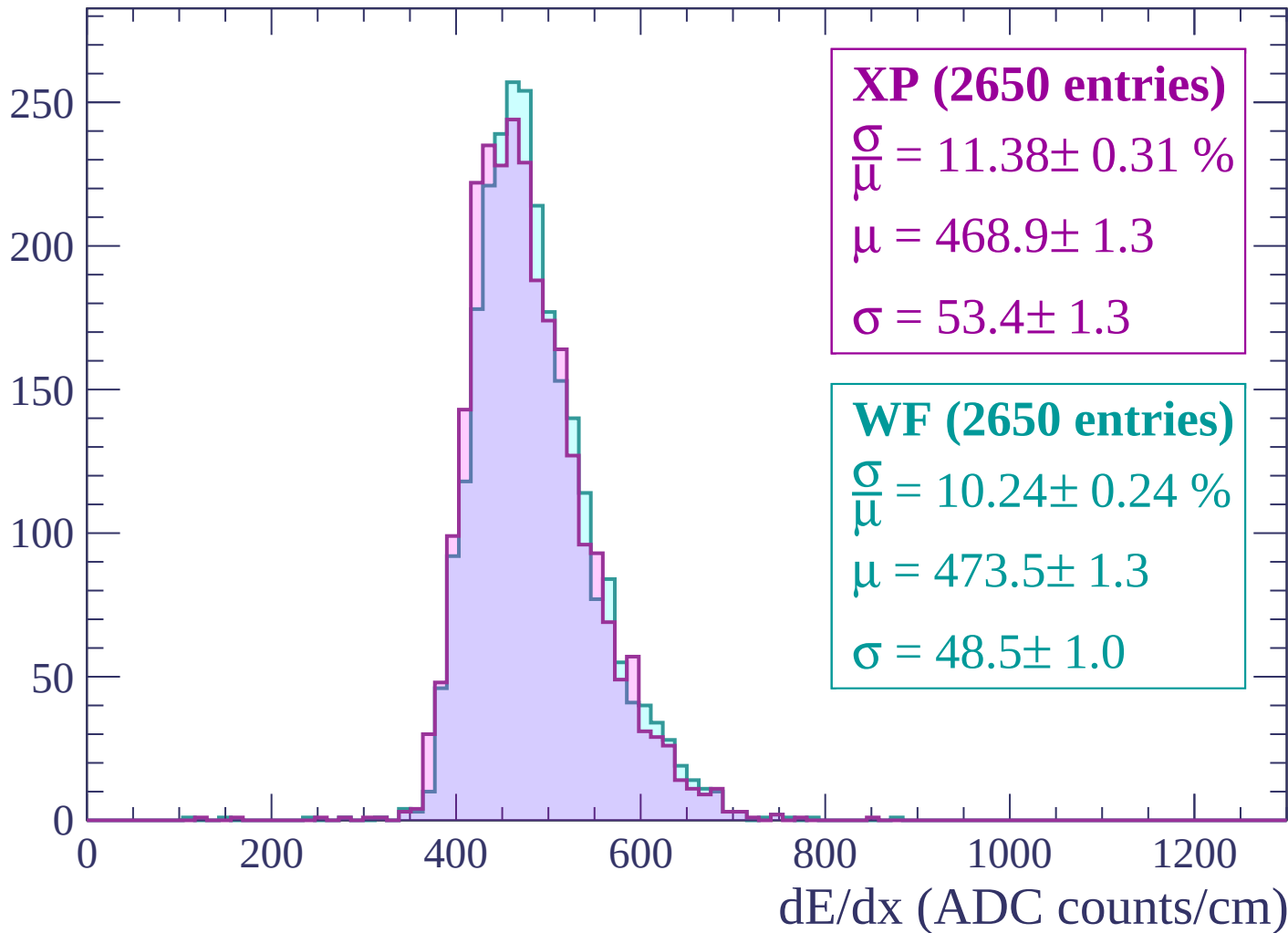
Energy loss | -1560 < p < -1500

Count



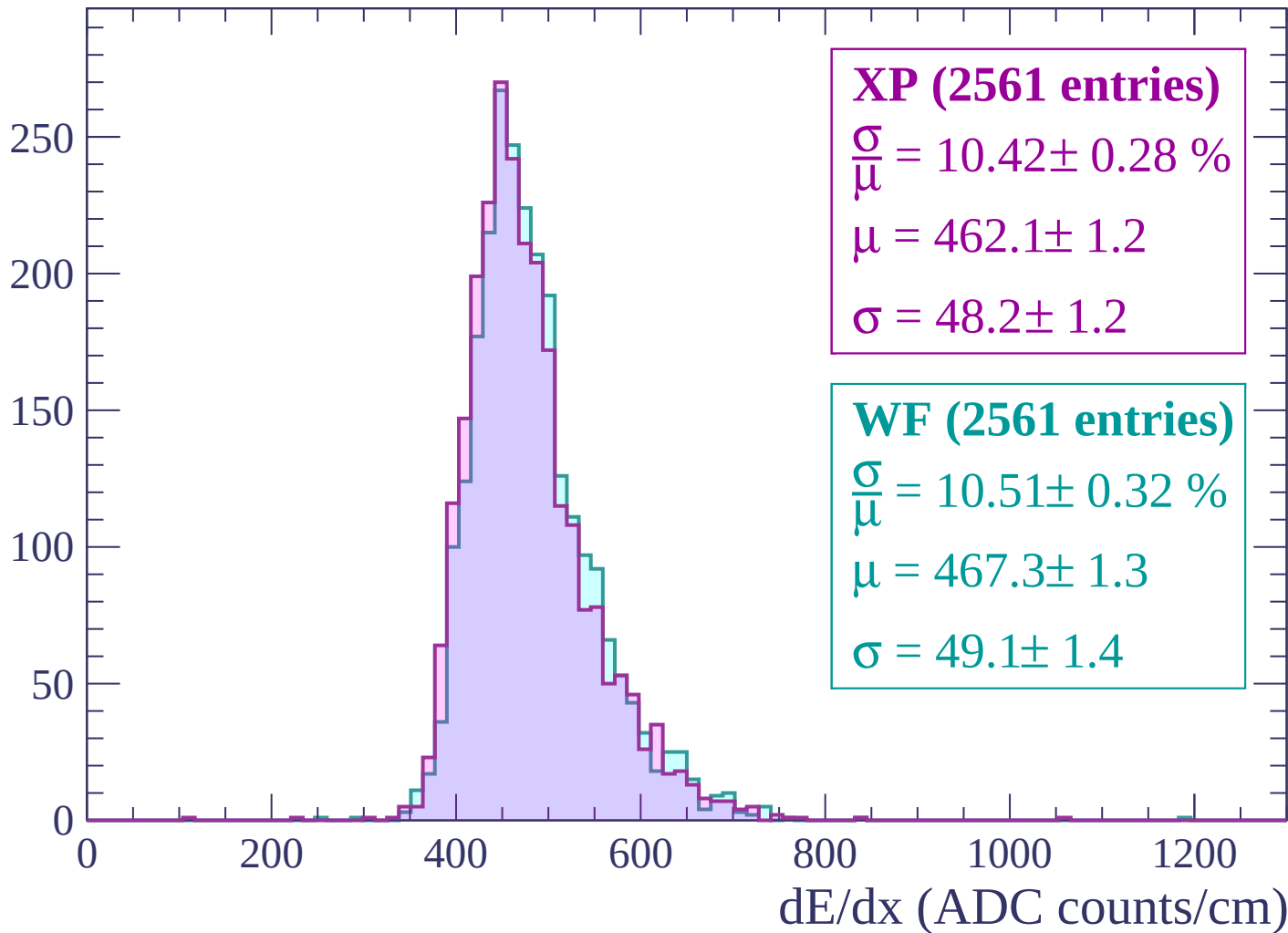
Energy loss | -1500 < p < -1440

Count



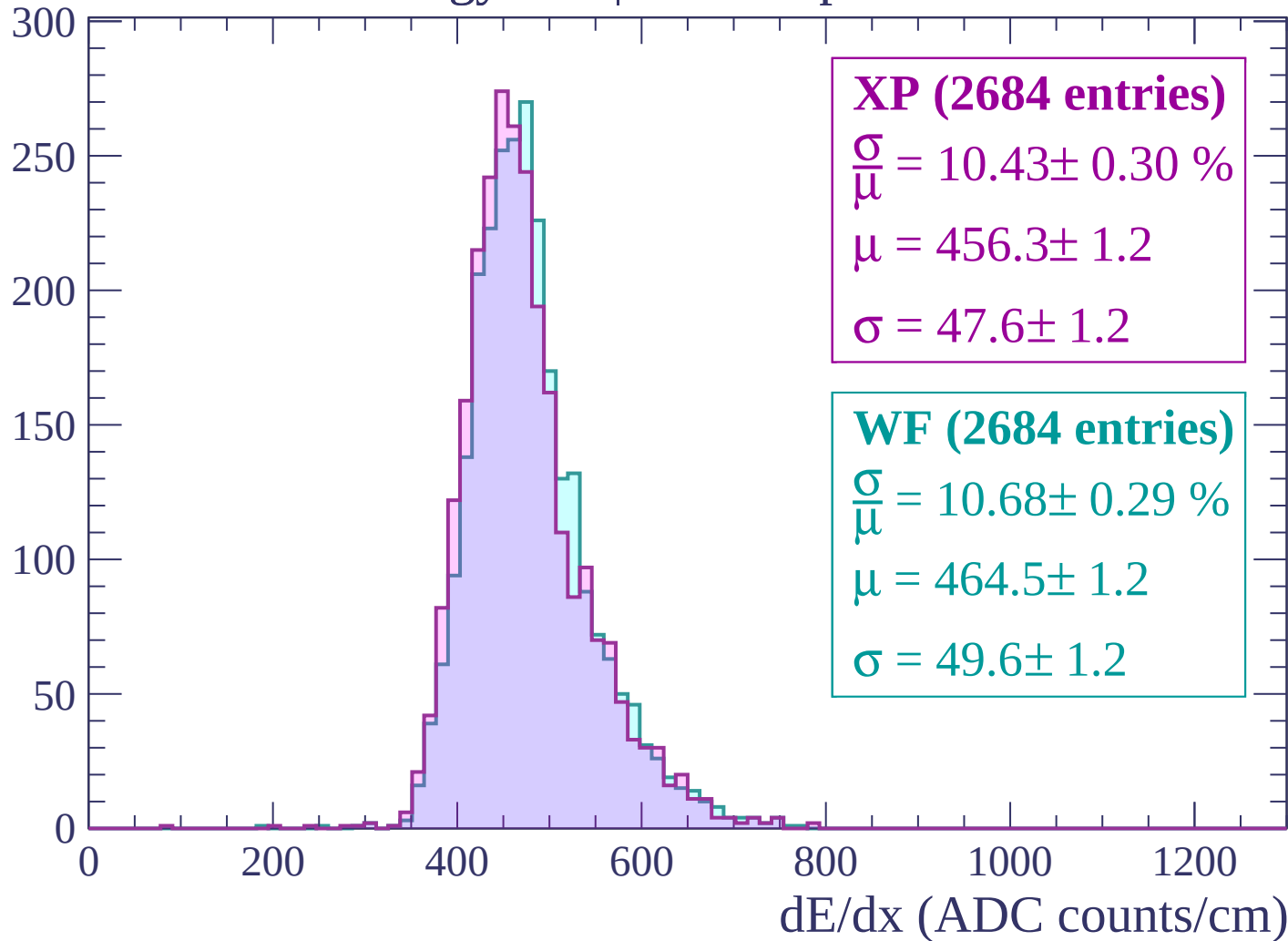
Energy loss | $-1440 < p < -1380$

Count



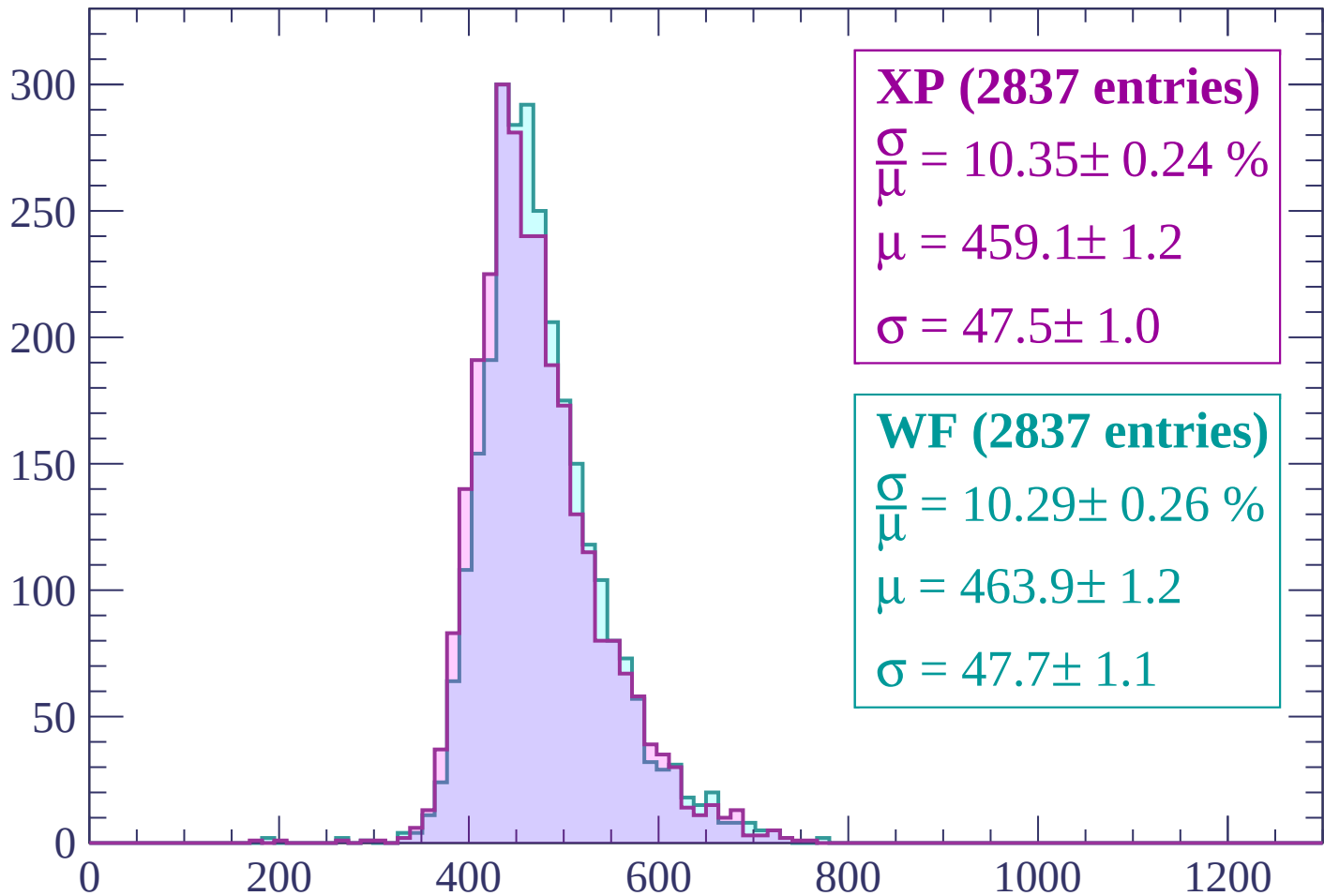
Energy loss | $-1380 < p < -1320$

Count



Energy loss | -1320 < p < -1260

Count



XP (2837 entries)

$\frac{\sigma}{\mu} = 10.35 \pm 0.24 \%$

$\mu = 459.1 \pm 1.2$

$\sigma = 47.5 \pm 1.0$

WF (2837 entries)

$\frac{\sigma}{\mu} = 10.29 \pm 0.26 \%$

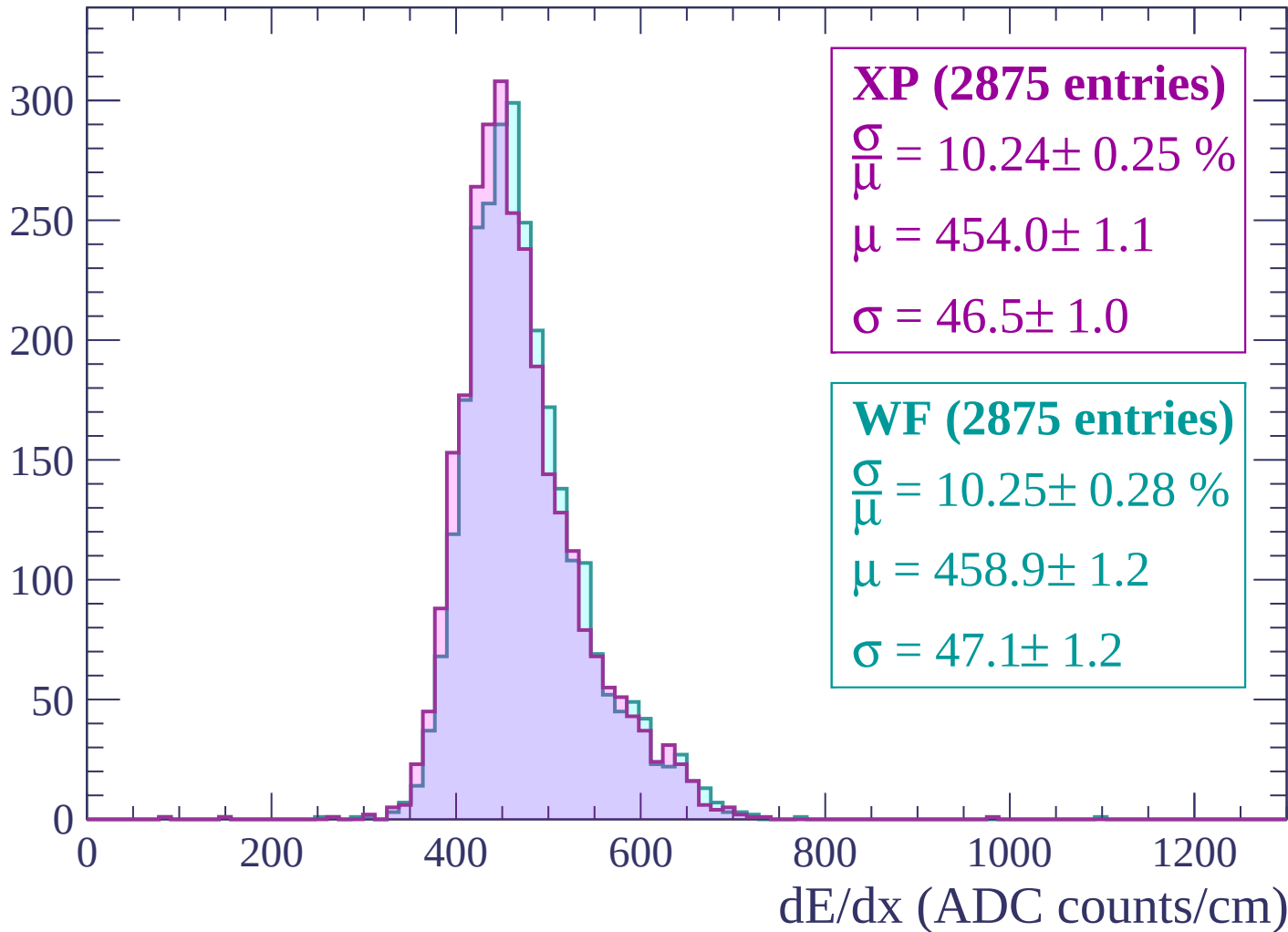
$\mu = 463.9 \pm 1.2$

$\sigma = 47.7 \pm 1.1$

dE/dx (ADC counts/cm)

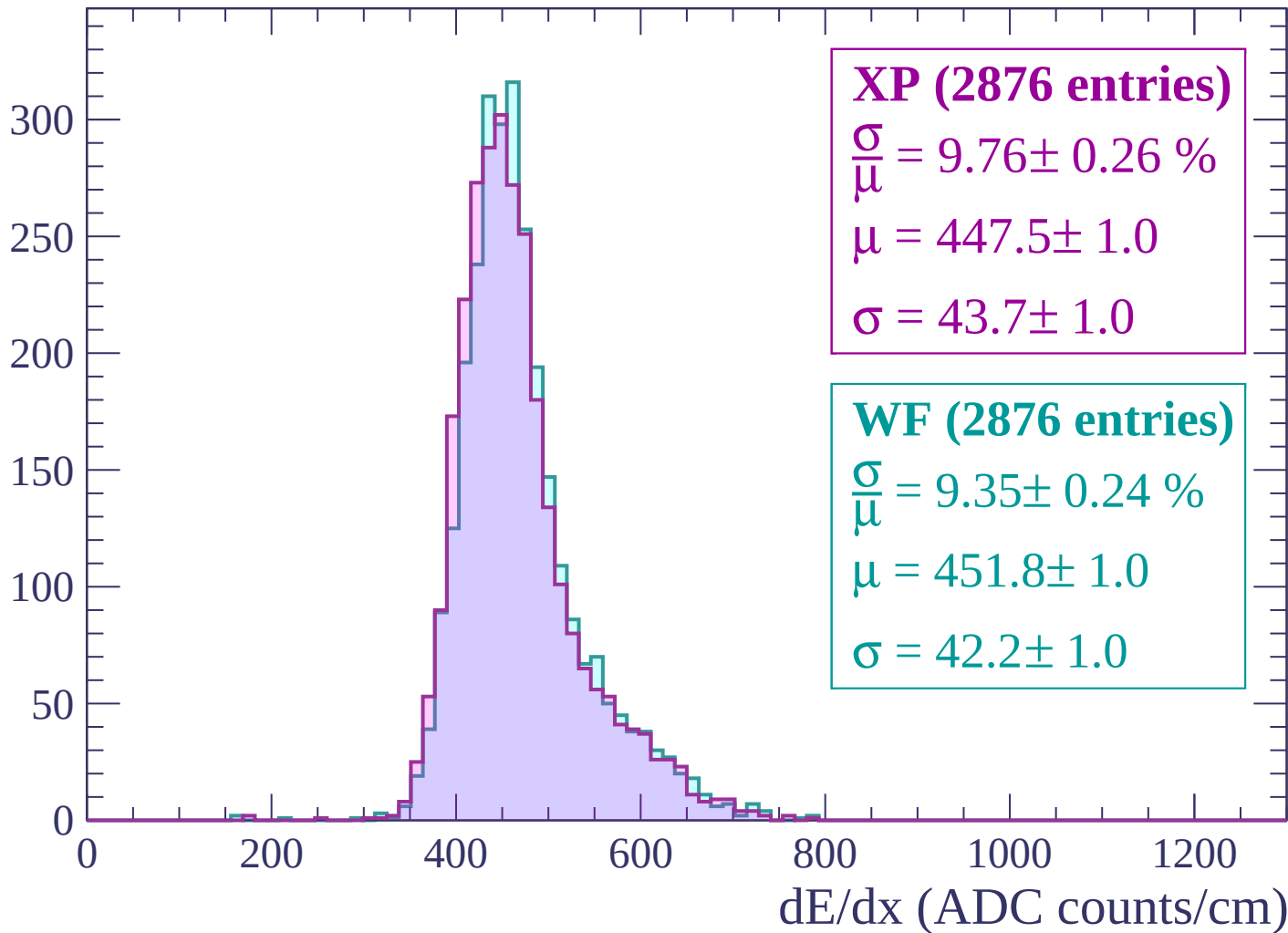
Energy loss | -1260 < p < -1200

Count



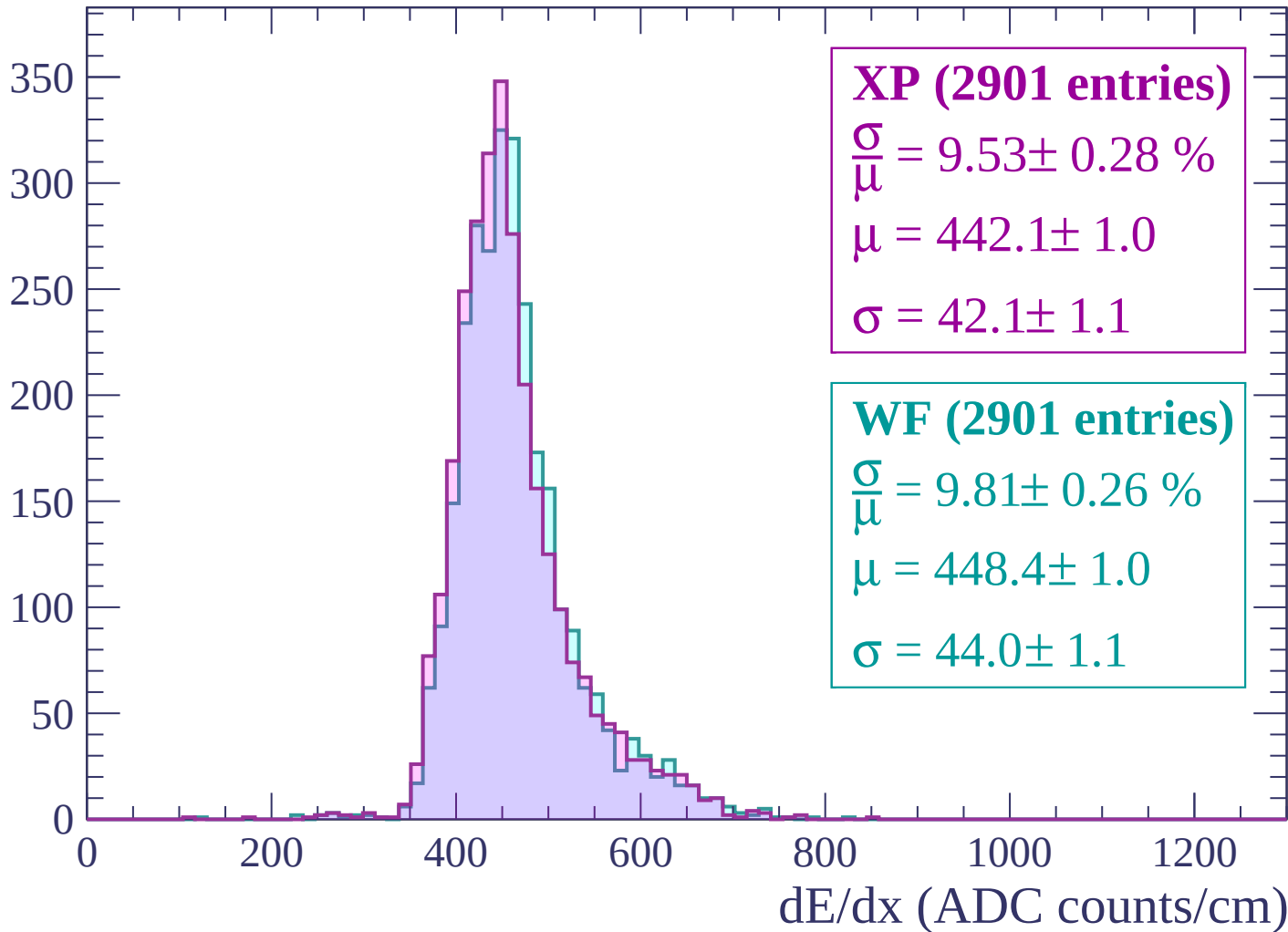
Energy loss | $-1200 < p < -1140$

Count



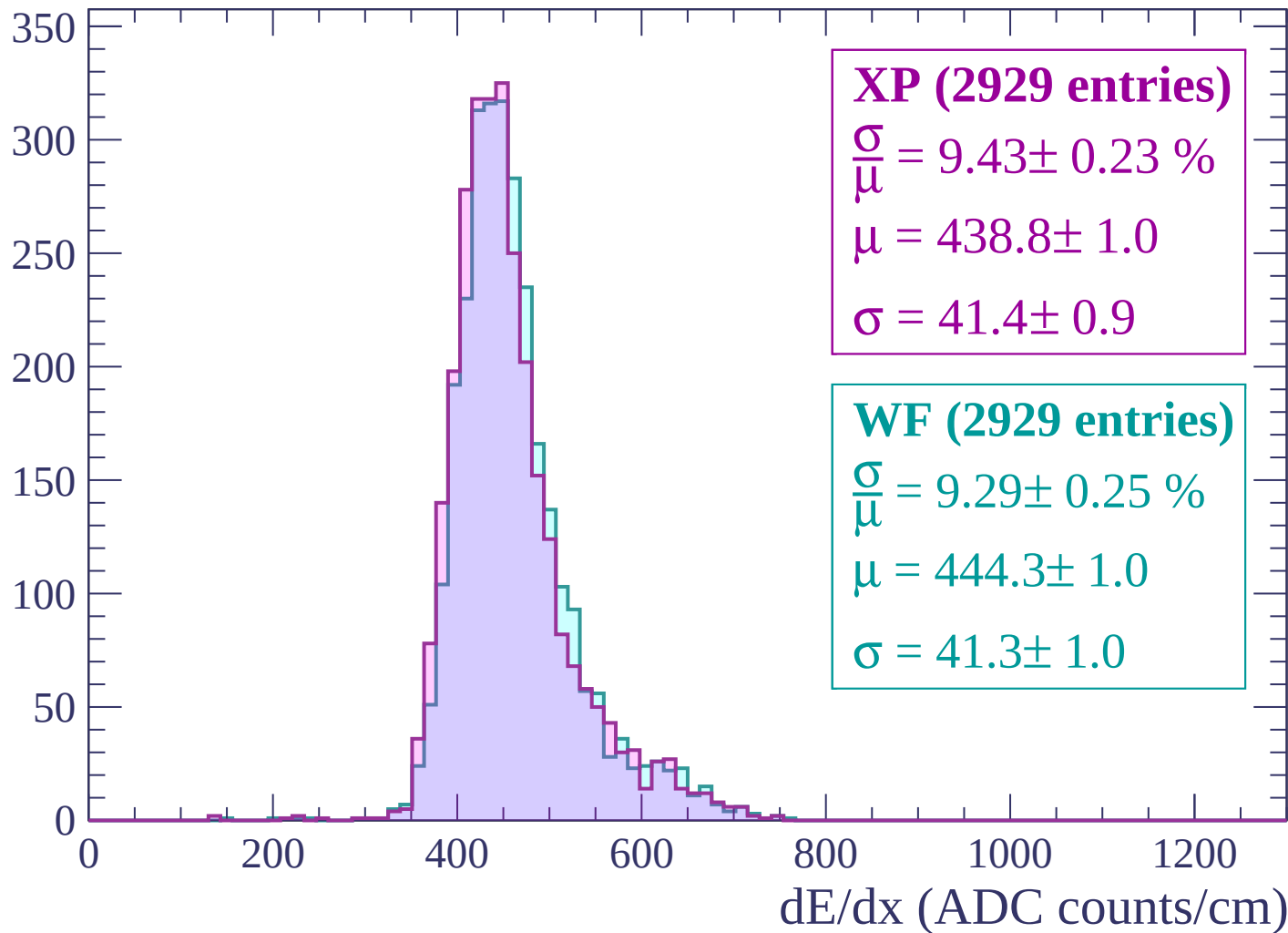
Energy loss | $-1140 < p < -1080$

Count



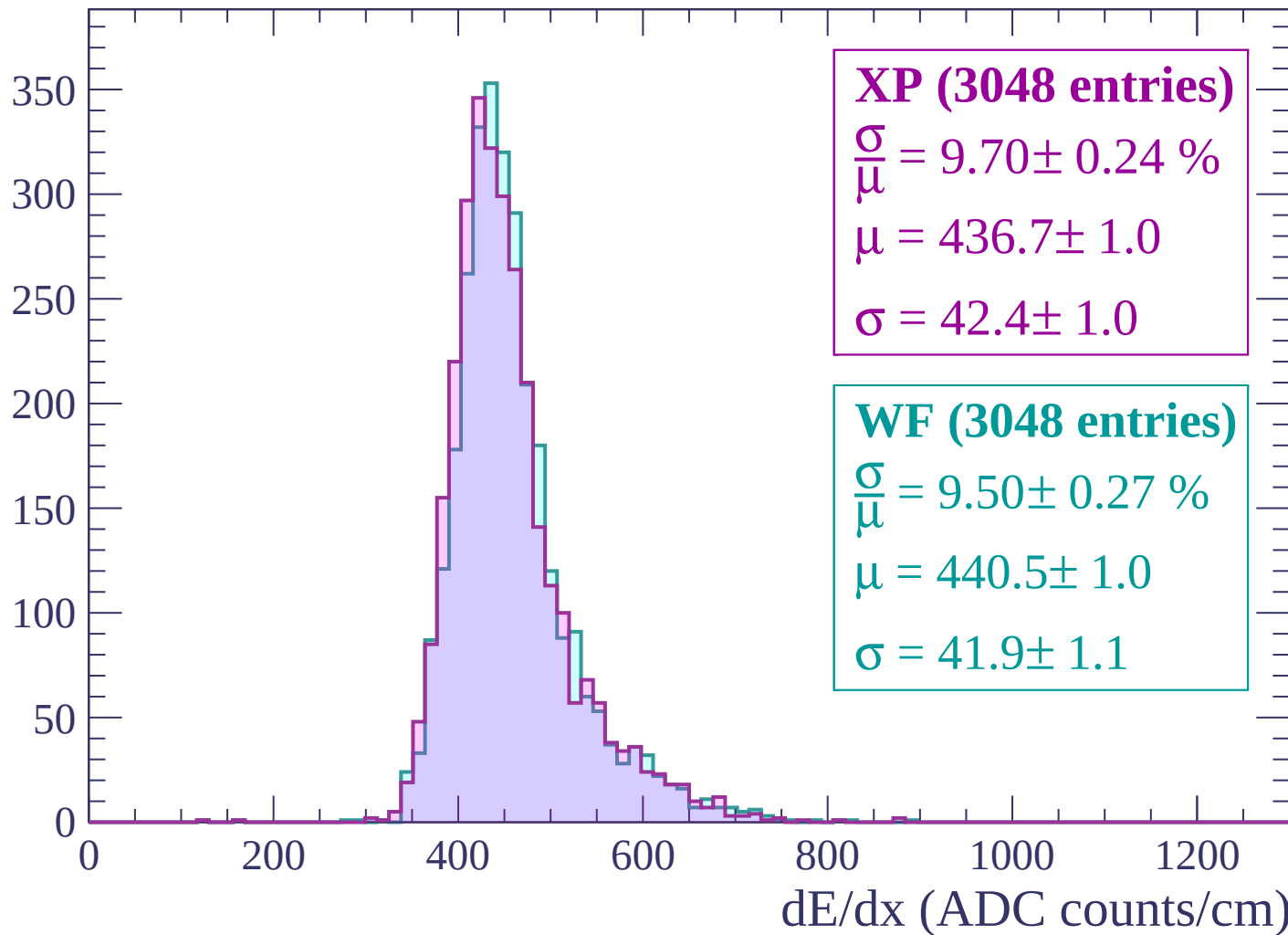
Energy loss | $-1080 < p < -1020$

Count



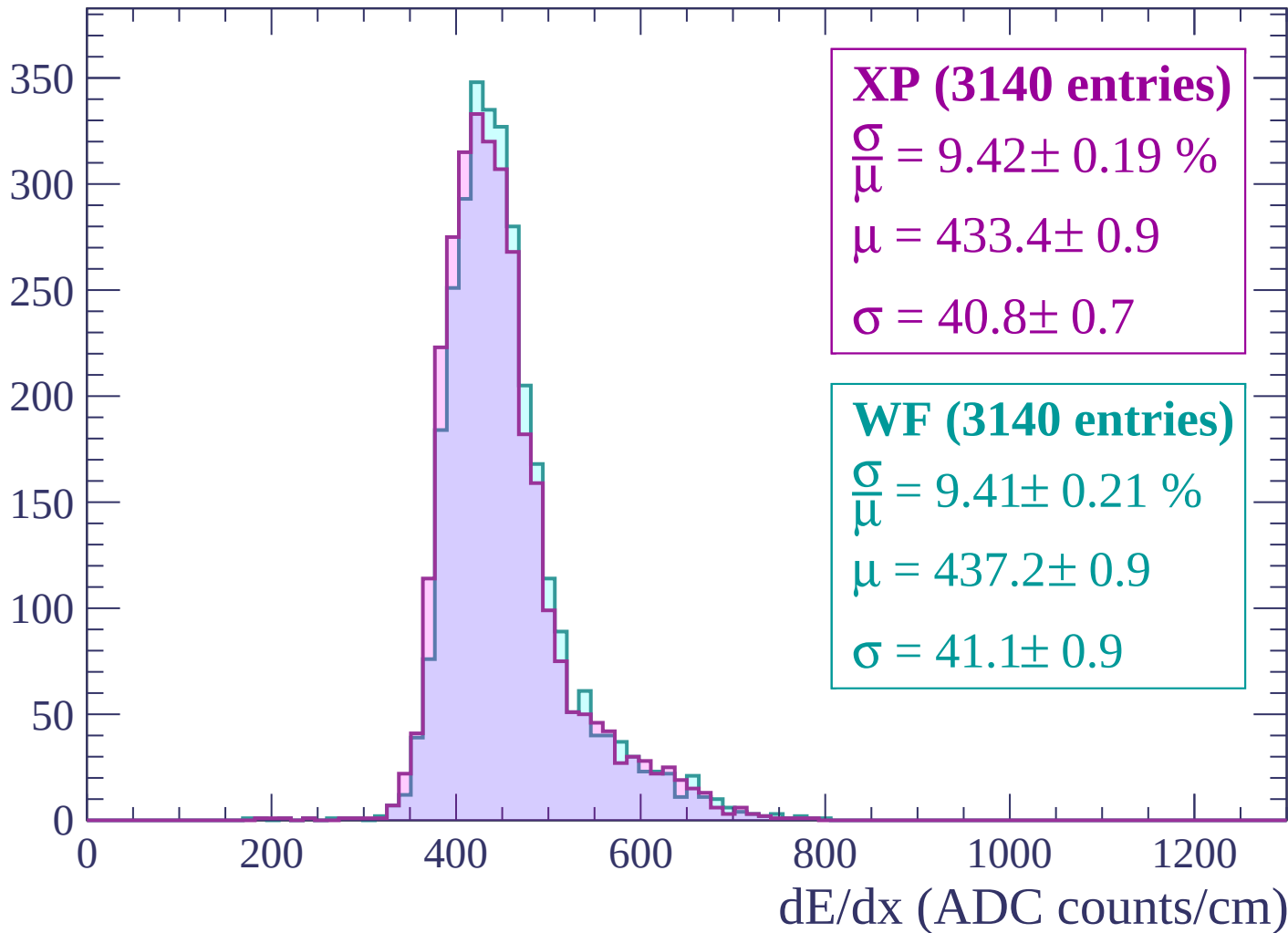
Energy loss | $-1020 < p < -960$

Count



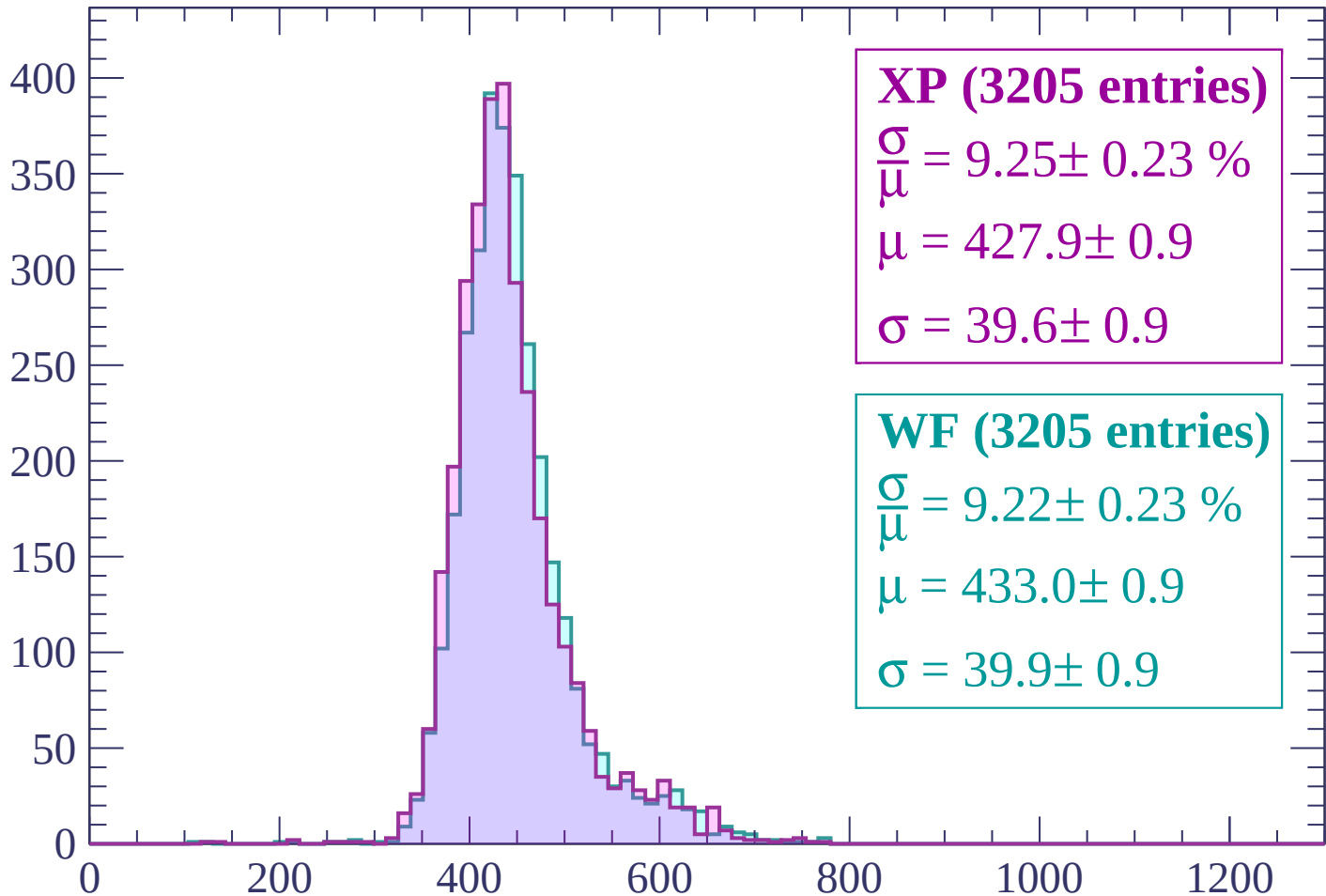
Energy loss | $-960 < p < -900$

Count



Energy loss | $-900 < p < -840$

Count



XP (3205 entries)

$$\frac{\sigma}{\mu} = 9.25 \pm 0.23 \%$$

$$\mu = 427.9 \pm 0.9$$

$$\sigma = 39.6 \pm 0.9$$

WF (3205 entries)

$$\frac{\sigma}{\mu} = 9.22 \pm 0.23 \%$$

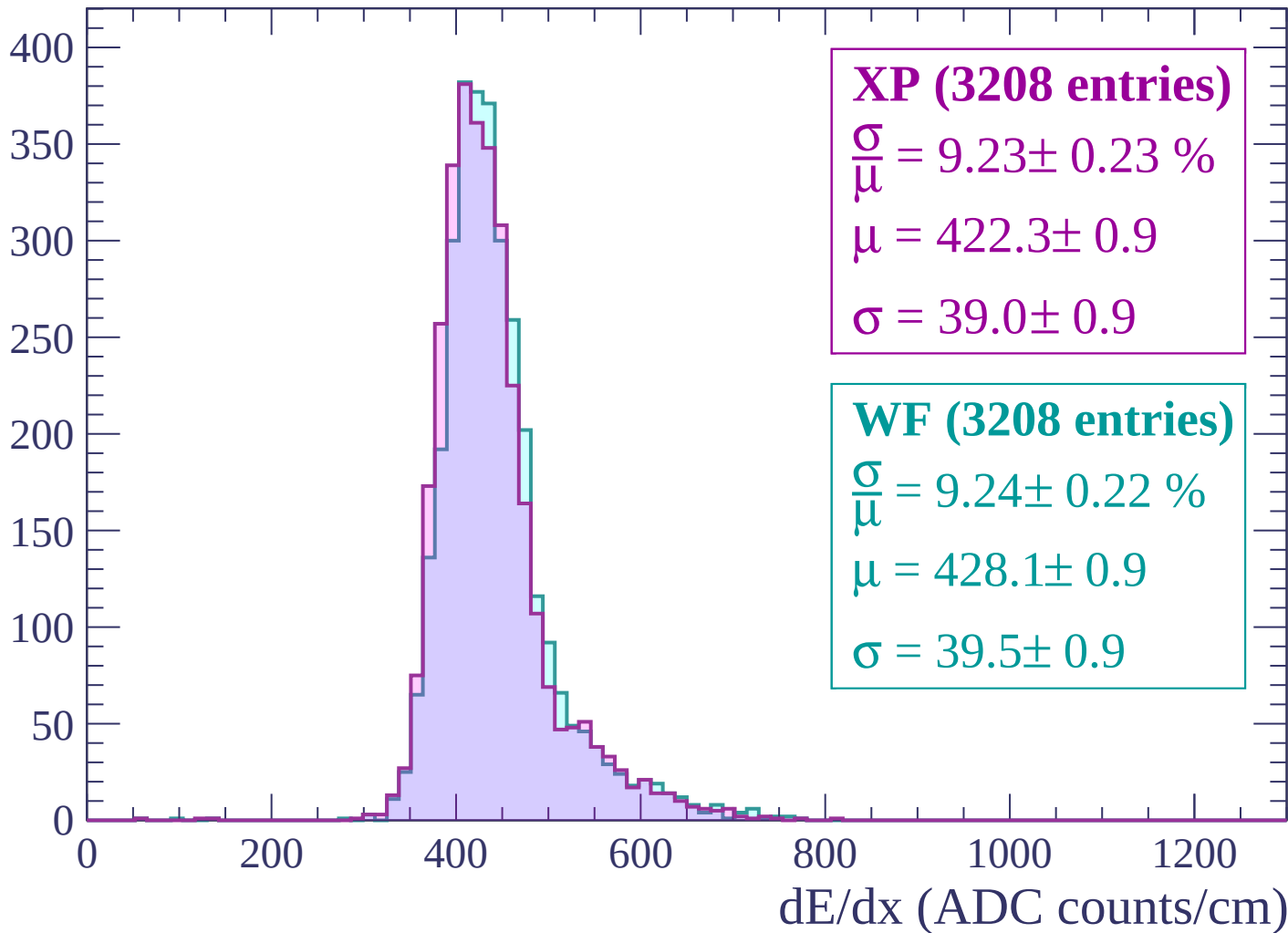
$$\mu = 433.0 \pm 0.9$$

$$\sigma = 39.9 \pm 0.9$$

dE/dx (ADC counts/cm)

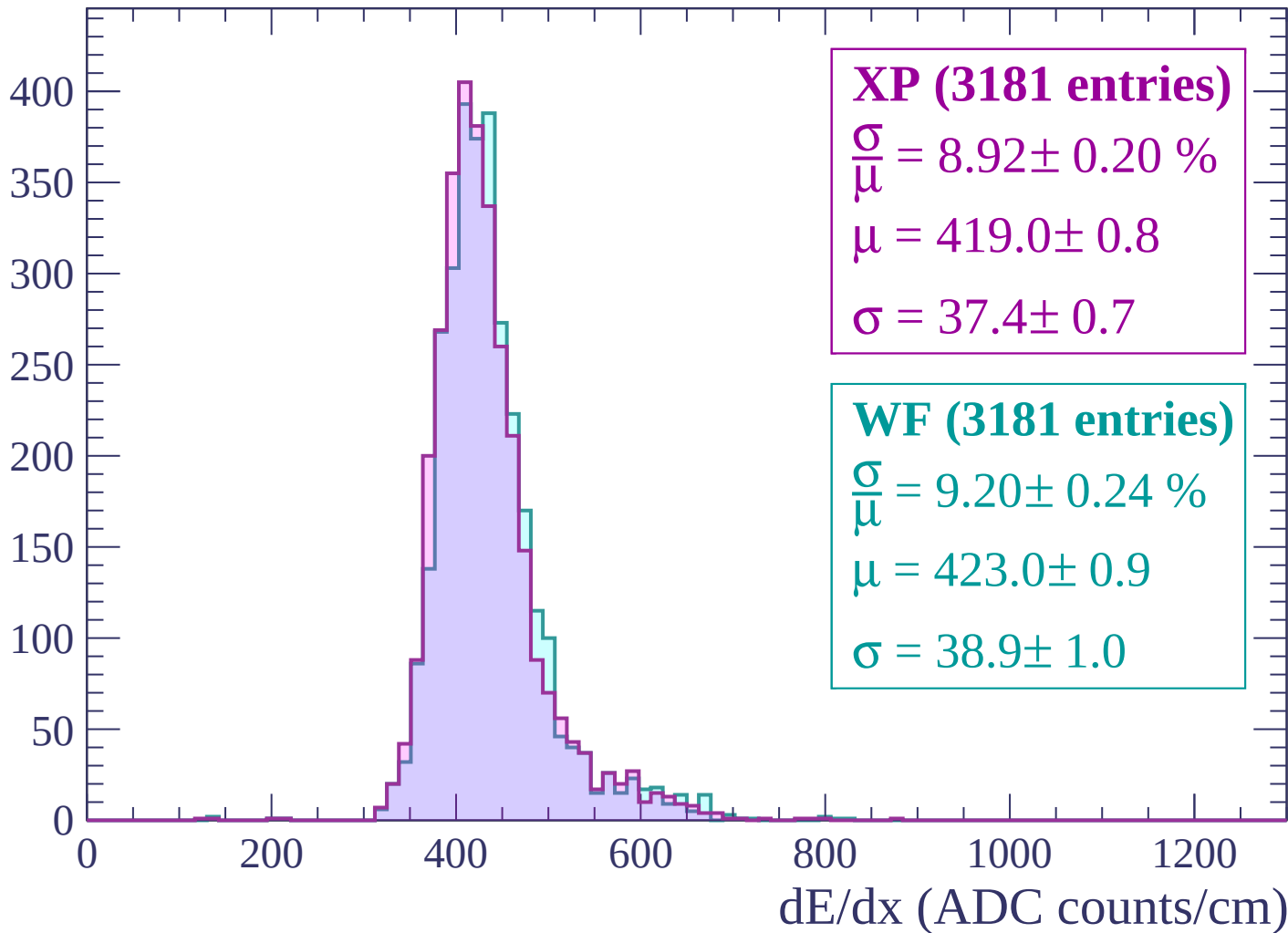
Energy loss | $-840 < p < -780$

Count



Energy loss | $-780 < p < -720$

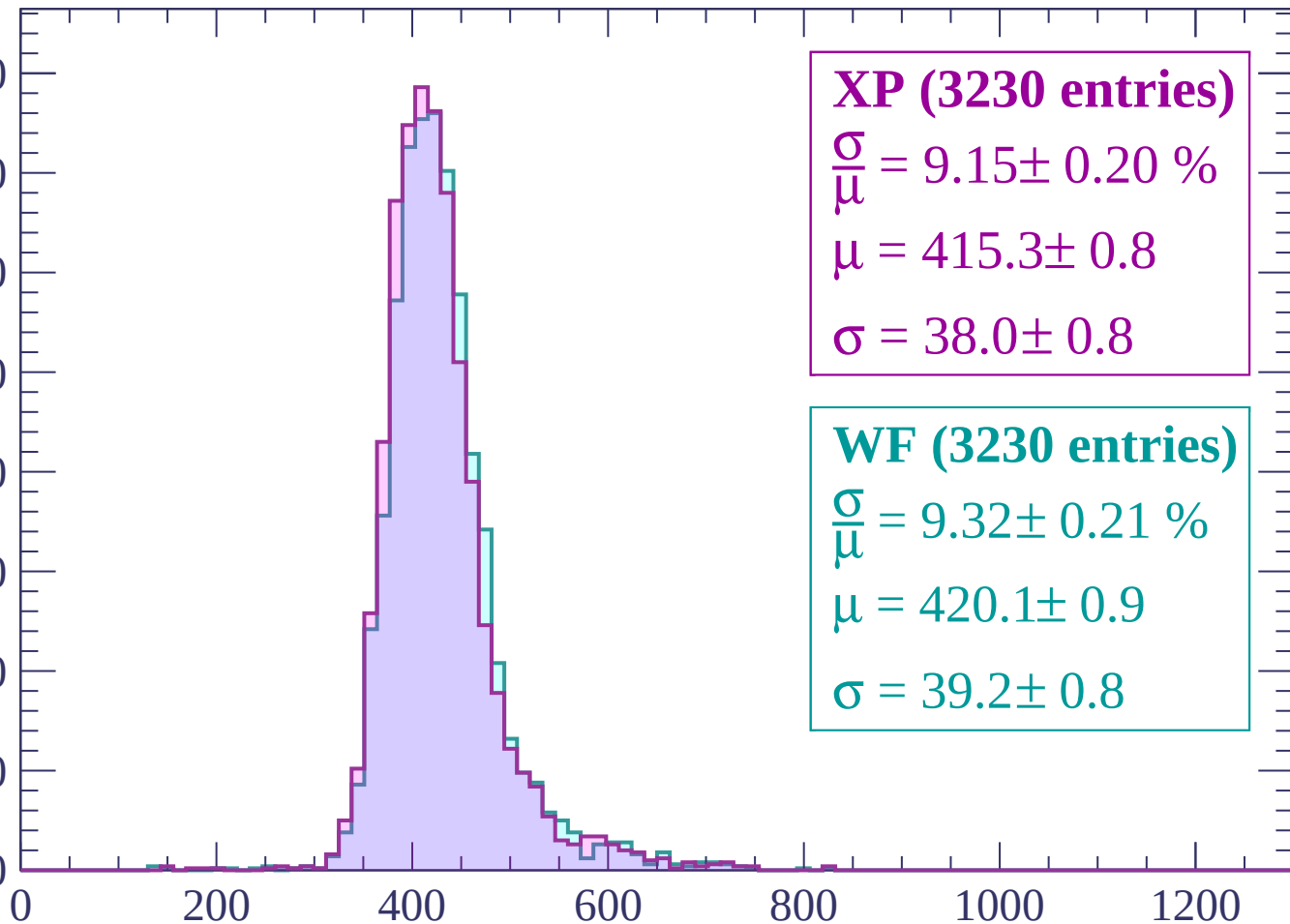
Count



Energy loss | $-720 < p < -660$

Count

400
350
300
250
200
150
100
50
0



XP (3230 entries)

$\frac{\sigma}{\mu} = 9.15 \pm 0.20 \%$

$\mu = 415.3 \pm 0.8$

$\sigma = 38.0 \pm 0.8$

WF (3230 entries)

$\frac{\sigma}{\mu} = 9.32 \pm 0.21 \%$

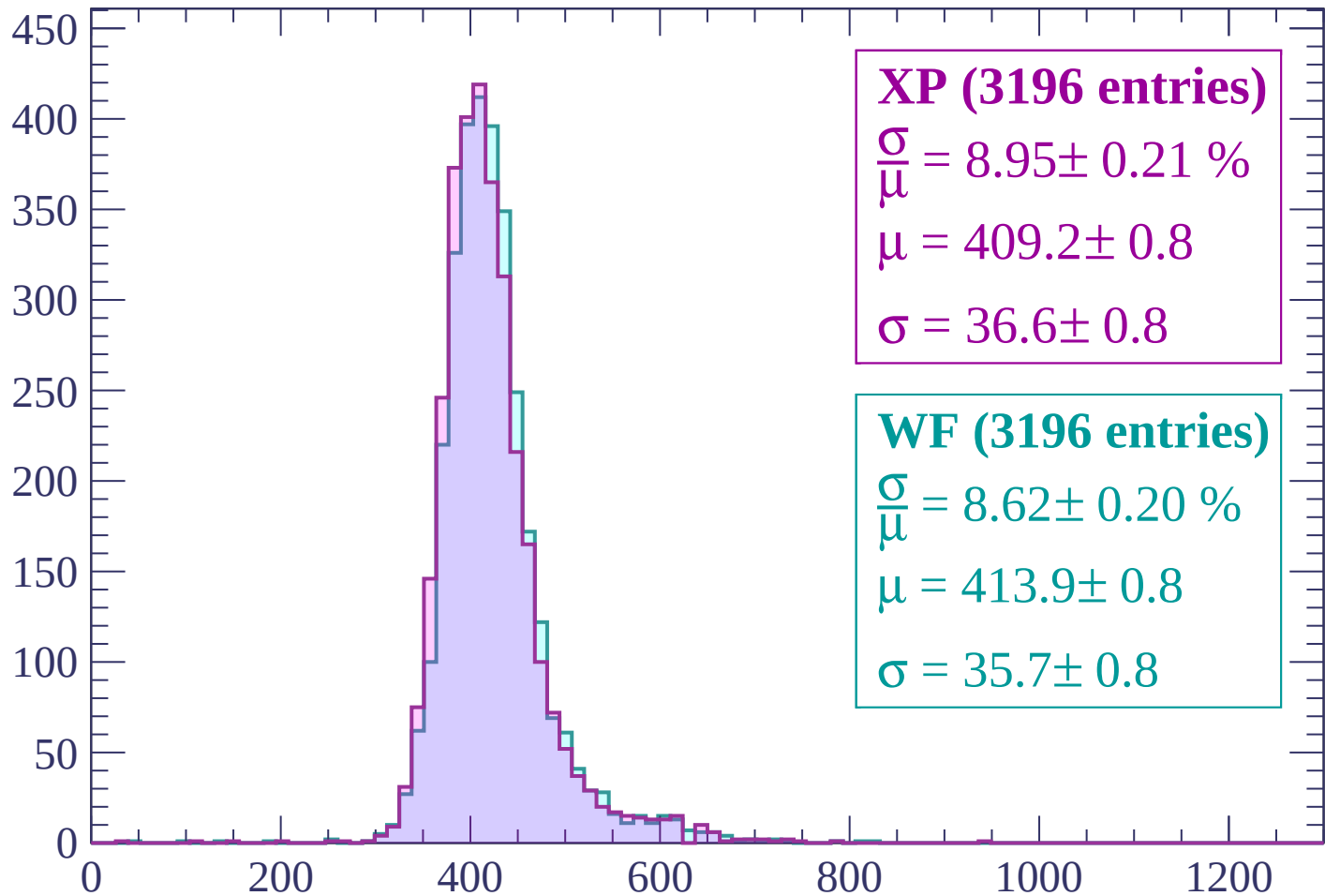
$\mu = 420.1 \pm 0.9$

$\sigma = 39.2 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count



XP (3196 entries)

$$\frac{\sigma}{\mu} = 8.95 \pm 0.21 \%$$

$$\mu = 409.2 \pm 0.8$$

$$\sigma = 36.6 \pm 0.8$$

WF (3196 entries)

$$\frac{\sigma}{\mu} = 8.62 \pm 0.20 \%$$

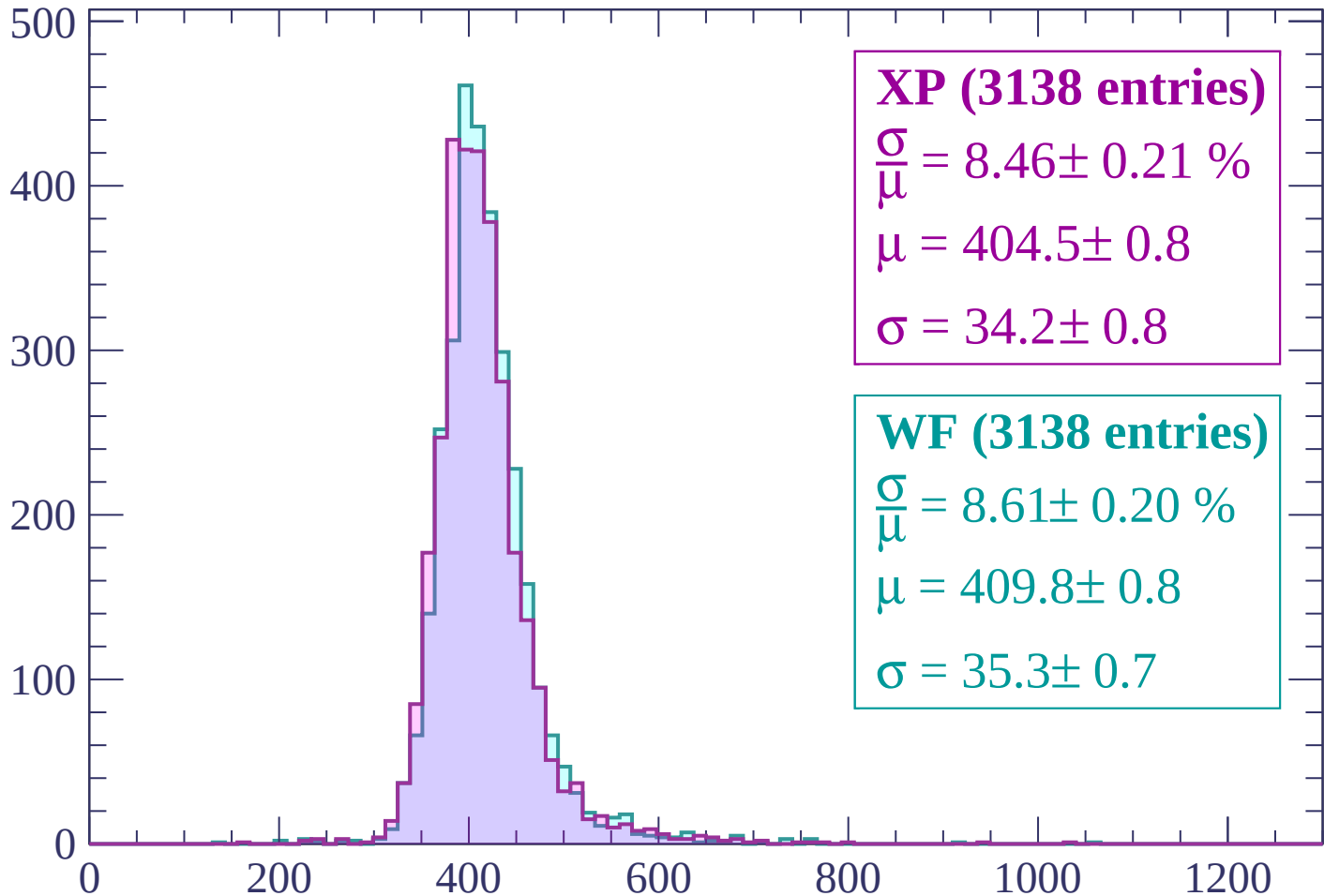
$$\mu = 413.9 \pm 0.8$$

$$\sigma = 35.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count



XP (3138 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.21 \%$$

$$\mu = 404.5 \pm 0.8$$

$$\sigma = 34.2 \pm 0.8$$

WF (3138 entries)

$$\frac{\sigma}{\mu} = 8.61 \pm 0.20 \%$$

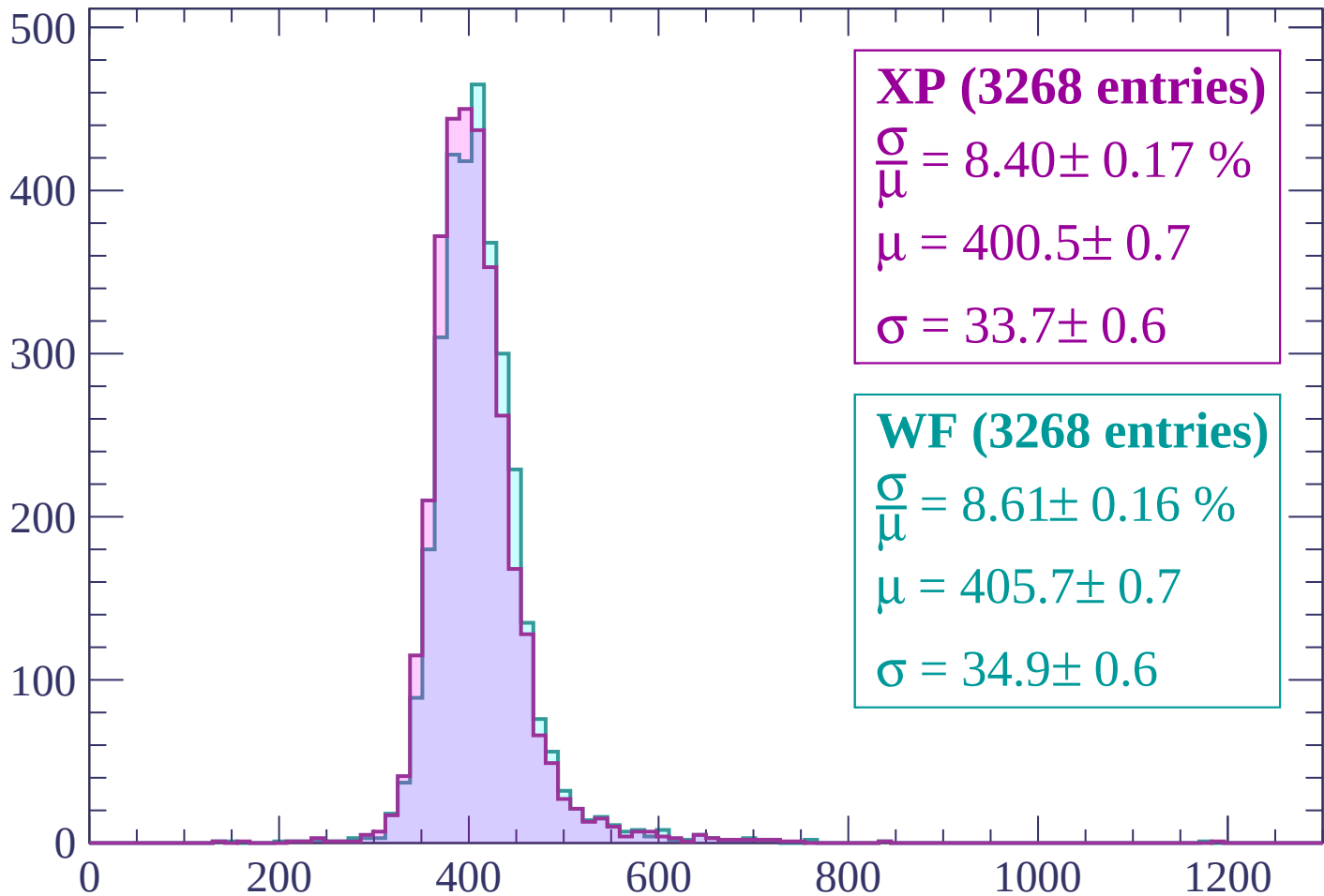
$$\mu = 409.8 \pm 0.8$$

$$\sigma = 35.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP (3268 entries)

$$\frac{\sigma}{\mu} = 8.40 \pm 0.17 \%$$

$$\mu = 400.5 \pm 0.7$$

$$\sigma = 33.7 \pm 0.6$$

WF (3268 entries)

$$\frac{\sigma}{\mu} = 8.61 \pm 0.16 \%$$

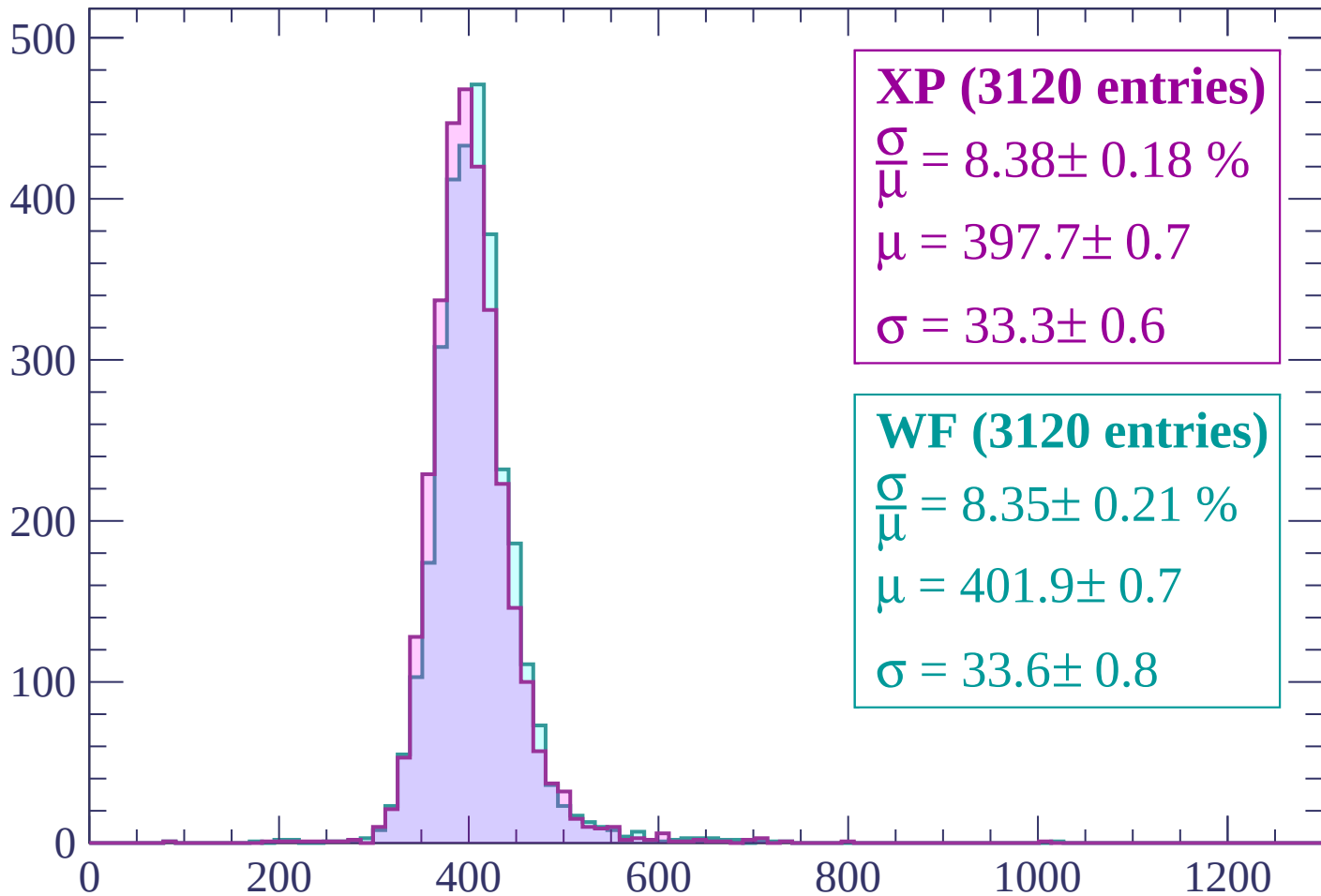
$$\mu = 405.7 \pm 0.7$$

$$\sigma = 34.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



XP (3120 entries)

$$\frac{\sigma}{\mu} = 8.38 \pm 0.18 \%$$

$$\mu = 397.7 \pm 0.7$$

$$\sigma = 33.3 \pm 0.6$$

WF (3120 entries)

$$\frac{\sigma}{\mu} = 8.35 \pm 0.21 \%$$

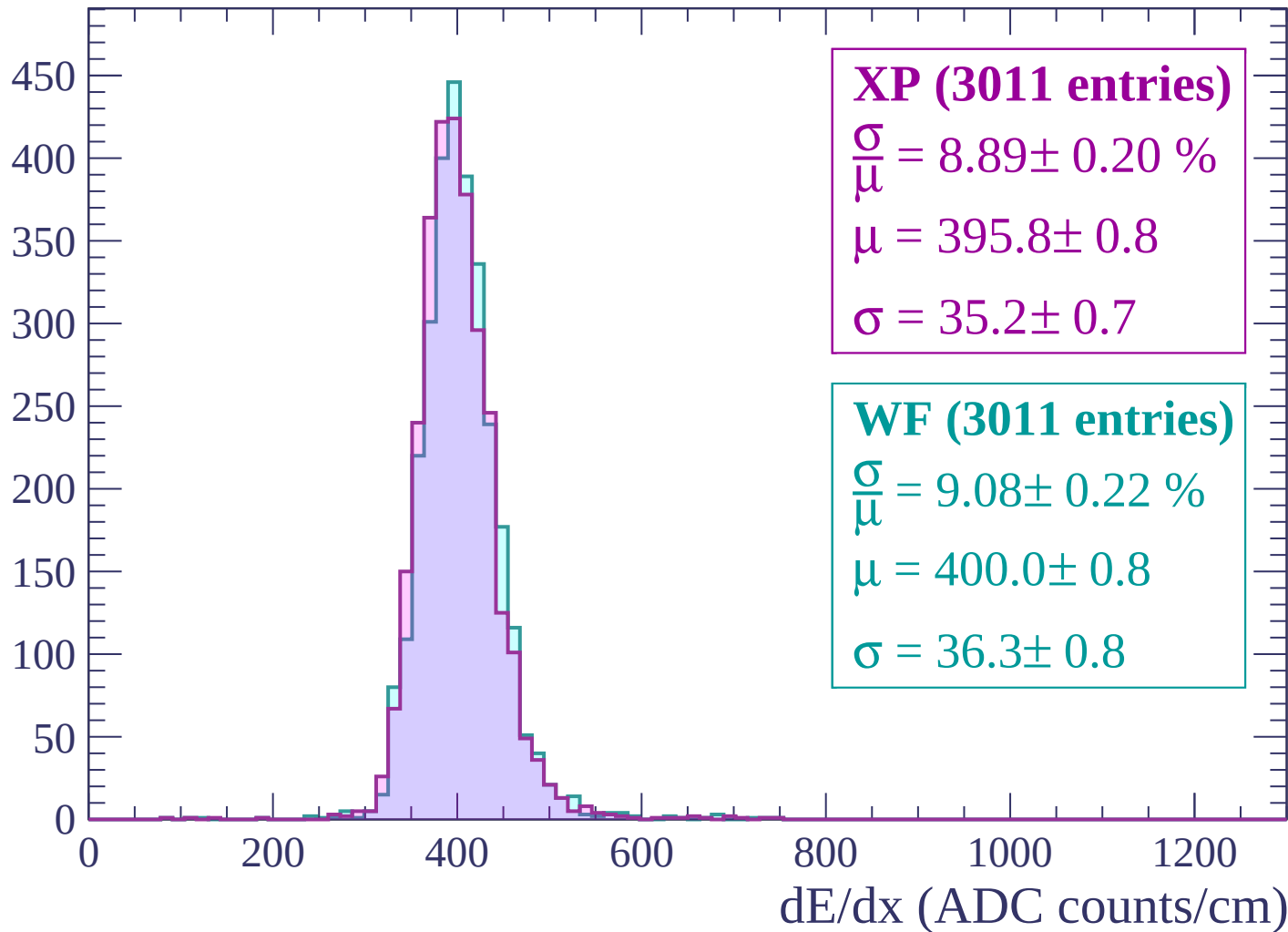
$$\mu = 401.9 \pm 0.7$$

$$\sigma = 33.6 \pm 0.8$$

dE/dx (ADC counts/cm)

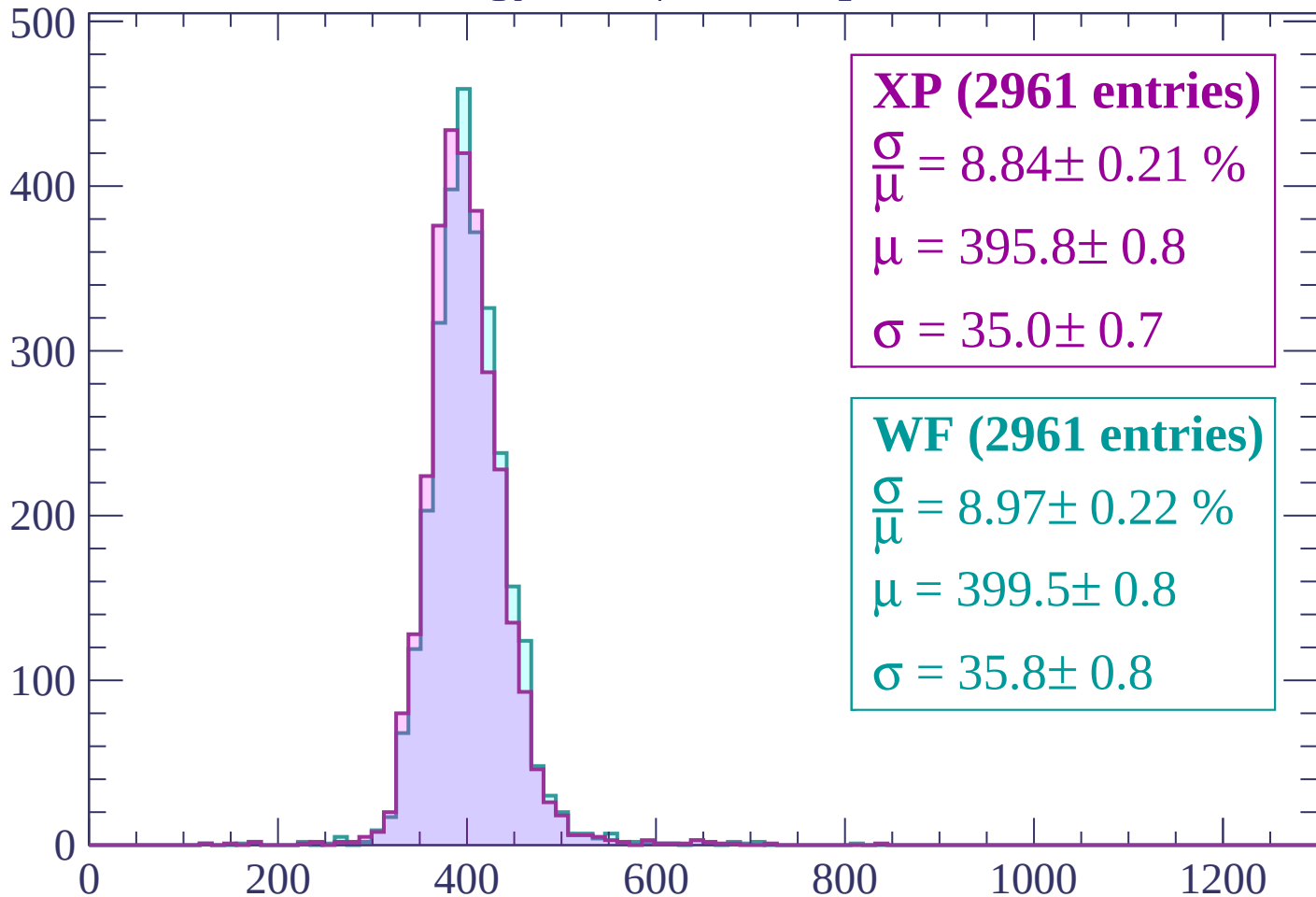
Energy loss | $-420 < p < -360$

Count



Energy loss | $-360 < p < -300$

Count



XP (2961 entries)

$$\frac{\sigma}{\mu} = 8.84 \pm 0.21 \%$$

$$\mu = 395.8 \pm 0.8$$

$$\sigma = 35.0 \pm 0.7$$

WF (2961 entries)

$$\frac{\sigma}{\mu} = 8.97 \pm 0.22 \%$$

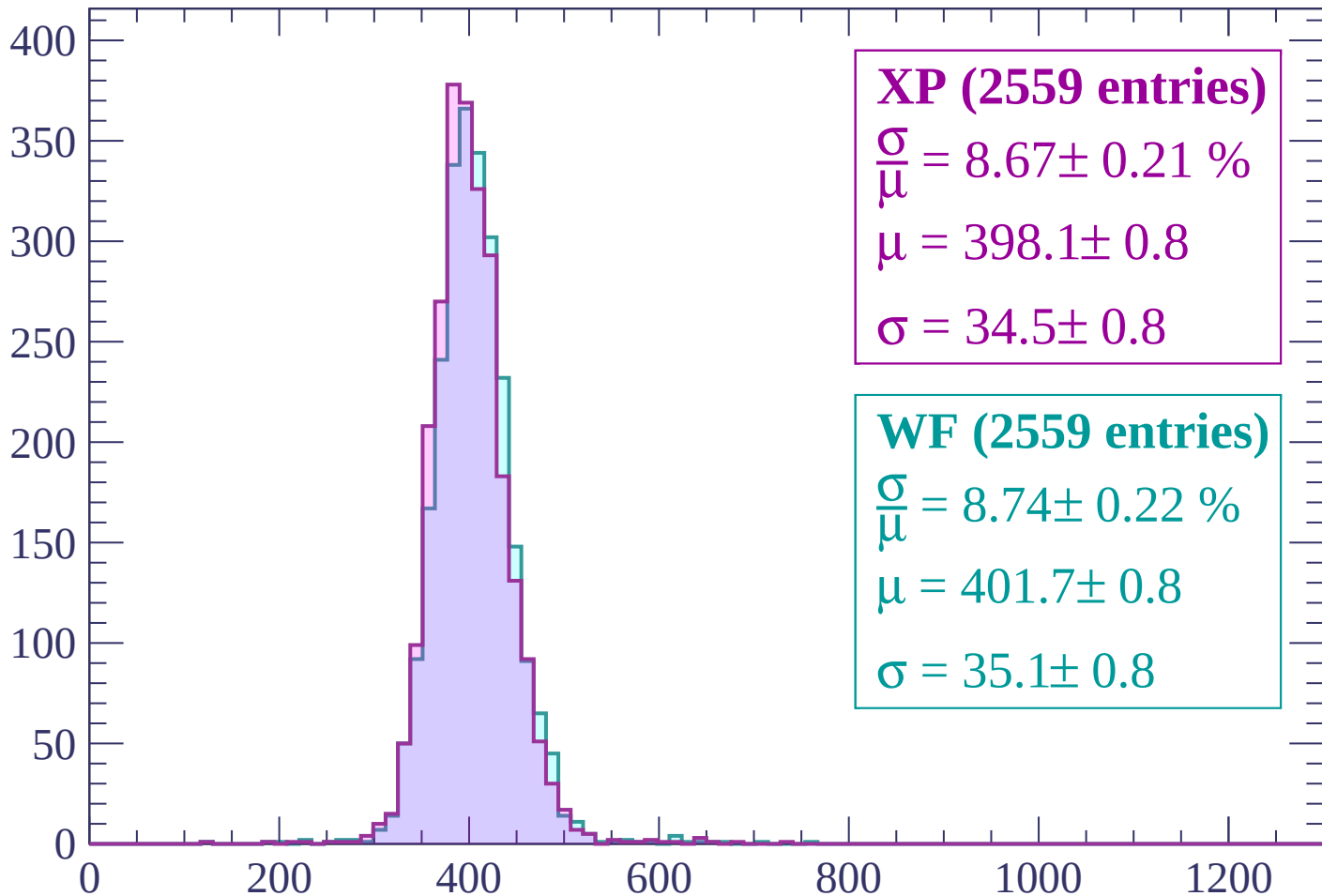
$$\mu = 399.5 \pm 0.8$$

$$\sigma = 35.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (2559 entries)

$\frac{\sigma}{\mu} = 8.67 \pm 0.21 \%$

$\mu = 398.1 \pm 0.8$

$\sigma = 34.5 \pm 0.8$

WF (2559 entries)

$\frac{\sigma}{\mu} = 8.74 \pm 0.22 \%$

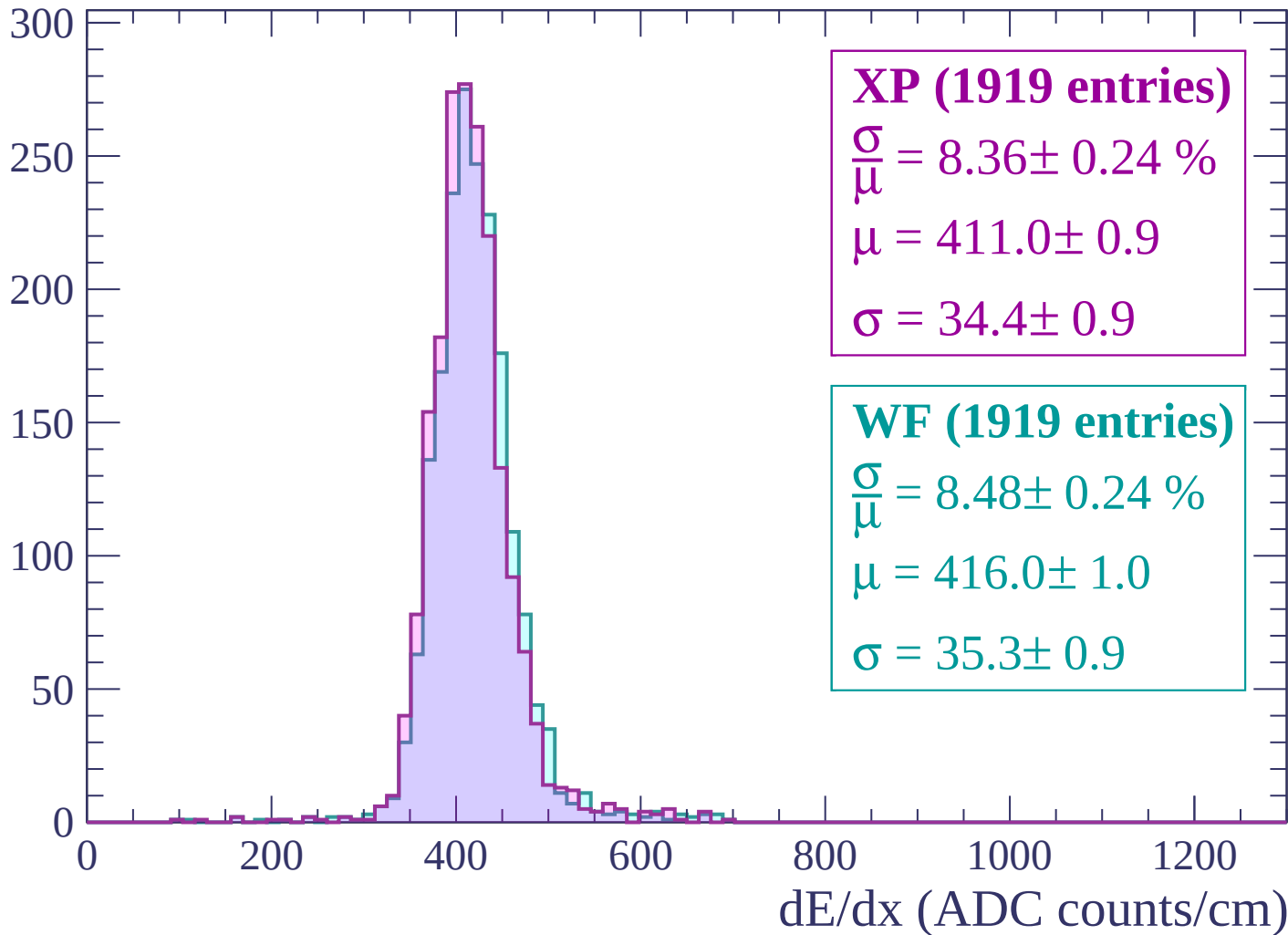
$\mu = 401.7 \pm 0.8$

$\sigma = 35.1 \pm 0.8$

dE/dx (ADC counts/cm)

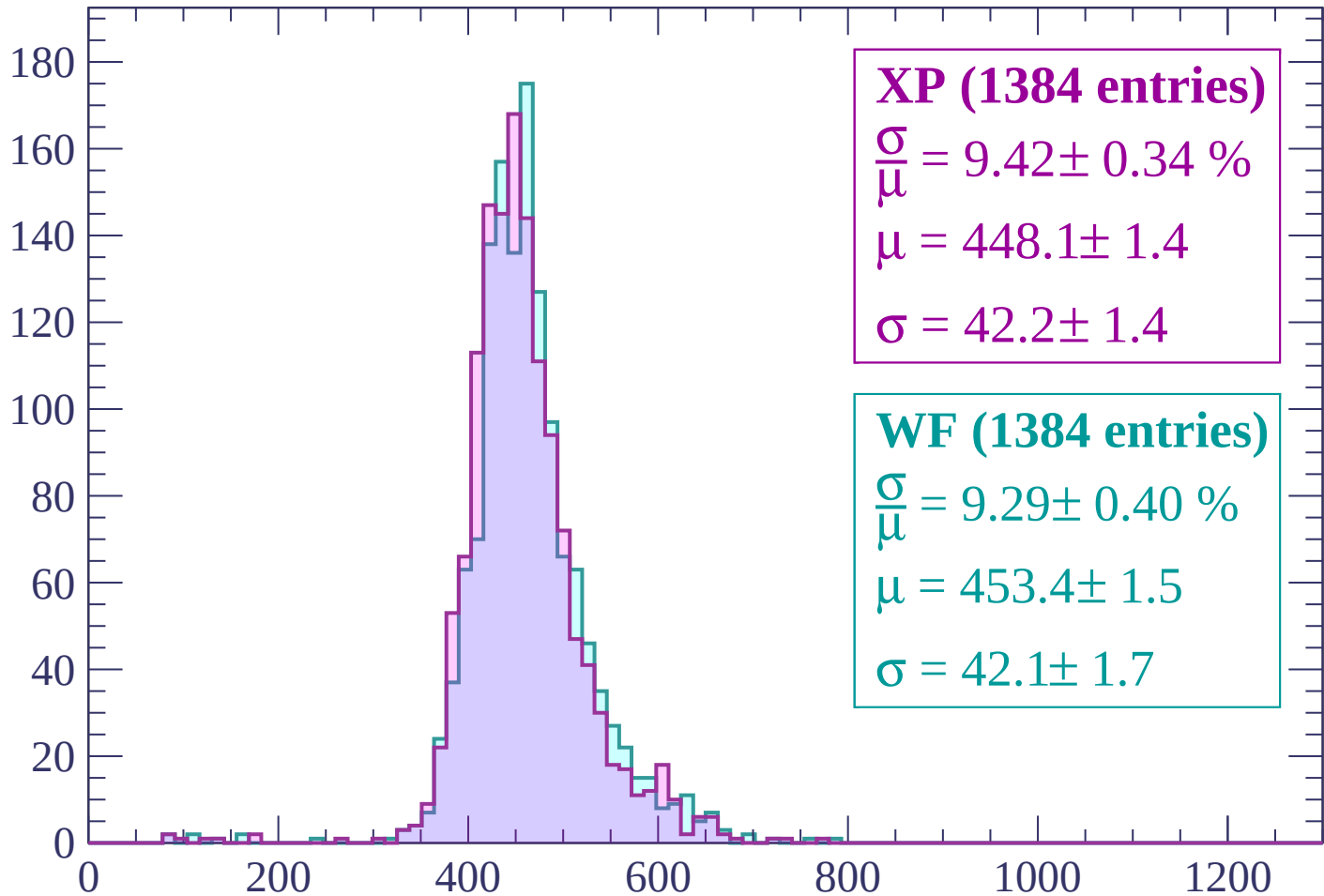
Energy loss | $-240 < p < -180$

Count



Energy loss | $-180 < p < -120$

Count



XP (1384 entries)

$\frac{\sigma}{\mu} = 9.42 \pm 0.34 \%$

$\mu = 448.1 \pm 1.4$

$\sigma = 42.2 \pm 1.4$

WF (1384 entries)

$\frac{\sigma}{\mu} = 9.29 \pm 0.40 \%$

$\mu = 453.4 \pm 1.5$

$\sigma = 42.1 \pm 1.7$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -60$

Count

XP (1008 entries)

$$\frac{\sigma}{\mu} = 11.18 \pm 0.38 \%$$

$$\mu = 560.8 \pm 2.5$$

$$\sigma = 62.7 \pm 1.9$$

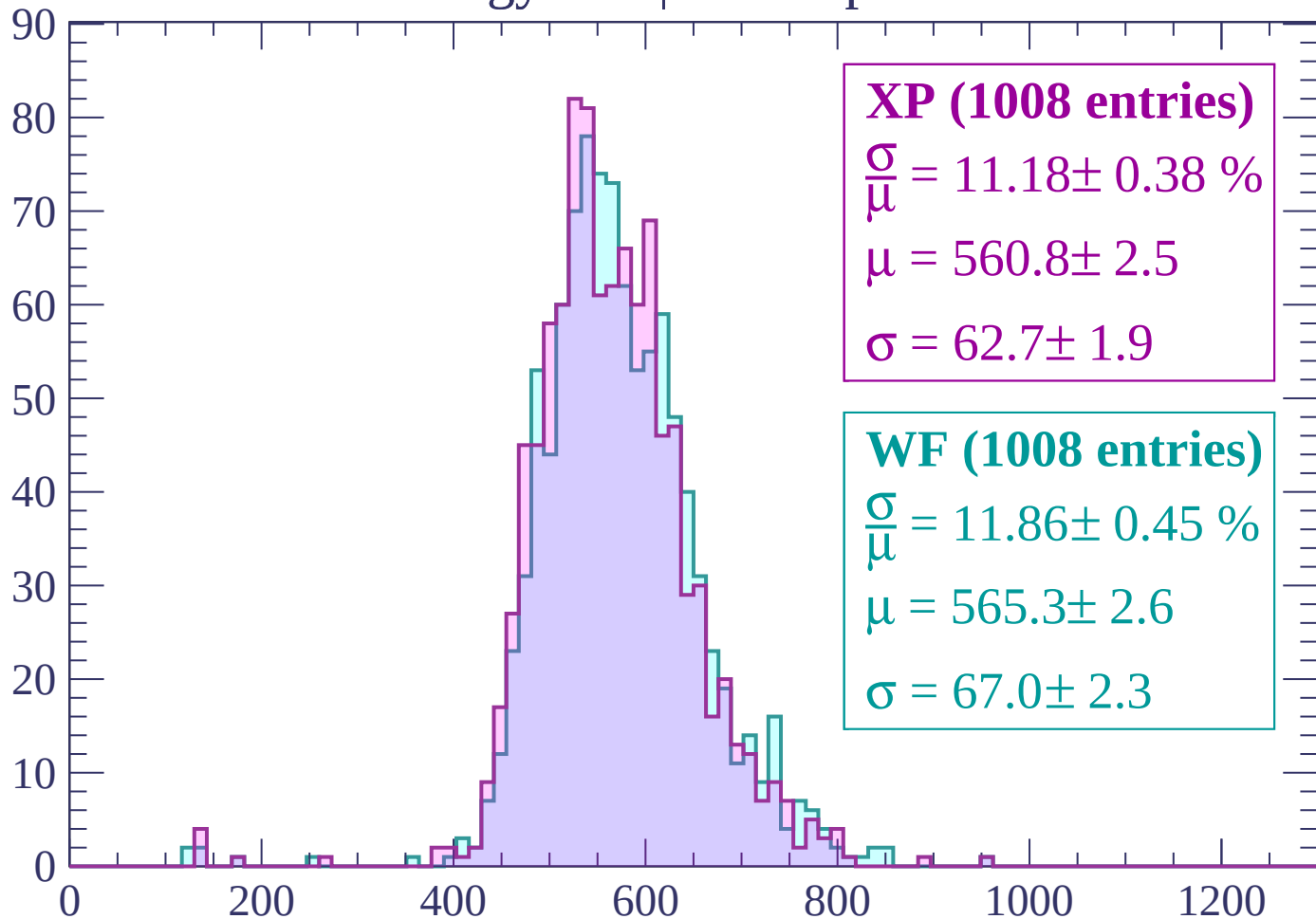
WF (1008 entries)

$$\frac{\sigma}{\mu} = 11.86 \pm 0.45 \%$$

$$\mu = 565.3 \pm 2.6$$

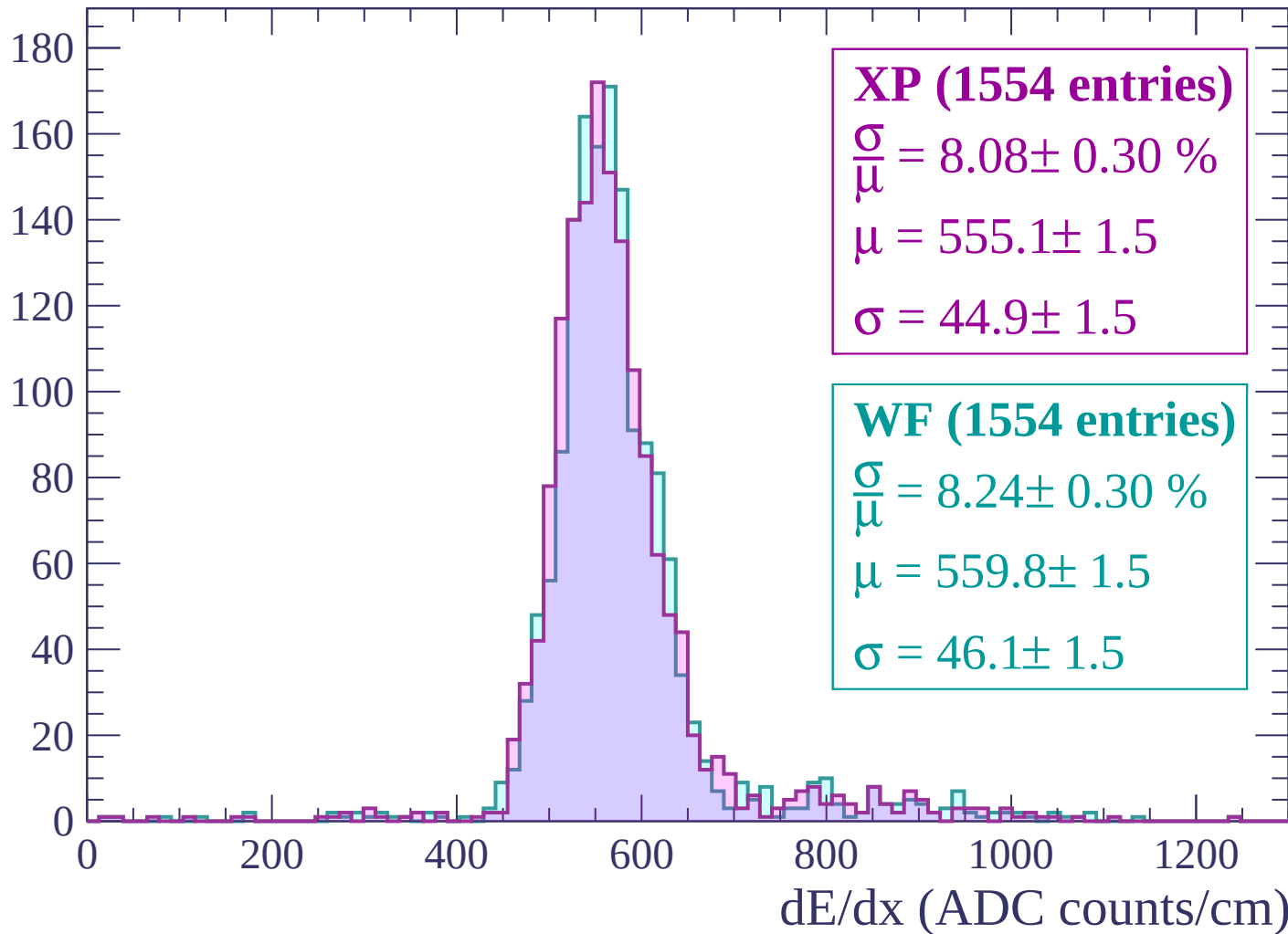
$$\sigma = 67.0 \pm 2.3$$

dE/dx (ADC counts/cm)



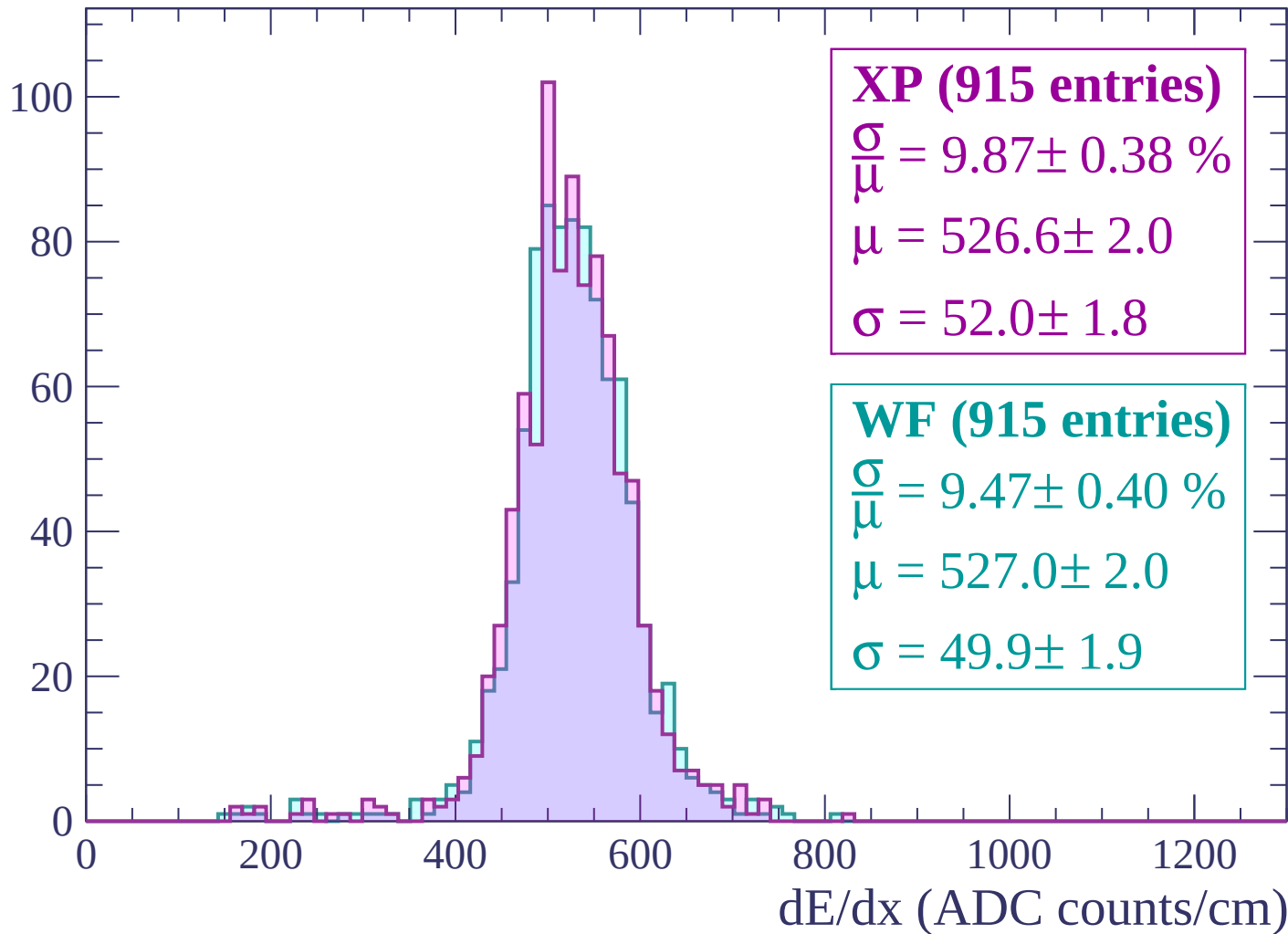
Energy loss | $-60 < p < 0$

Count



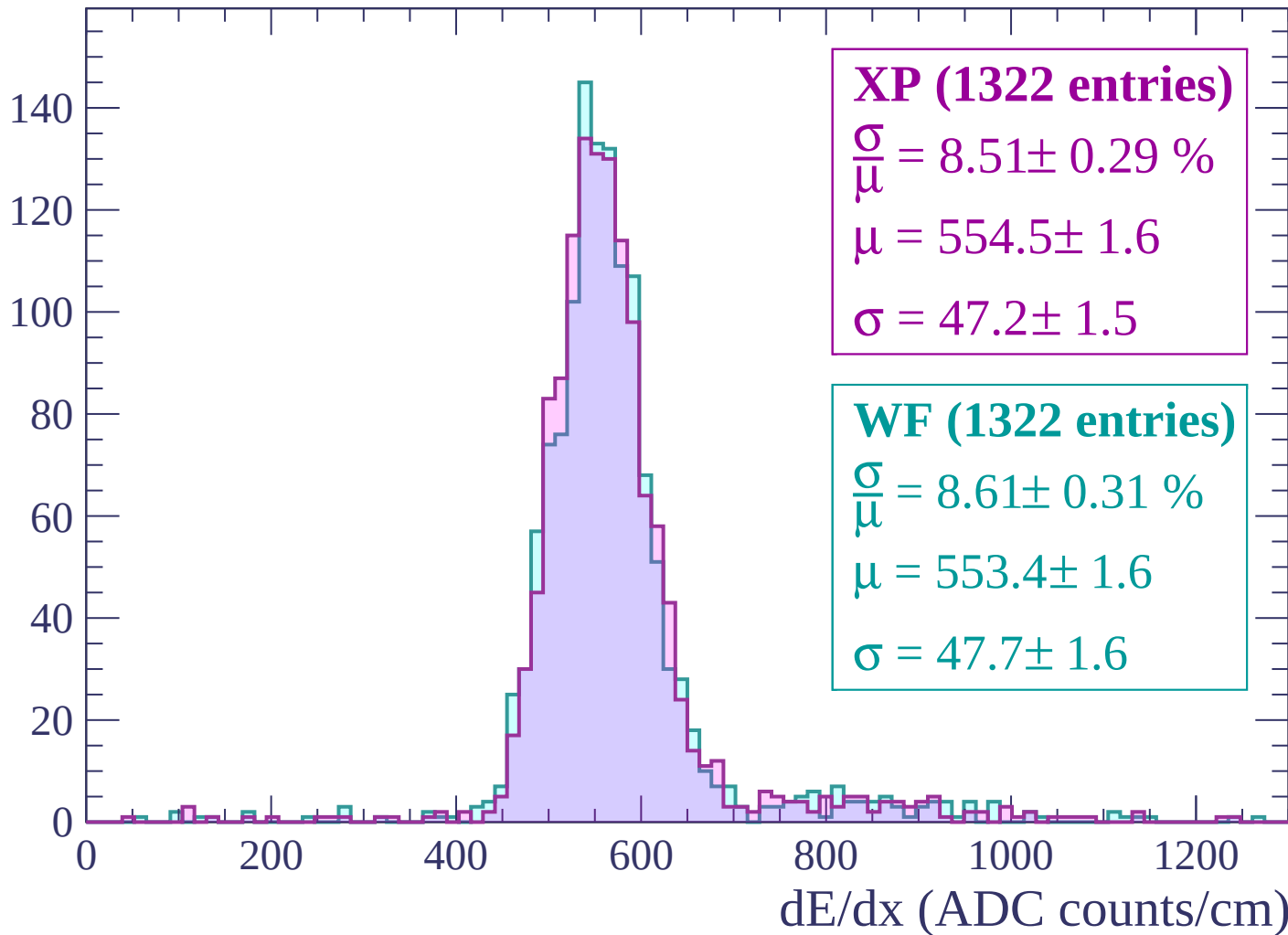
Energy loss | $0 < p < 60$

Count



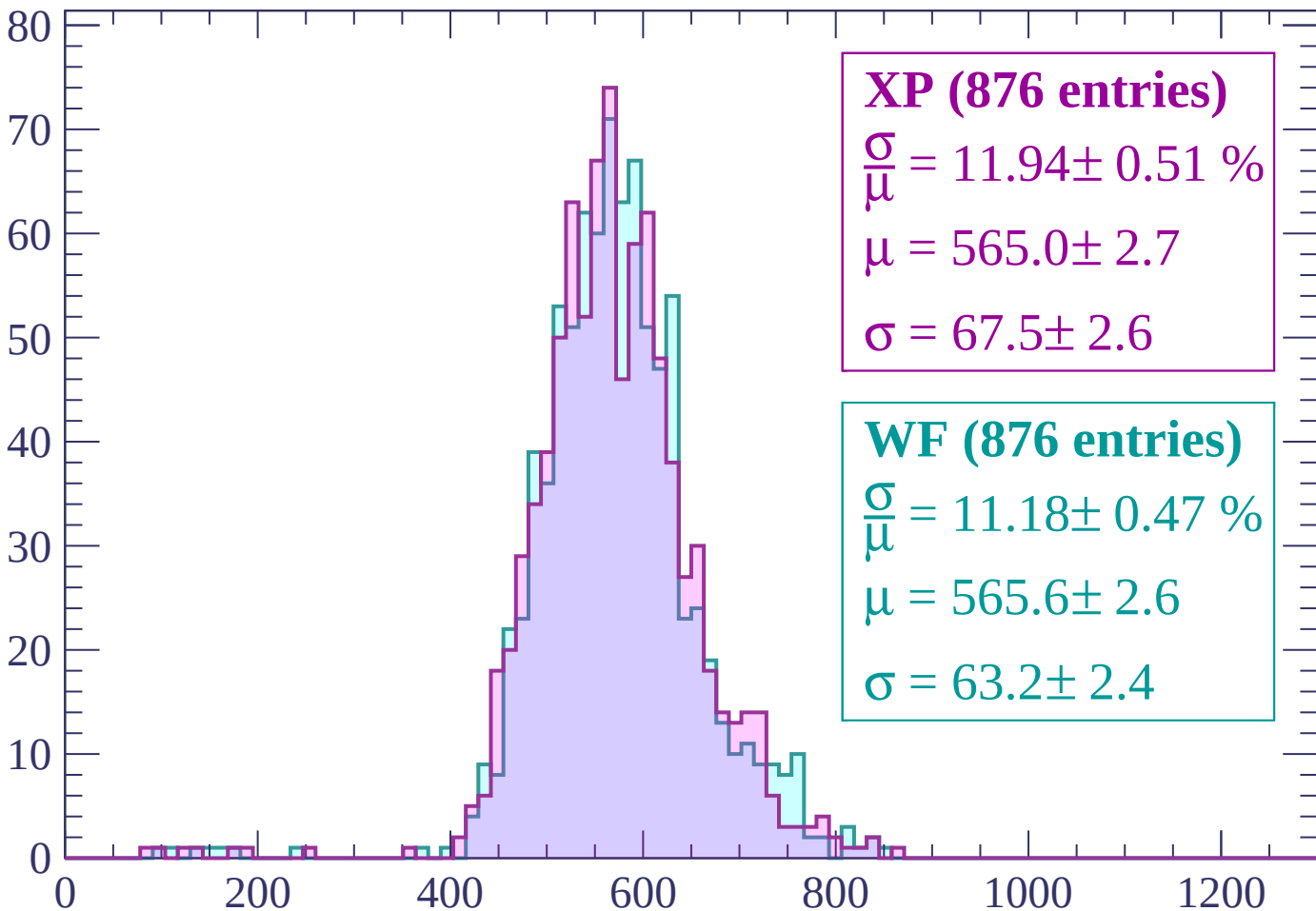
Energy loss | $60 < p < 120$

Count



Energy loss | $120 < p < 180$

Count



XP (876 entries)

$$\frac{\sigma}{\mu} = 11.94 \pm 0.51 \%$$

$$\mu = 565.0 \pm 2.7$$

$$\sigma = 67.5 \pm 2.6$$

WF (876 entries)

$$\frac{\sigma}{\mu} = 11.18 \pm 0.47 \%$$

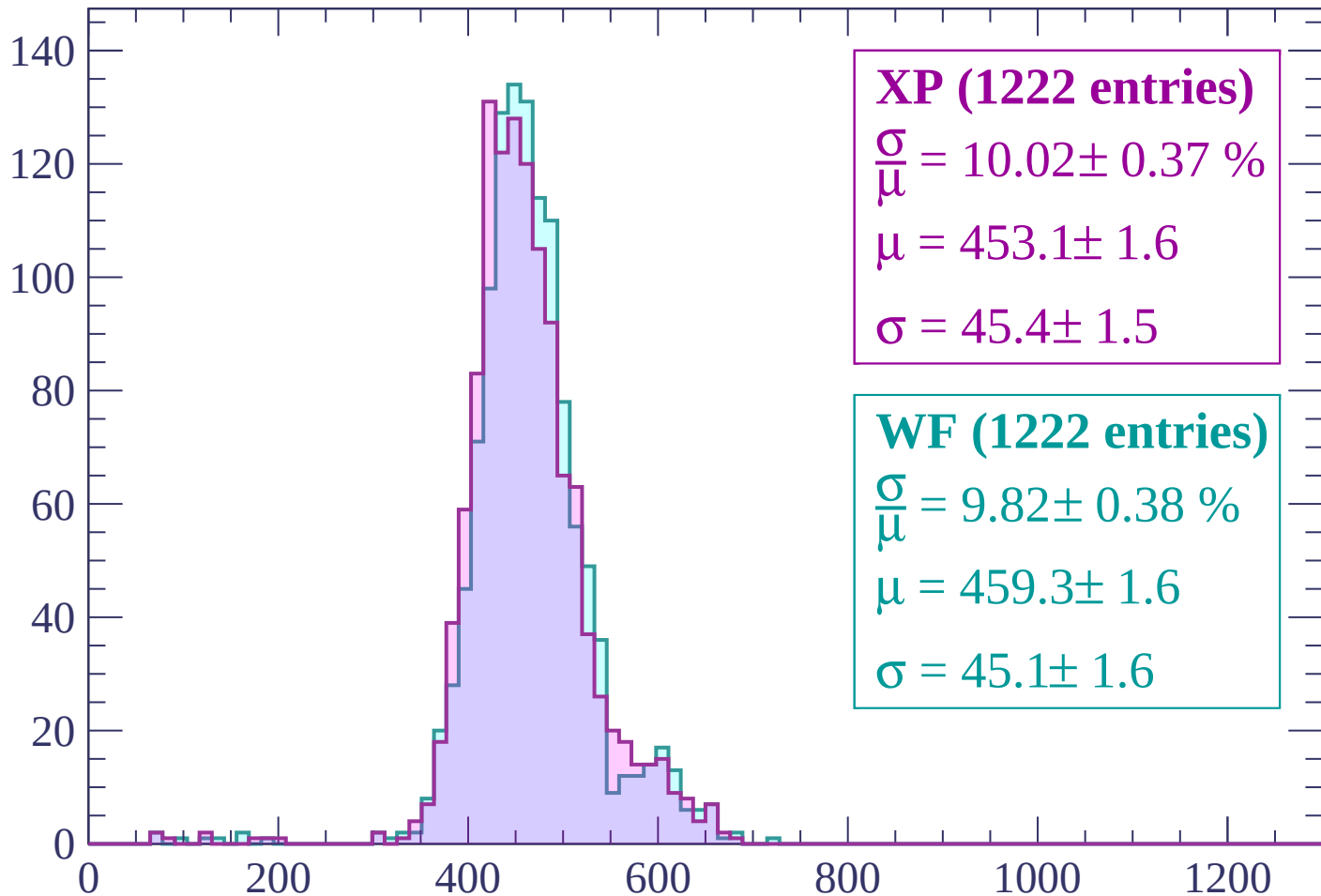
$$\mu = 565.6 \pm 2.6$$

$$\sigma = 63.2 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $180 < p < 240$

Count



XP (1222 entries)

$$\frac{\sigma}{\mu} = 10.02 \pm 0.37 \%$$

$$\mu = 453.1 \pm 1.6$$

$$\sigma = 45.4 \pm 1.5$$

WF (1222 entries)

$$\frac{\sigma}{\mu} = 9.82 \pm 0.38 \%$$

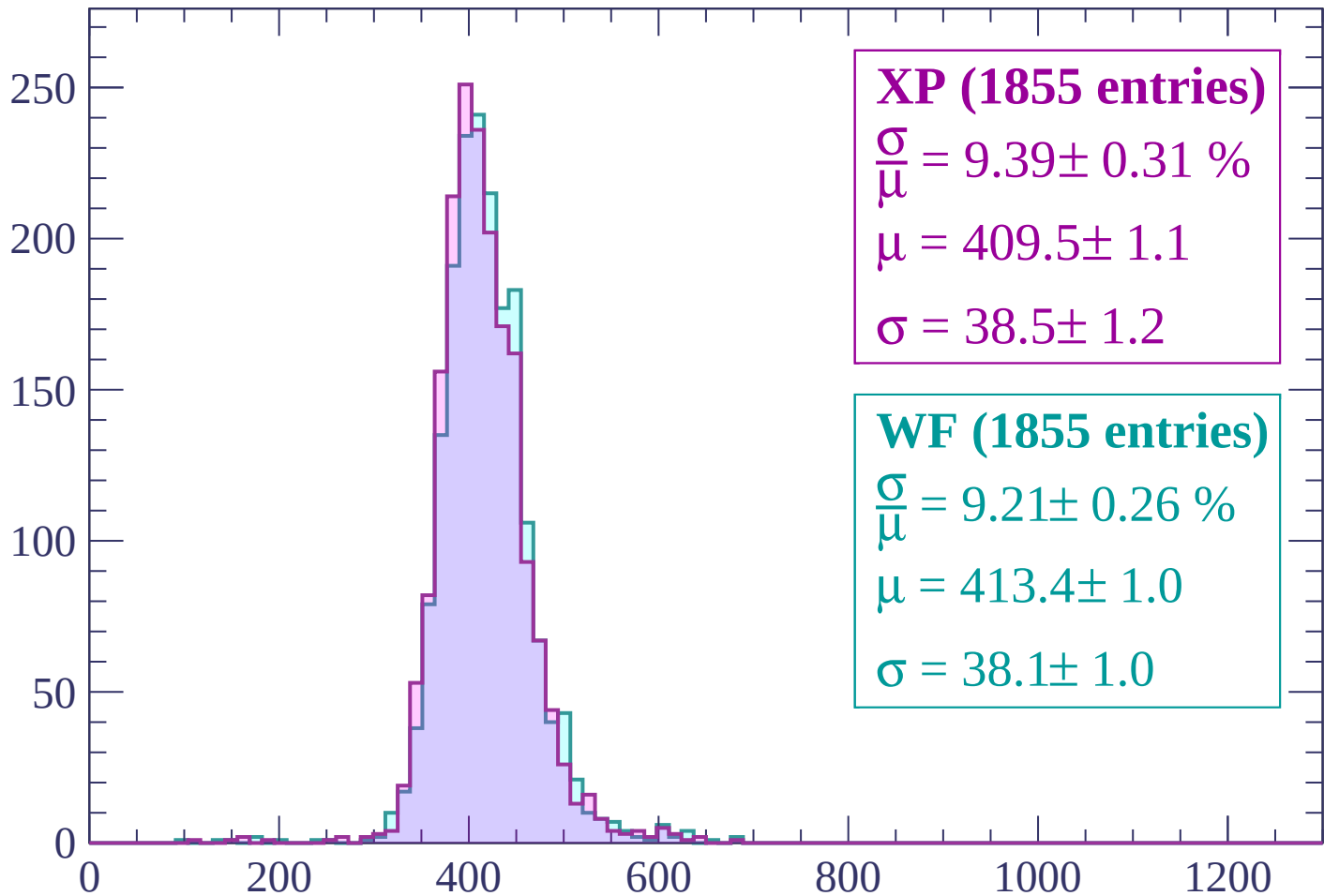
$$\mu = 459.3 \pm 1.6$$

$$\sigma = 45.1 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (1855 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.31 \%$$

$$\mu = 409.5 \pm 1.1$$

$$\sigma = 38.5 \pm 1.2$$

WF (1855 entries)

$$\frac{\sigma}{\mu} = 9.21 \pm 0.26 \%$$

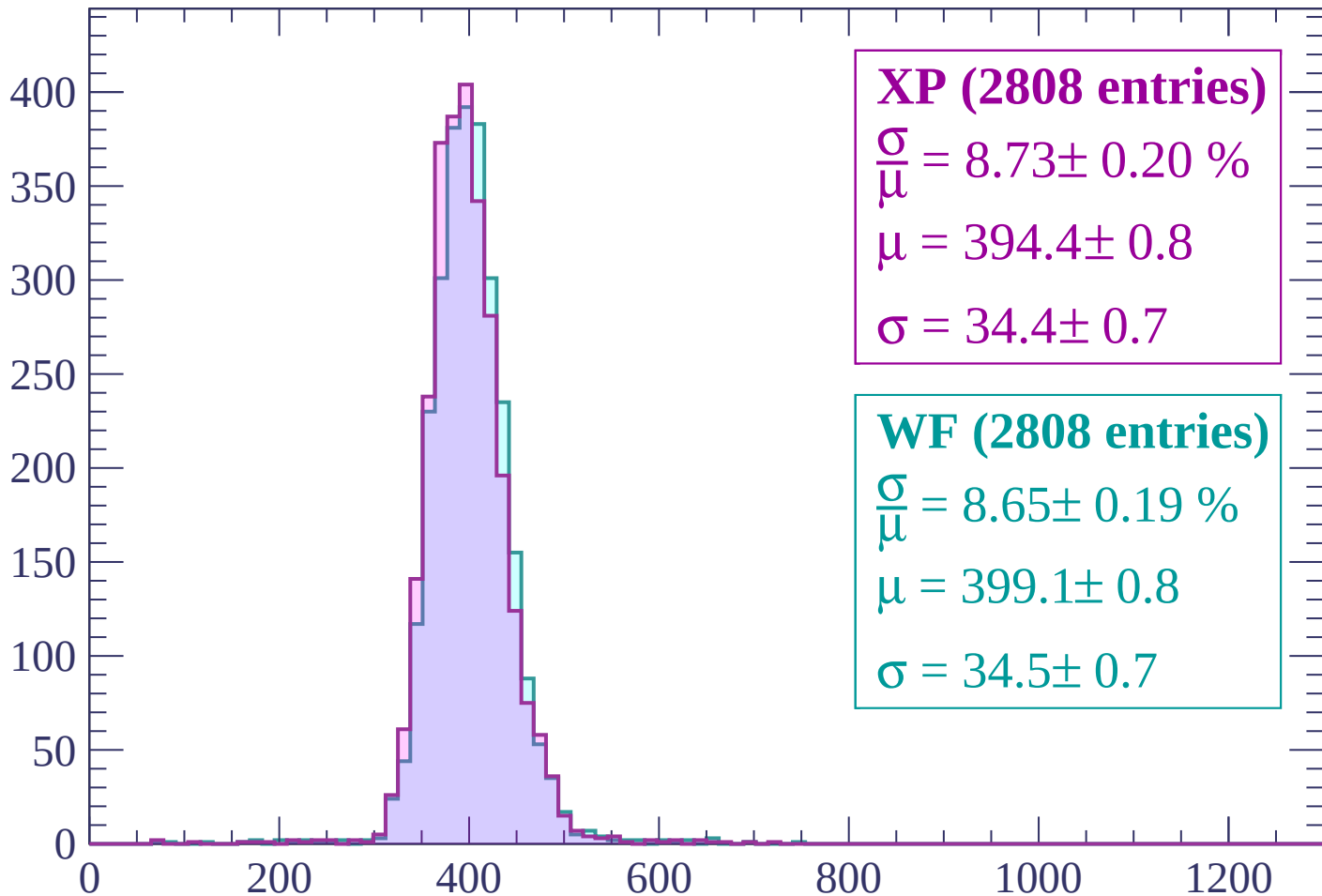
$$\mu = 413.4 \pm 1.0$$

$$\sigma = 38.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (2808 entries)

$$\frac{\sigma}{\mu} = 8.73 \pm 0.20 \%$$

$$\mu = 394.4 \pm 0.8$$

$$\sigma = 34.4 \pm 0.7$$

WF (2808 entries)

$$\frac{\sigma}{\mu} = 8.65 \pm 0.19 \%$$

$$\mu = 399.1 \pm 0.8$$

$$\sigma = 34.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (2942 entries)

$\frac{\sigma}{\mu} = 8.47 \pm 0.20 \%$

$\mu = 390.0 \pm 0.7$

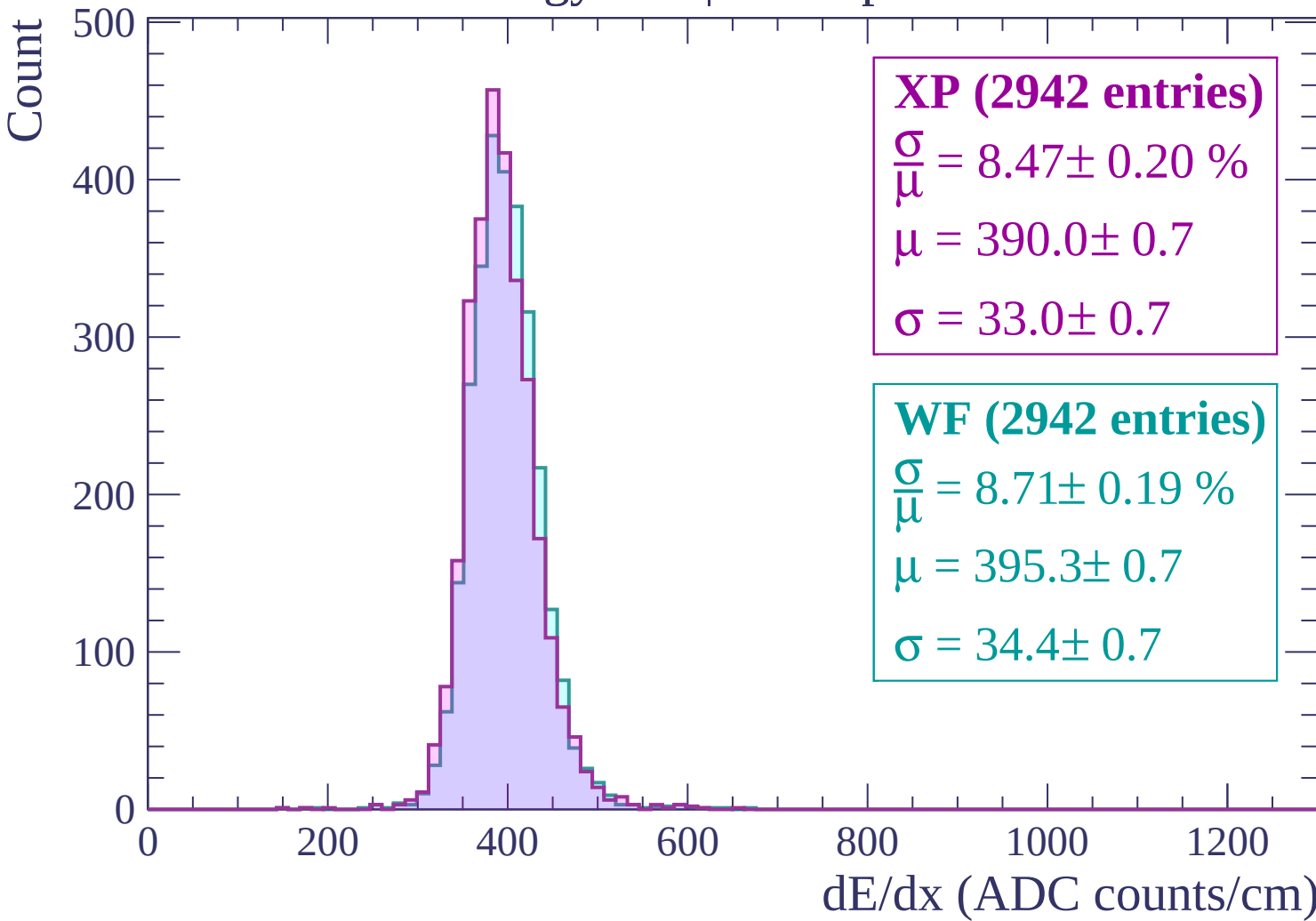
$\sigma = 33.0 \pm 0.7$

WF (2942 entries)

$\frac{\sigma}{\mu} = 8.71 \pm 0.19 \%$

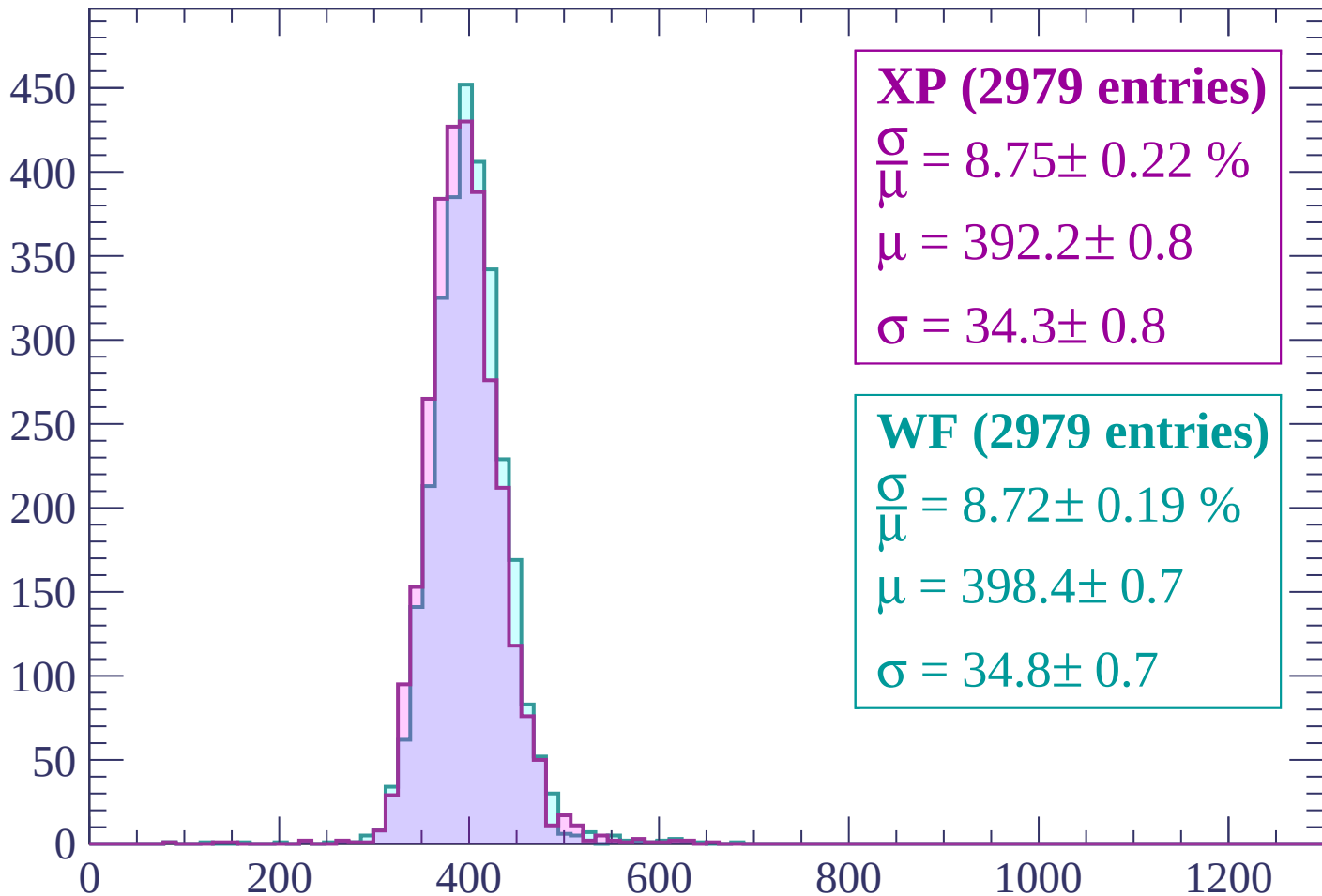
$\mu = 395.3 \pm 0.7$

$\sigma = 34.4 \pm 0.7$



Energy loss | $420 < p < 480$

Count



XP (2979 entries)

$$\frac{\sigma}{\mu} = 8.75 \pm 0.22 \%$$

$$\mu = 392.2 \pm 0.8$$

$$\sigma = 34.3 \pm 0.8$$

WF (2979 entries)

$$\frac{\sigma}{\mu} = 8.72 \pm 0.19 \%$$

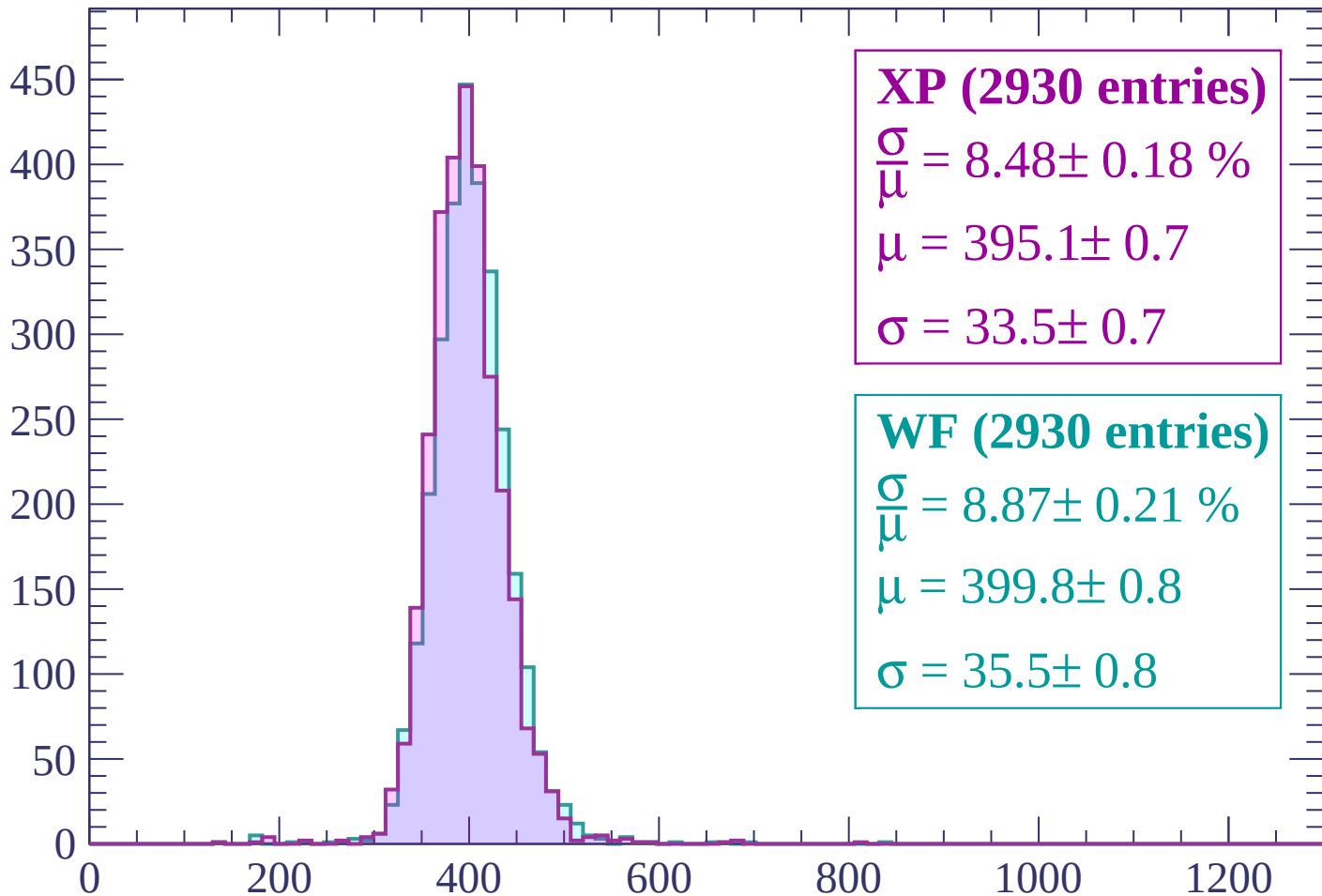
$$\mu = 398.4 \pm 0.7$$

$$\sigma = 34.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (2930 entries)

$$\frac{\sigma}{\mu} = 8.48 \pm 0.18 \%$$

$$\mu = 395.1 \pm 0.7$$

$$\sigma = 33.5 \pm 0.7$$

WF (2930 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.21 \%$$

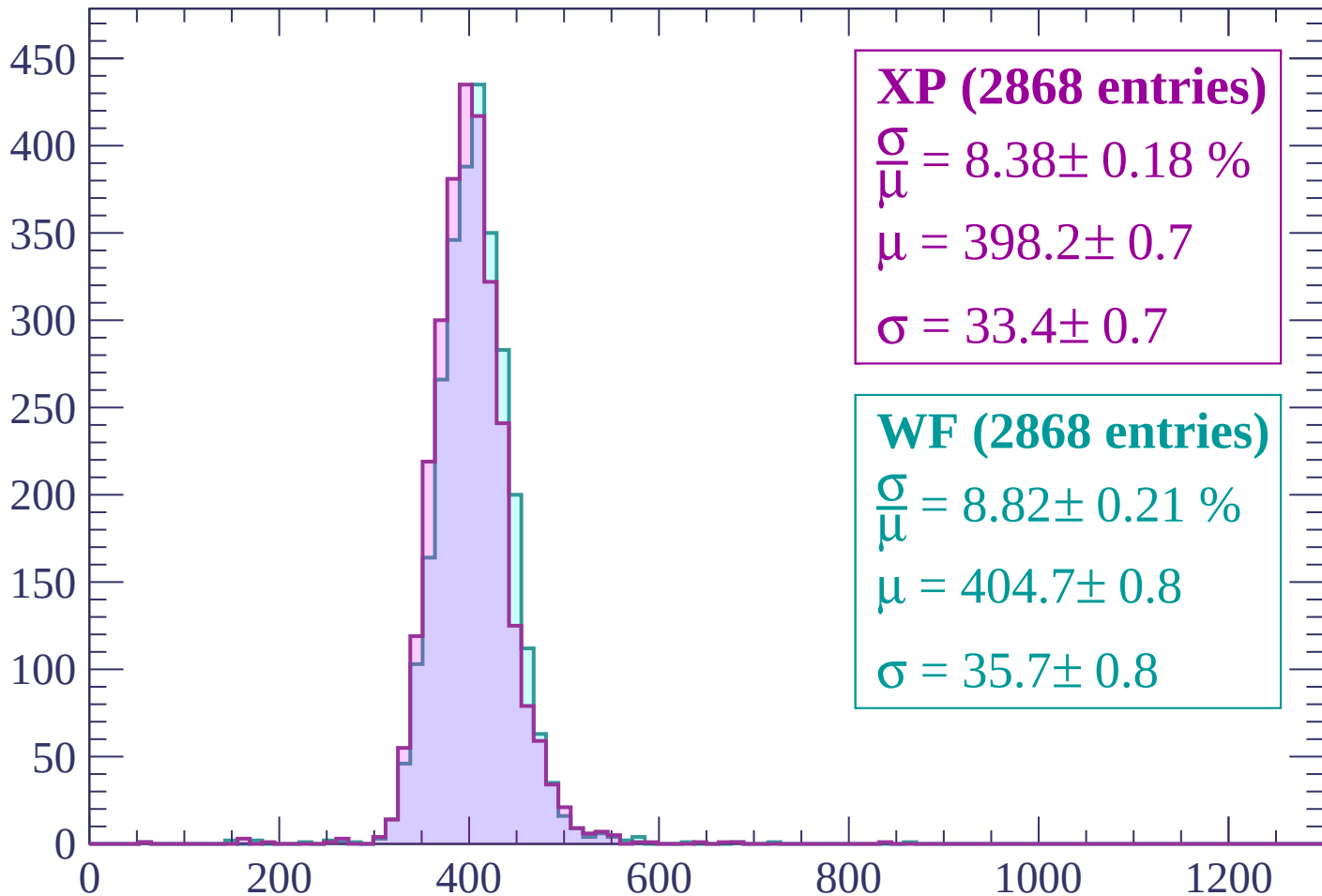
$$\mu = 399.8 \pm 0.8$$

$$\sigma = 35.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (2868 entries)

$$\frac{\sigma}{\mu} = 8.38 \pm 0.18 \%$$

$$\mu = 398.2 \pm 0.7$$

$$\sigma = 33.4 \pm 0.7$$

WF (2868 entries)

$$\frac{\sigma}{\mu} = 8.82 \pm 0.21 \%$$

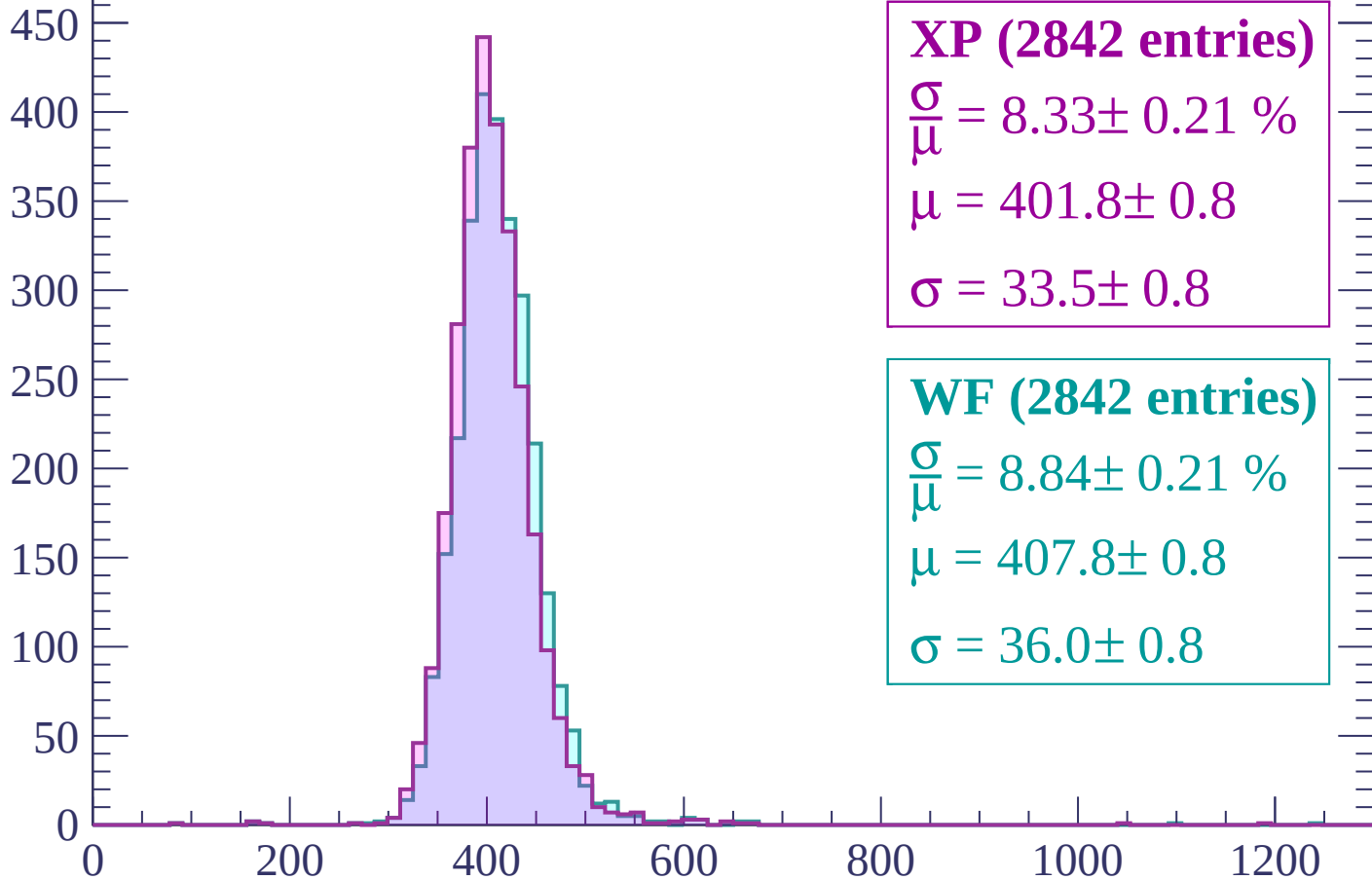
$$\mu = 404.7 \pm 0.8$$

$$\sigma = 35.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (2842 entries)

$$\frac{\sigma}{\mu} = 8.33 \pm 0.21 \%$$

$$\mu = 401.8 \pm 0.8$$

$$\sigma = 33.5 \pm 0.8$$

WF (2842 entries)

$$\frac{\sigma}{\mu} = 8.84 \pm 0.21 \%$$

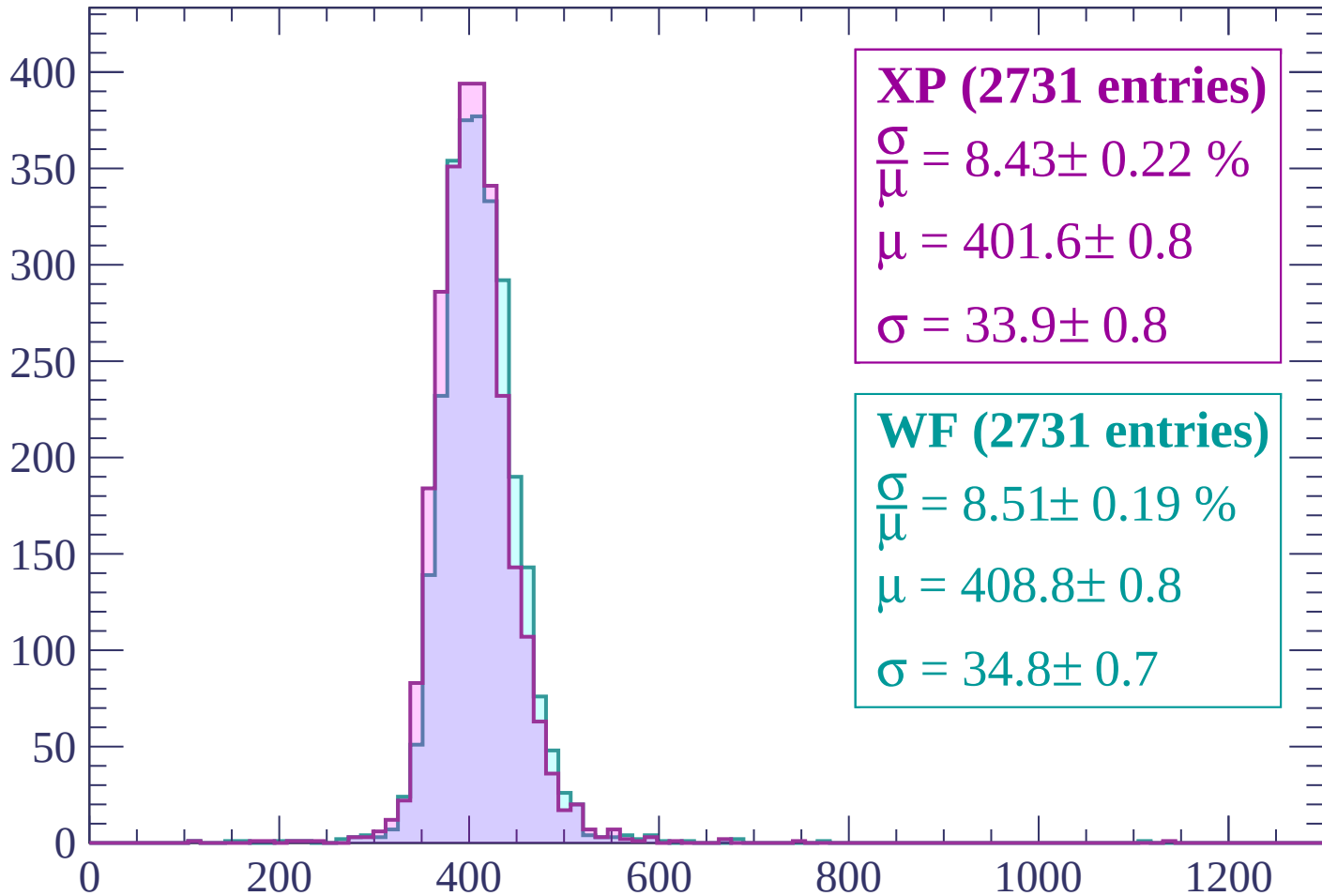
$$\mu = 407.8 \pm 0.8$$

$$\sigma = 36.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (2731 entries)

$$\frac{\sigma}{\mu} = 8.43 \pm 0.22 \%$$

$$\mu = 401.6 \pm 0.8$$

$$\sigma = 33.9 \pm 0.8$$

WF (2731 entries)

$$\frac{\sigma}{\mu} = 8.51 \pm 0.19 \%$$

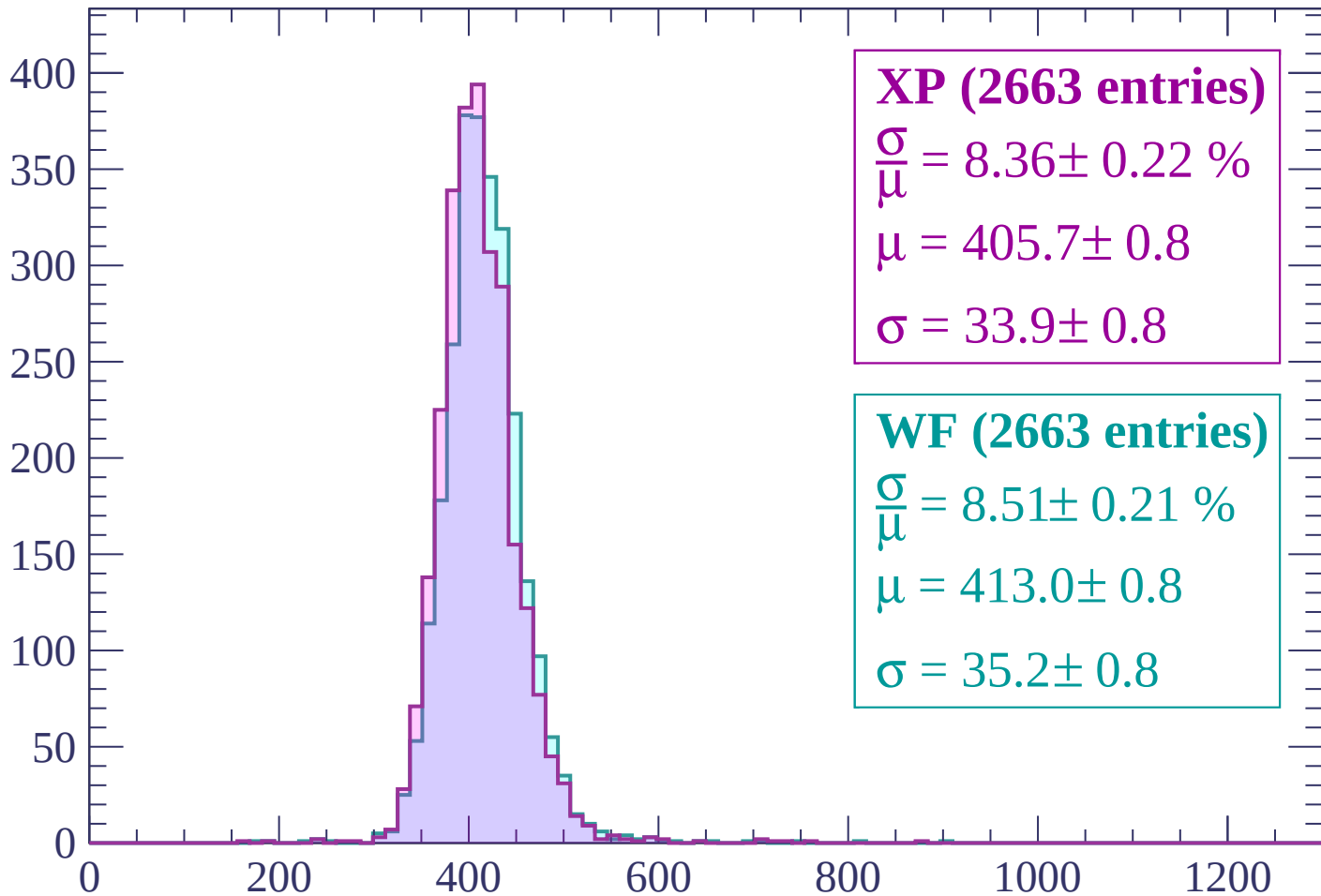
$$\mu = 408.8 \pm 0.8$$

$$\sigma = 34.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (2663 entries)

$\frac{\sigma}{\mu} = 8.36 \pm 0.22 \%$

$\mu = 405.7 \pm 0.8$

$\sigma = 33.9 \pm 0.8$

WF (2663 entries)

$\frac{\sigma}{\mu} = 8.51 \pm 0.21 \%$

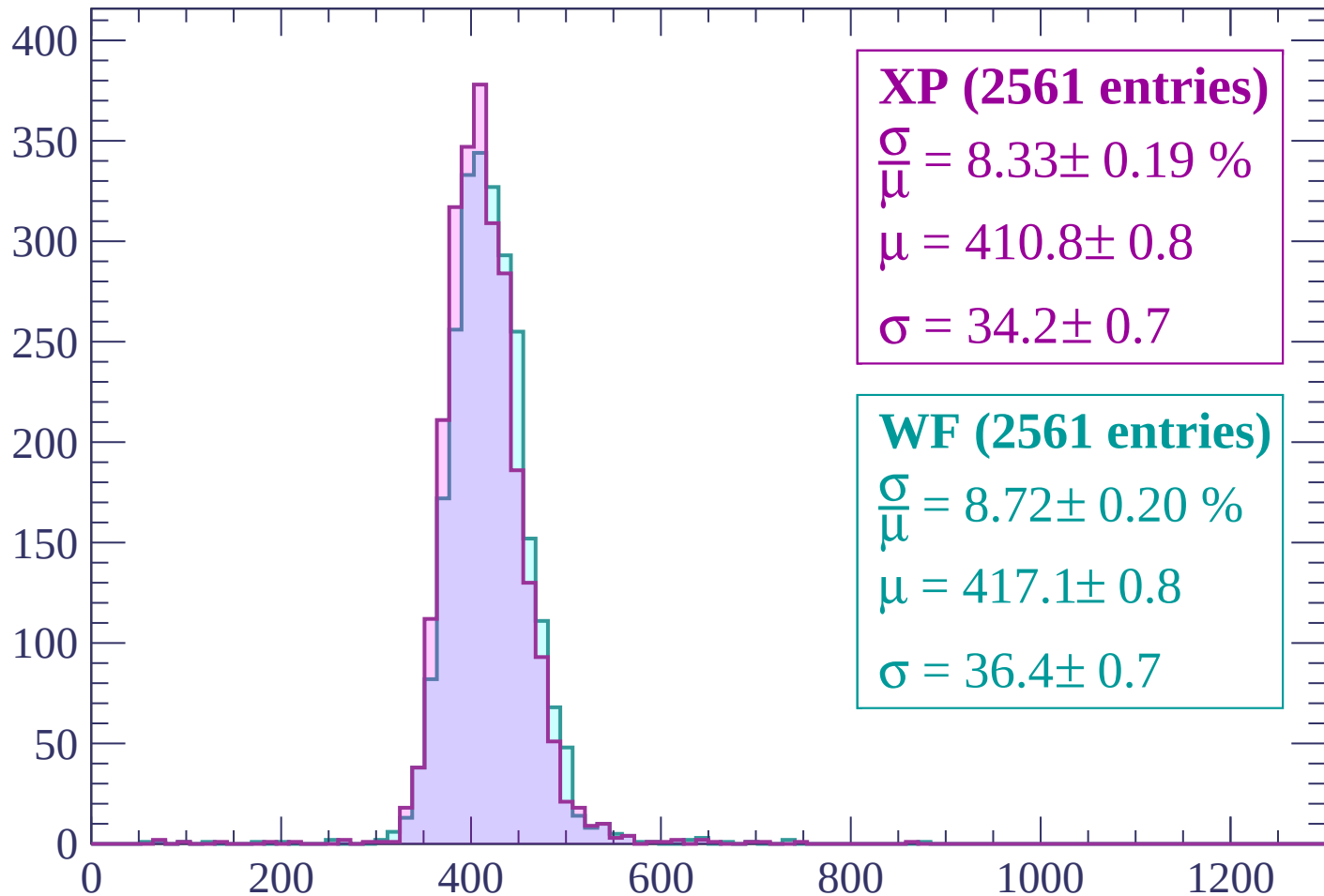
$\mu = 413.0 \pm 0.8$

$\sigma = 35.2 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (2561 entries)

$$\frac{\sigma}{\mu} = 8.33 \pm 0.19 \%$$

$$\mu = 410.8 \pm 0.8$$

$$\sigma = 34.2 \pm 0.7$$

WF (2561 entries)

$$\frac{\sigma}{\mu} = 8.72 \pm 0.20 \%$$

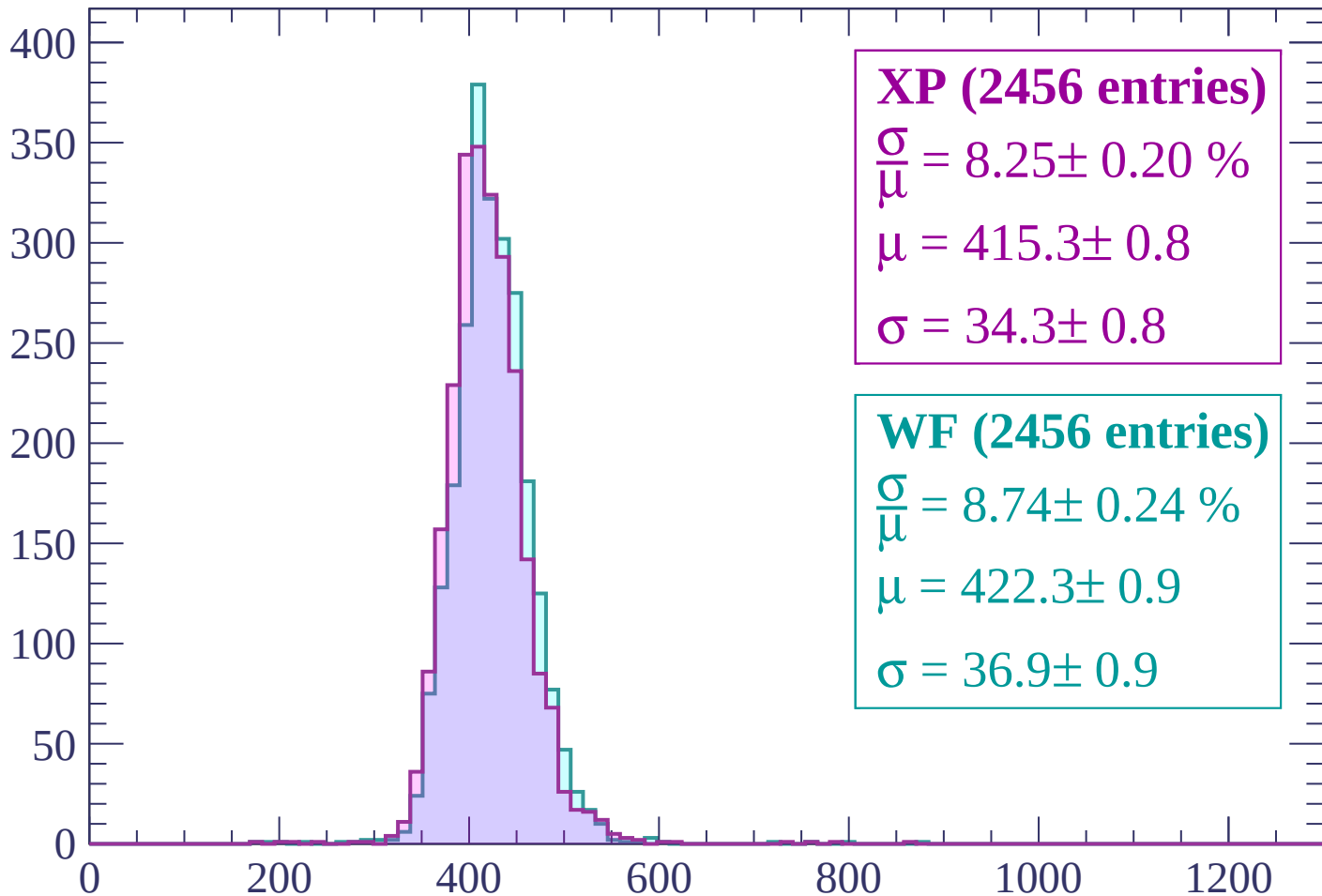
$$\mu = 417.1 \pm 0.8$$

$$\sigma = 36.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (2456 entries)

$$\frac{\sigma}{\mu} = 8.25 \pm 0.20 \%$$

$$\mu = 415.3 \pm 0.8$$

$$\sigma = 34.3 \pm 0.8$$

WF (2456 entries)

$$\frac{\sigma}{\mu} = 8.74 \pm 0.24 \%$$

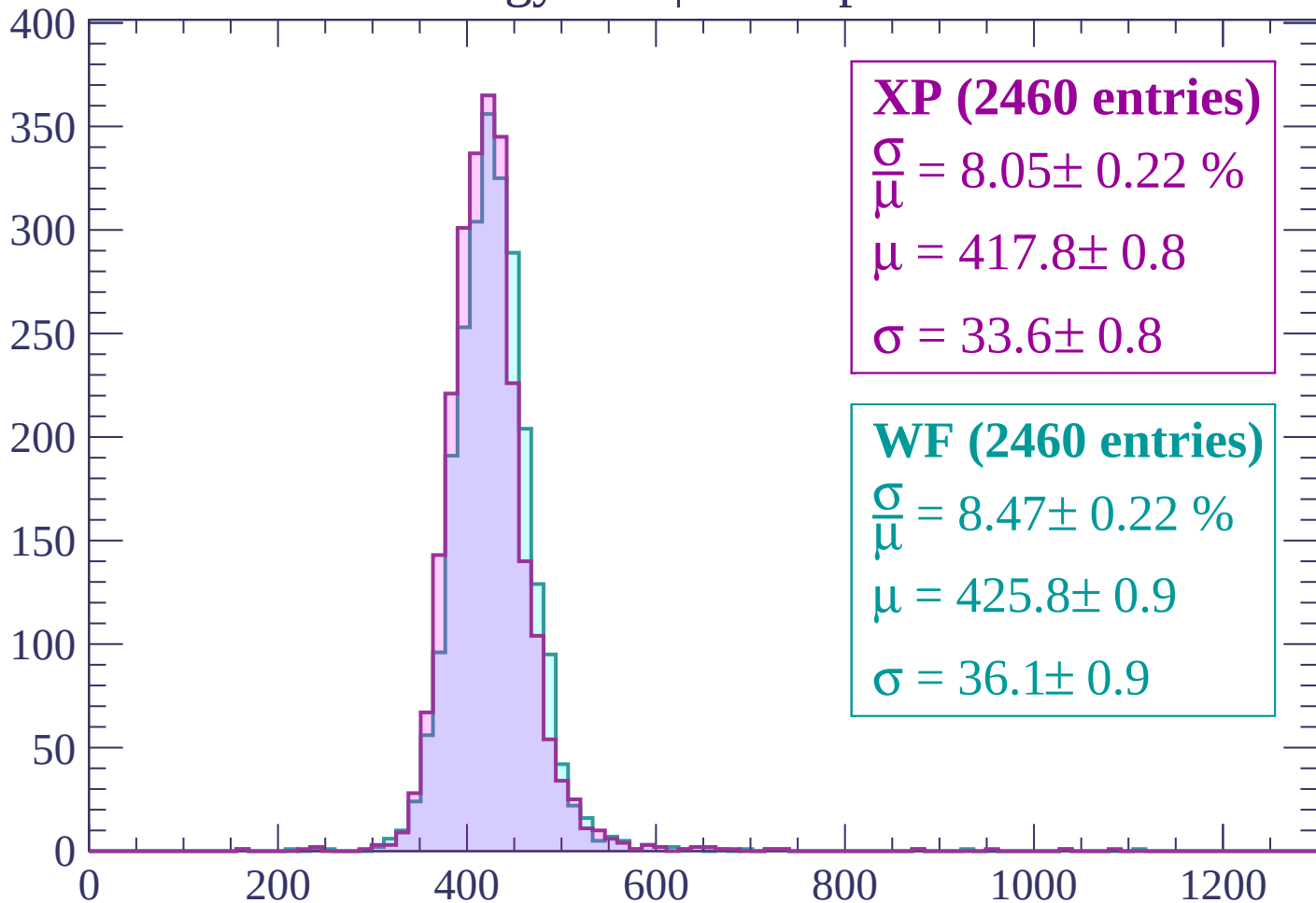
$$\mu = 422.3 \pm 0.9$$

$$\sigma = 36.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (2460 entries)

$$\frac{\sigma}{\mu} = 8.05 \pm 0.22 \%$$

$$\mu = 417.8 \pm 0.8$$

$$\sigma = 33.6 \pm 0.8$$

WF (2460 entries)

$$\frac{\sigma}{\mu} = 8.47 \pm 0.22 \%$$

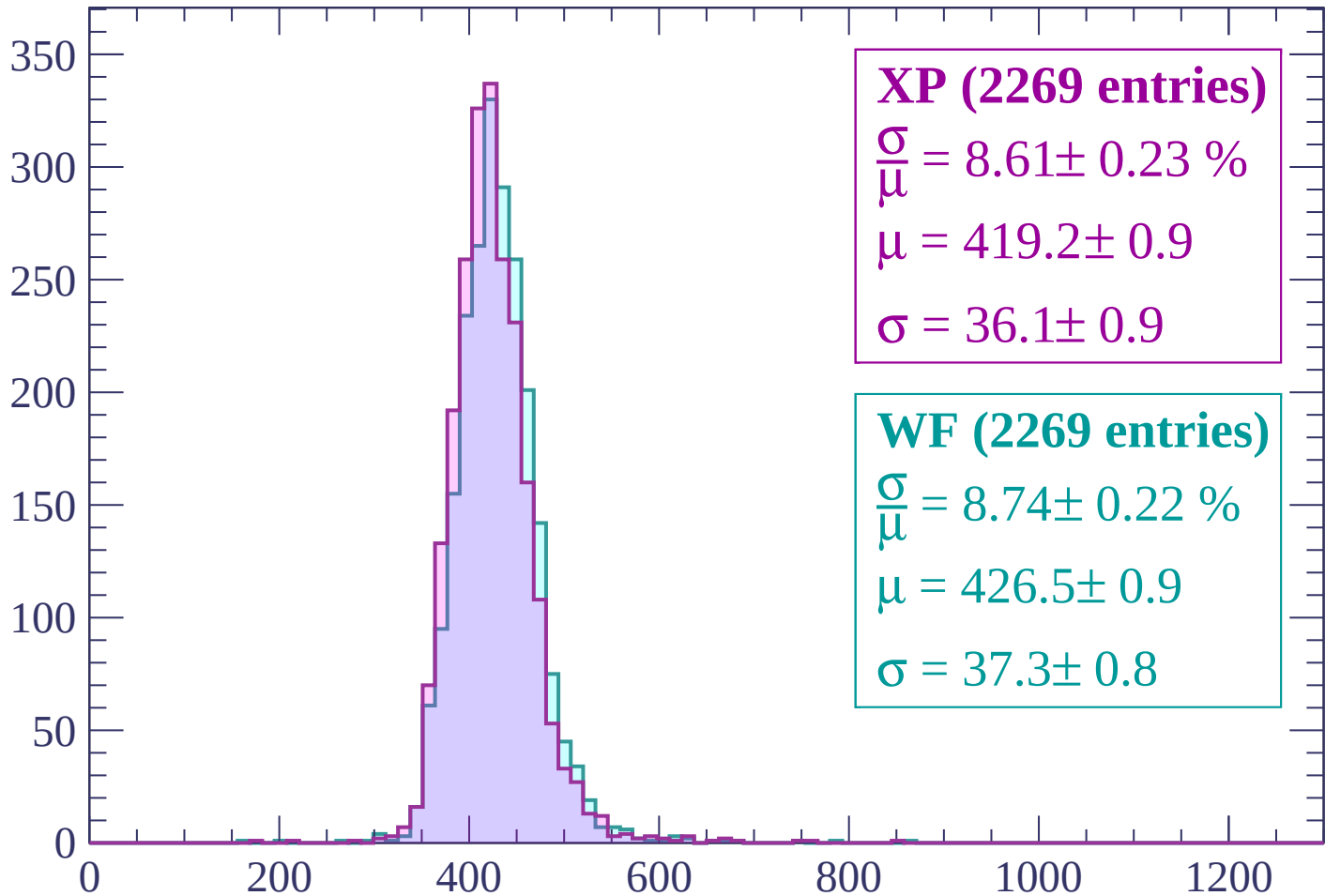
$$\mu = 425.8 \pm 0.9$$

$$\sigma = 36.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (2269 entries)

$$\frac{\sigma}{\mu} = 8.61 \pm 0.23 \%$$

$$\mu = 419.2 \pm 0.9$$

$$\sigma = 36.1 \pm 0.9$$

WF (2269 entries)

$$\frac{\sigma}{\mu} = 8.74 \pm 0.22 \%$$

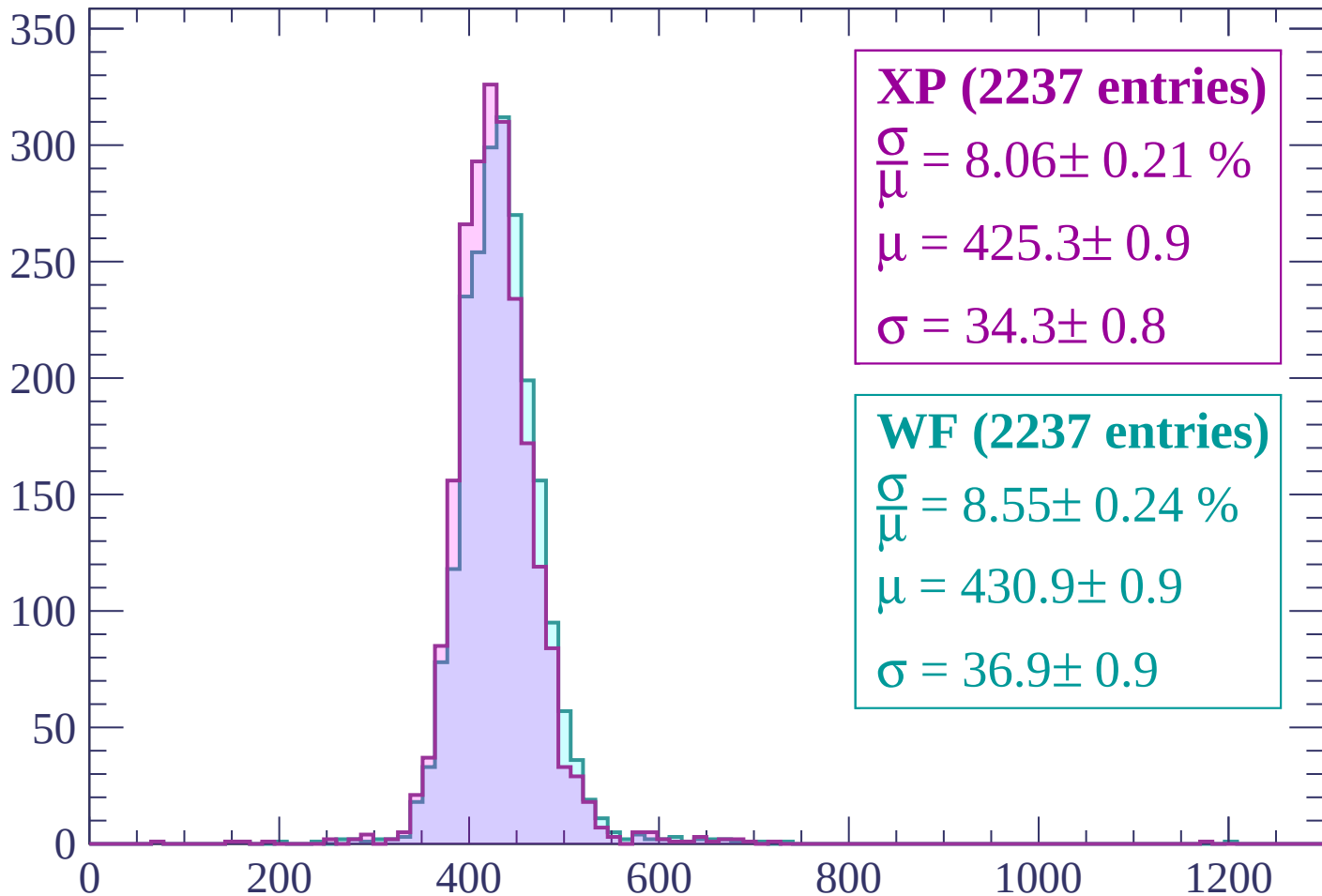
$$\mu = 426.5 \pm 0.9$$

$$\sigma = 37.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (2237 entries)

$\frac{\sigma}{\mu} = 8.06 \pm 0.21 \%$

$\mu = 425.3 \pm 0.9$

$\sigma = 34.3 \pm 0.8$

WF (2237 entries)

$\frac{\sigma}{\mu} = 8.55 \pm 0.24 \%$

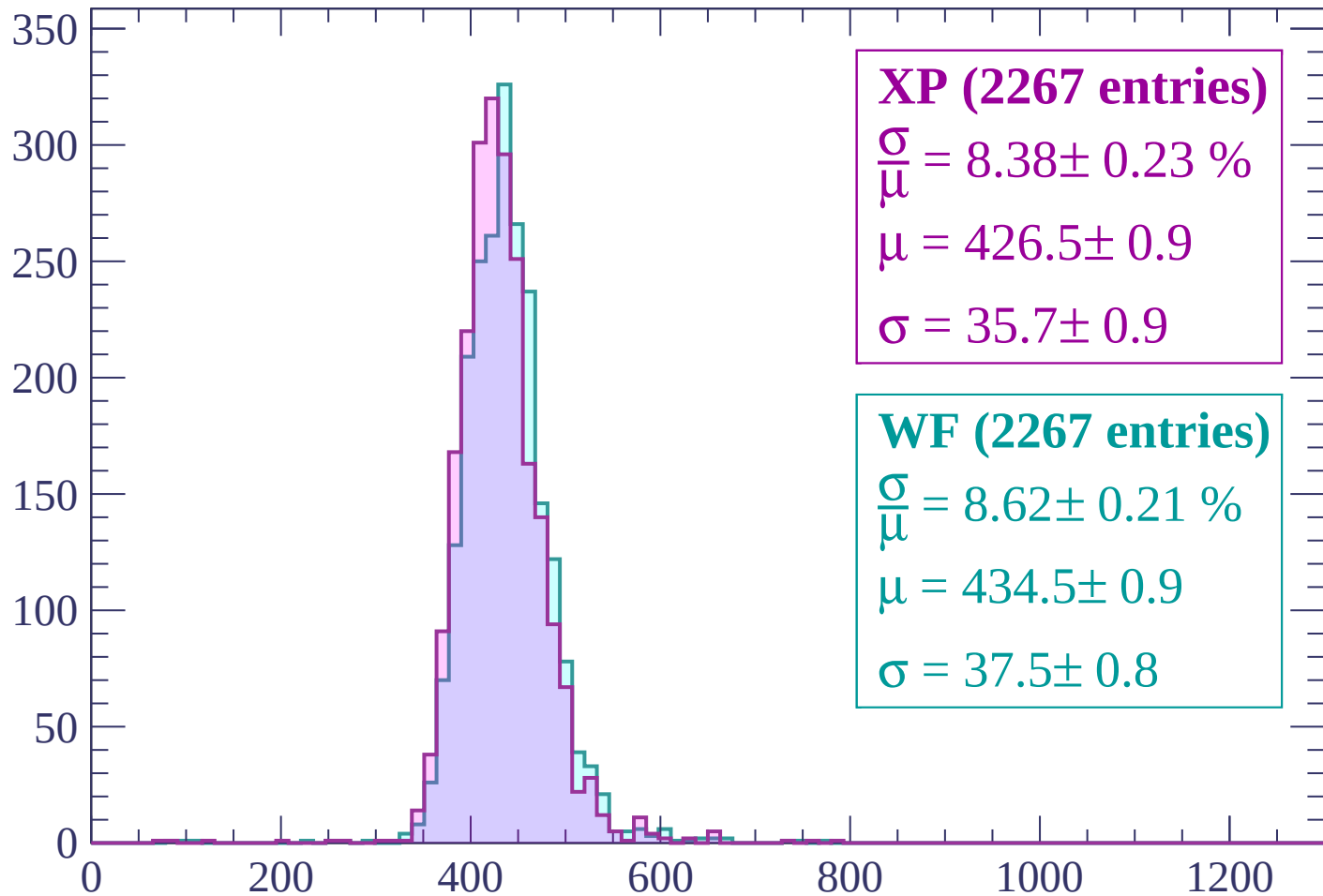
$\mu = 430.9 \pm 0.9$

$\sigma = 36.9 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (2267 entries)

$$\frac{\sigma}{\mu} = 8.38 \pm 0.23 \%$$

$$\mu = 426.5 \pm 0.9$$

$$\sigma = 35.7 \pm 0.9$$

WF (2267 entries)

$$\frac{\sigma}{\mu} = 8.62 \pm 0.21 \%$$

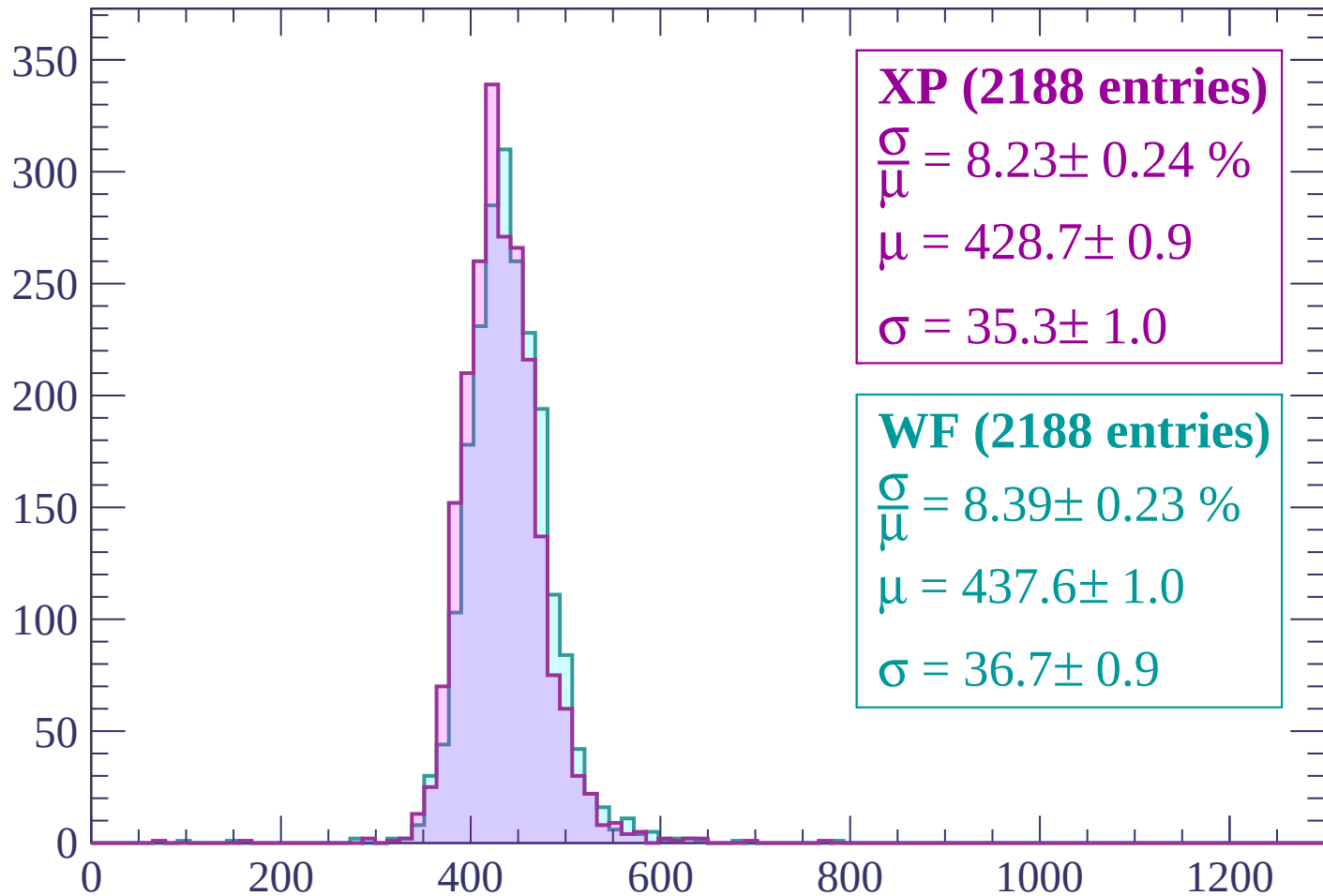
$$\mu = 434.5 \pm 0.9$$

$$\sigma = 37.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (2188 entries)

$$\frac{\sigma}{\mu} = 8.23 \pm 0.24 \%$$

$$\mu = 428.7 \pm 0.9$$

$$\sigma = 35.3 \pm 1.0$$

WF (2188 entries)

$$\frac{\sigma}{\mu} = 8.39 \pm 0.23 \%$$

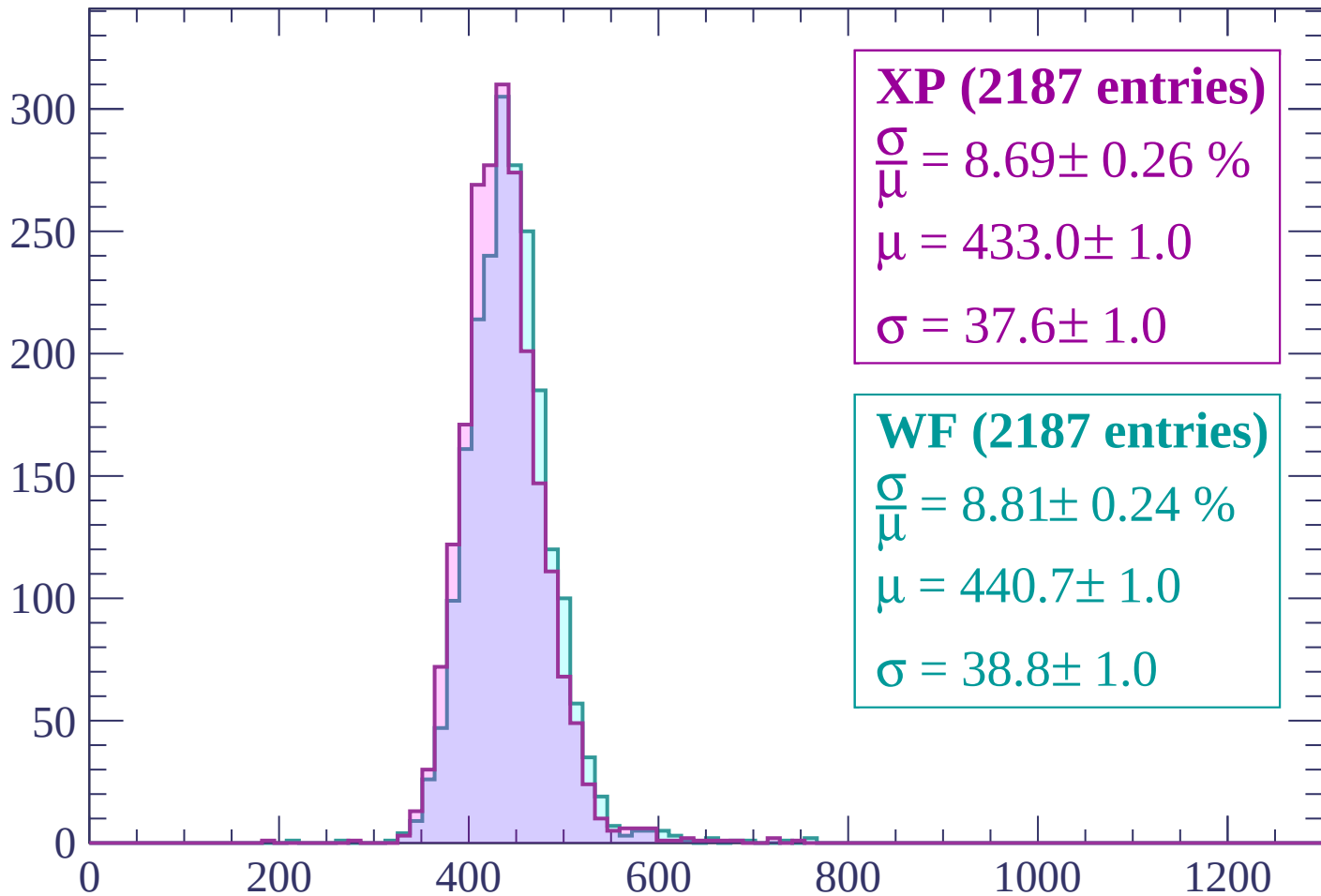
$$\mu = 437.6 \pm 1.0$$

$$\sigma = 36.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (2187 entries)

$$\frac{\sigma}{\mu} = 8.69 \pm 0.26 \%$$

$$\mu = 433.0 \pm 1.0$$

$$\sigma = 37.6 \pm 1.0$$

WF (2187 entries)

$$\frac{\sigma}{\mu} = 8.81 \pm 0.24 \%$$

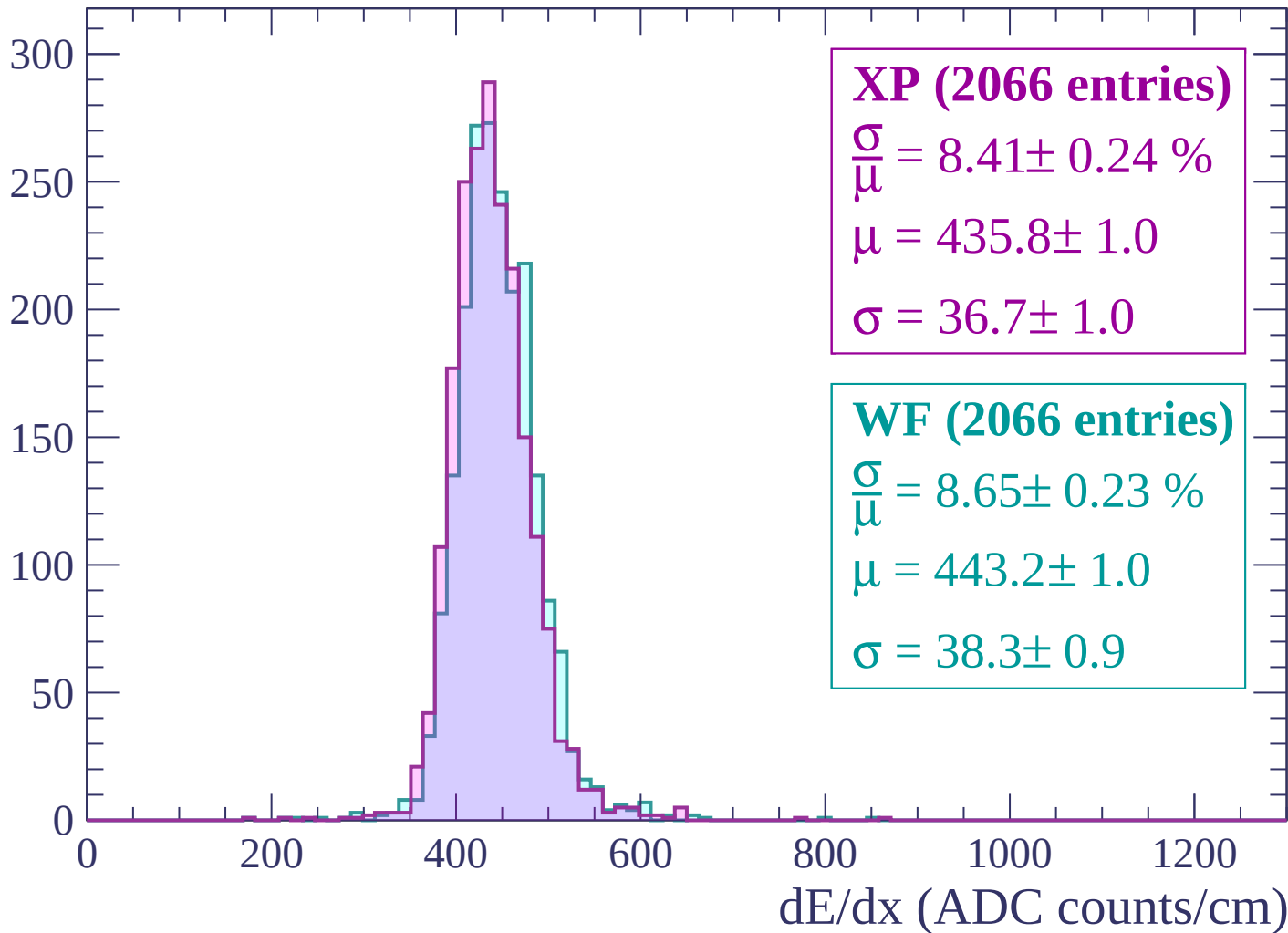
$$\mu = 440.7 \pm 1.0$$

$$\sigma = 38.8 \pm 1.0$$

dE/dx (ADC counts/cm)

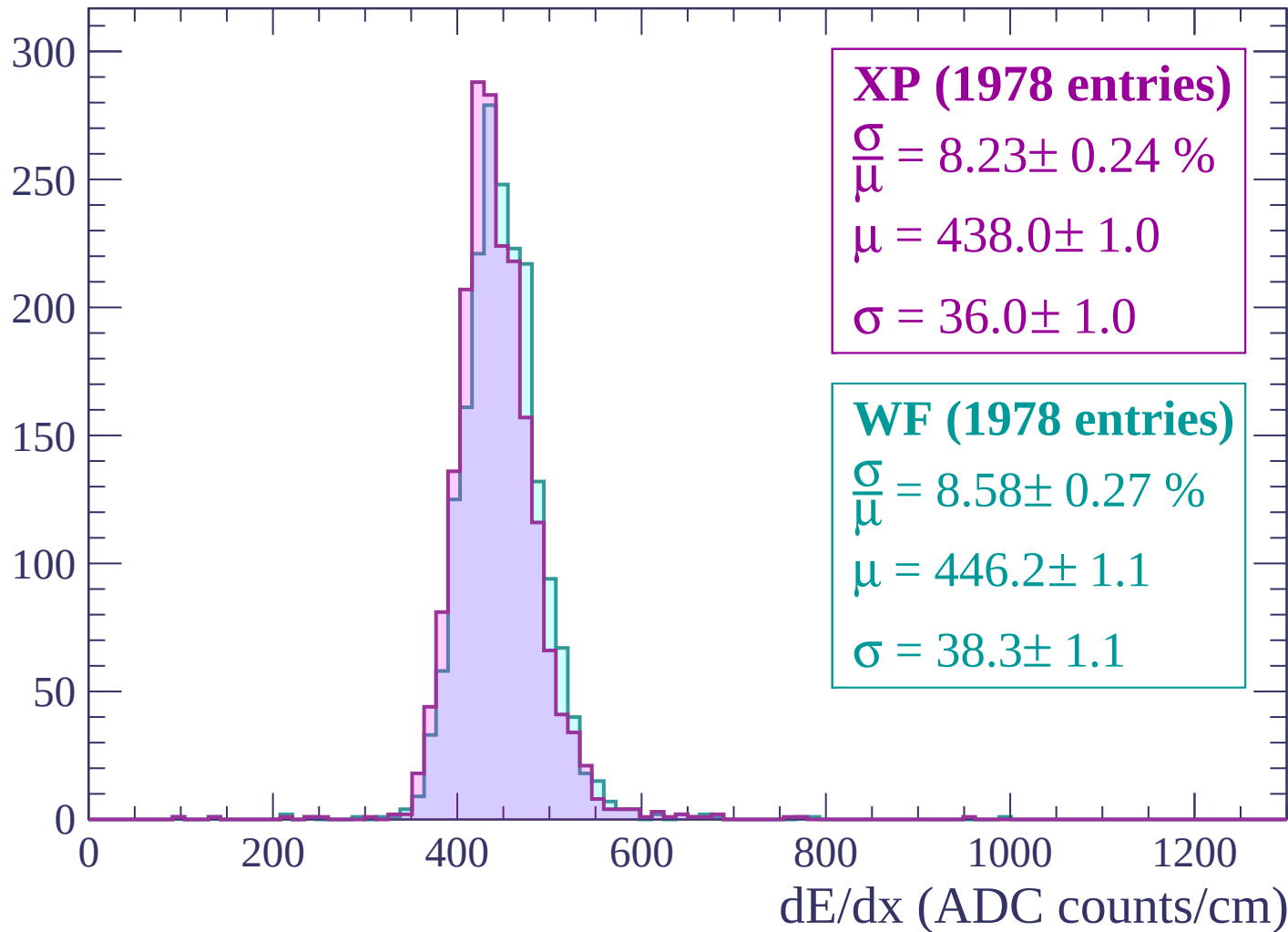
Energy loss | $1260 < p < 1320$

Count



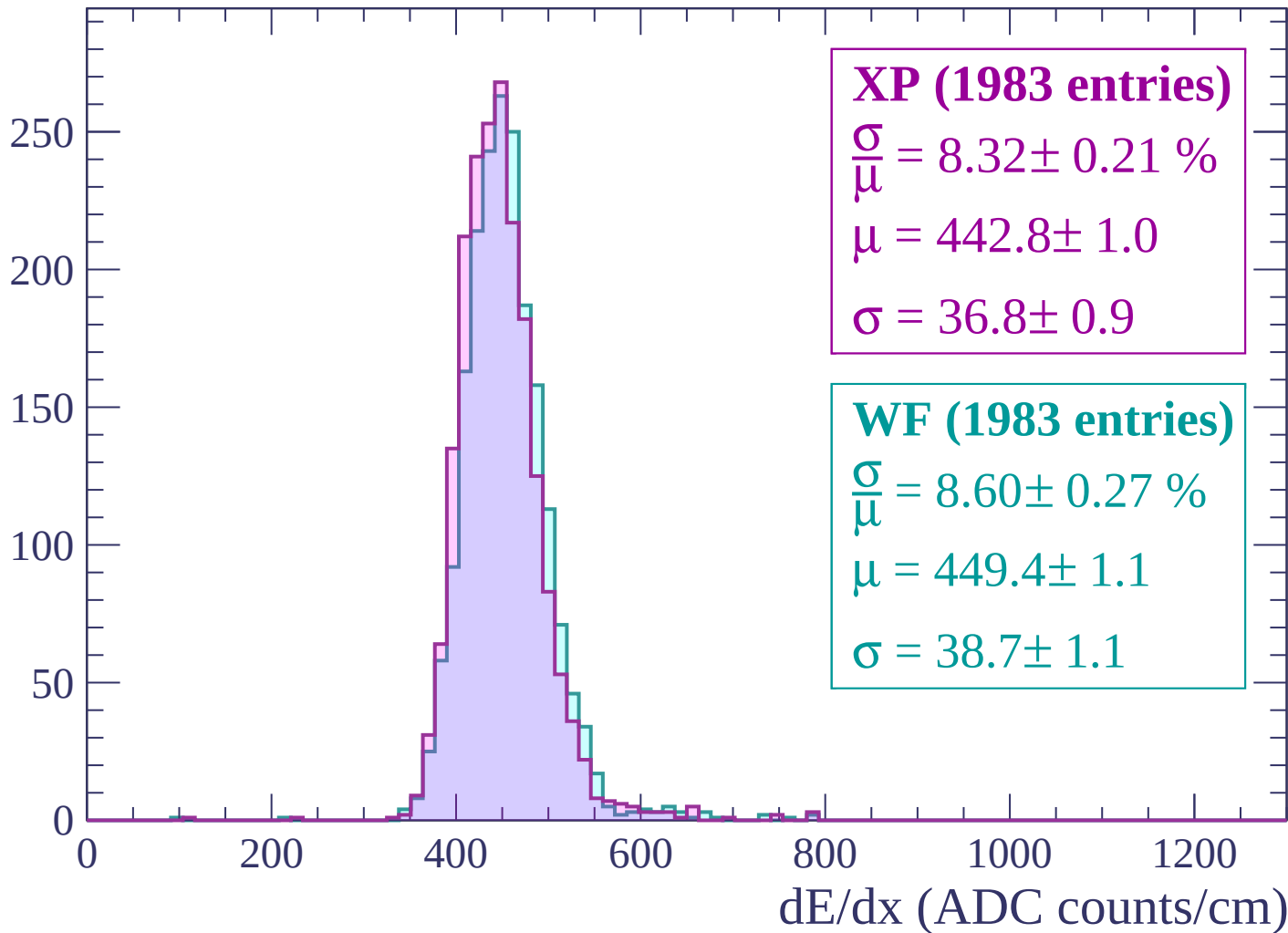
Energy loss | $1320 < p < 1380$

Count



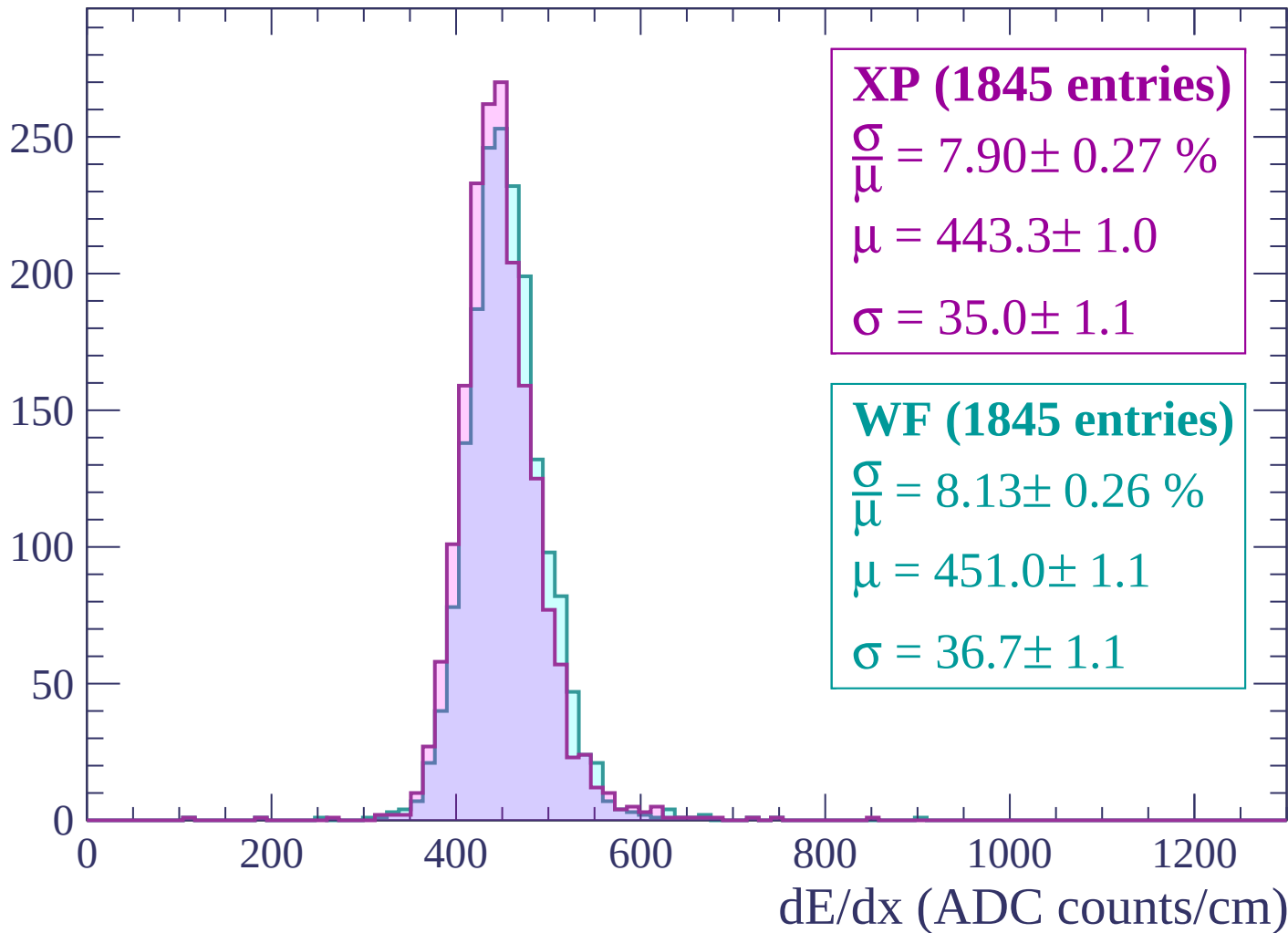
Energy loss | $1380 < p < 1440$

Count



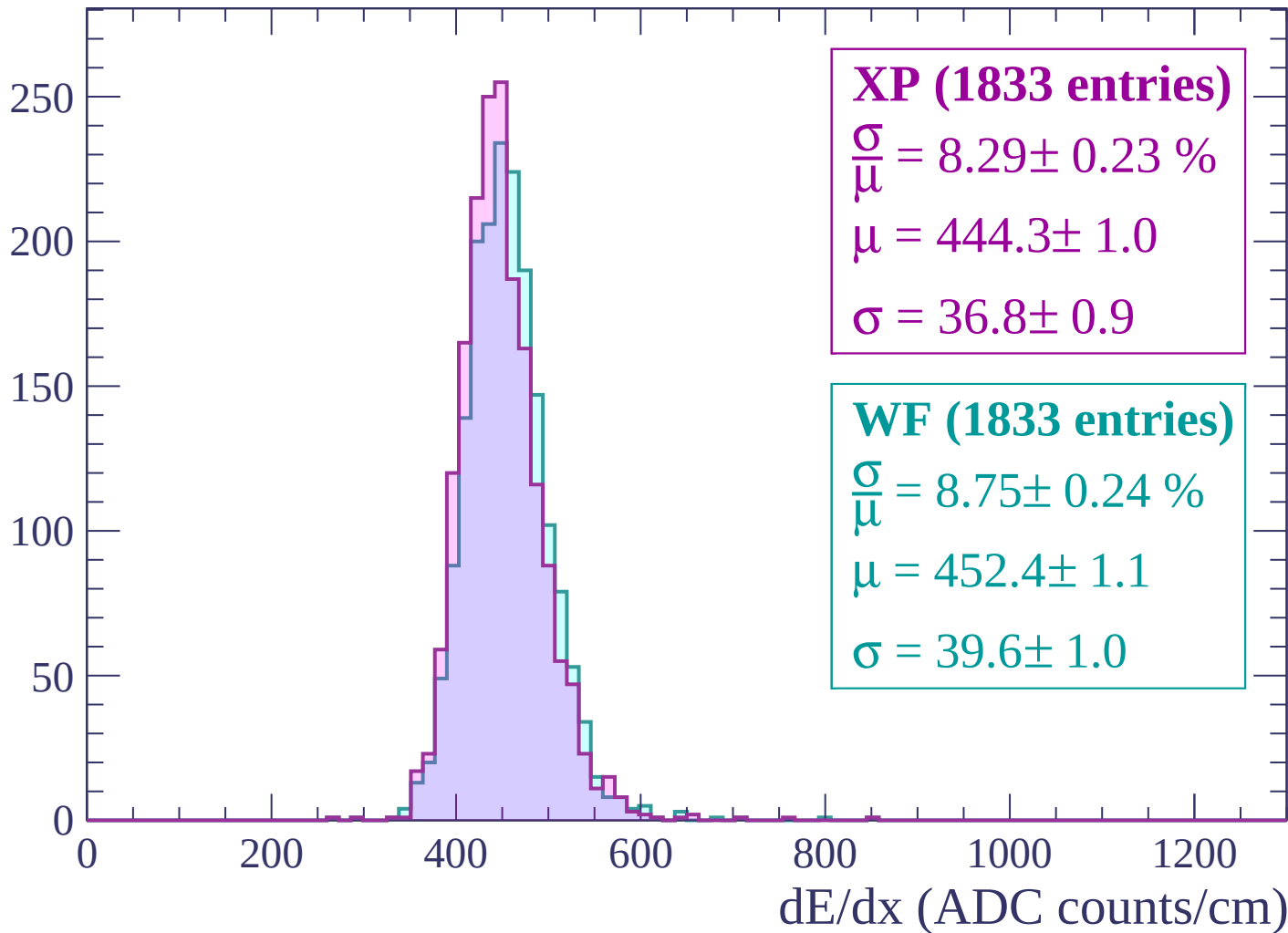
Energy loss | $1440 < p < 1500$

Count



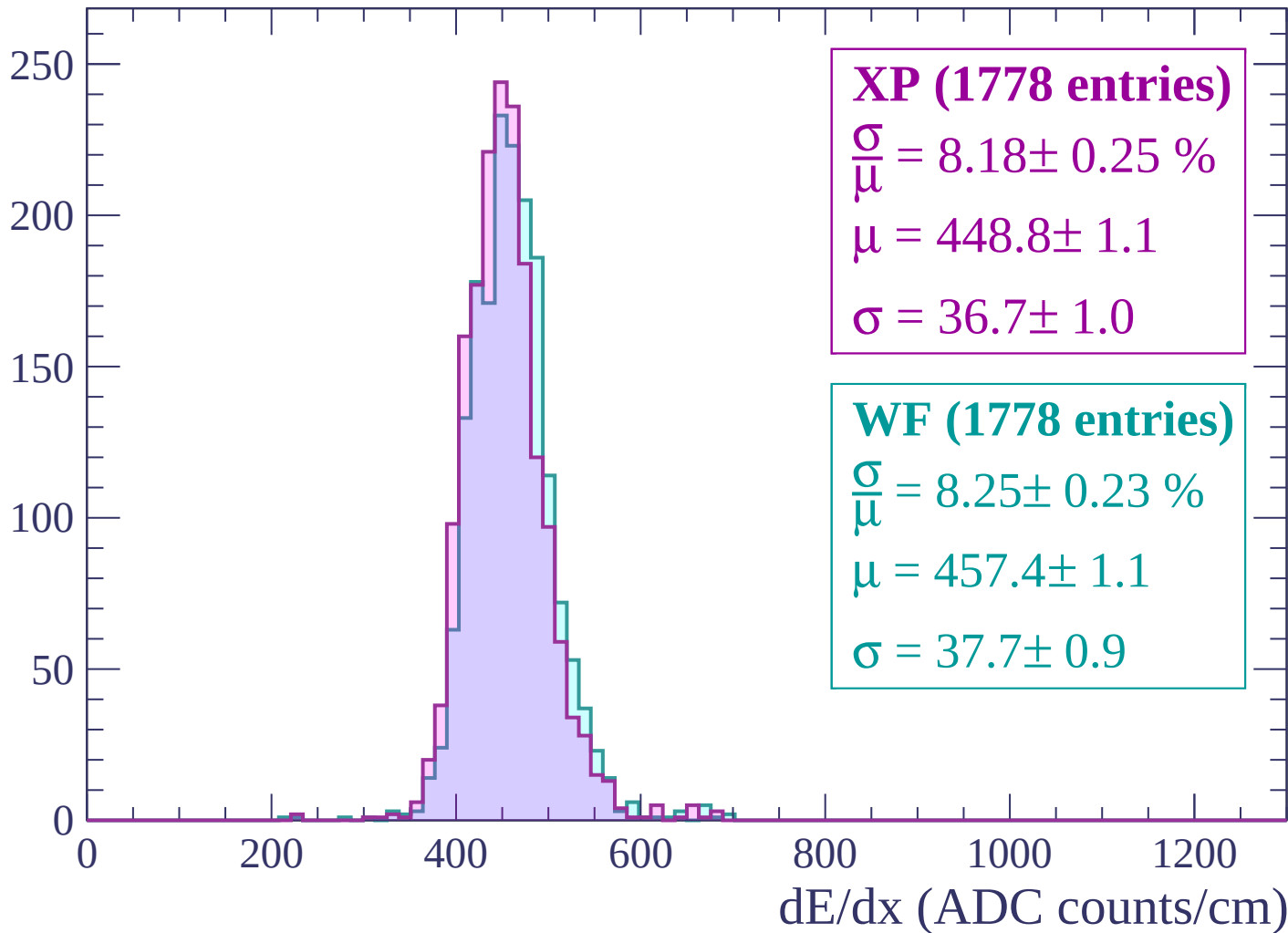
Energy loss | $1500 < p < 1560$

Count



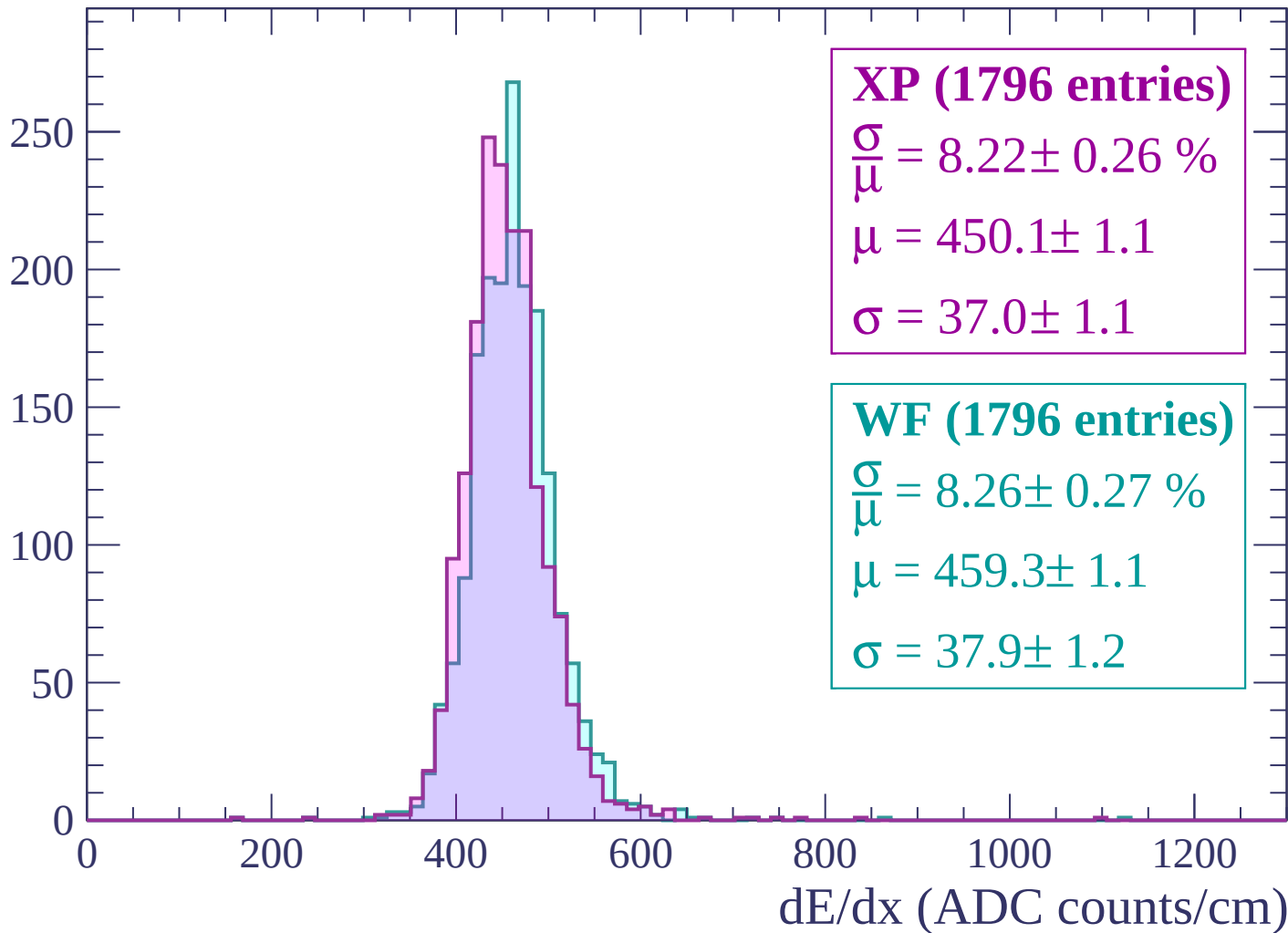
Energy loss | $1560 < p < 1620$

Count



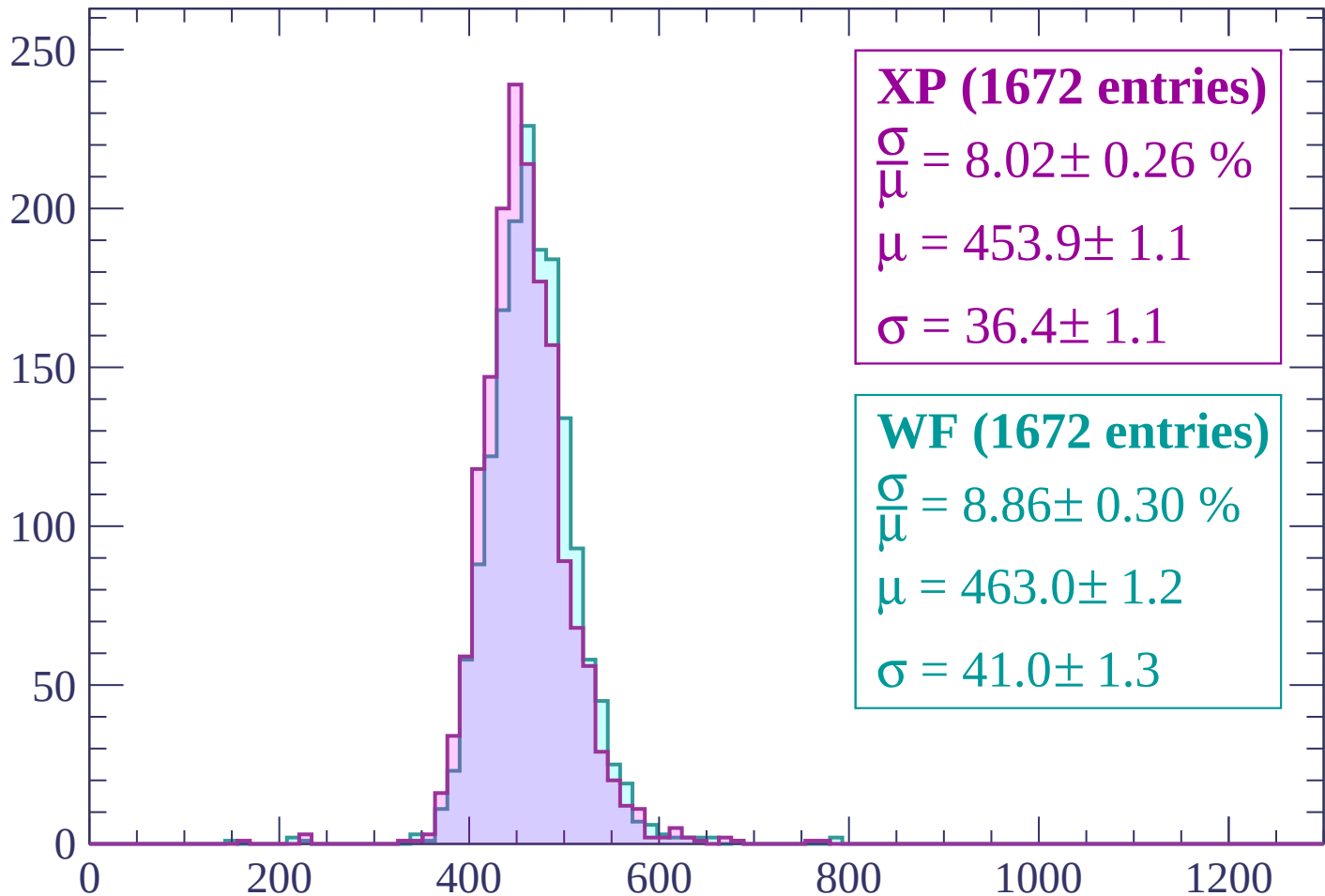
Energy loss | $1620 < p < 1680$

Count



Energy loss | $1680 < p < 1740$

Count



XP (1672 entries)

$$\frac{\sigma}{\mu} = 8.02 \pm 0.26 \%$$

$$\mu = 453.9 \pm 1.1$$

$$\sigma = 36.4 \pm 1.1$$

WF (1672 entries)

$$\frac{\sigma}{\mu} = 8.86 \pm 0.30 \%$$

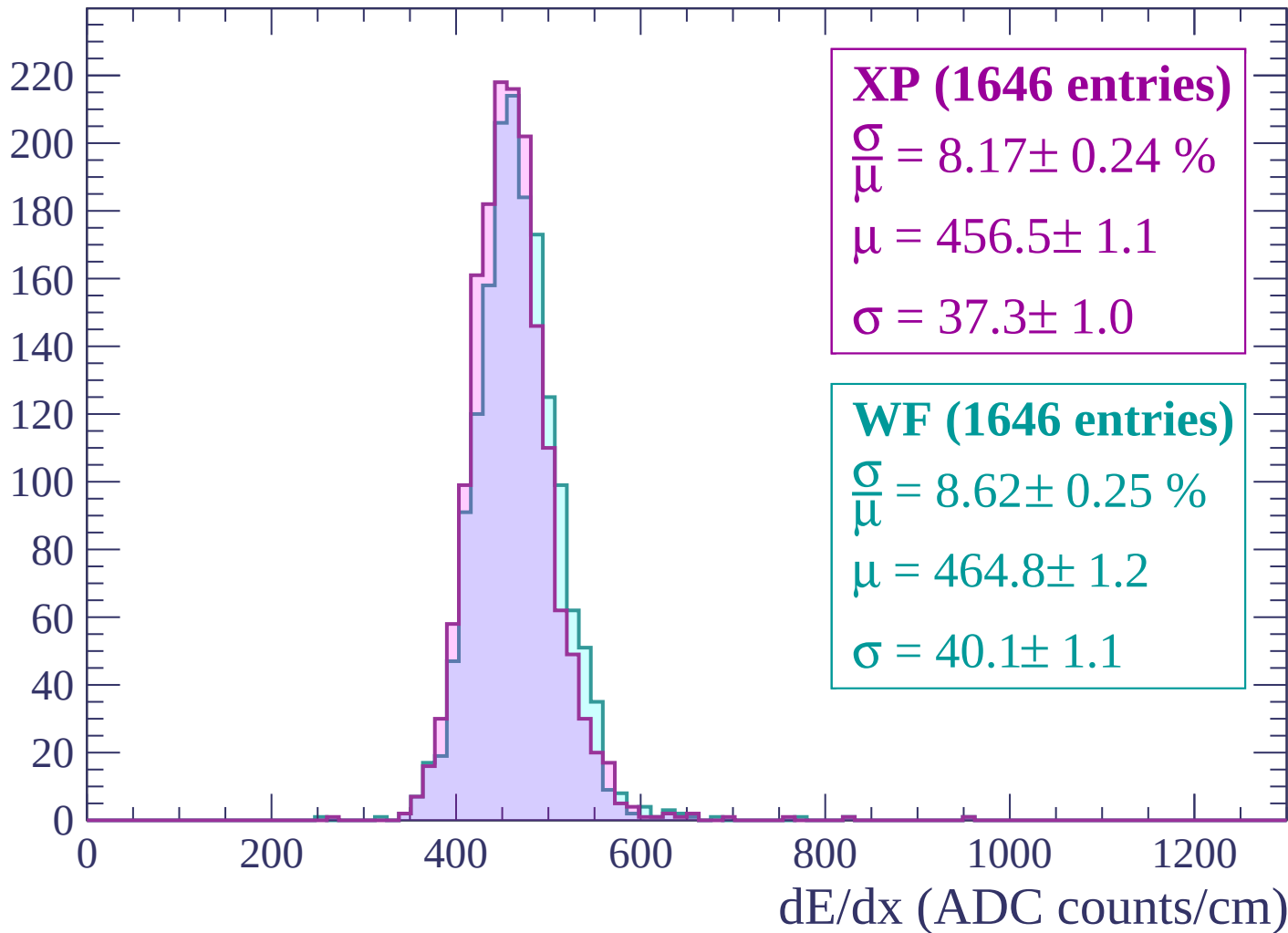
$$\mu = 463.0 \pm 1.2$$

$$\sigma = 41.0 \pm 1.3$$

dE/dx (ADC counts/cm)

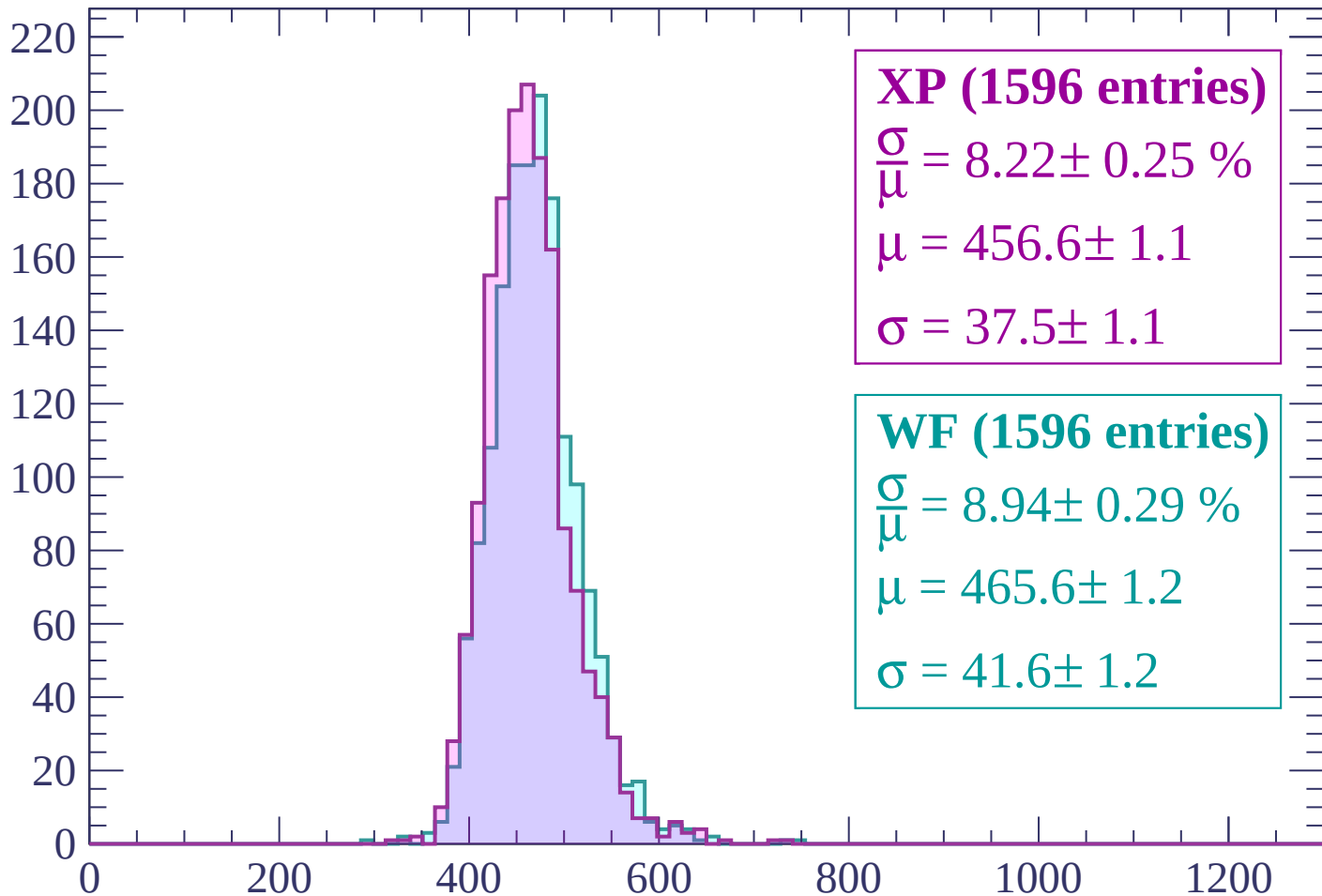
Energy loss | $1740 < p < 1800$

Count



Energy loss | $1800 < p < 1860$

Count



XP (1596 entries)

$$\frac{\sigma}{\mu} = 8.22 \pm 0.25 \%$$

$$\mu = 456.6 \pm 1.1$$

$$\sigma = 37.5 \pm 1.1$$

WF (1596 entries)

$$\frac{\sigma}{\mu} = 8.94 \pm 0.29 \%$$

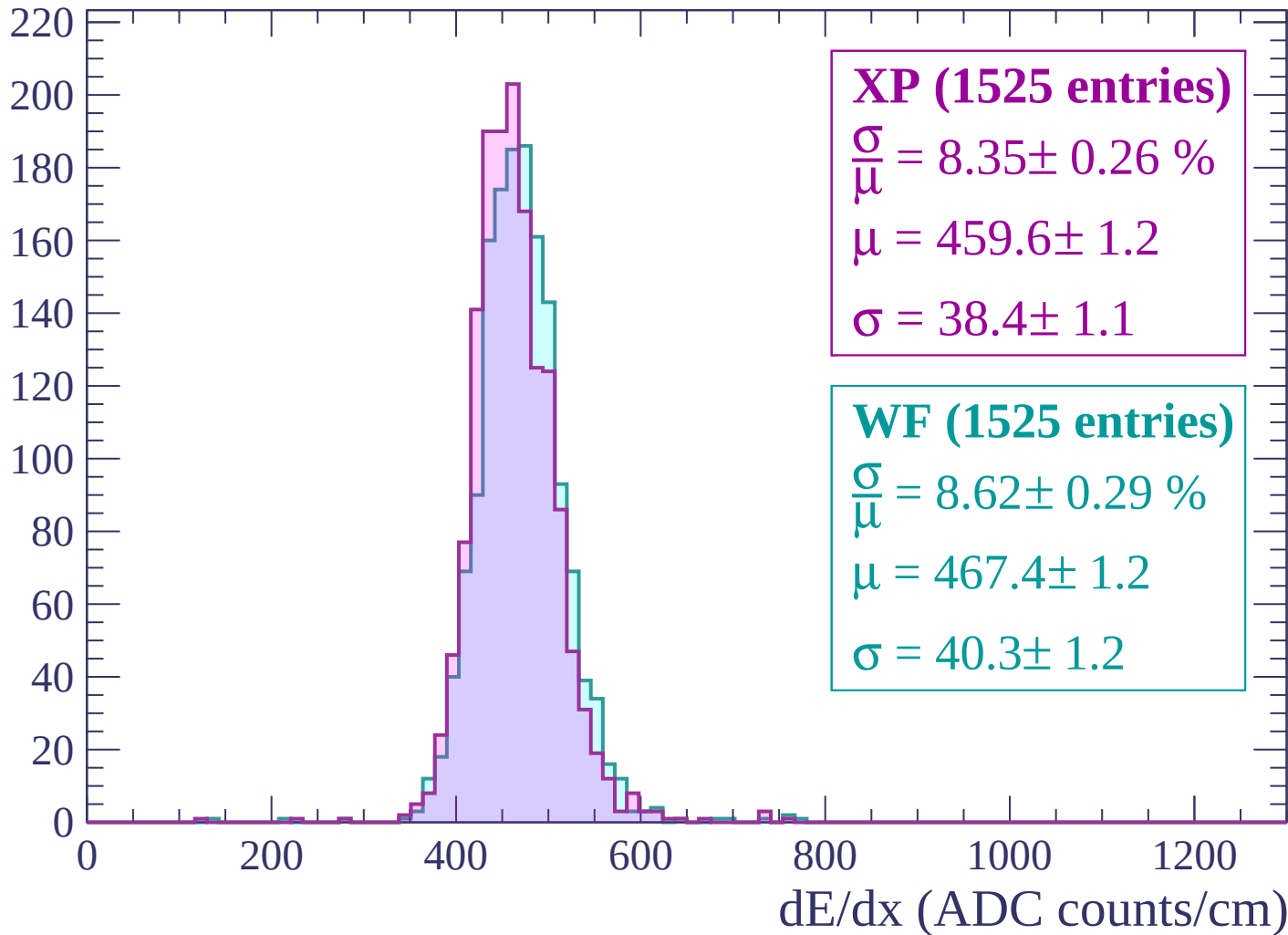
$$\mu = 465.6 \pm 1.2$$

$$\sigma = 41.6 \pm 1.2$$

dE/dx (ADC counts/cm)

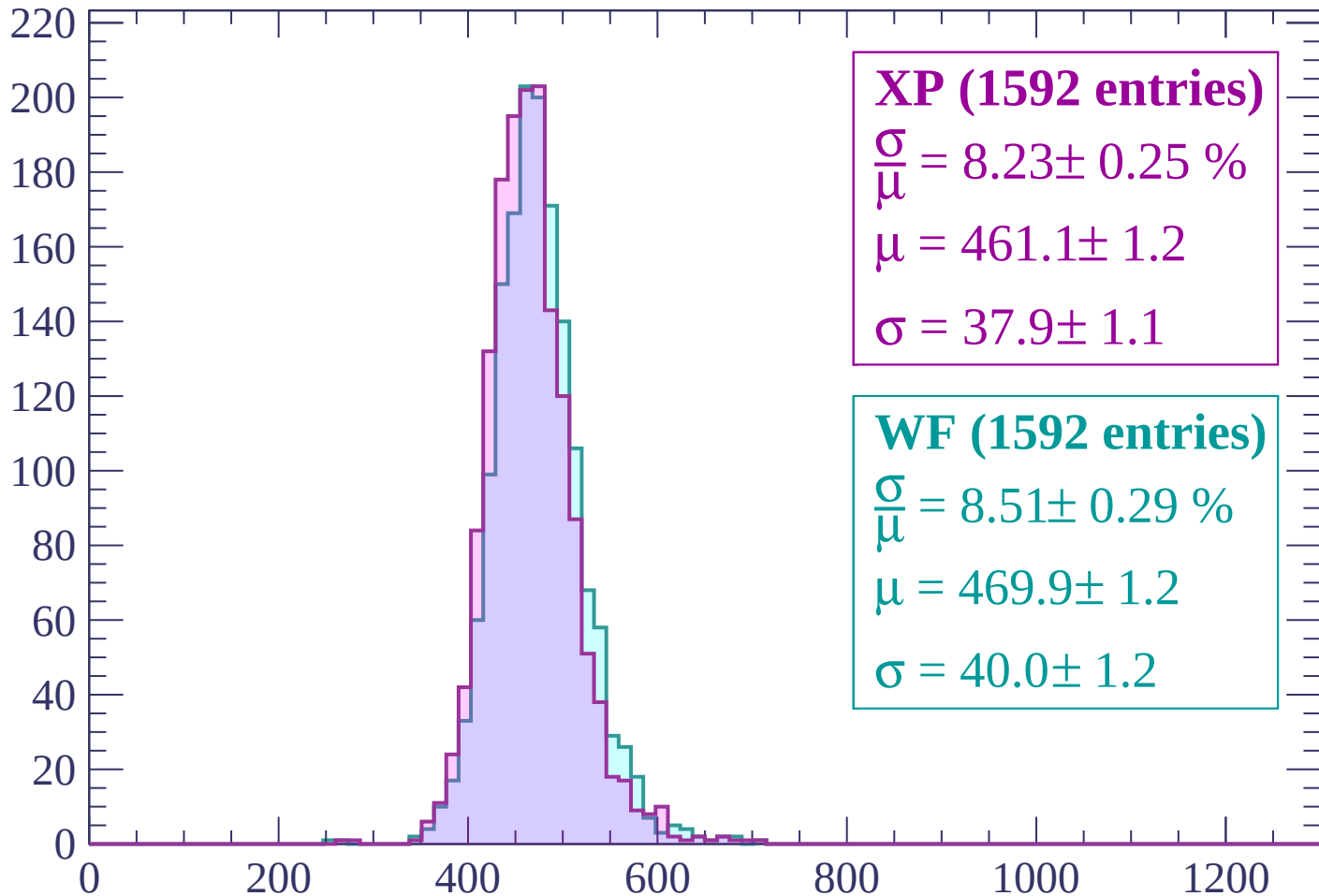
Energy loss | $1860 < p < 1920$

Count



Energy loss | $1920 < p < 1980$

Count



XP (1592 entries)

$$\frac{\sigma}{\mu} = 8.23 \pm 0.25 \%$$

$$\mu = 461.1 \pm 1.2$$

$$\sigma = 37.9 \pm 1.1$$

WF (1592 entries)

$$\frac{\sigma}{\mu} = 8.51 \pm 0.29 \%$$

$$\mu = 469.9 \pm 1.2$$

$$\sigma = 40.0 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1980 < p < 2040$

Count

200
180
160
140
120
100
80
60
40
20
0

XP (1486 entries)

$$\frac{\sigma}{\mu} = 9.01 \pm 0.34 \%$$

$$\mu = 465.5 \pm 1.3$$

$$\sigma = 41.9 \pm 1.5$$

WF (1486 entries)

$$\frac{\sigma}{\mu} = 8.51 \pm 0.23 \%$$

$$\mu = 474.5 \pm 1.2$$

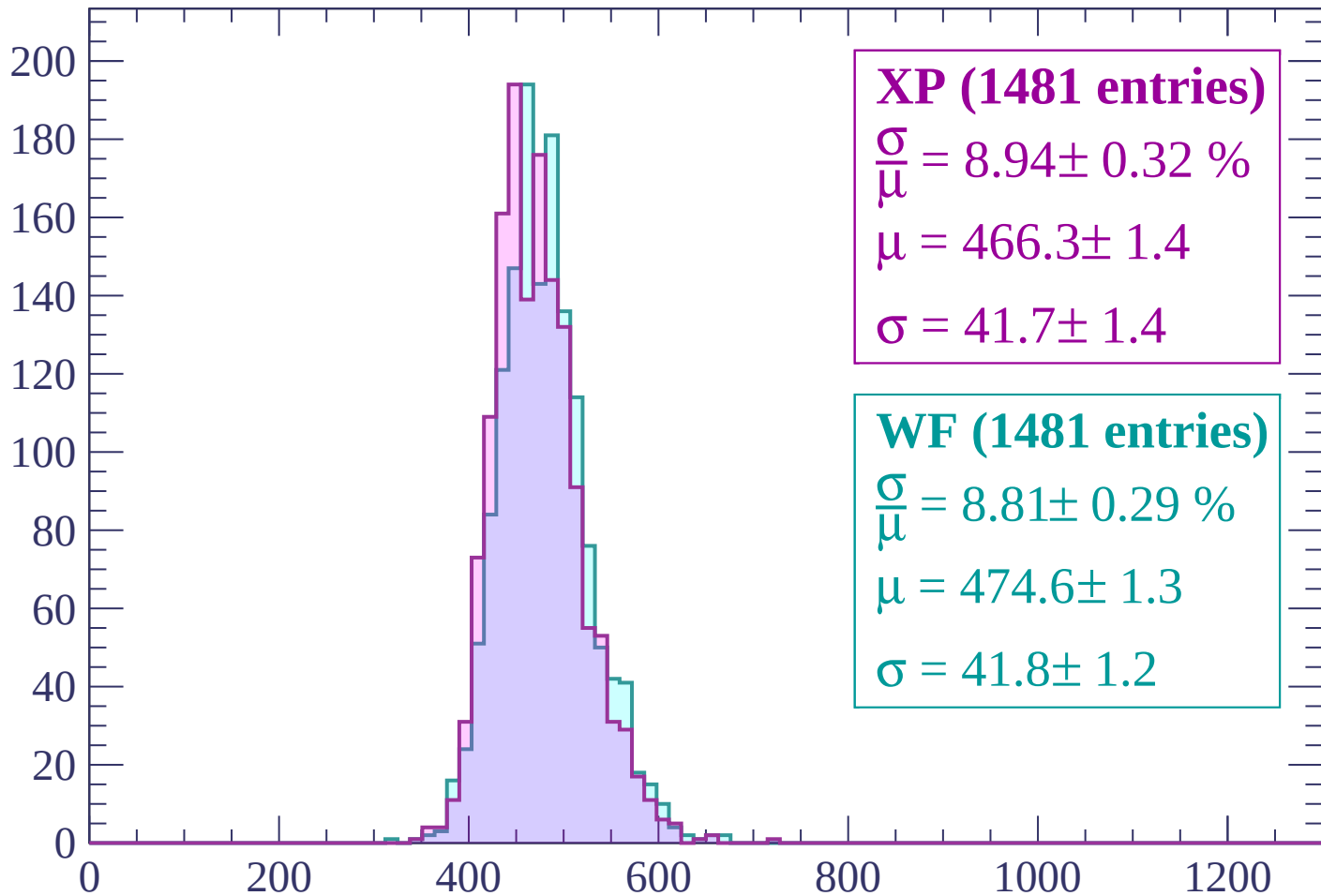
$$\sigma = 40.4 \pm 1.0$$

0 200 400 600 800 1000 1200

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP (1481 entries)

$$\frac{\sigma}{\mu} = 8.94 \pm 0.32 \%$$

$$\mu = 466.3 \pm 1.4$$

$$\sigma = 41.7 \pm 1.4$$

WF (1481 entries)

$$\frac{\sigma}{\mu} = 8.81 \pm 0.29 \%$$

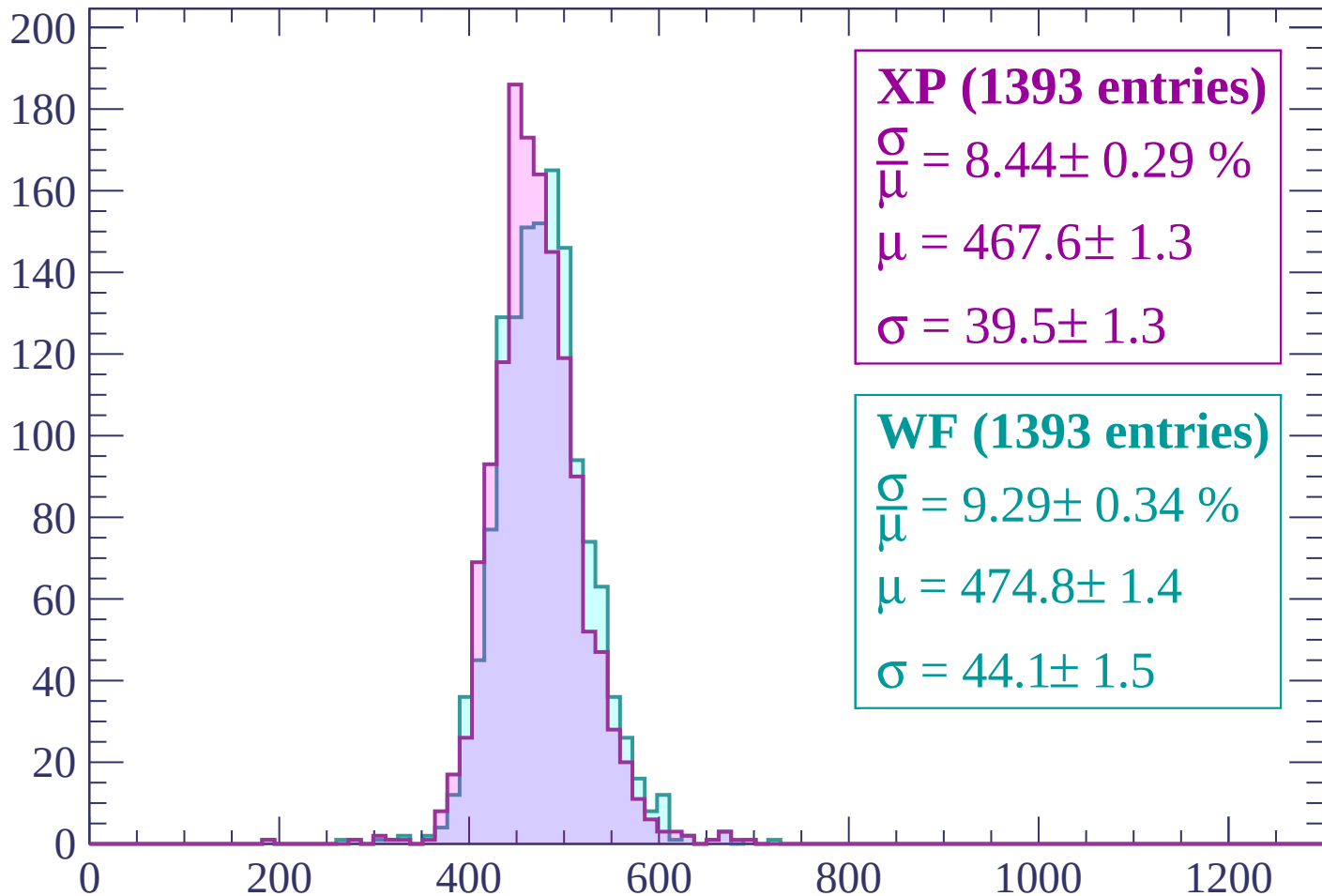
$$\mu = 474.6 \pm 1.3$$

$$\sigma = 41.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (1393 entries)

$\frac{\sigma}{\mu} = 8.44 \pm 0.29 \%$

$\mu = 467.6 \pm 1.3$

$\sigma = 39.5 \pm 1.3$

WF (1393 entries)

$\frac{\sigma}{\mu} = 9.29 \pm 0.34 \%$

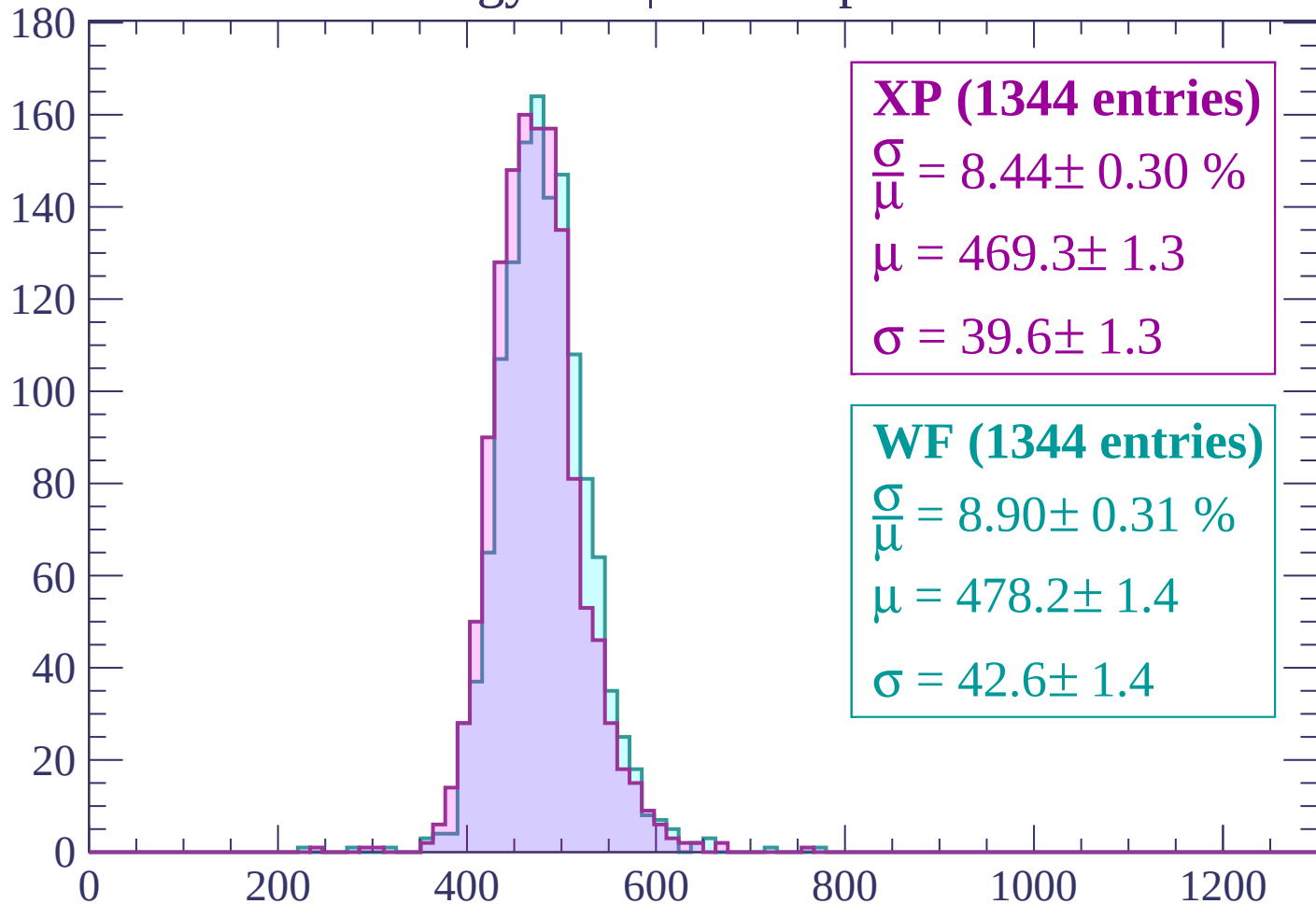
$\mu = 474.8 \pm 1.4$

$\sigma = 44.1 \pm 1.5$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (1344 entries)

$$\frac{\sigma}{\mu} = 8.44 \pm 0.30 \%$$

$$\mu = 469.3 \pm 1.3$$

$$\sigma = 39.6 \pm 1.3$$

WF (1344 entries)

$$\frac{\sigma}{\mu} = 8.90 \pm 0.31 \%$$

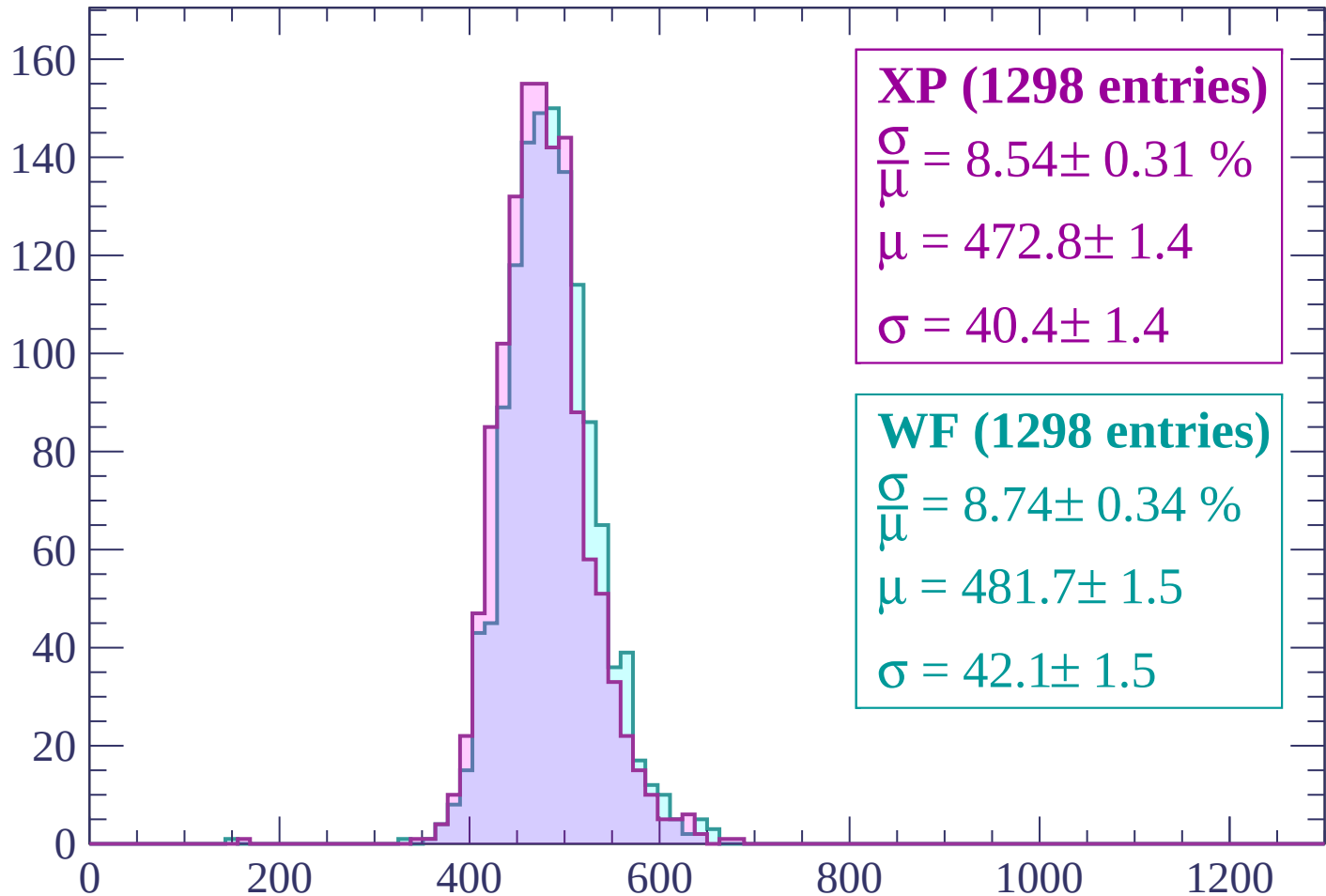
$$\mu = 478.2 \pm 1.4$$

$$\sigma = 42.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (1298 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.31 \%$$

$$\mu = 472.8 \pm 1.4$$

$$\sigma = 40.4 \pm 1.4$$

WF (1298 entries)

$$\frac{\sigma}{\mu} = 8.74 \pm 0.34 \%$$

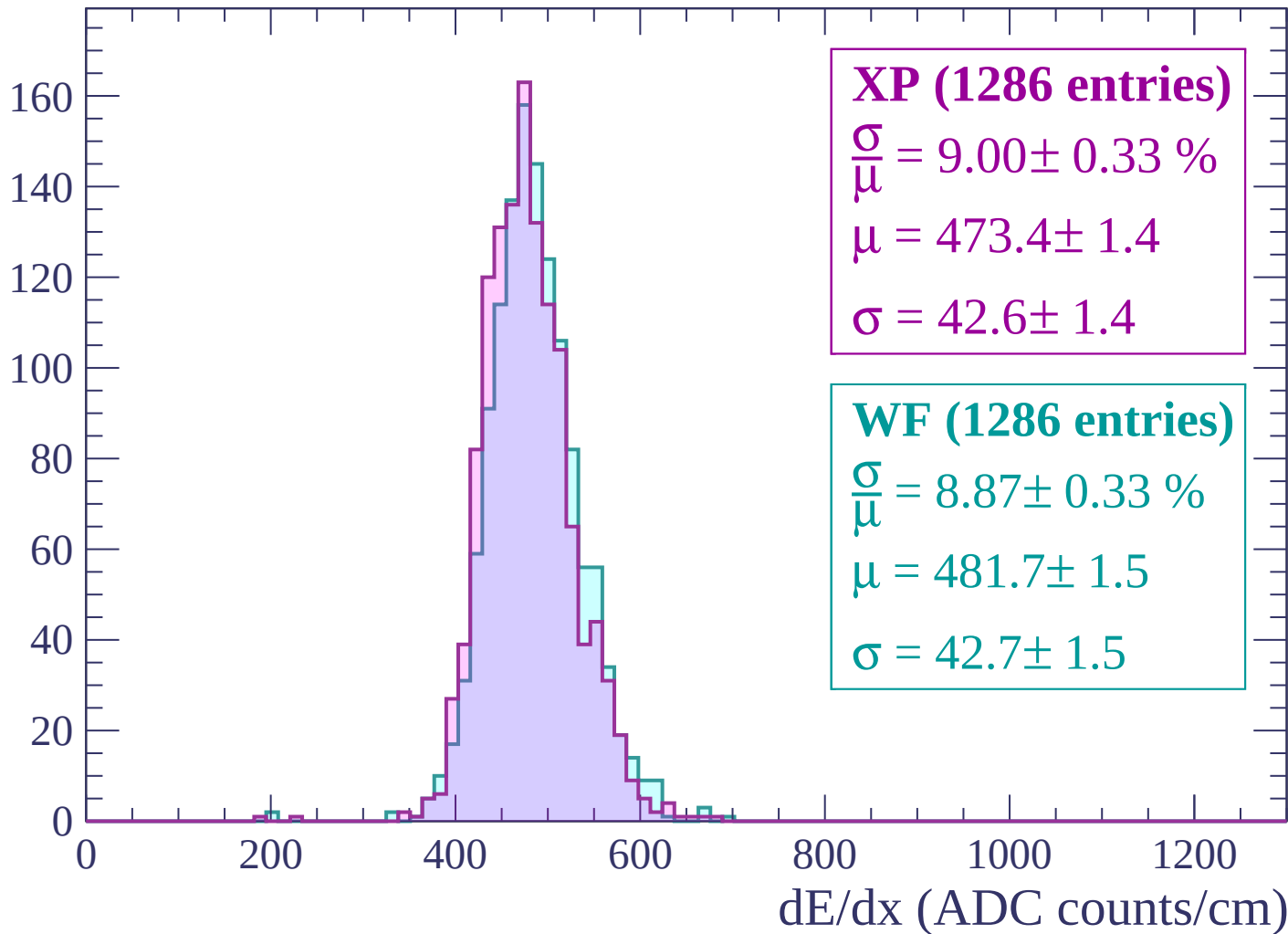
$$\mu = 481.7 \pm 1.5$$

$$\sigma = 42.1 \pm 1.5$$

dE/dx (ADC counts/cm)

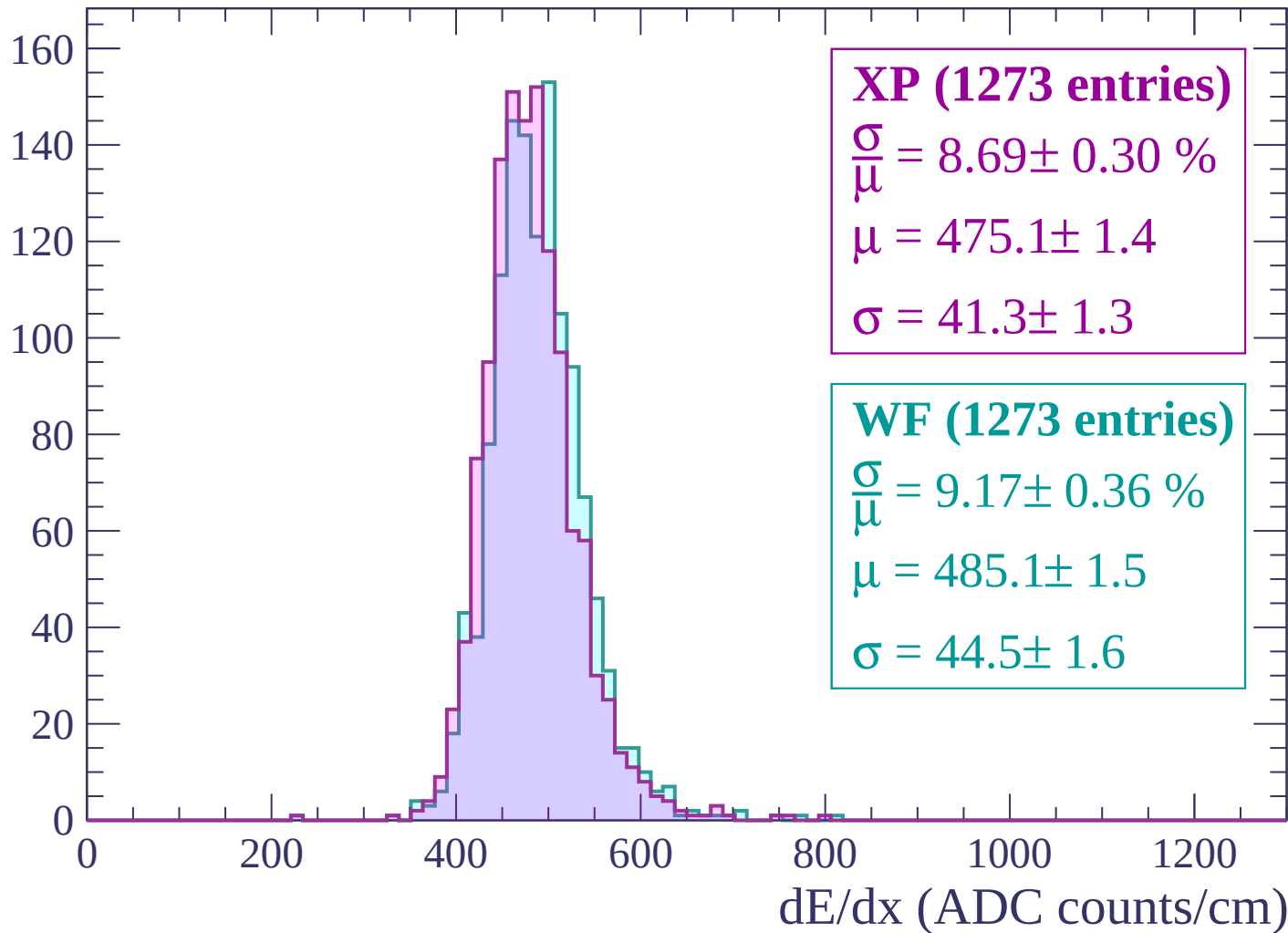
Energy loss | $2280 < p < 2340$

Count



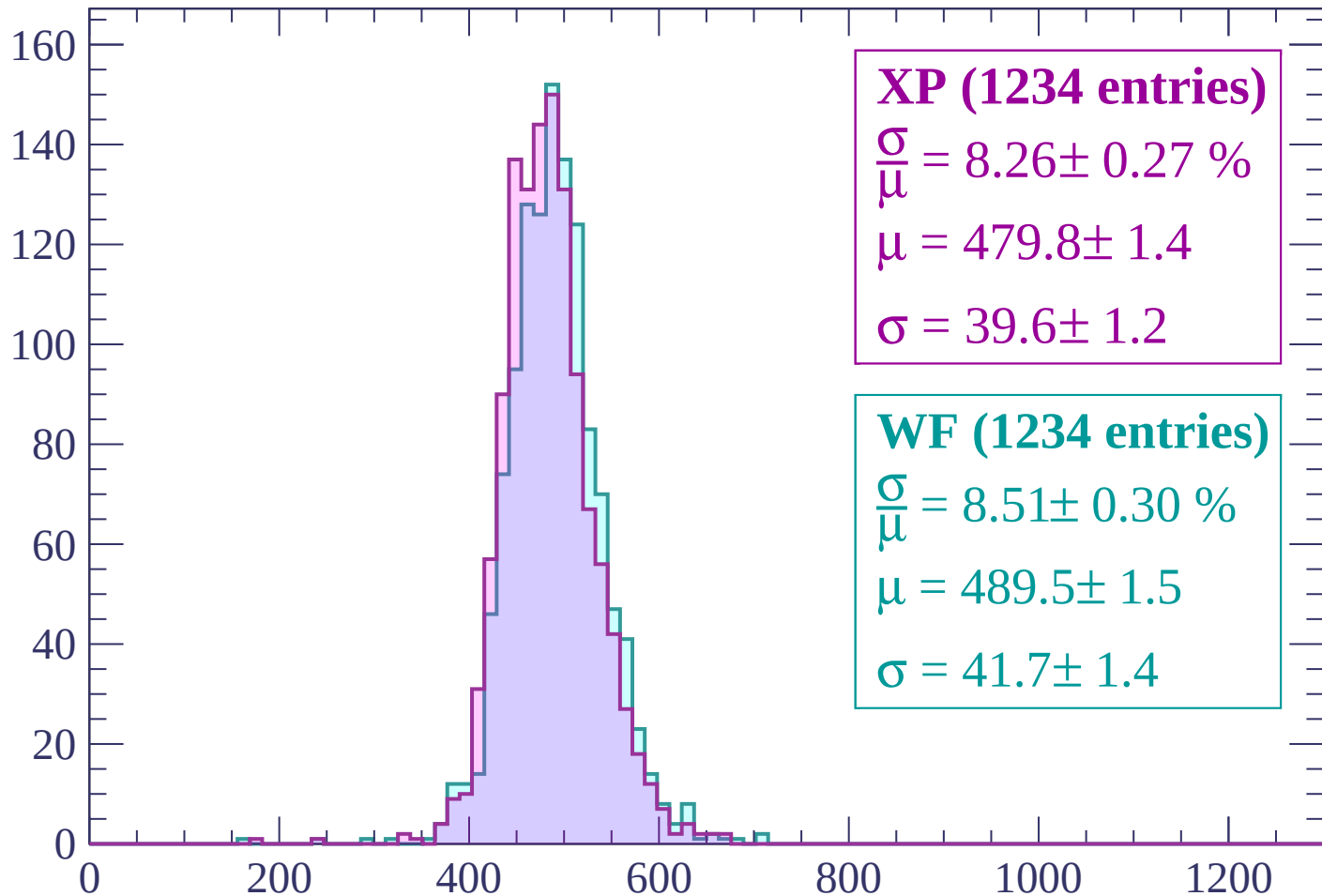
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP (1234 entries)

$$\frac{\sigma}{\mu} = 8.26 \pm 0.27 \%$$

$$\mu = 479.8 \pm 1.4$$

$$\sigma = 39.6 \pm 1.2$$

WF (1234 entries)

$$\frac{\sigma}{\mu} = 8.51 \pm 0.30 \%$$

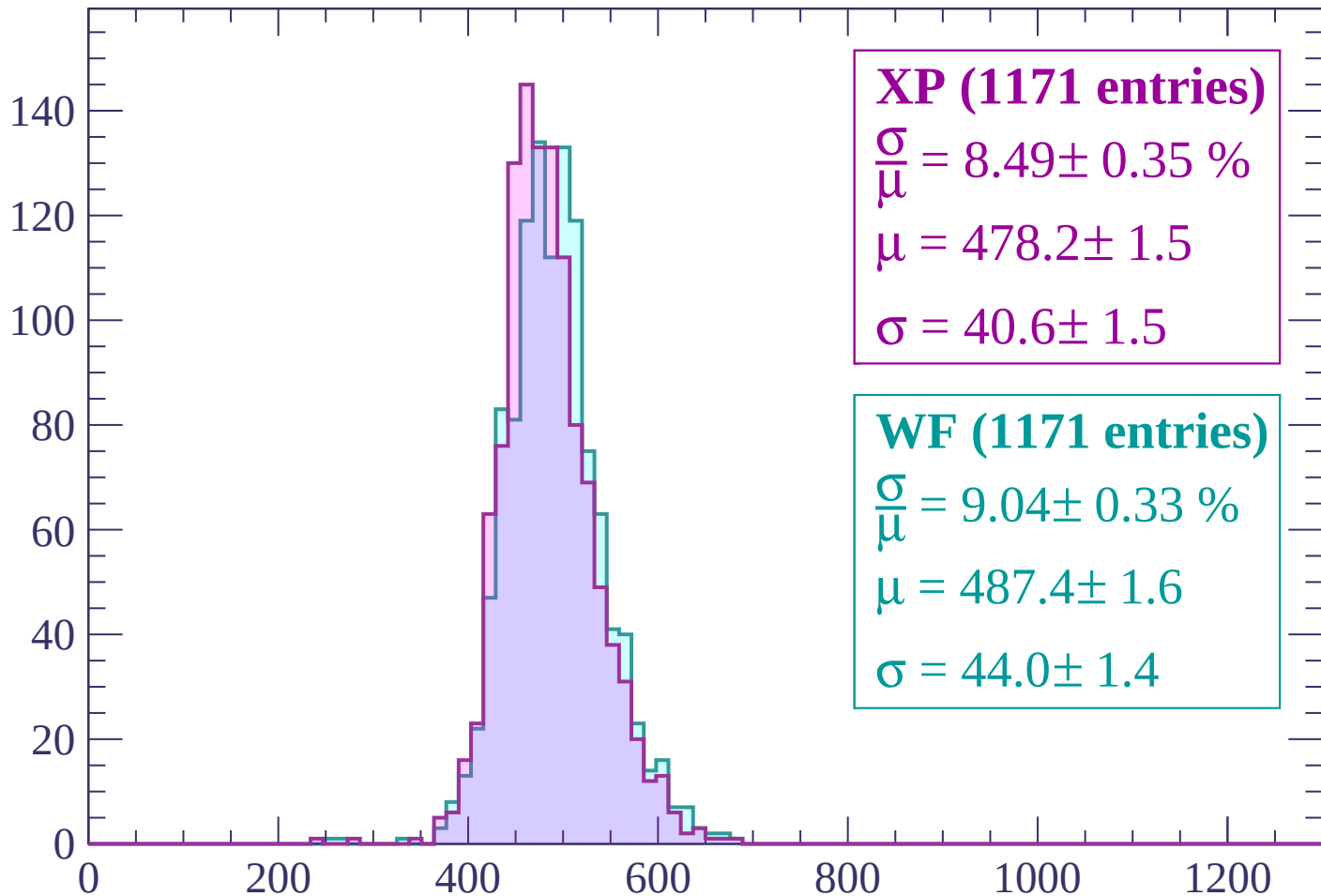
$$\mu = 489.5 \pm 1.5$$

$$\sigma = 41.7 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2460 < p < 2520$

Count



XP (1171 entries)

$$\frac{\sigma}{\mu} = 8.49 \pm 0.35 \%$$

$$\mu = 478.2 \pm 1.5$$

$$\sigma = 40.6 \pm 1.5$$

WF (1171 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.33 \%$$

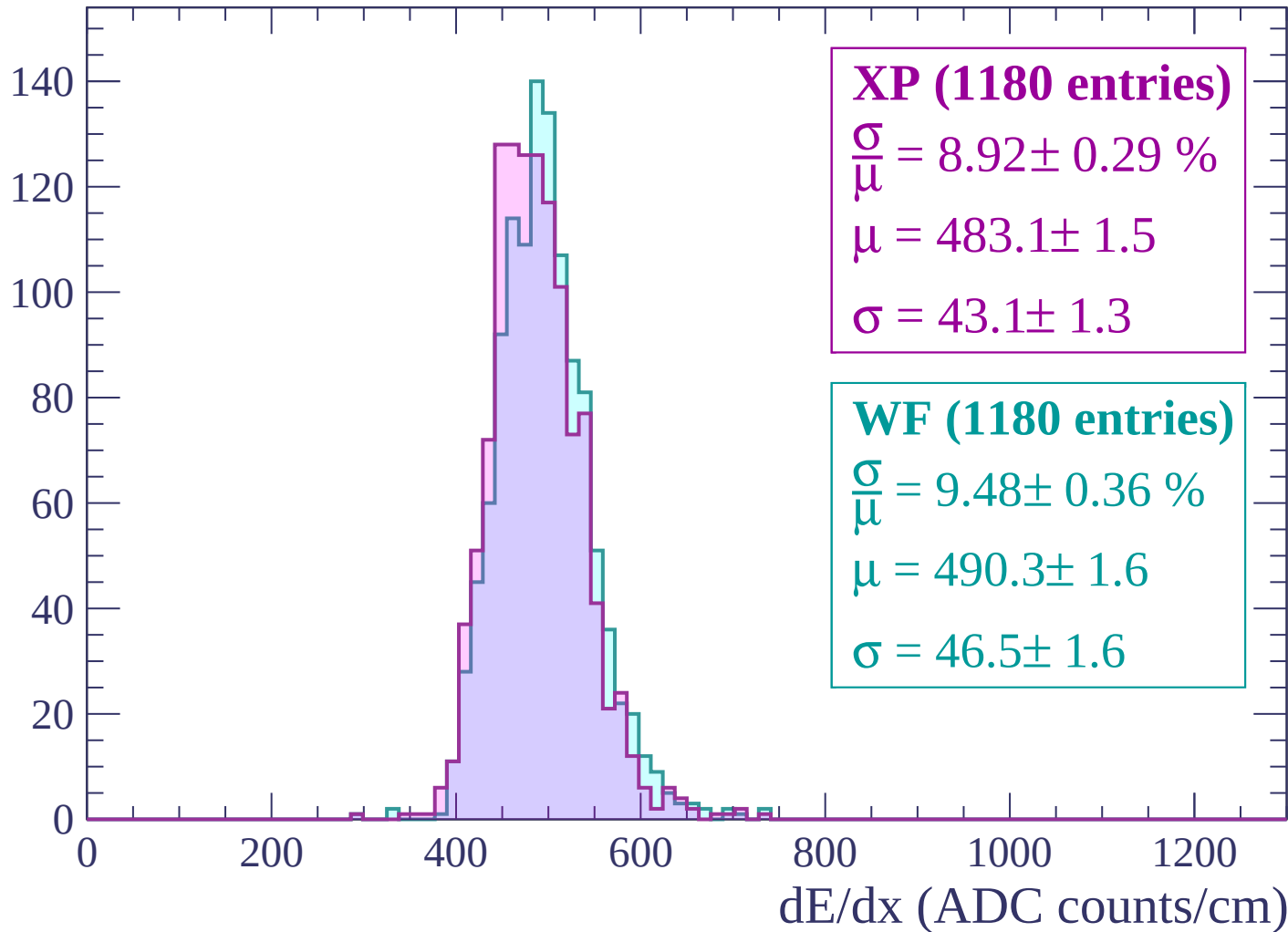
$$\mu = 487.4 \pm 1.6$$

$$\sigma = 44.0 \pm 1.4$$

dE/dx (ADC counts/cm)

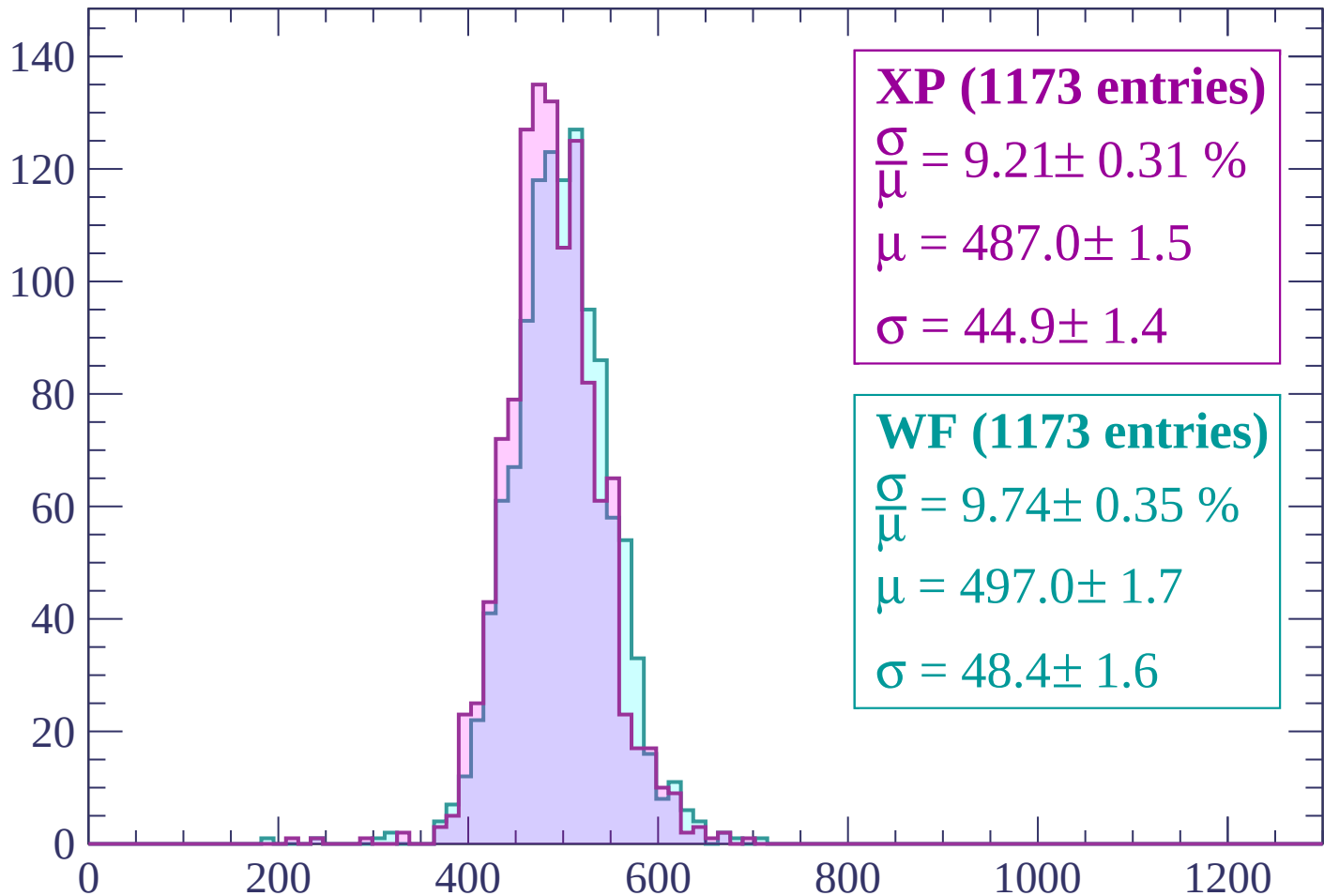
Energy loss | $2520 < p < 2580$

Count



Energy loss | $2580 < p < 2640$

Count



XP (1173 entries)

$$\frac{\sigma}{\mu} = 9.21 \pm 0.31 \%$$

$$\mu = 487.0 \pm 1.5$$

$$\sigma = 44.9 \pm 1.4$$

WF (1173 entries)

$$\frac{\sigma}{\mu} = 9.74 \pm 0.35 \%$$

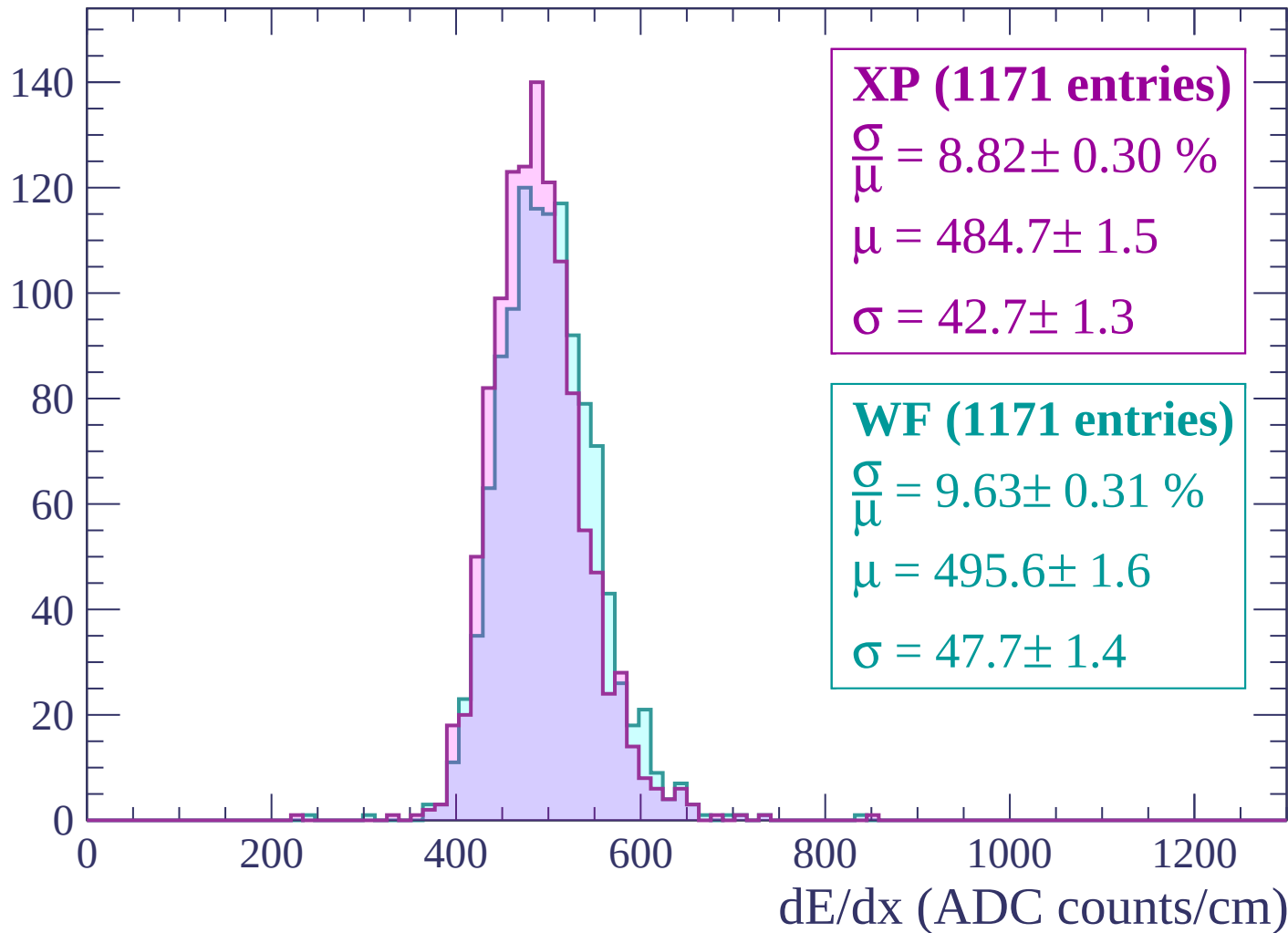
$$\mu = 497.0 \pm 1.7$$

$$\sigma = 48.4 \pm 1.6$$

dE/dx (ADC counts/cm)

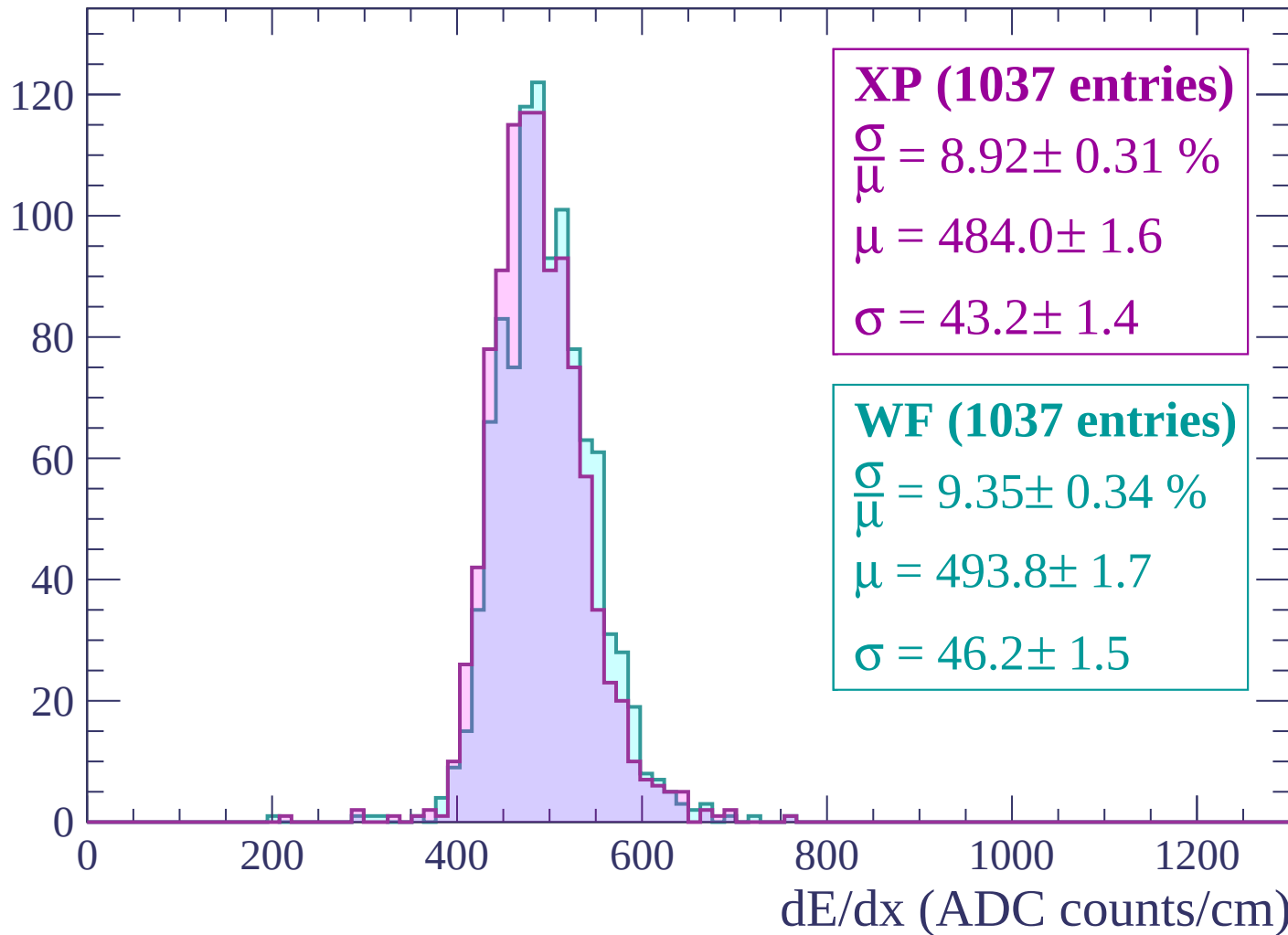
Energy loss | $2640 < p < 2700$

Count



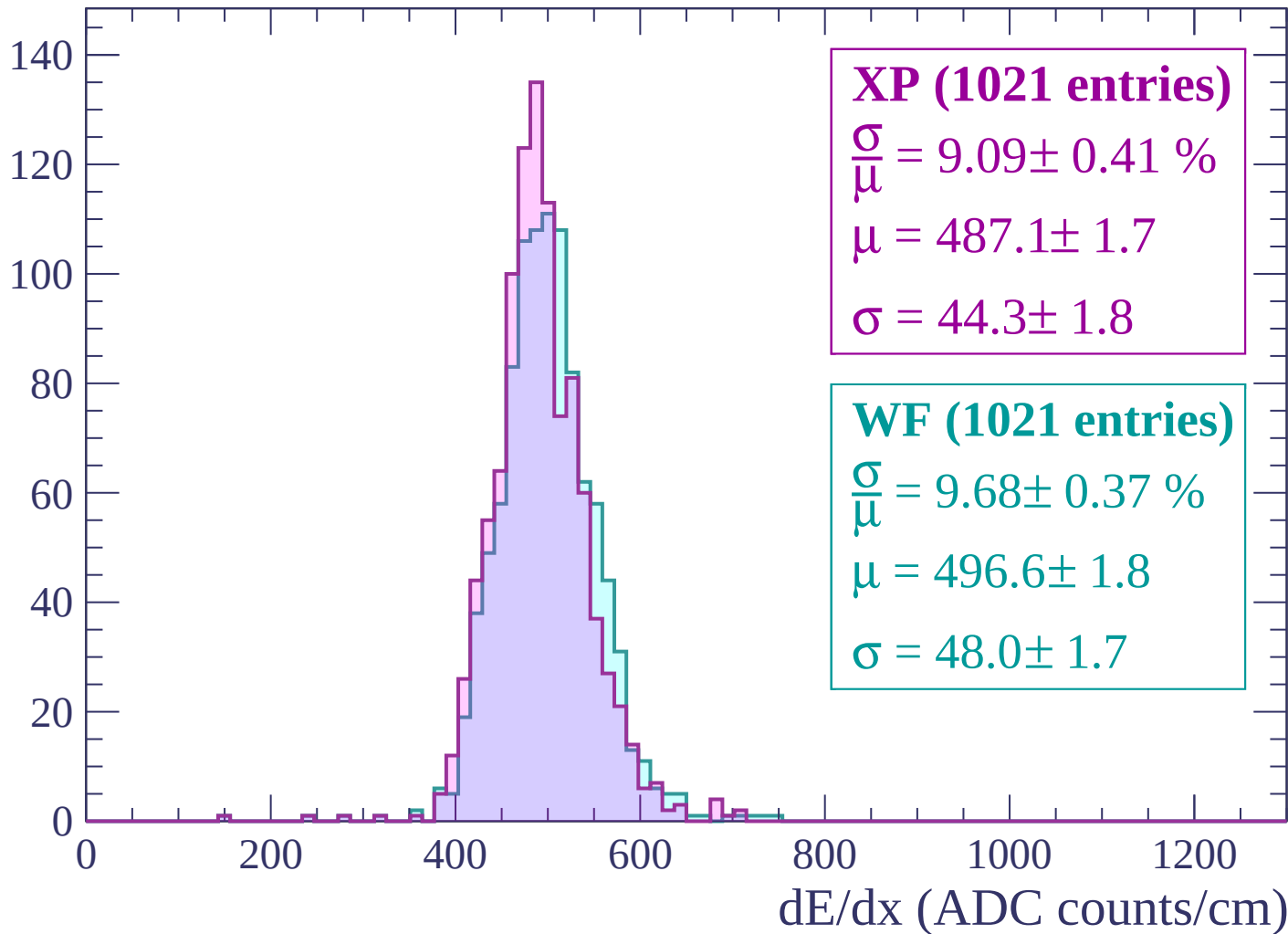
Energy loss | $2700 < p < 2760$

Count



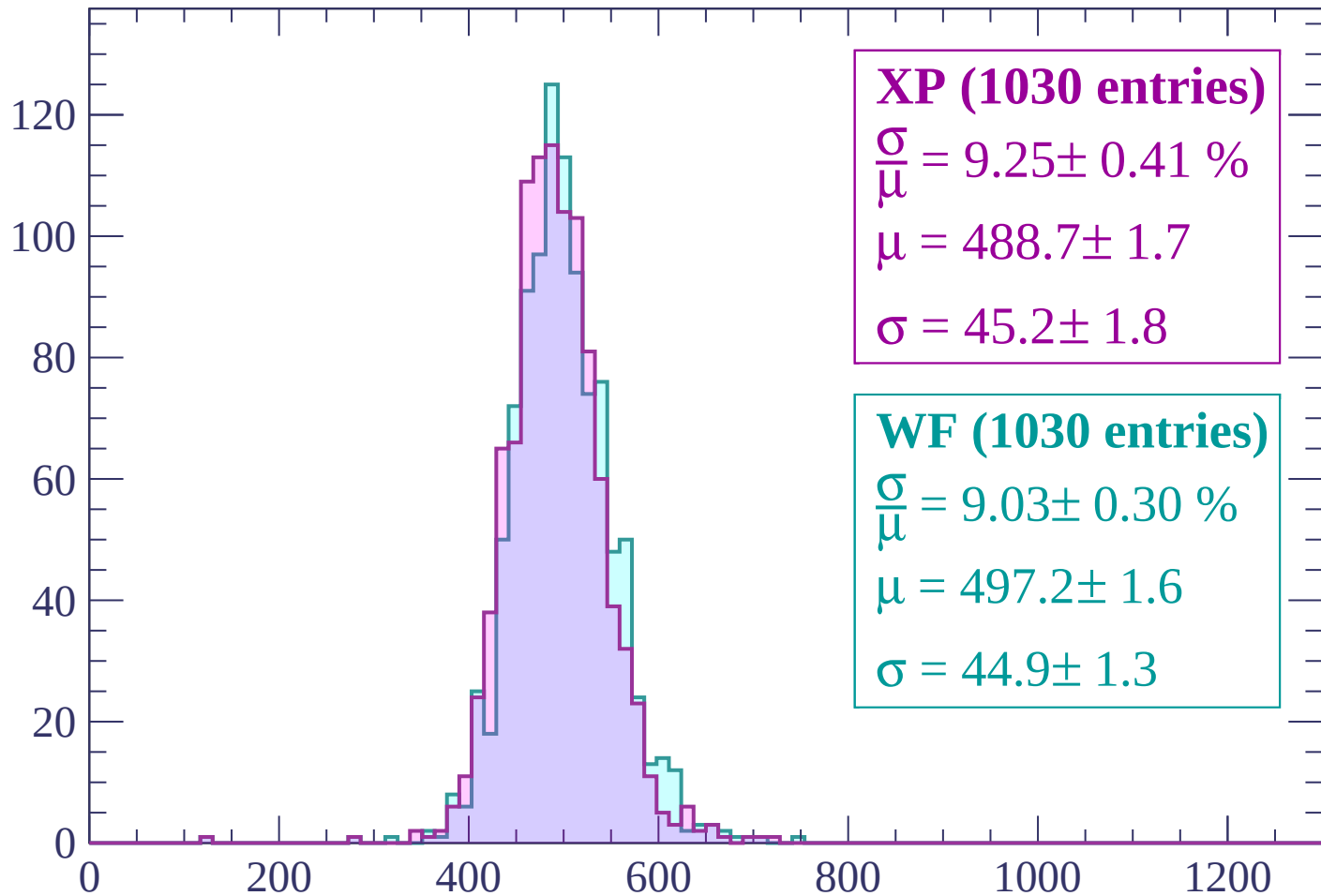
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (1030 entries)

$$\frac{\sigma}{\mu} = 9.25 \pm 0.41 \%$$

$$\mu = 488.7 \pm 1.7$$

$$\sigma = 45.2 \pm 1.8$$

WF (1030 entries)

$$\frac{\sigma}{\mu} = 9.03 \pm 0.30 \%$$

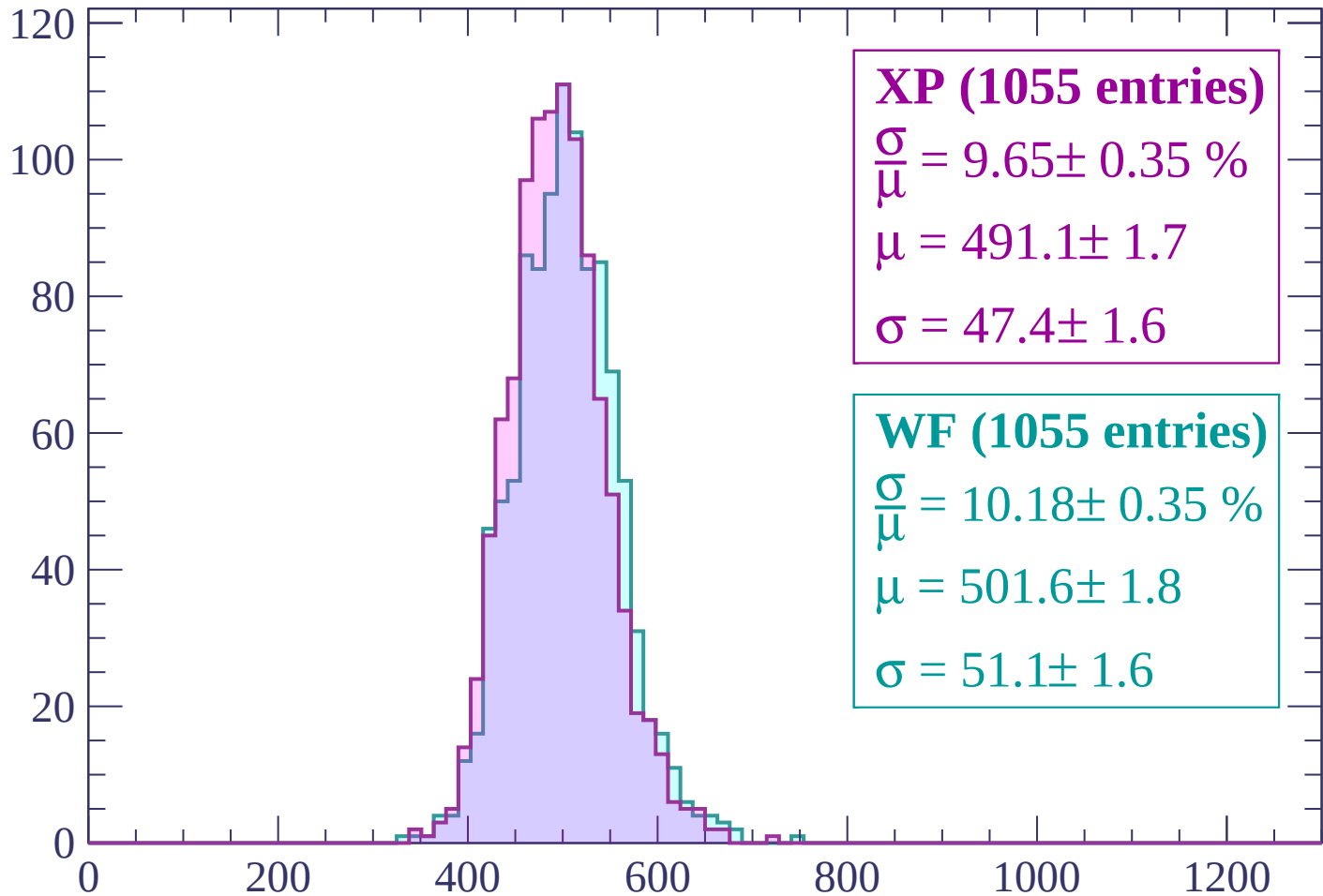
$$\mu = 497.2 \pm 1.6$$

$$\sigma = 44.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2880 < p < 2940$

Count



XP (1055 entries)

$$\frac{\sigma}{\mu} = 9.65 \pm 0.35 \%$$

$$\mu = 491.1 \pm 1.7$$

$$\sigma = 47.4 \pm 1.6$$

WF (1055 entries)

$$\frac{\sigma}{\mu} = 10.18 \pm 0.35 \%$$

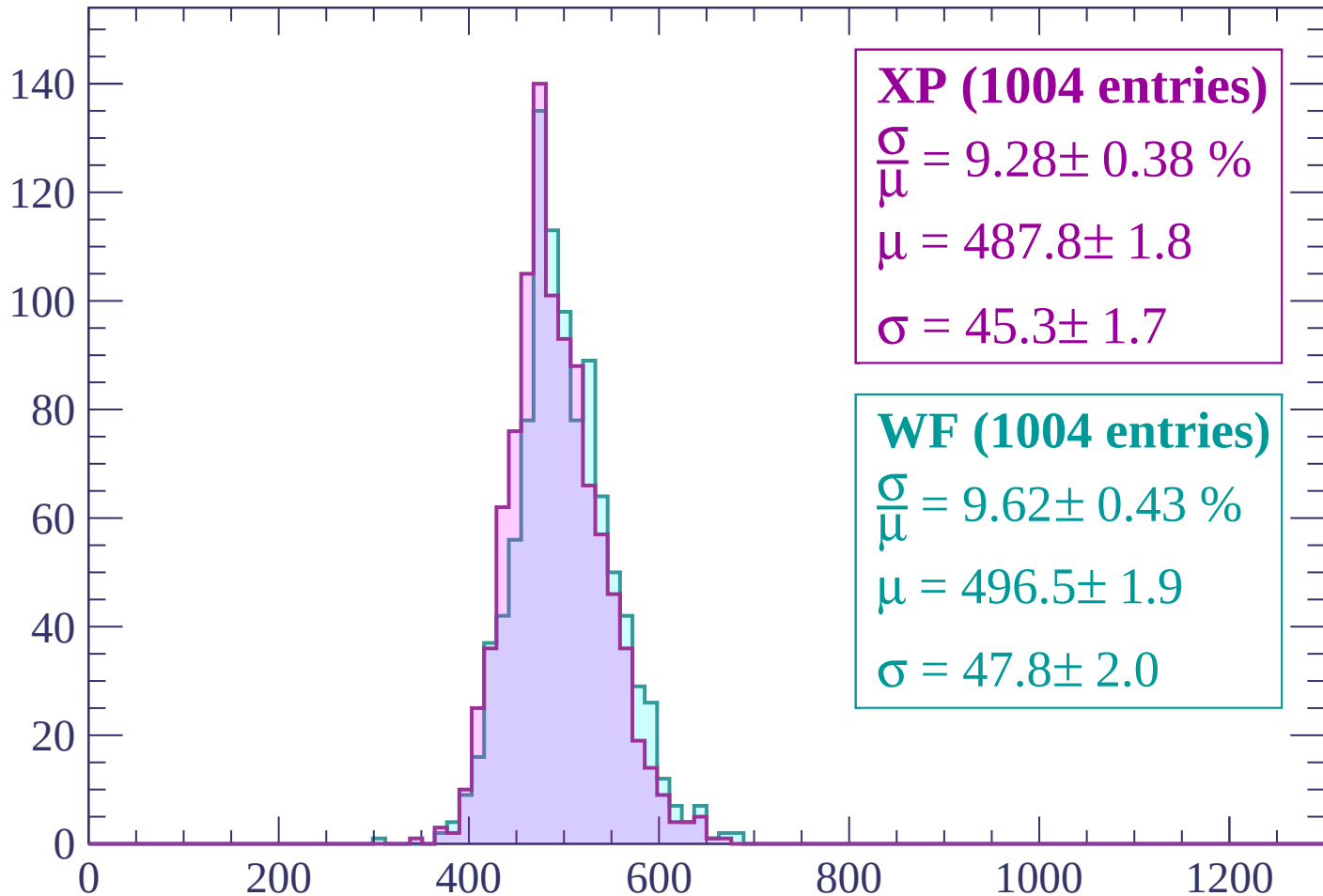
$$\mu = 501.6 \pm 1.8$$

$$\sigma = 51.1 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$

Count



XP (1004 entries)

$$\frac{\sigma}{\mu} = 9.28 \pm 0.38 \%$$

$$\mu = 487.8 \pm 1.8$$

$$\sigma = 45.3 \pm 1.7$$

WF (1004 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.43 \%$$

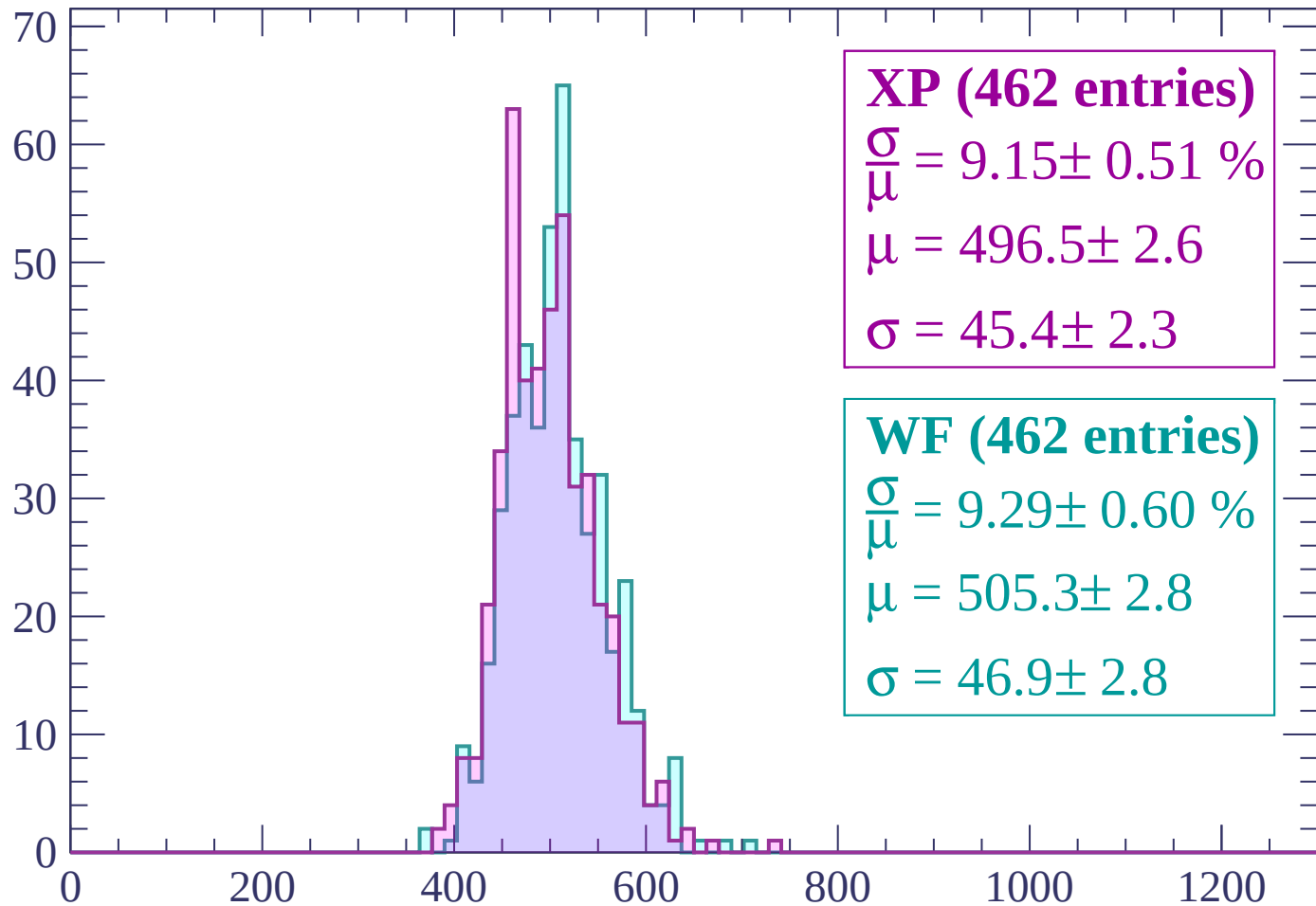
$$\mu = 496.5 \pm 1.9$$

$$\sigma = 47.8 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count



XP (462 entries)

$$\frac{\sigma}{\mu} = 9.15 \pm 0.51 \%$$

$$\mu = 496.5 \pm 2.6$$

$$\sigma = 45.4 \pm 2.3$$

WF (462 entries)

$$\frac{\sigma}{\mu} = 9.29 \pm 0.60 \%$$

$$\mu = 505.3 \pm 2.8$$

$$\sigma = 46.9 \pm 2.8$$

dE/dx (ADC counts/cm)

